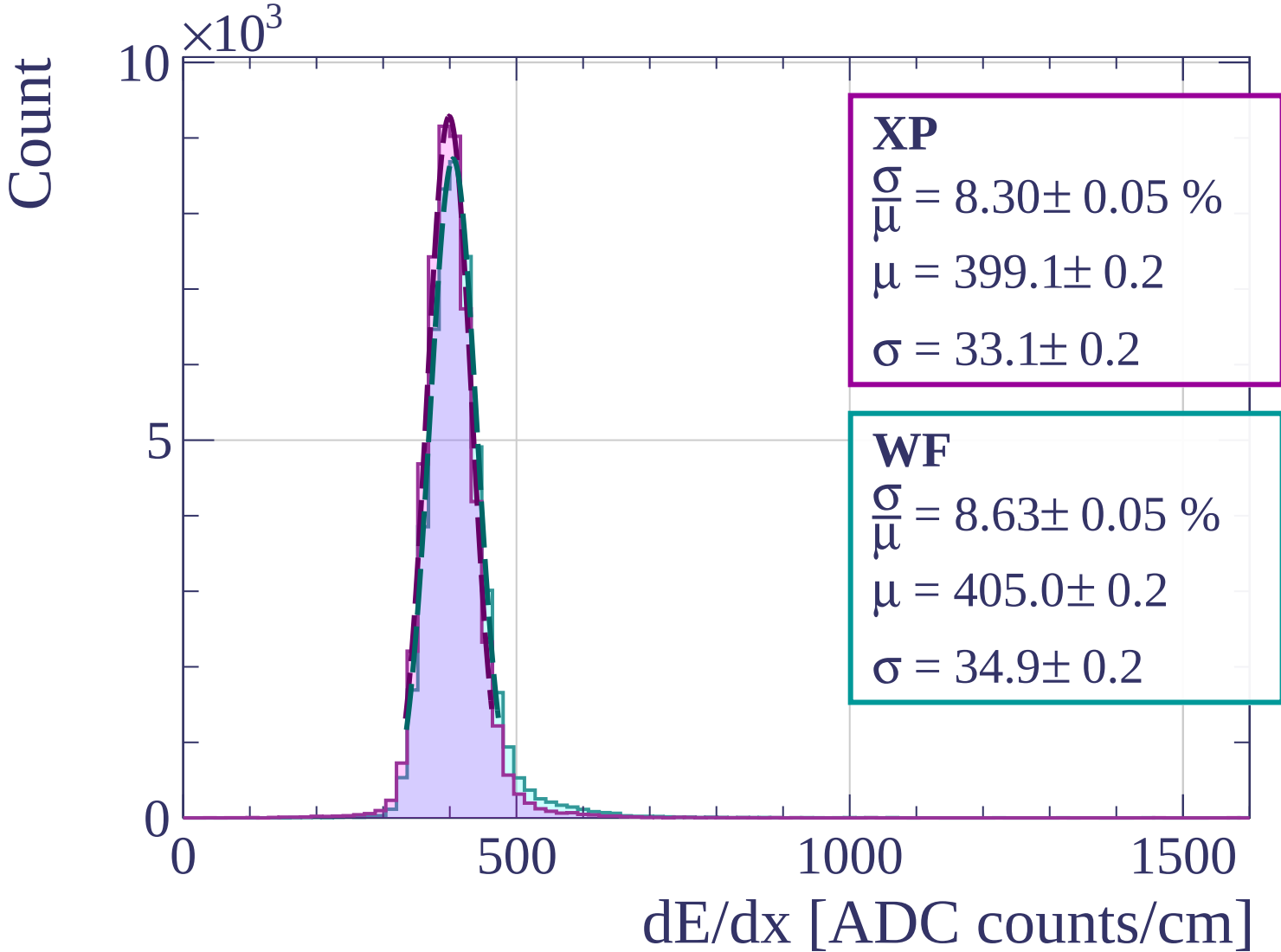
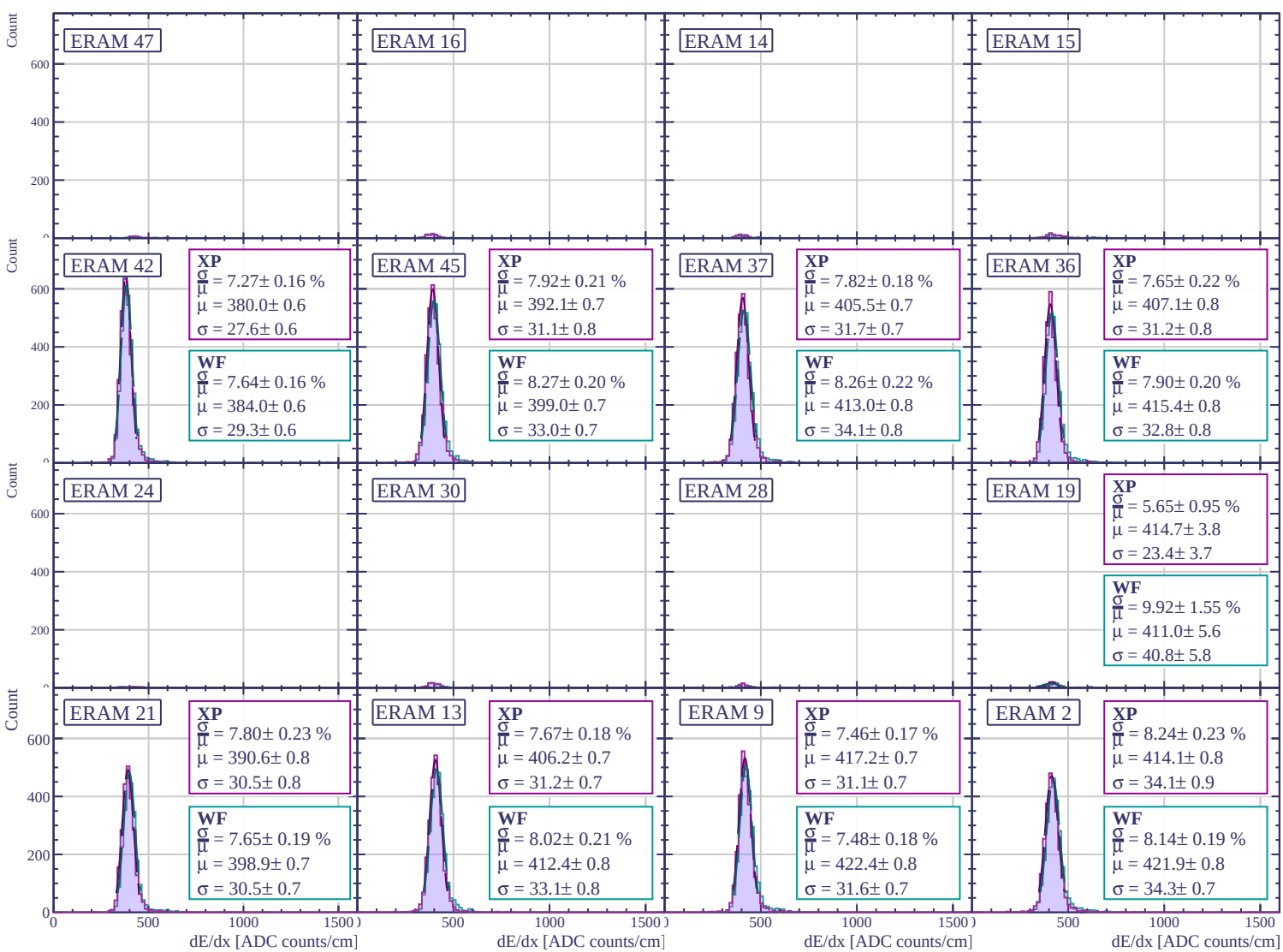
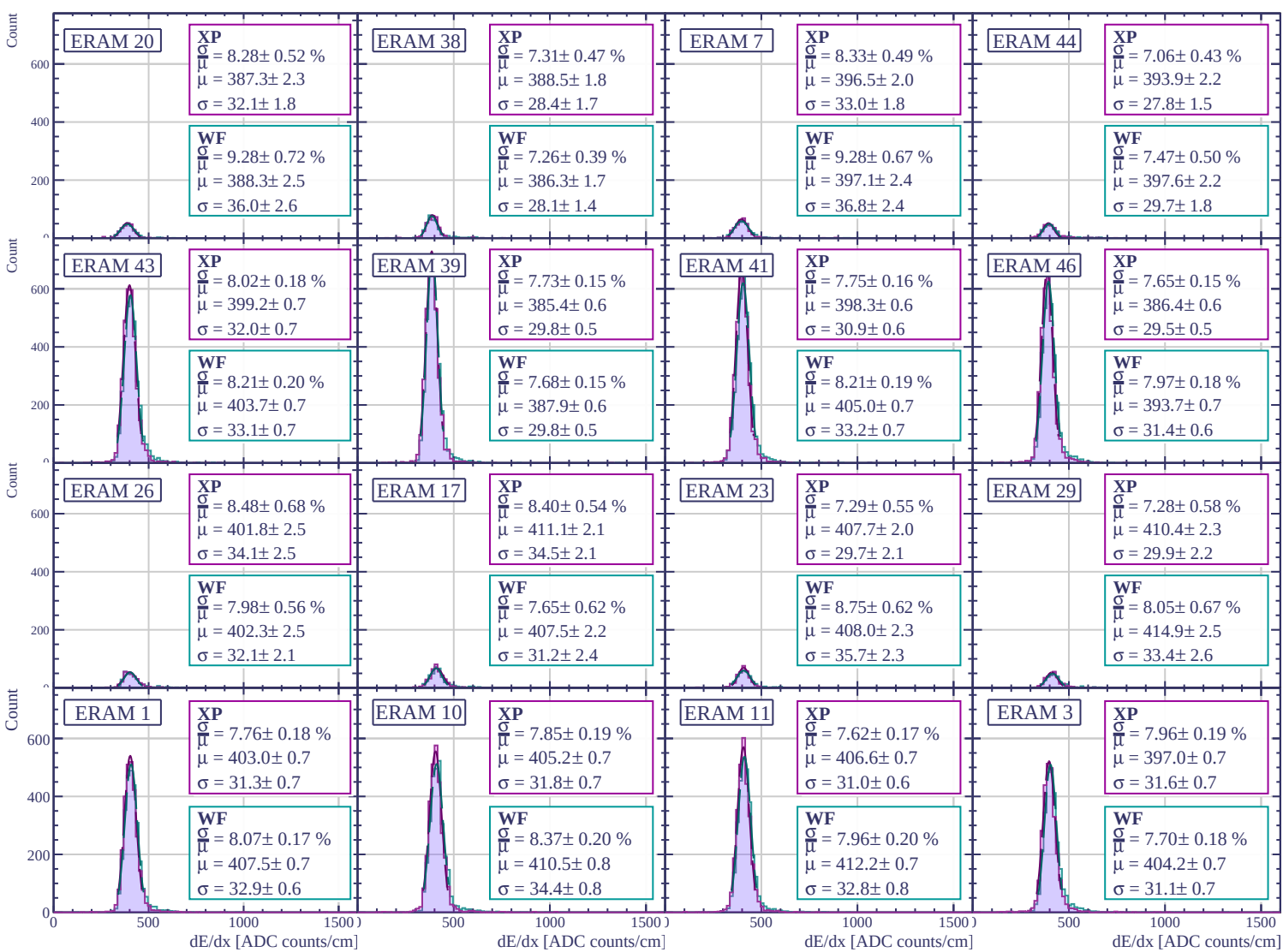
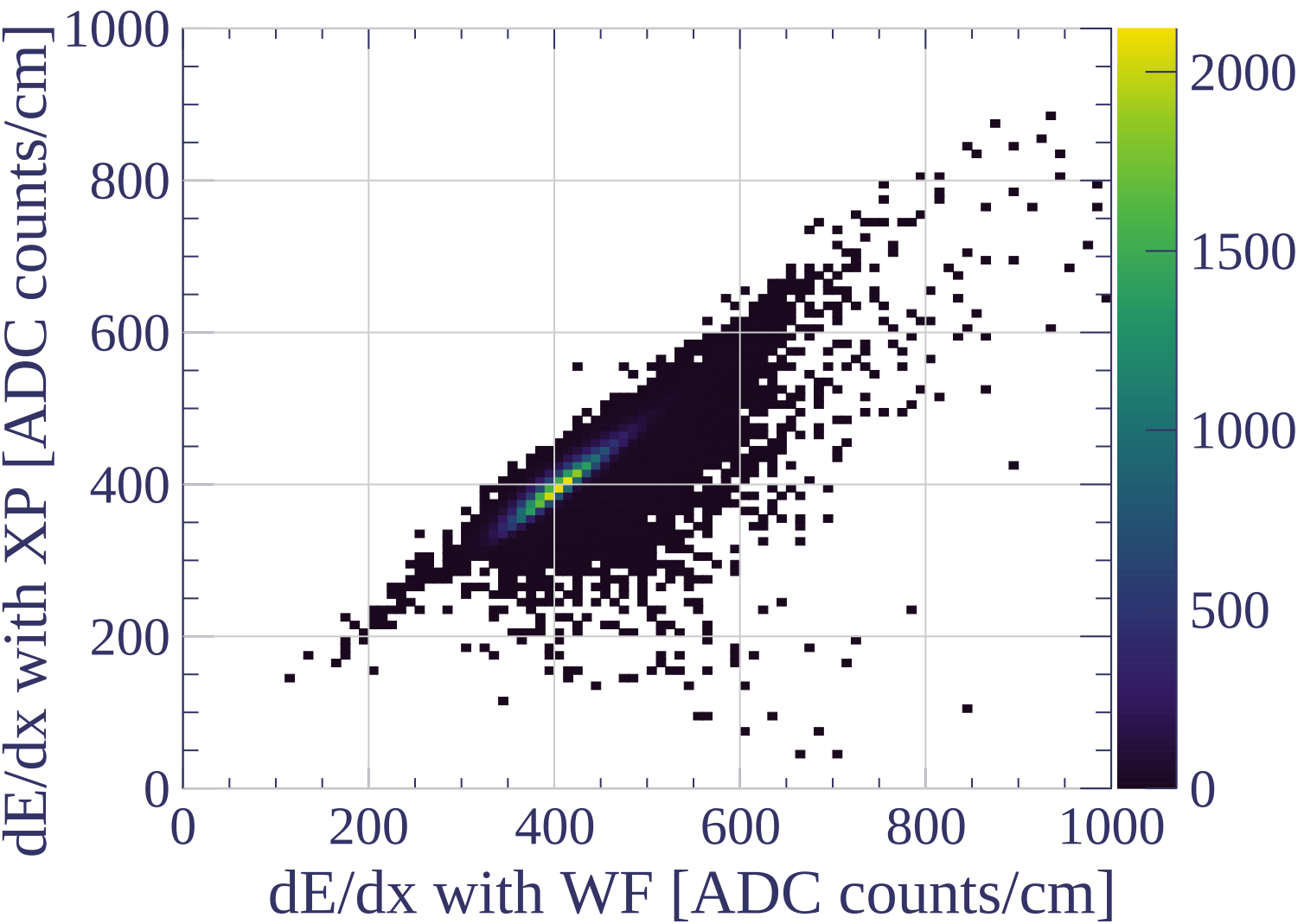
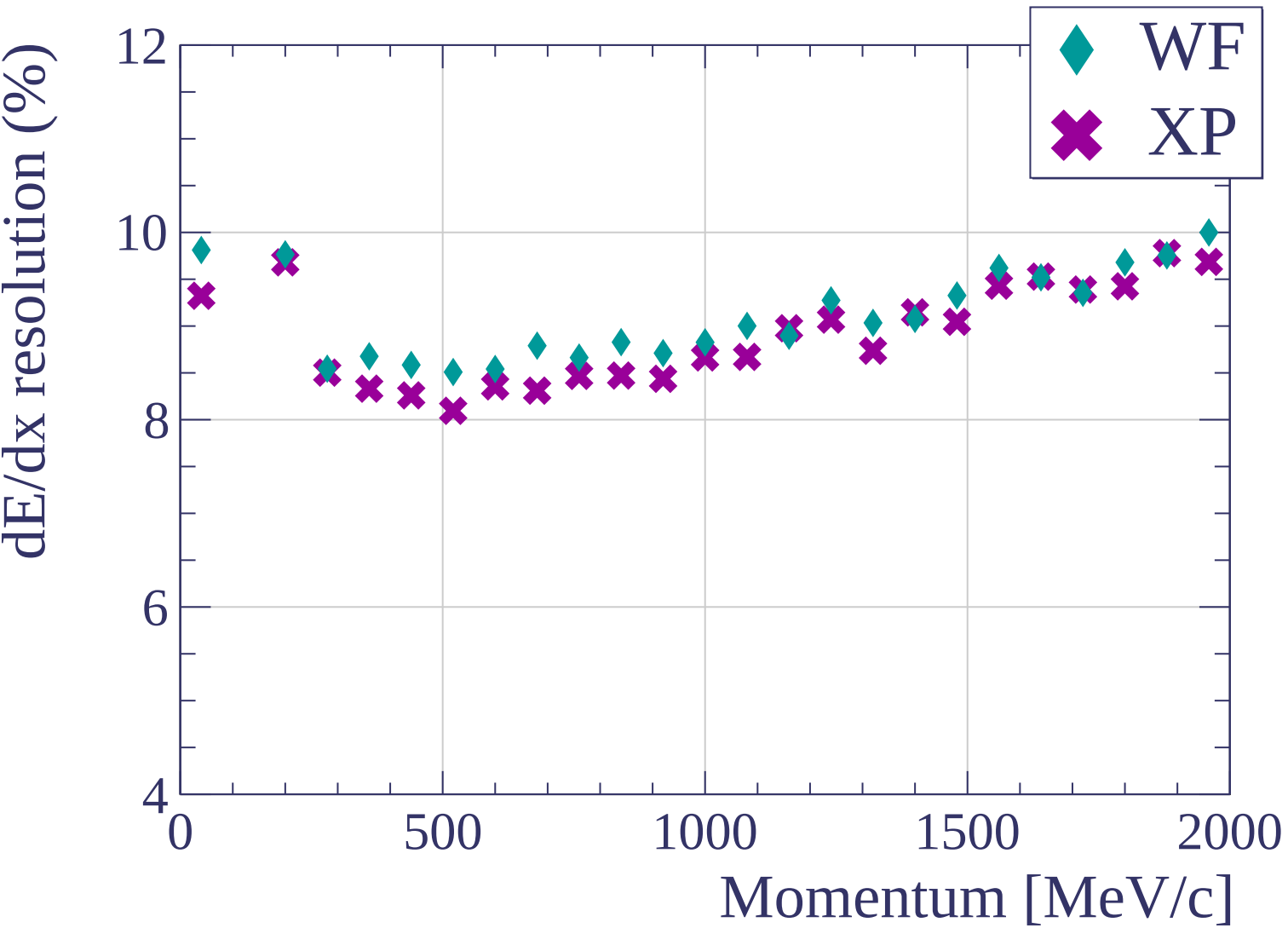


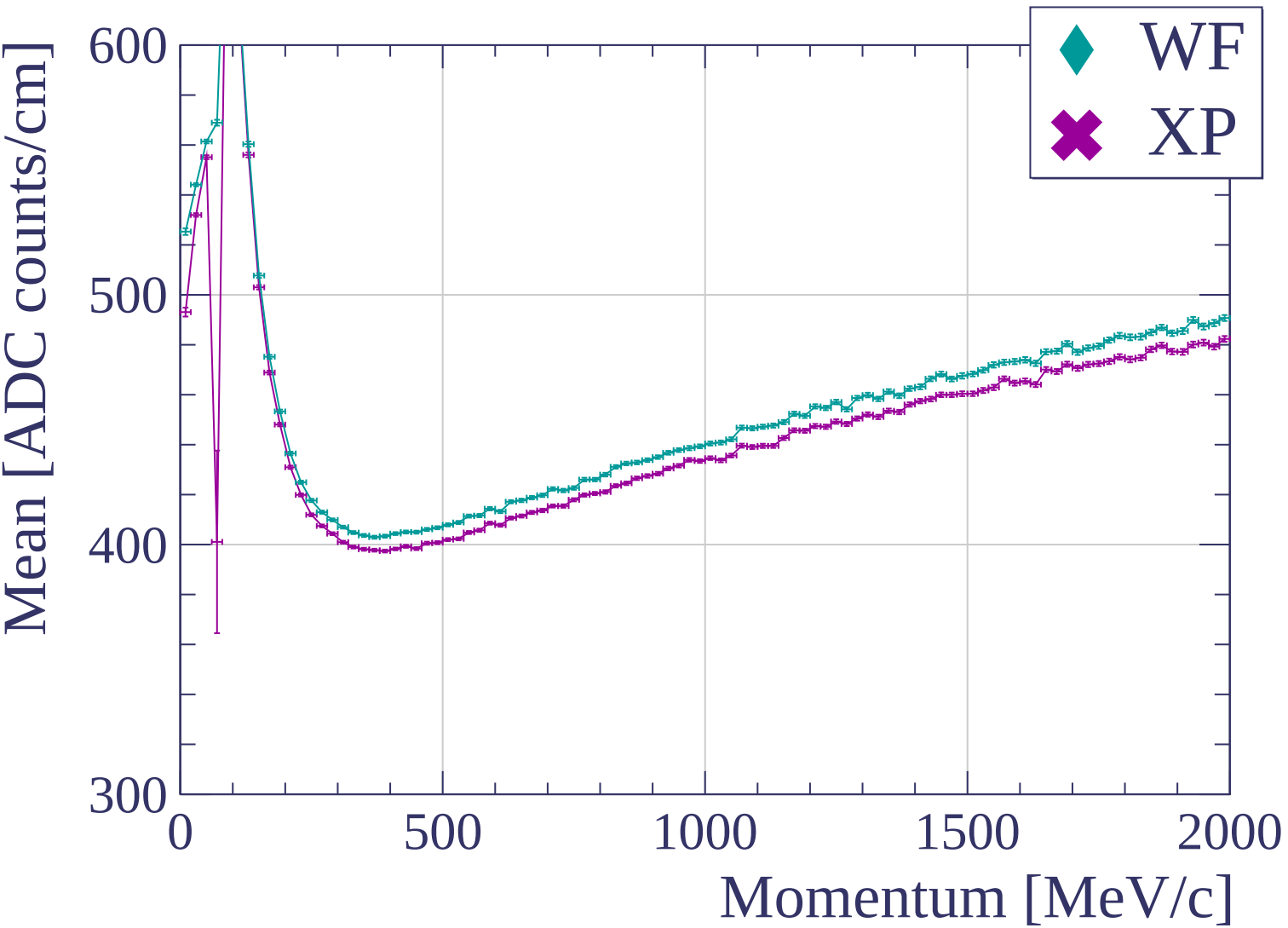


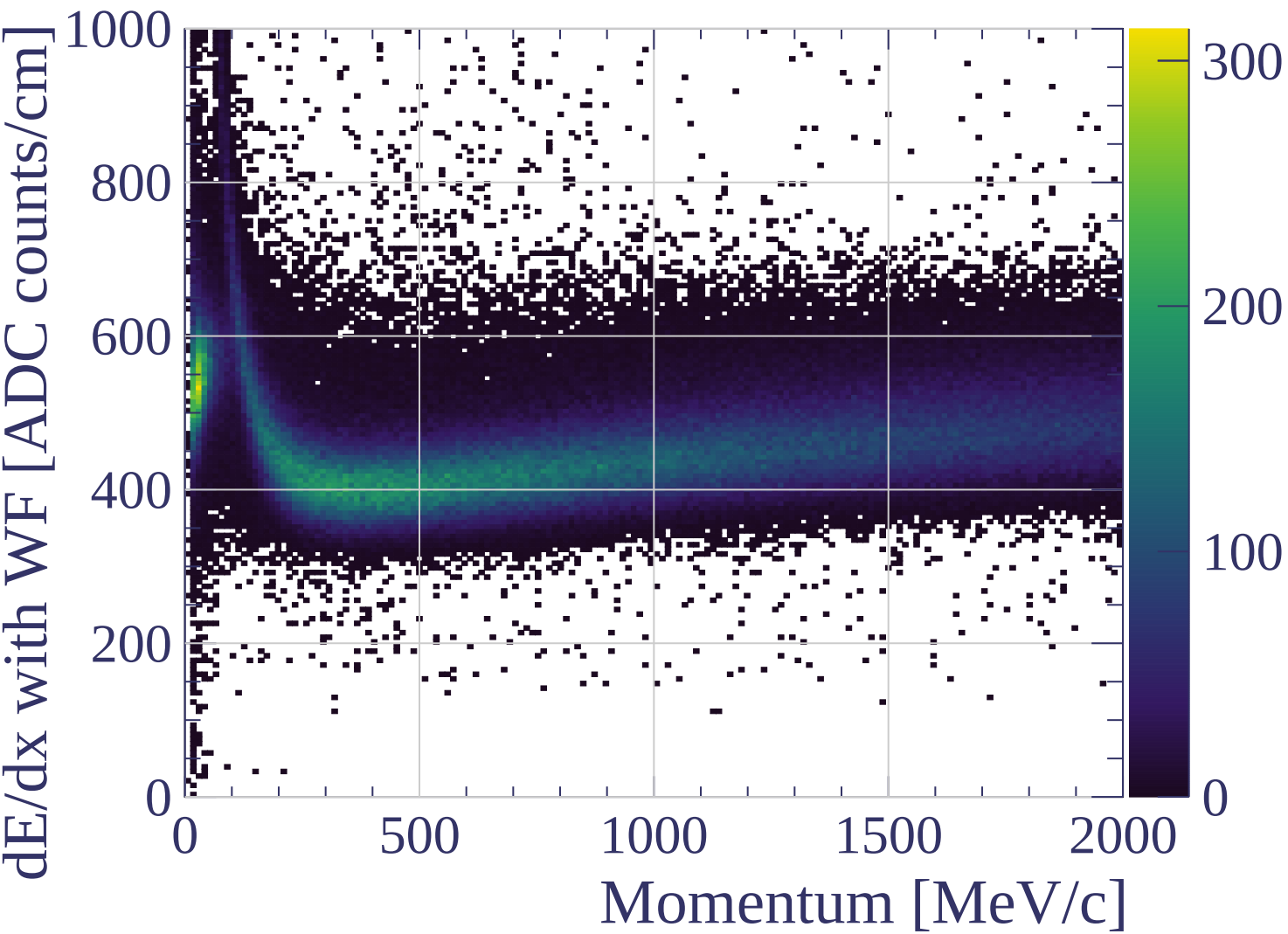


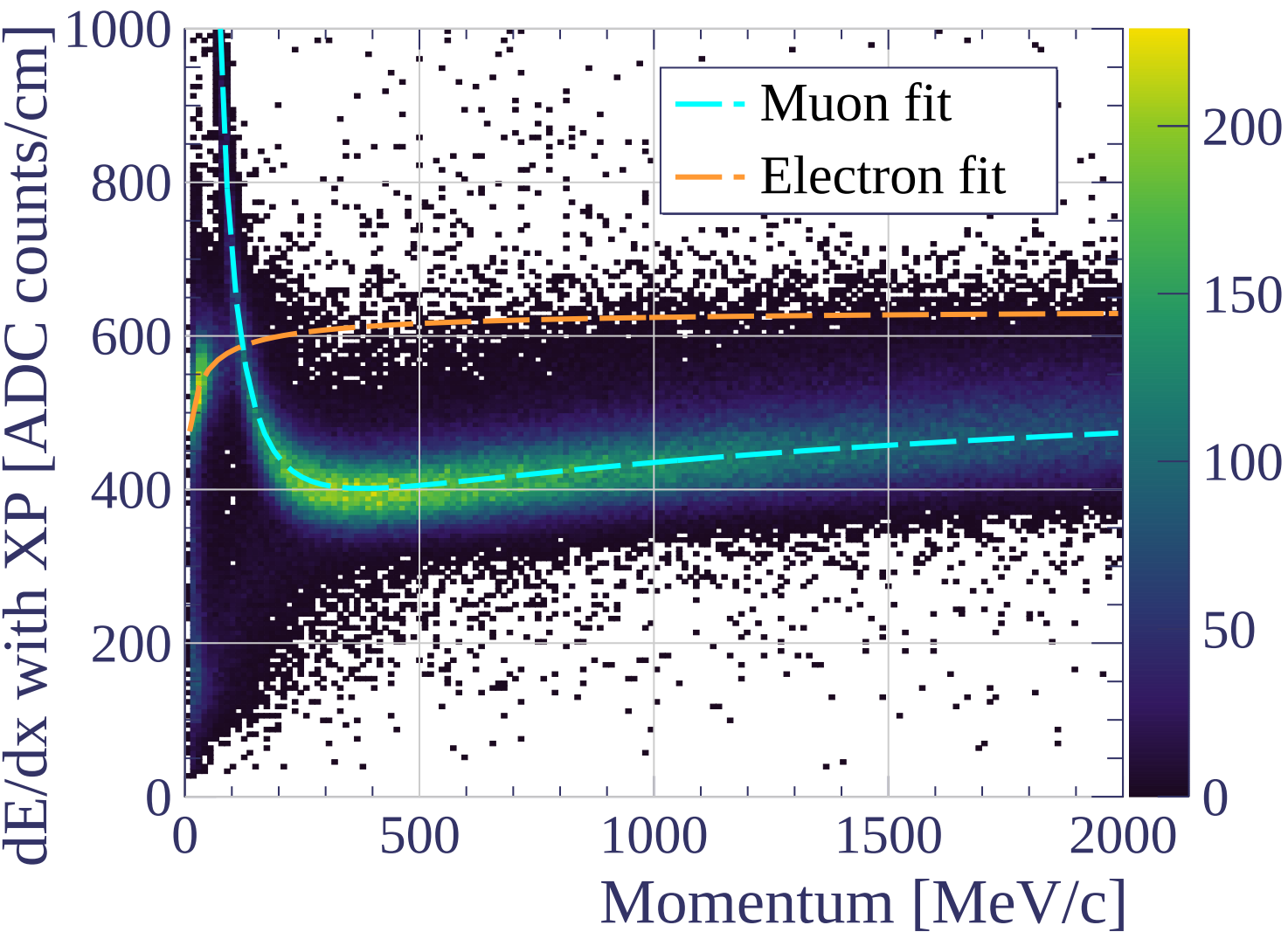






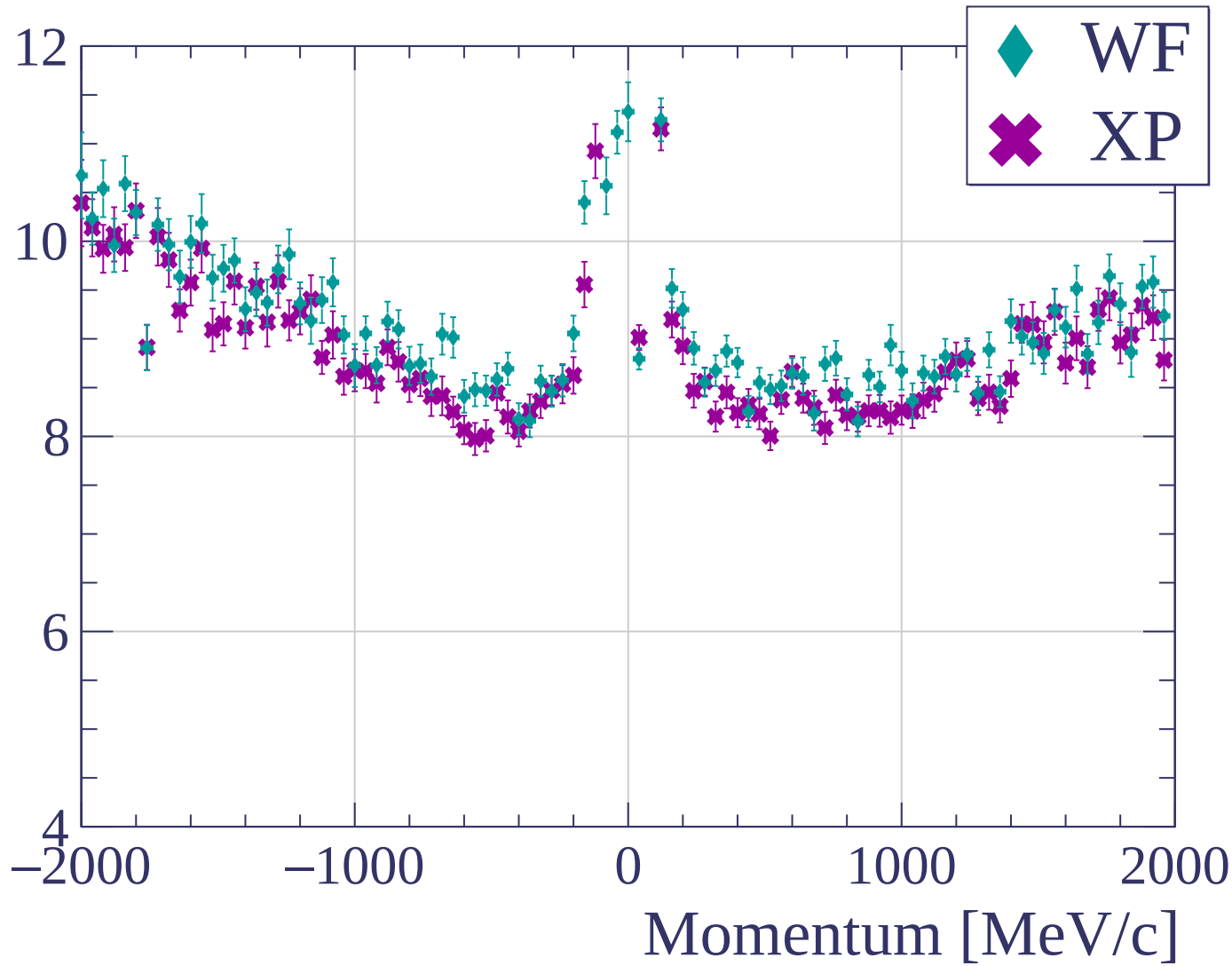


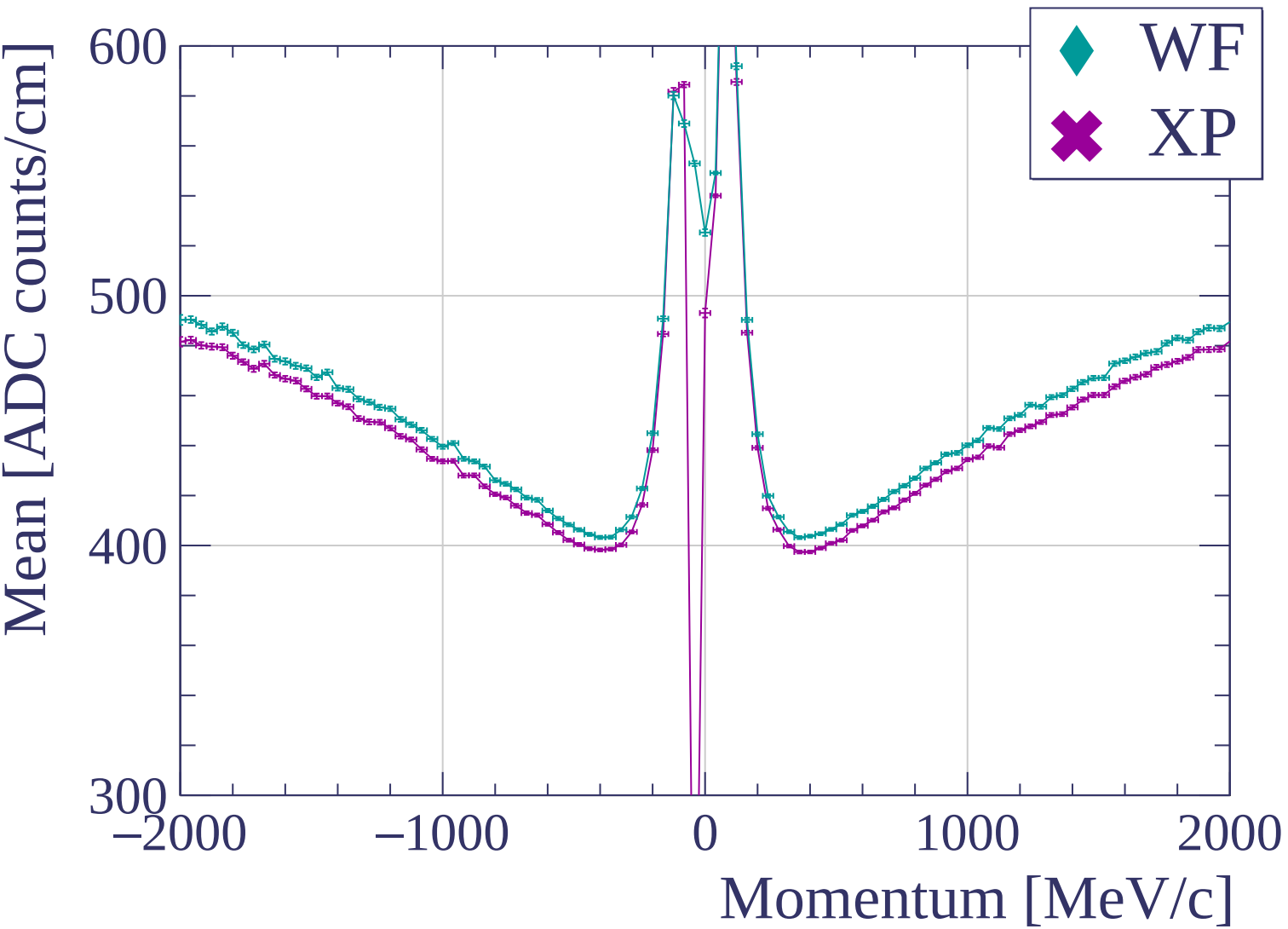


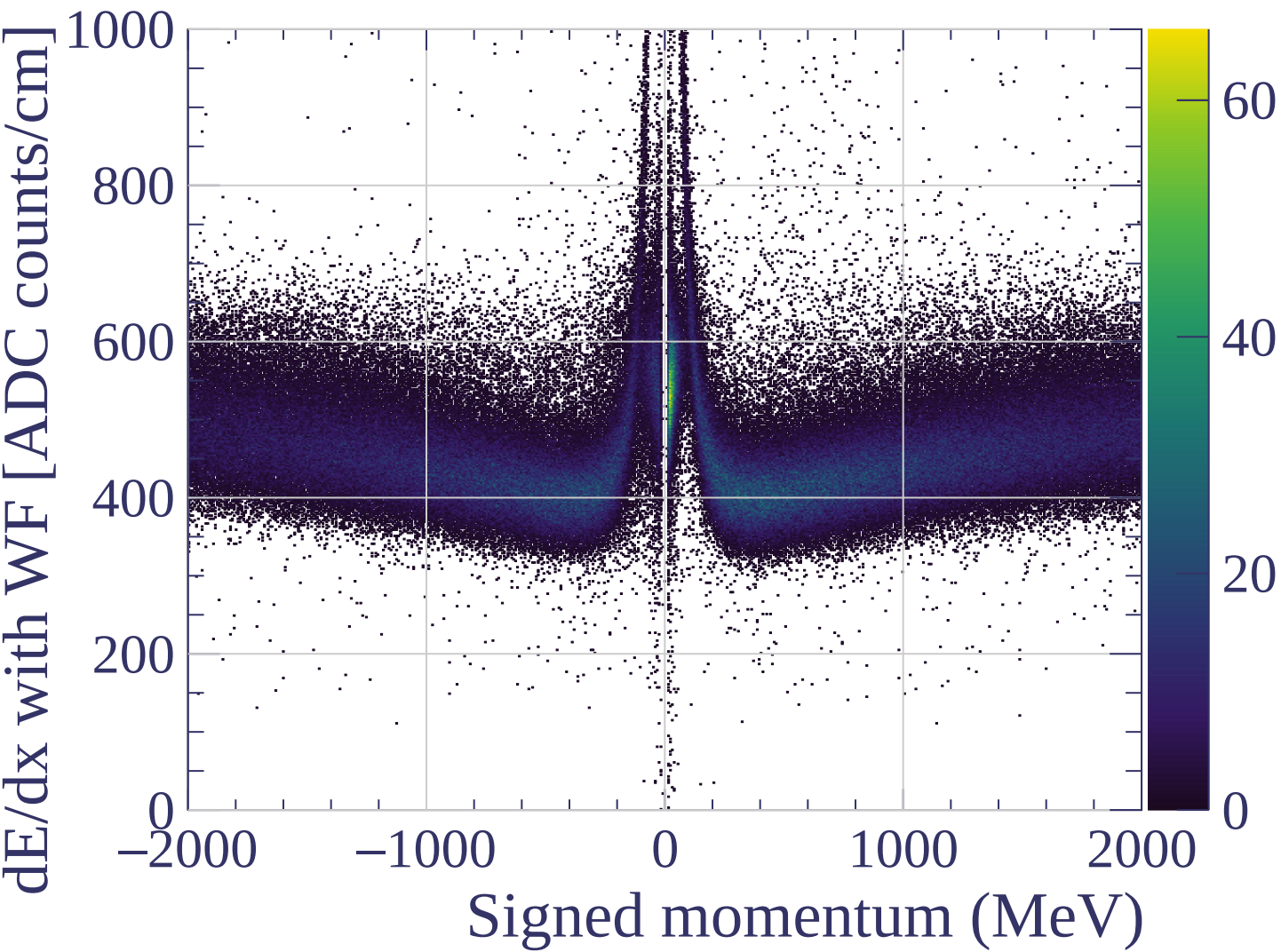


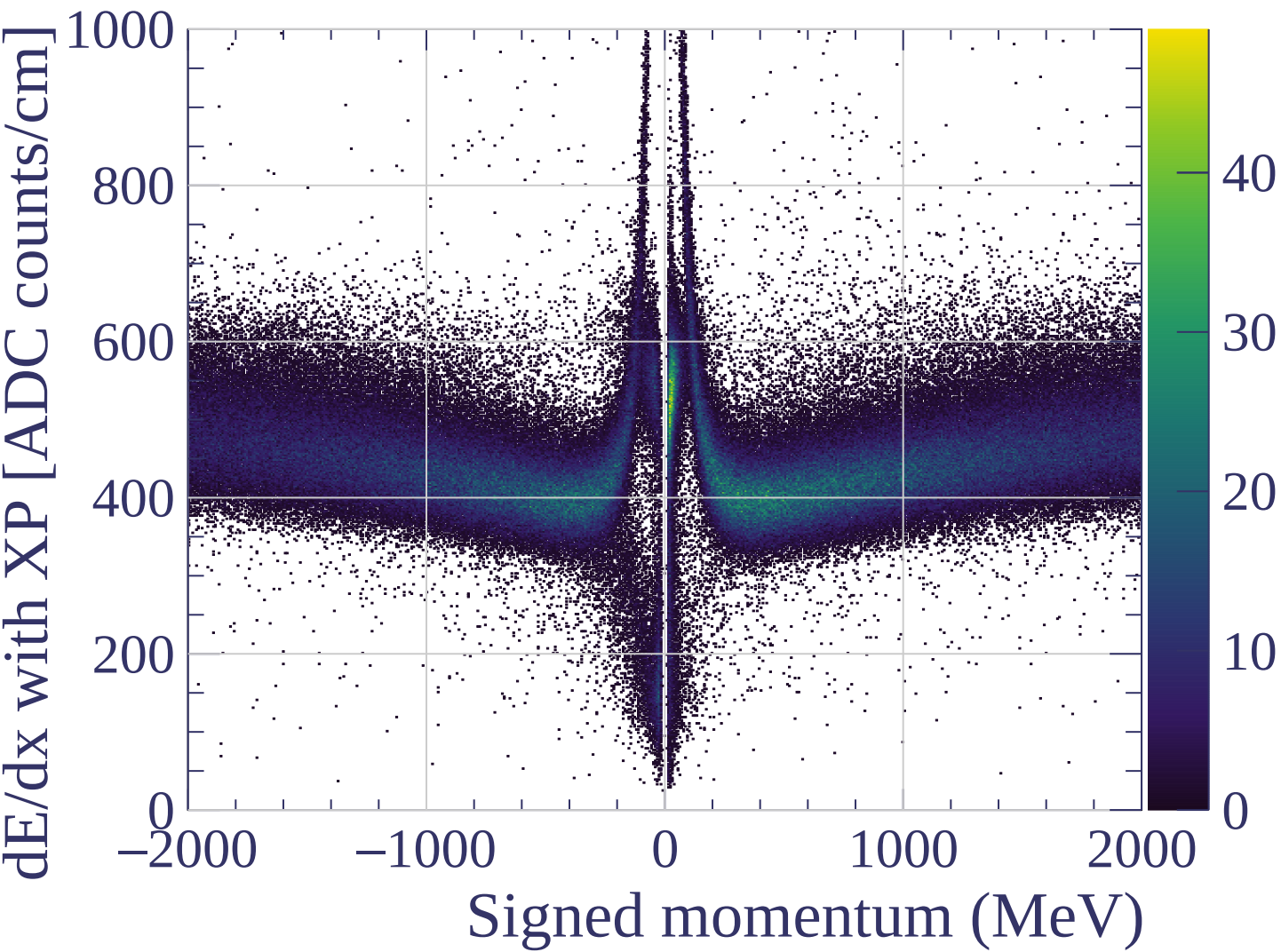


$dE/dx$  resolution (%)

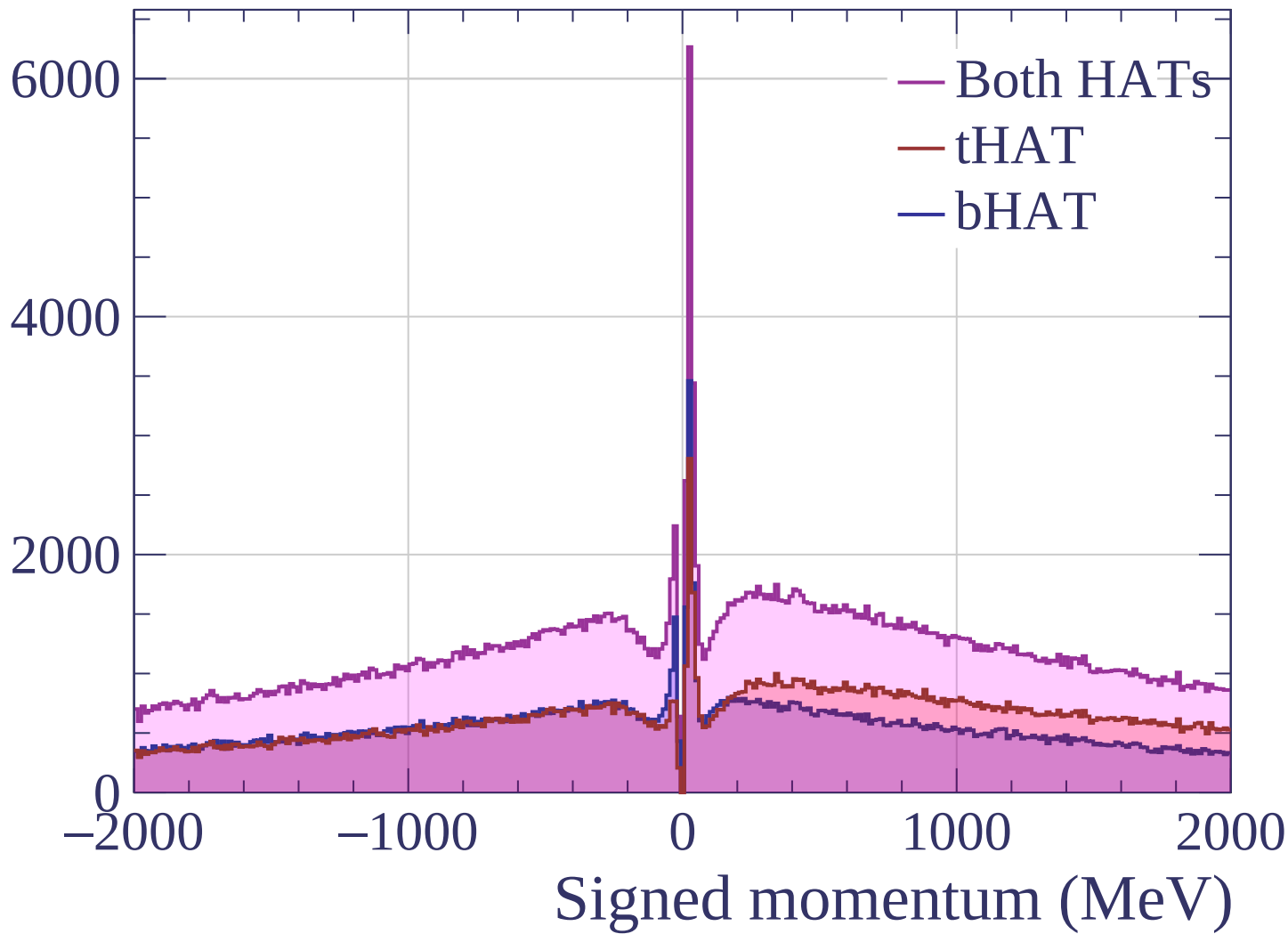


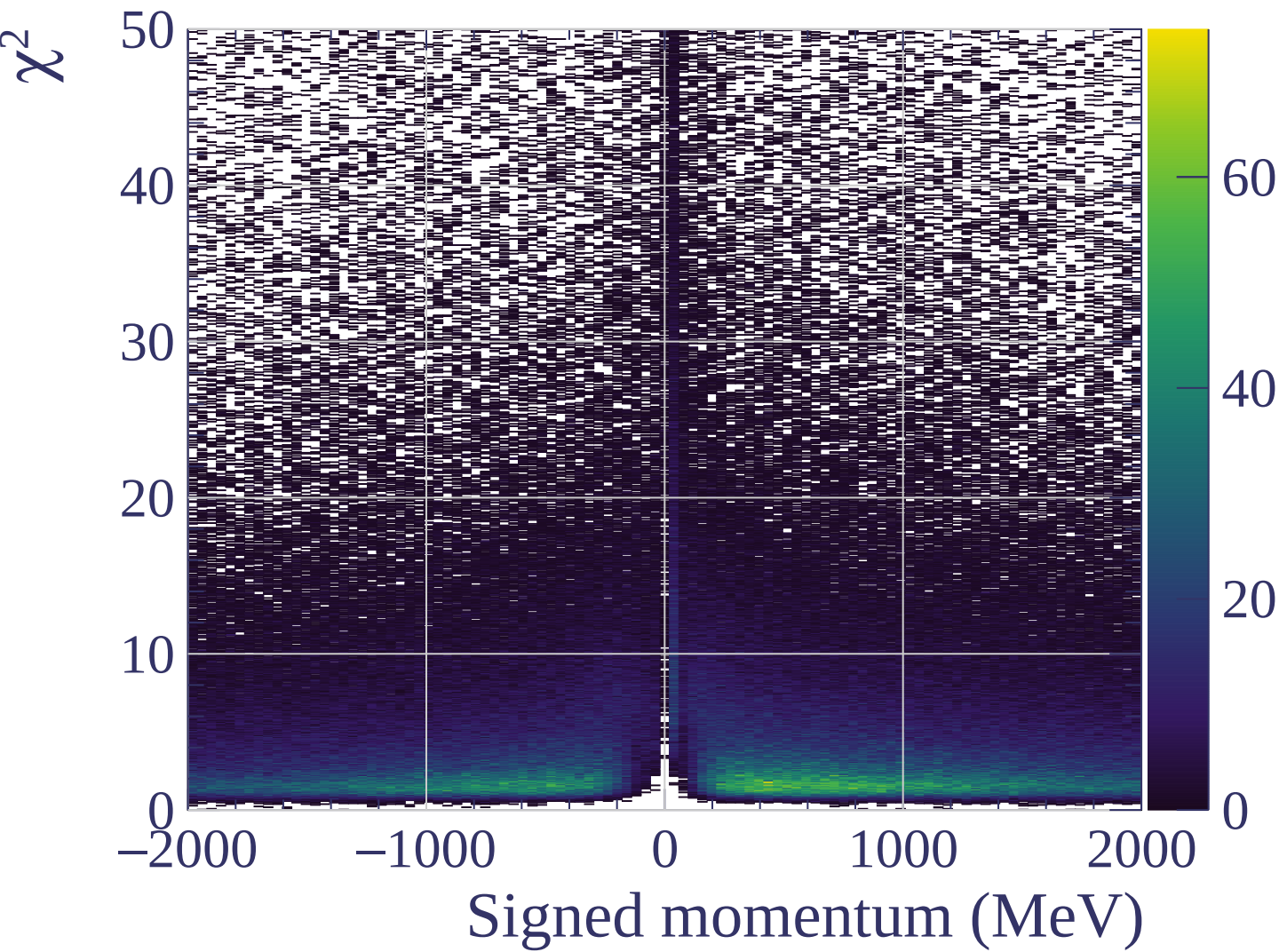


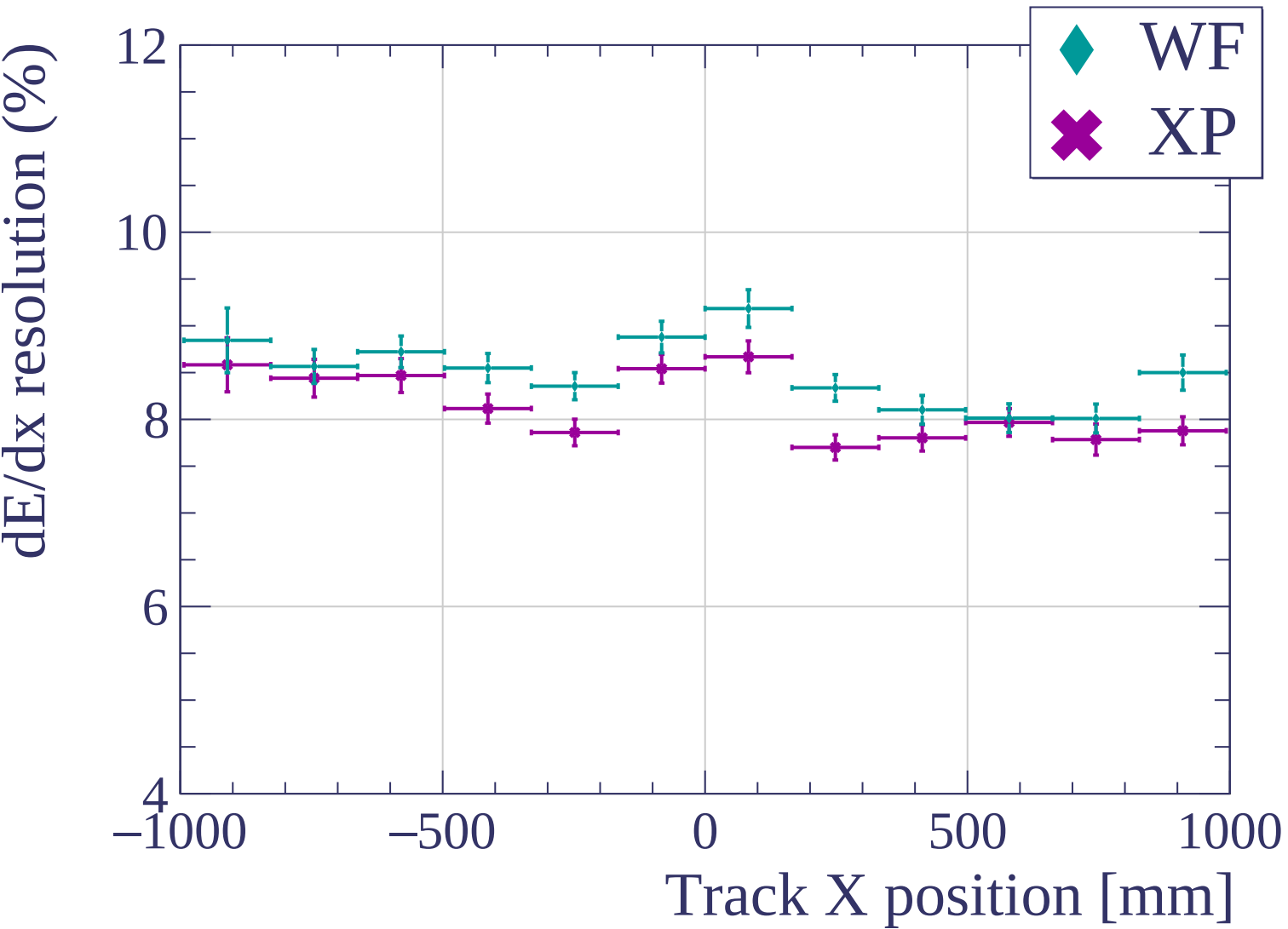


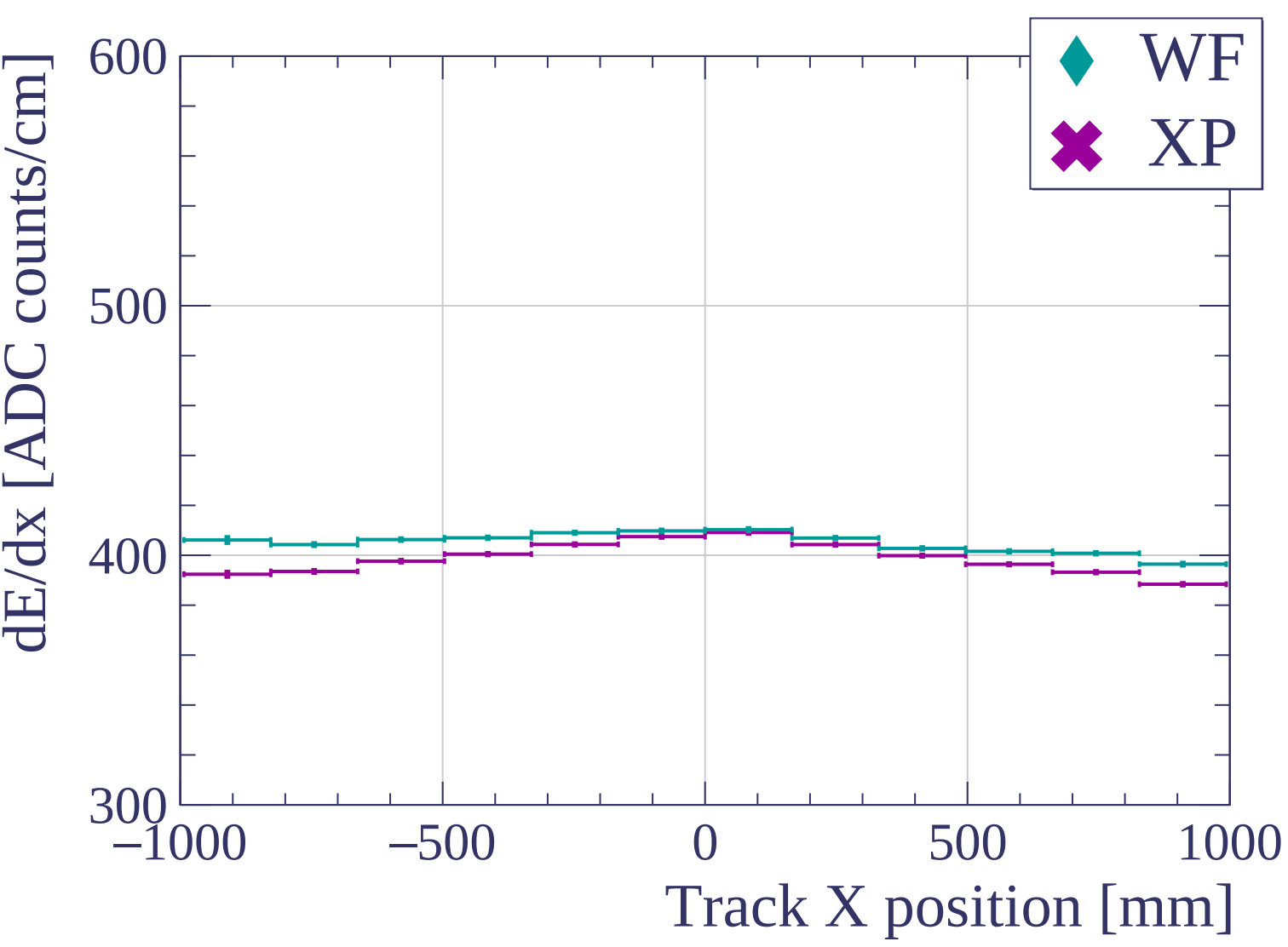


Count

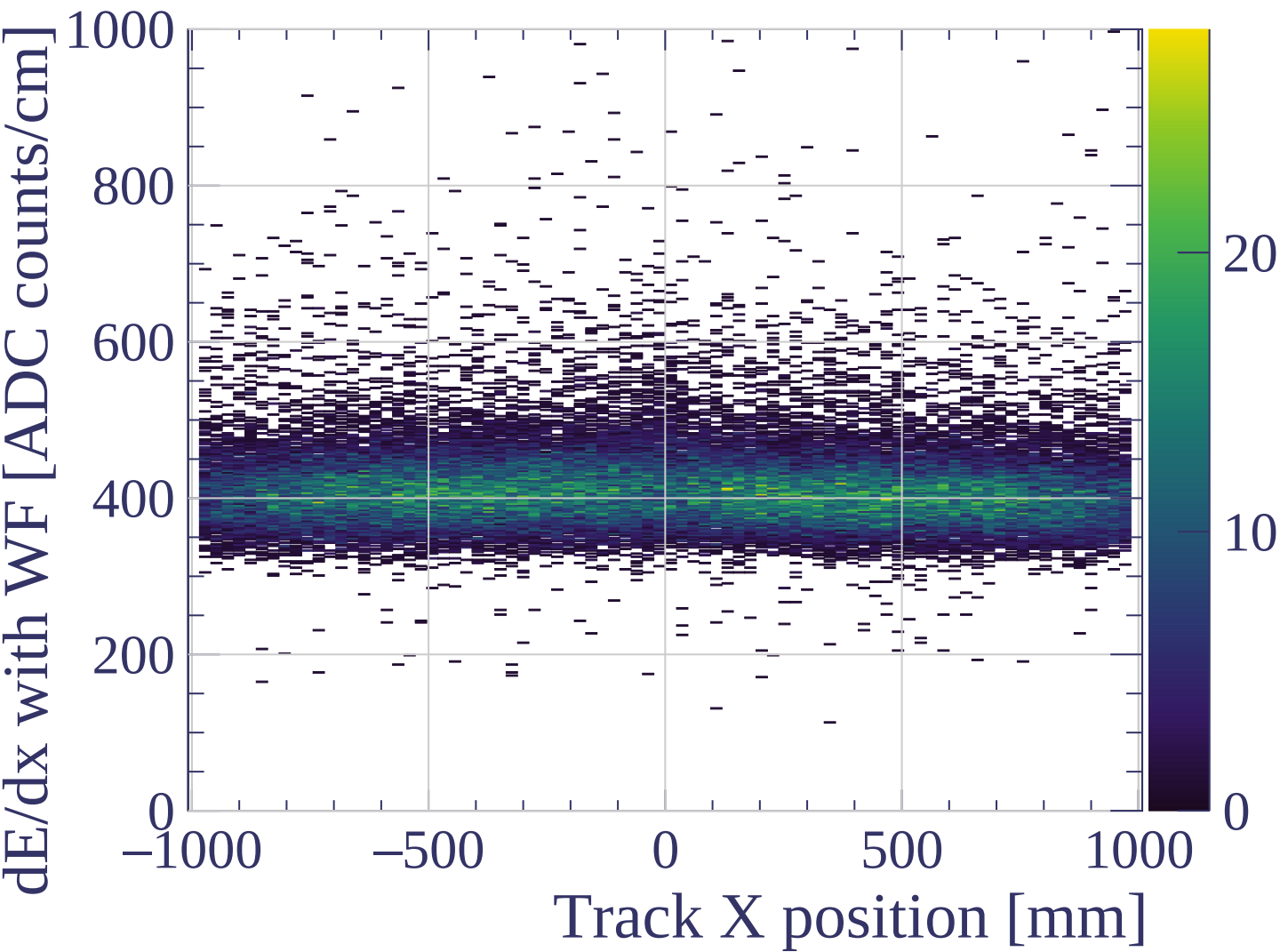


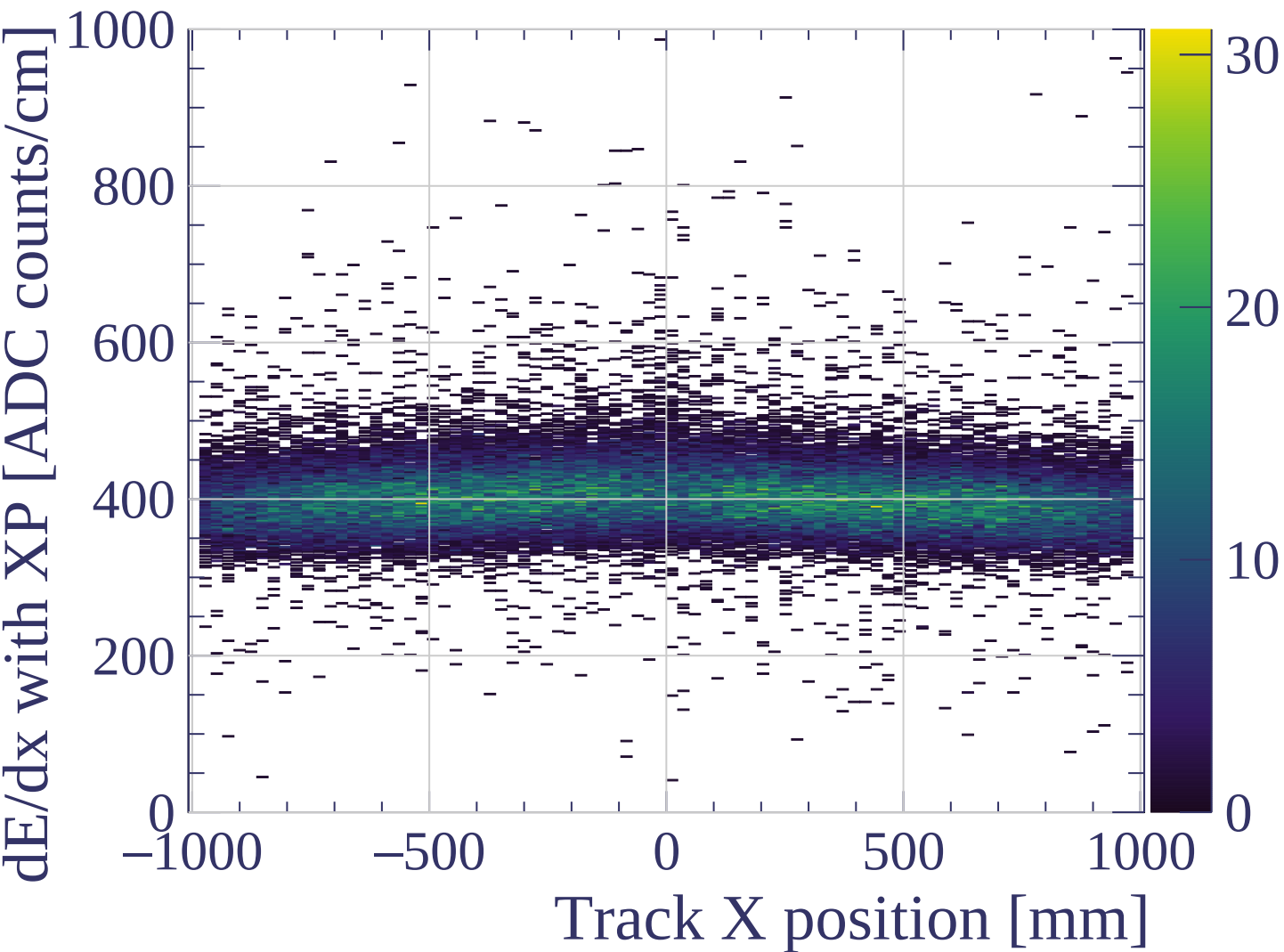


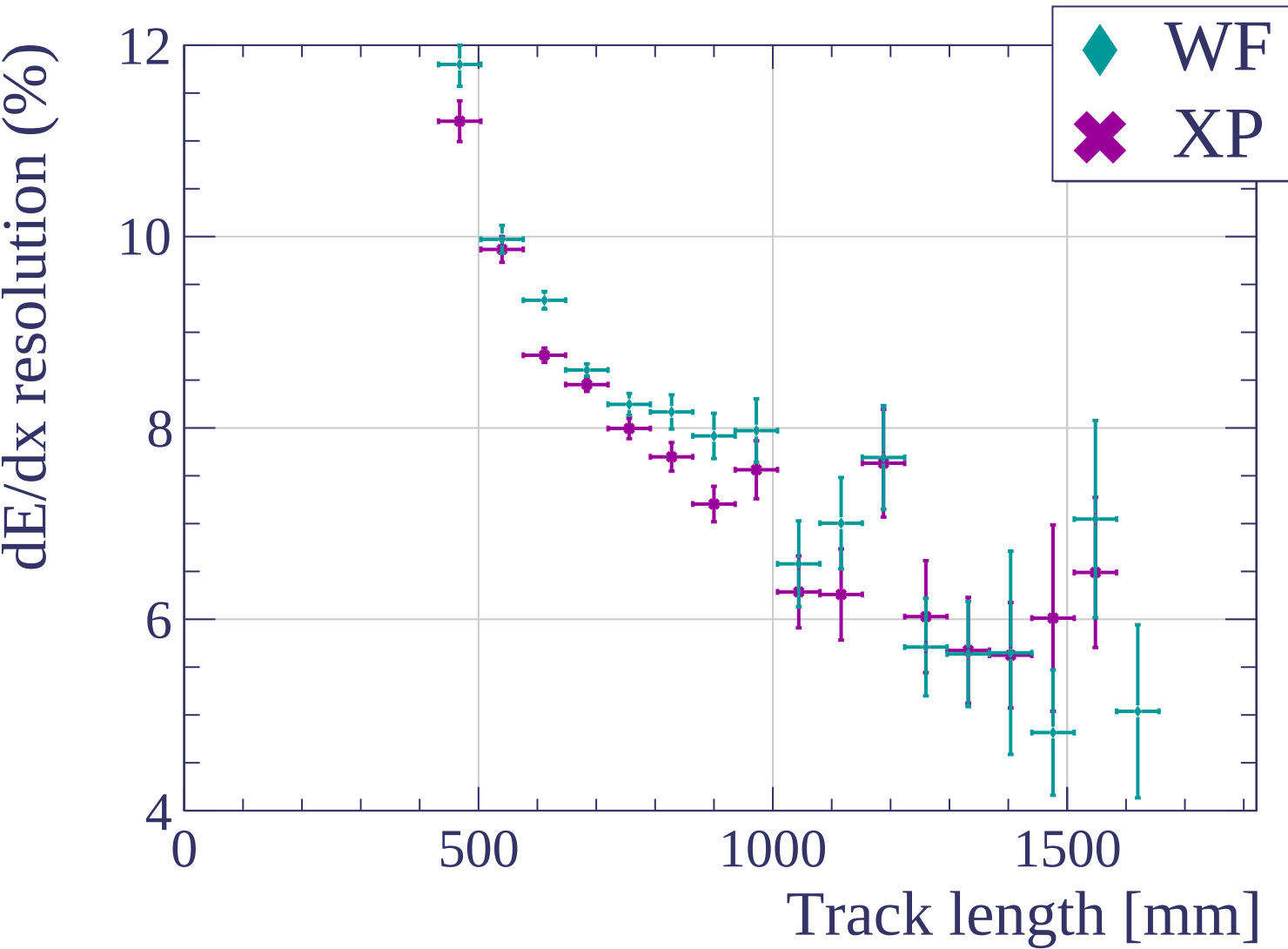


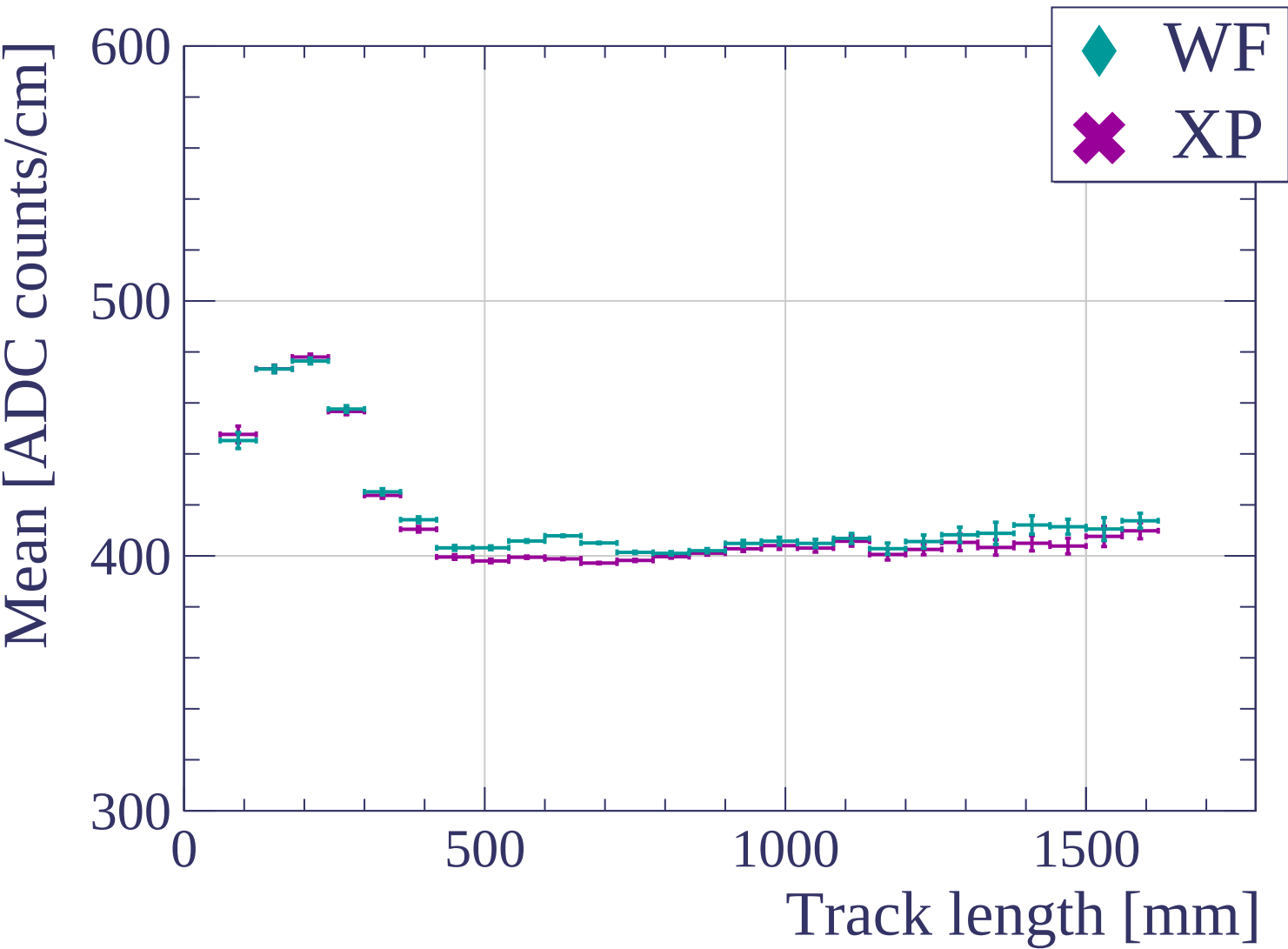


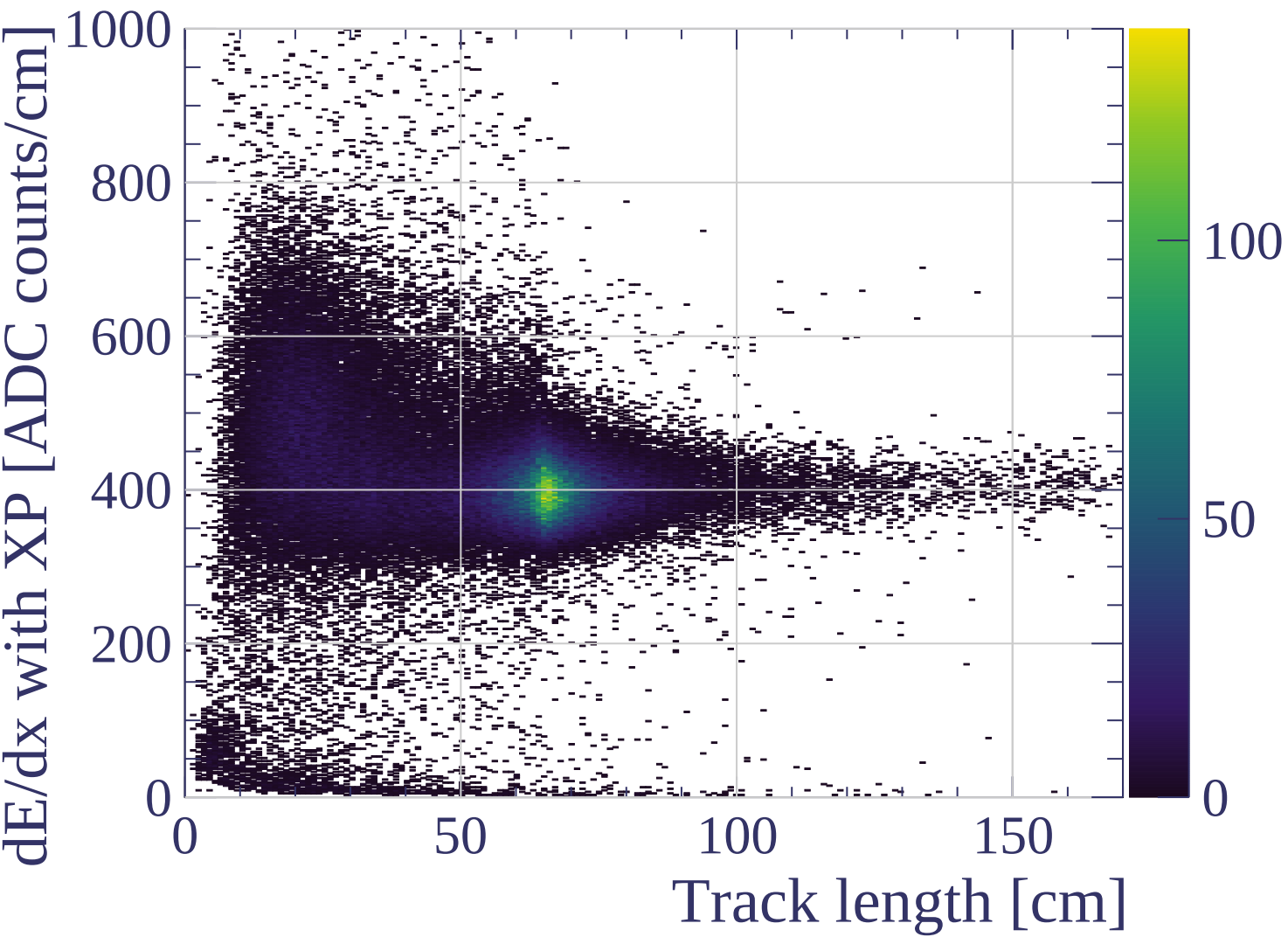


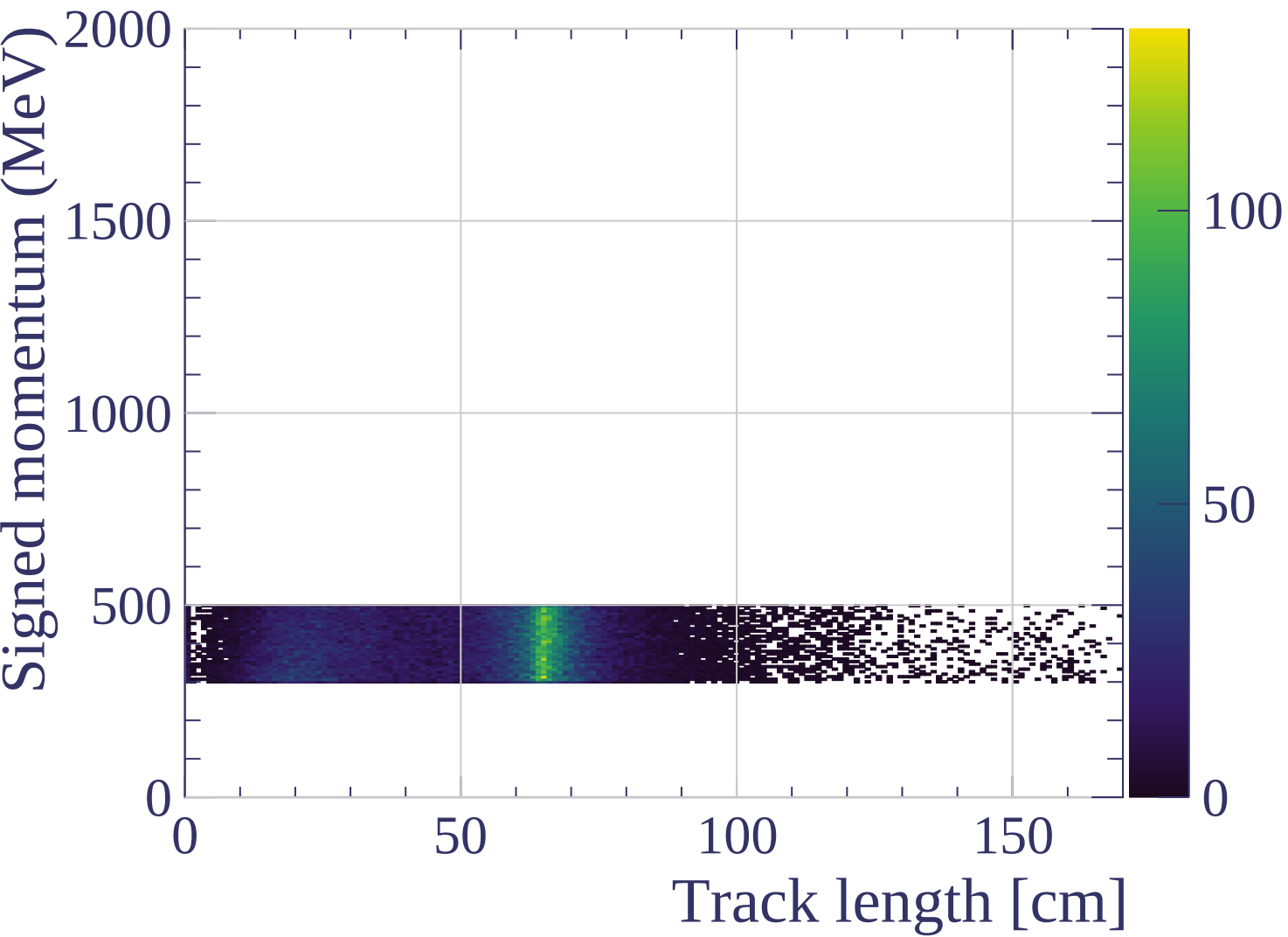


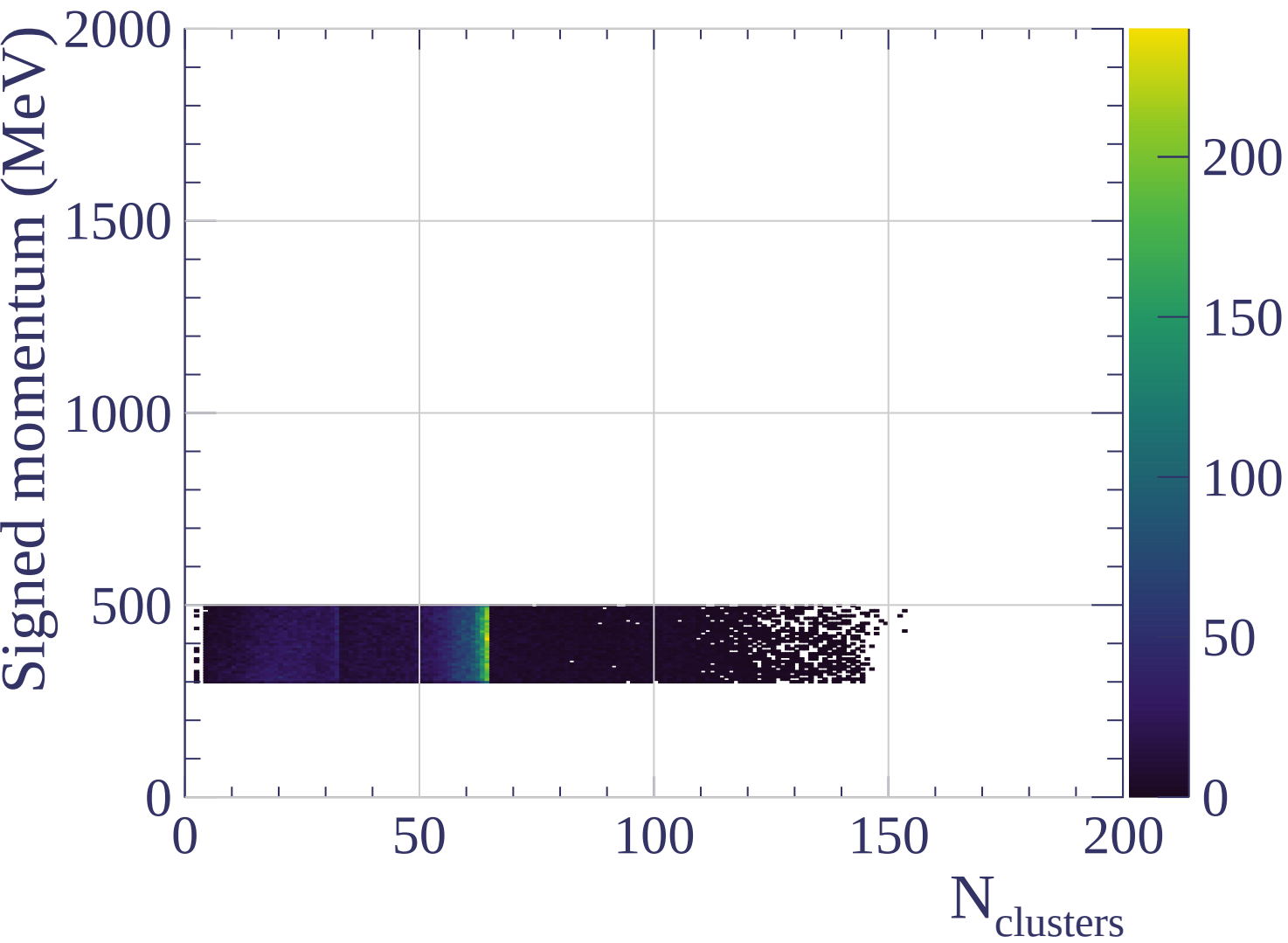


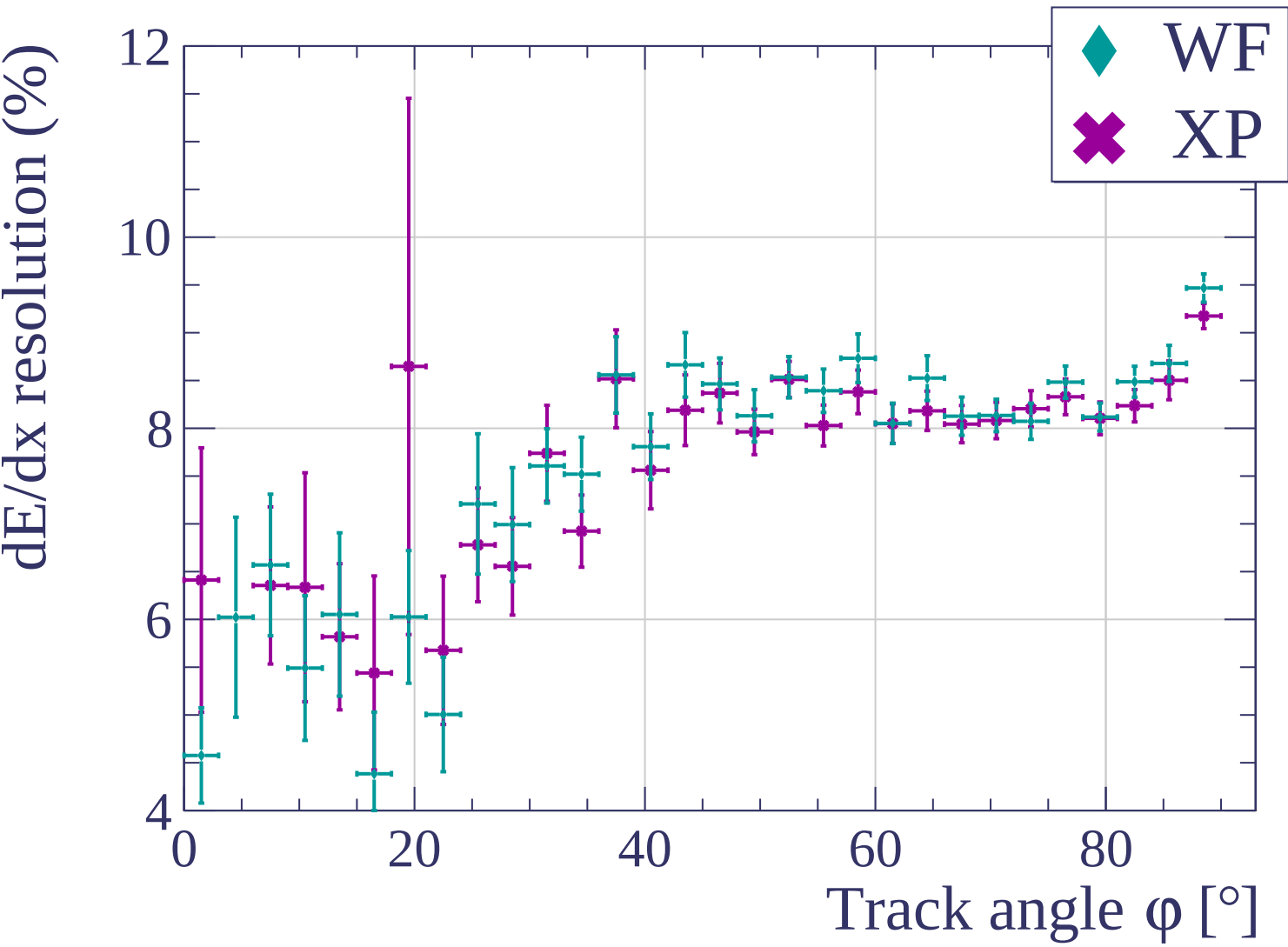




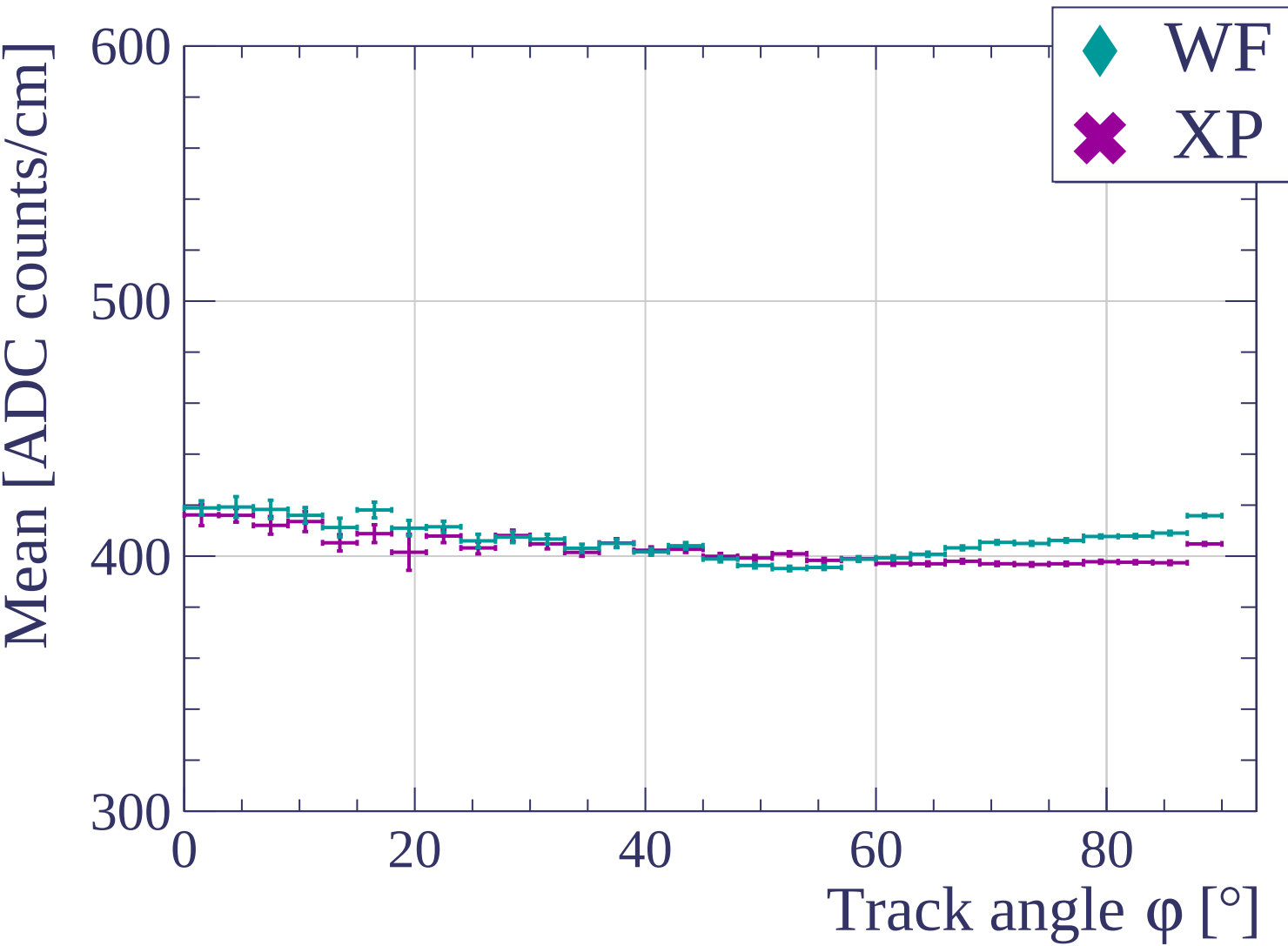


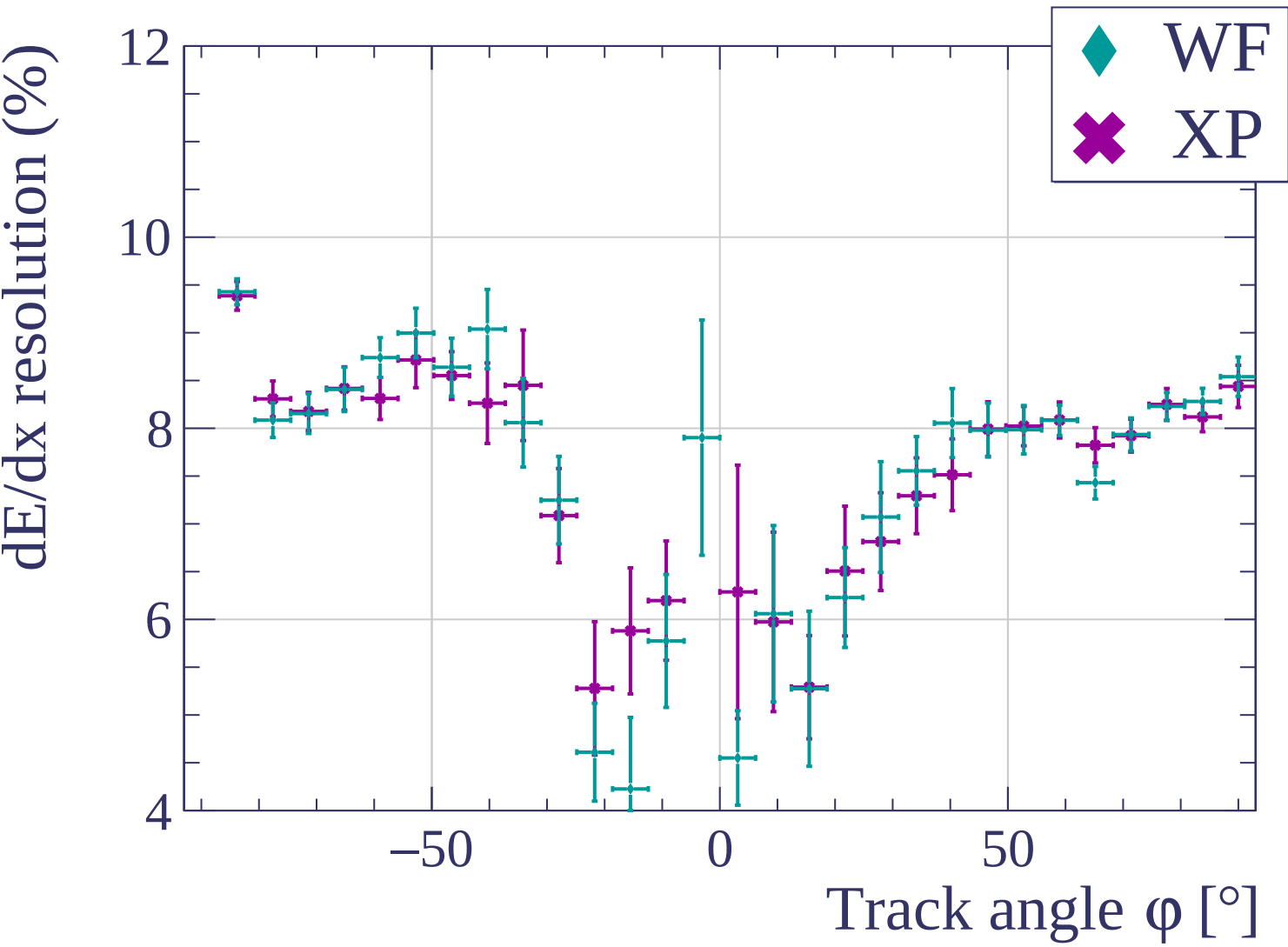


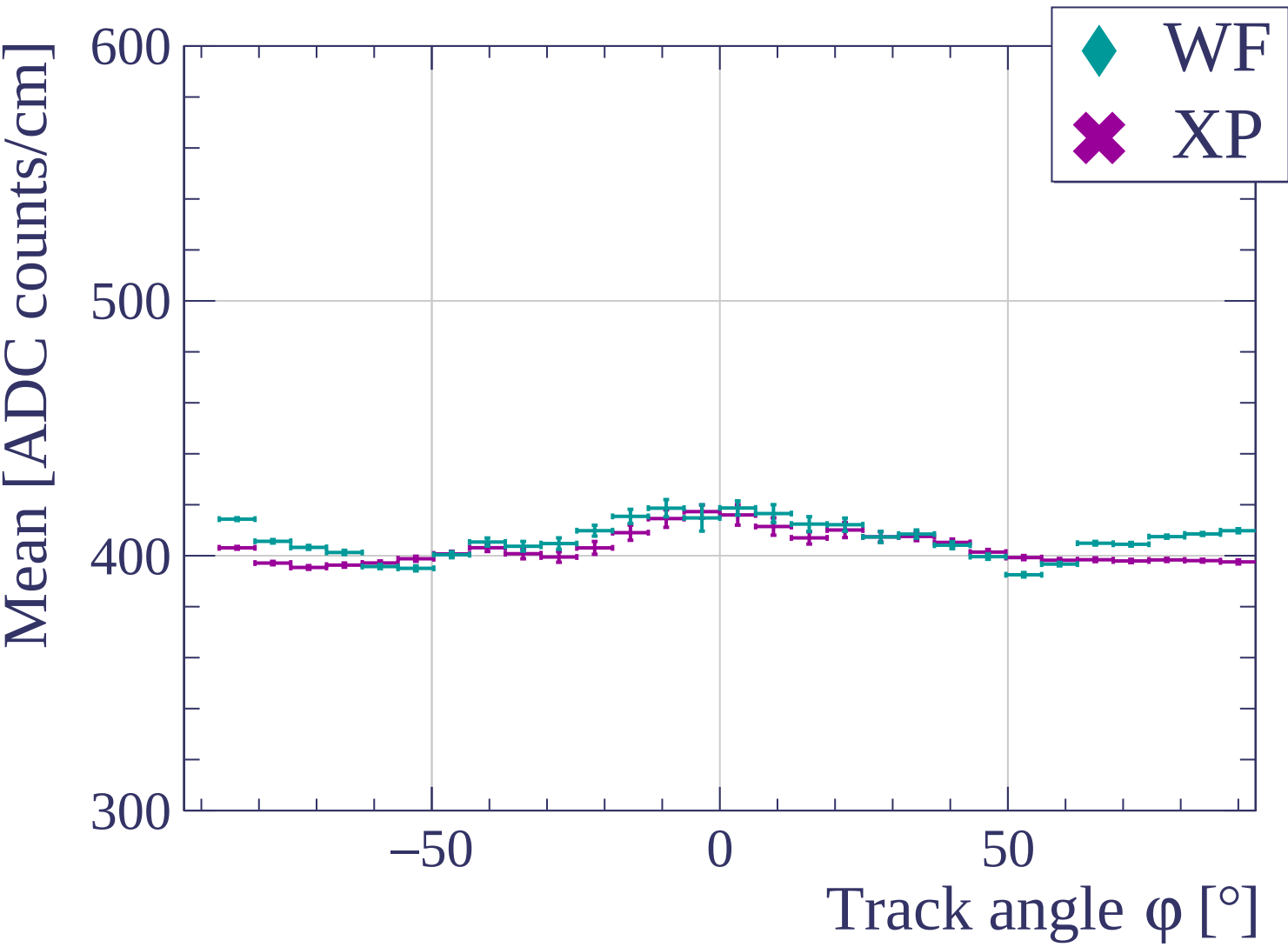


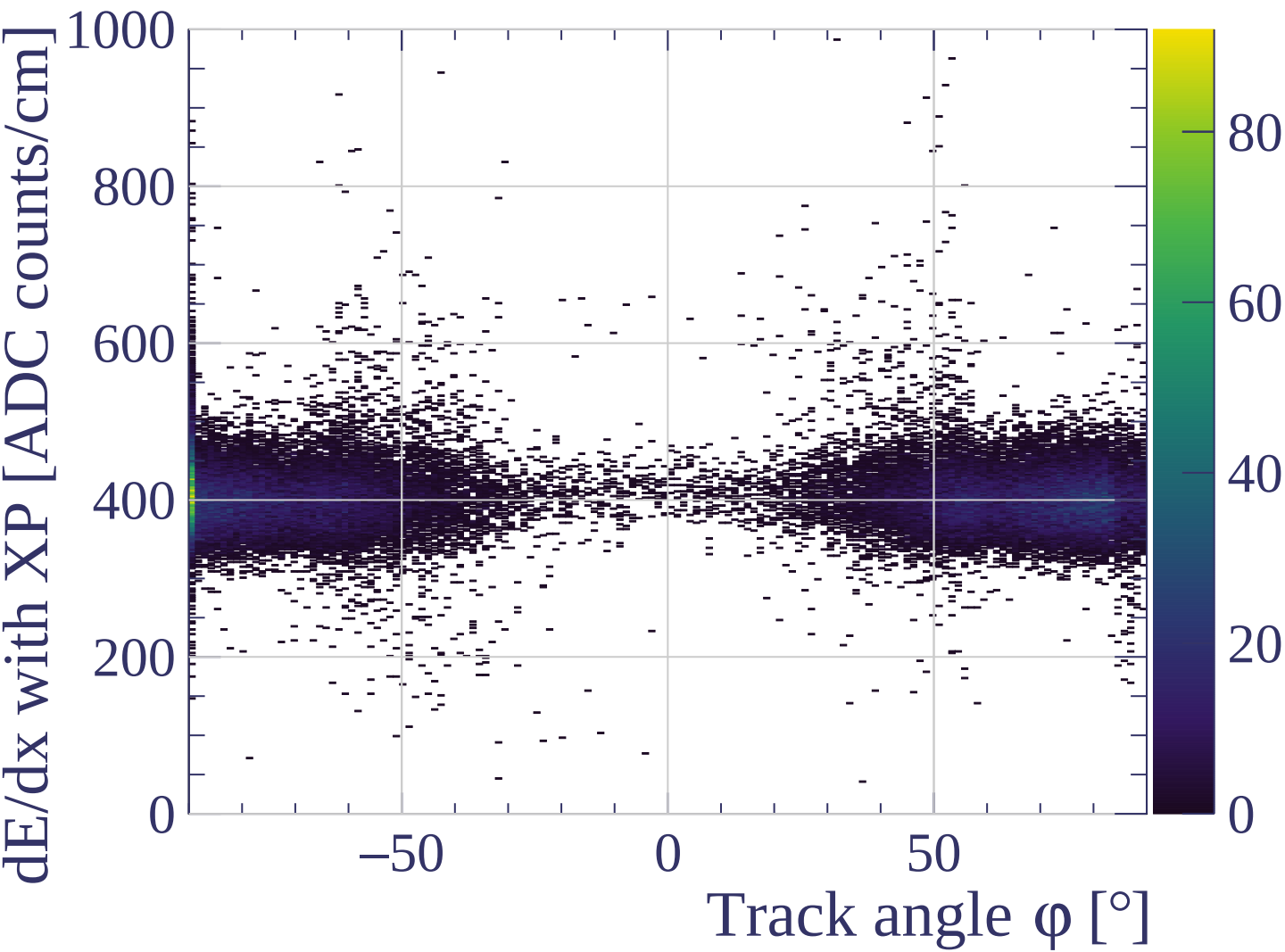




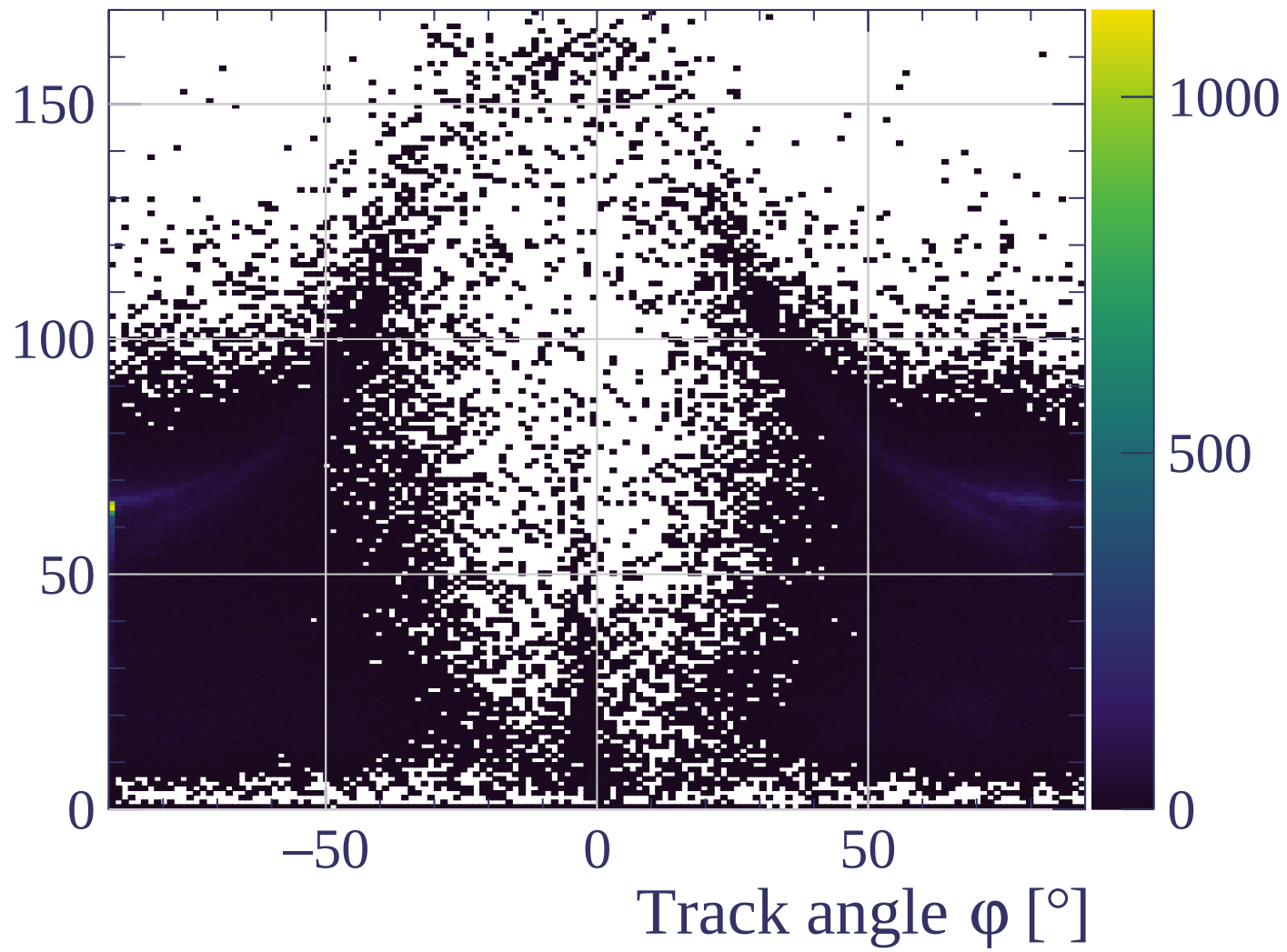


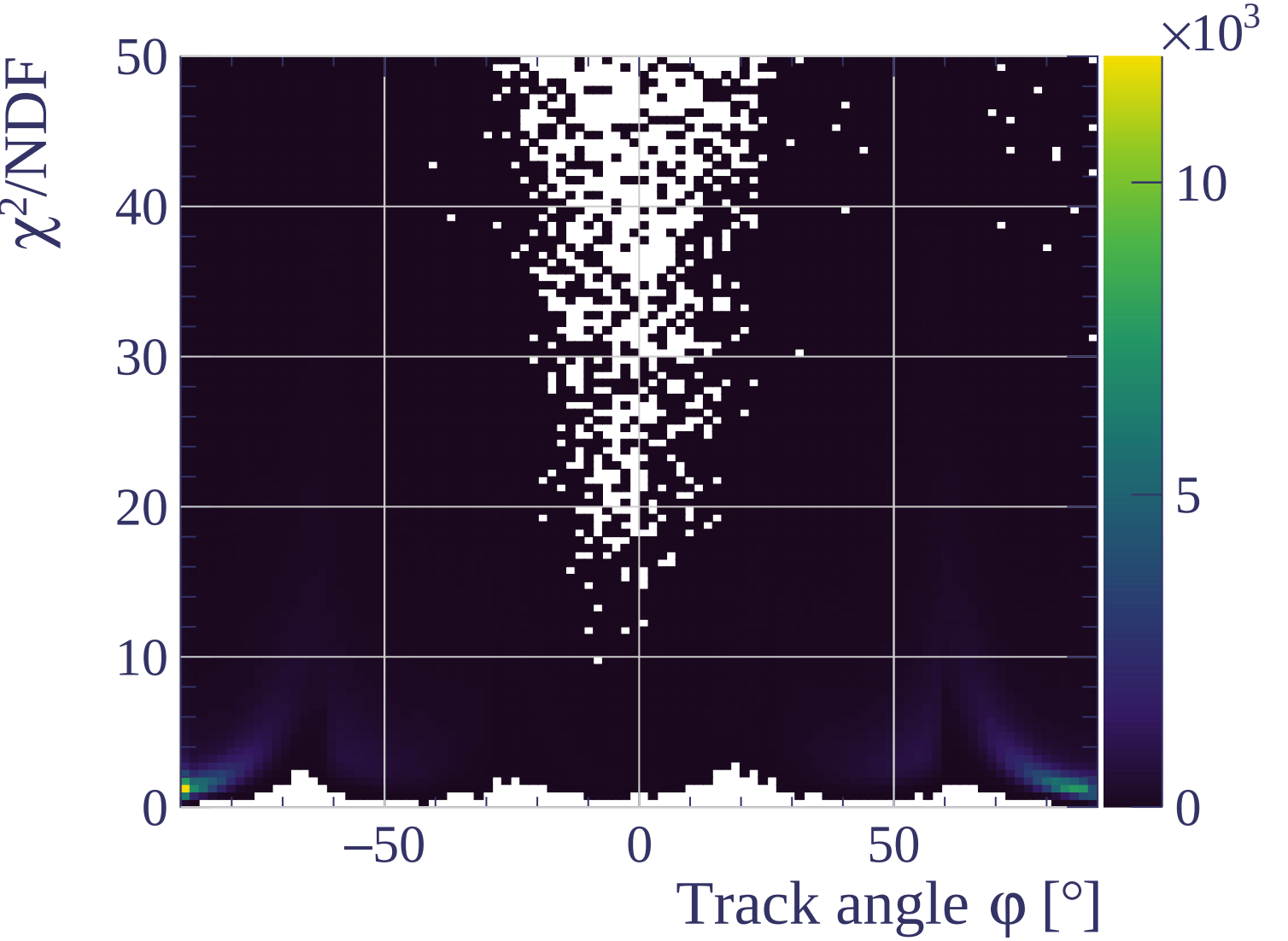


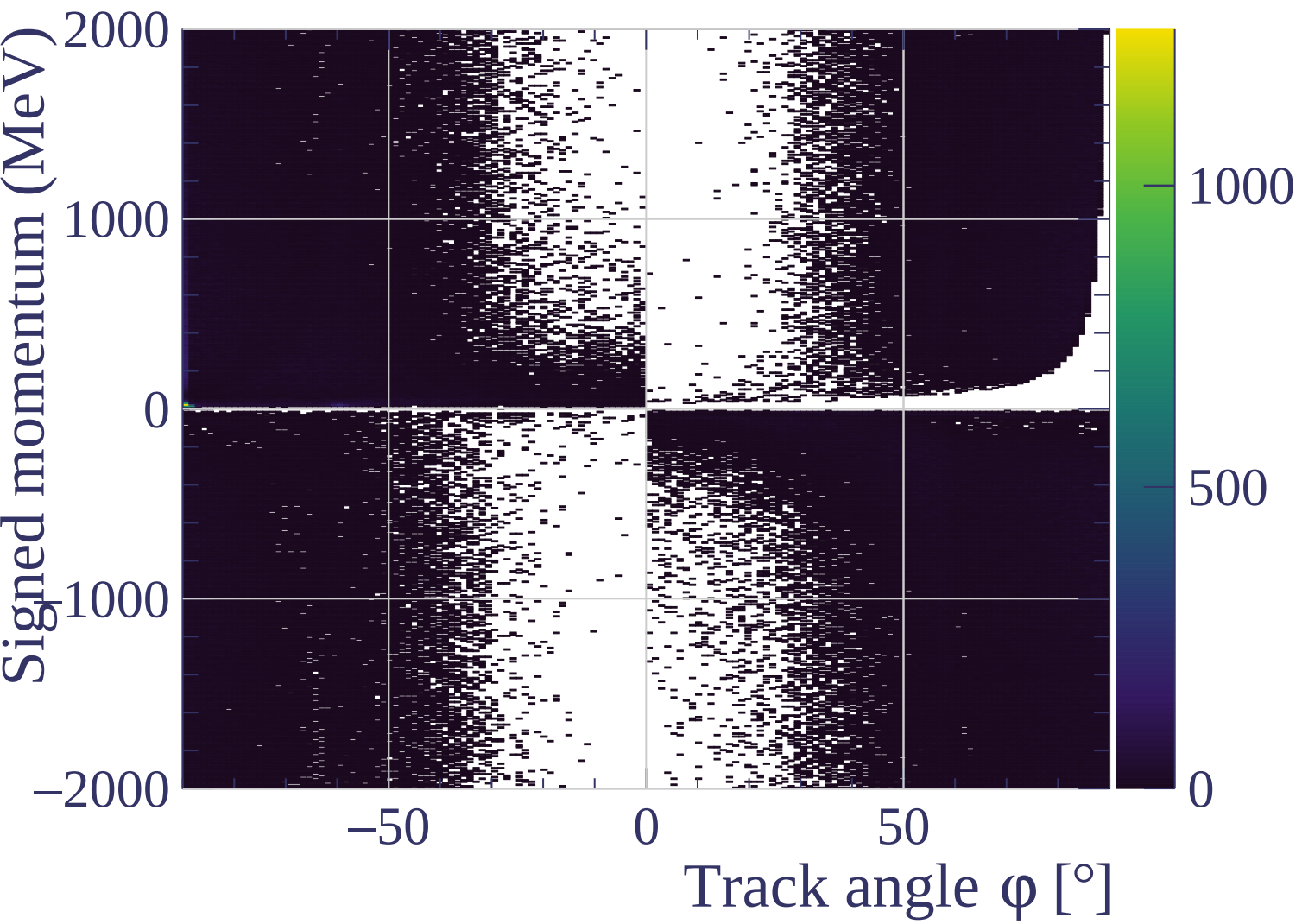


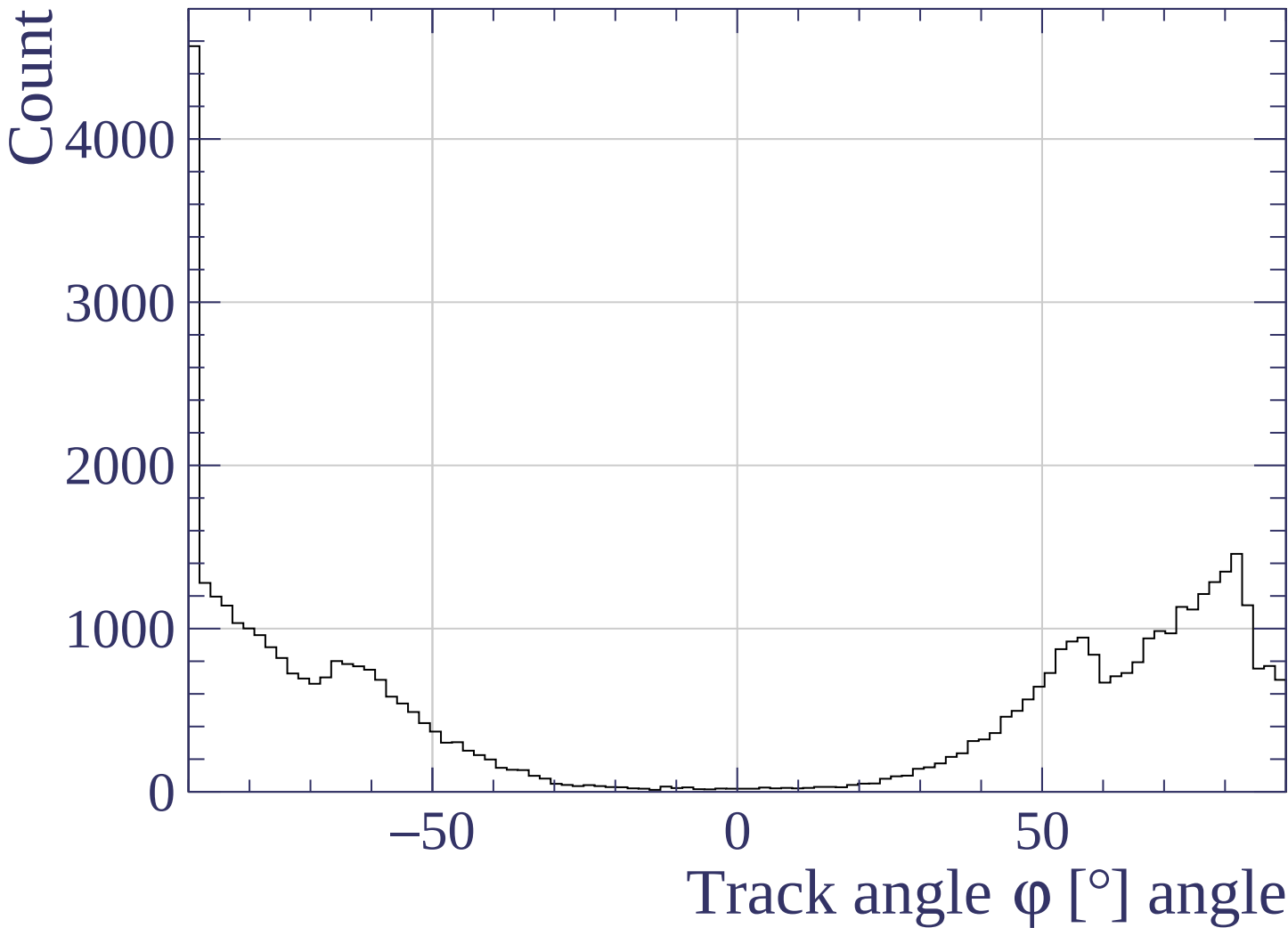


Track length [cm]

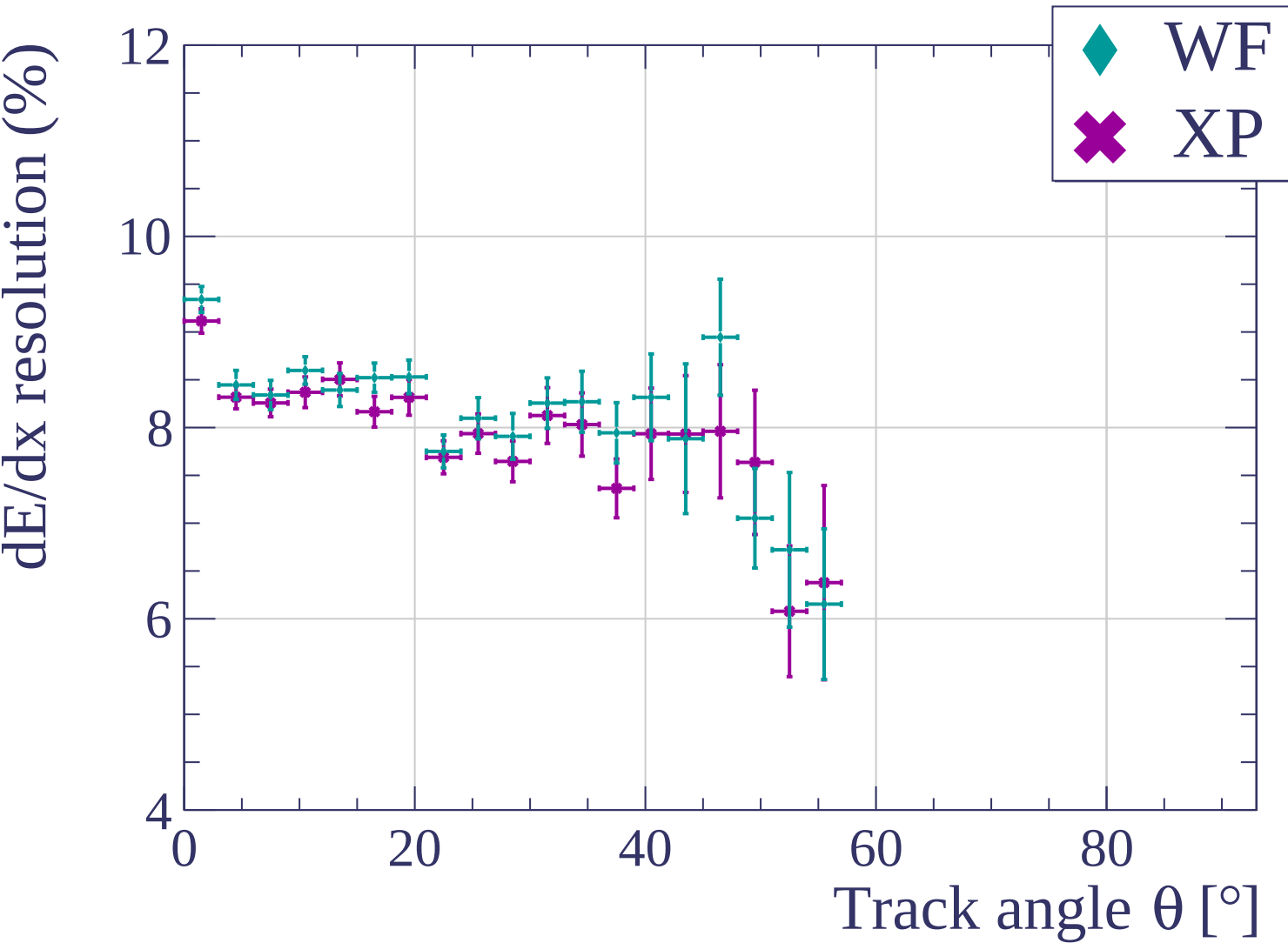


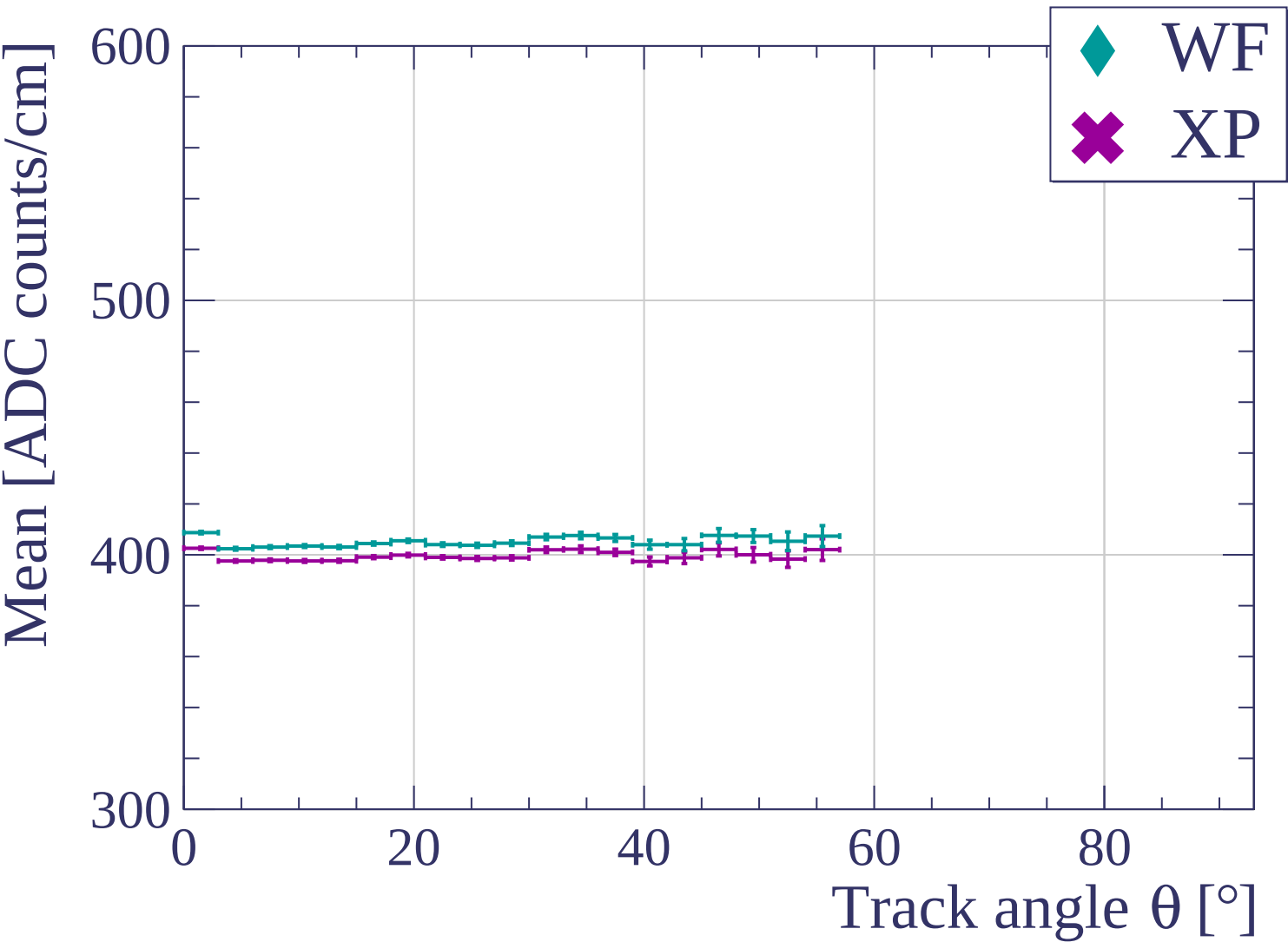


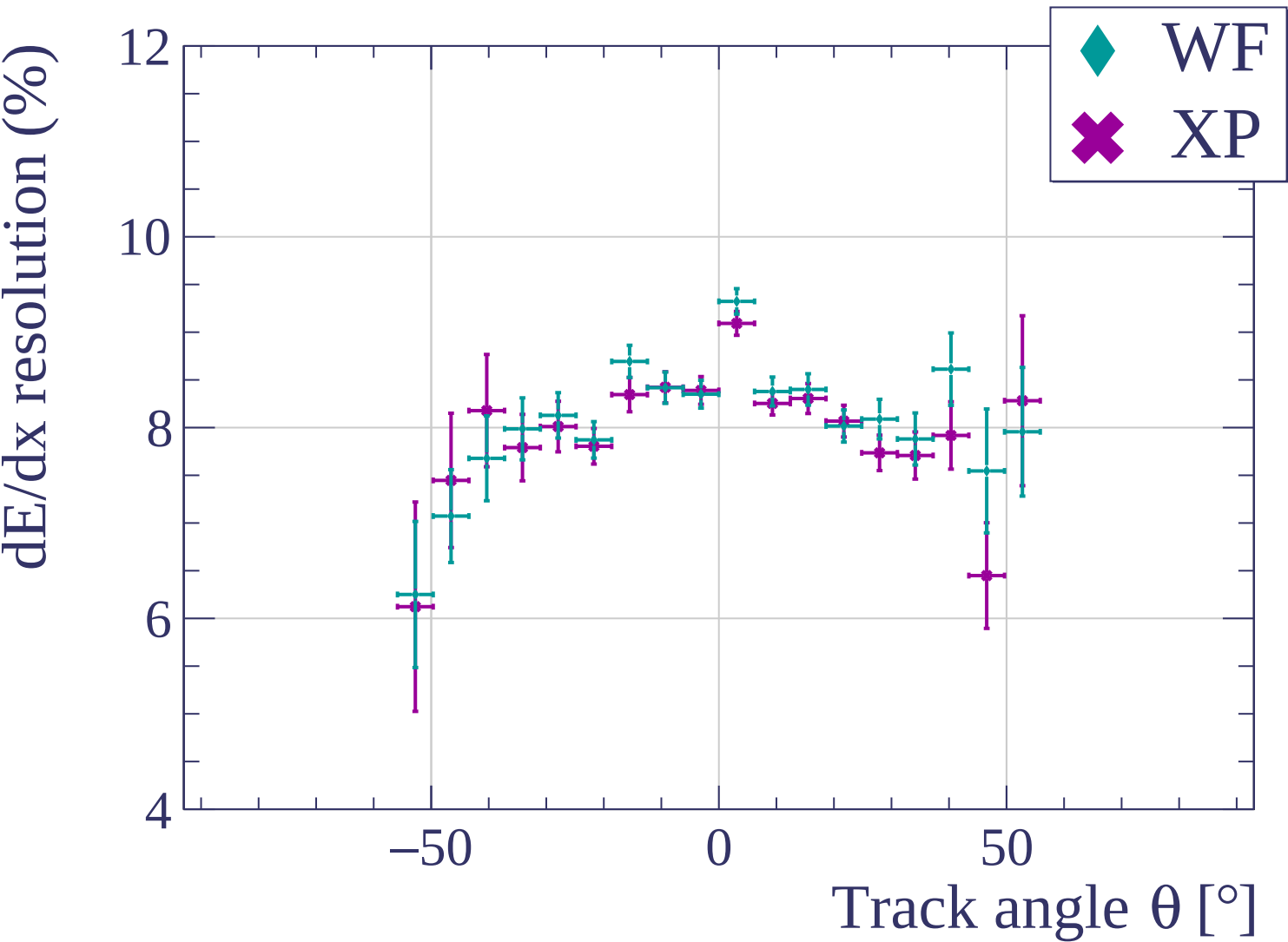


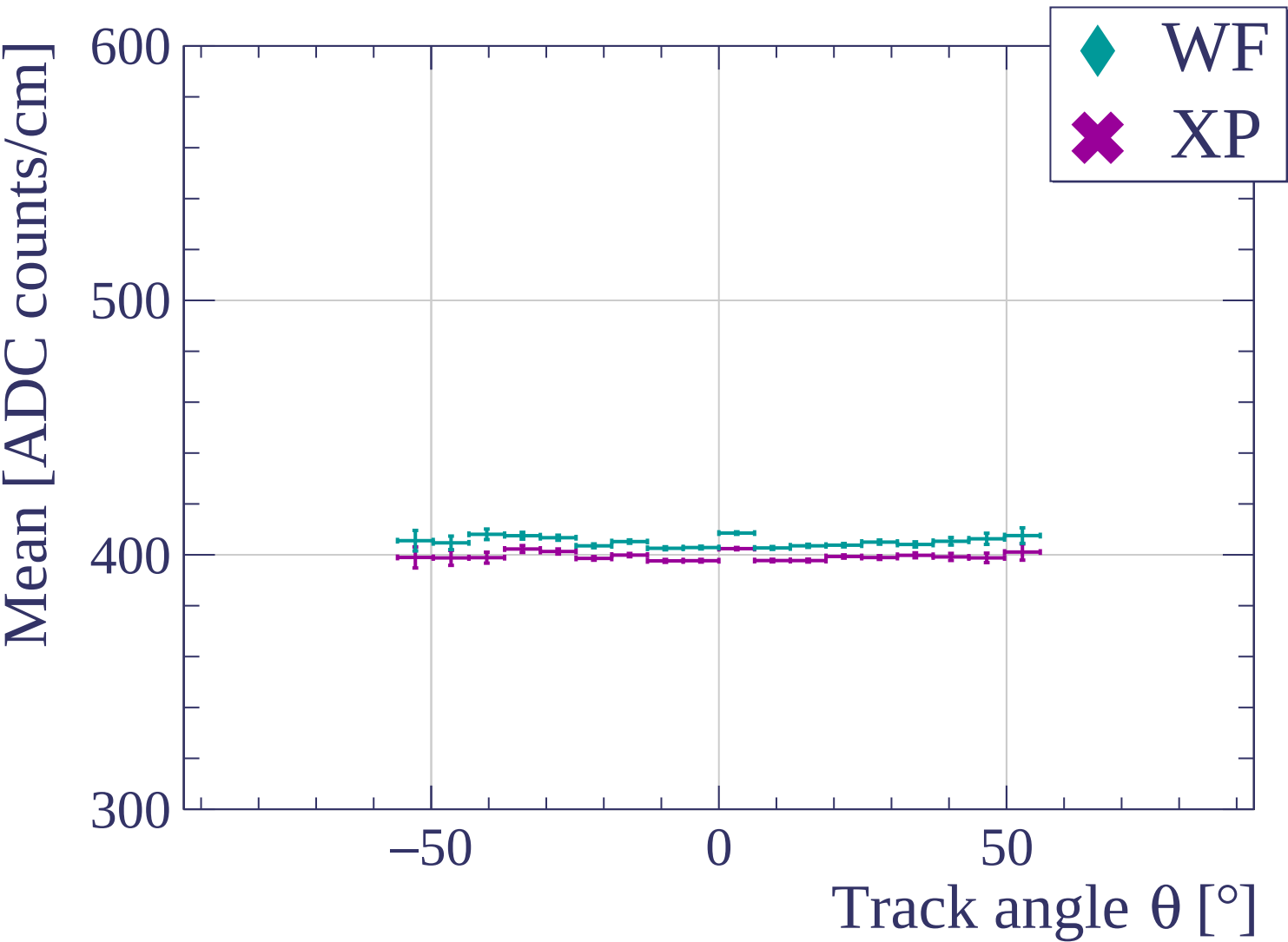


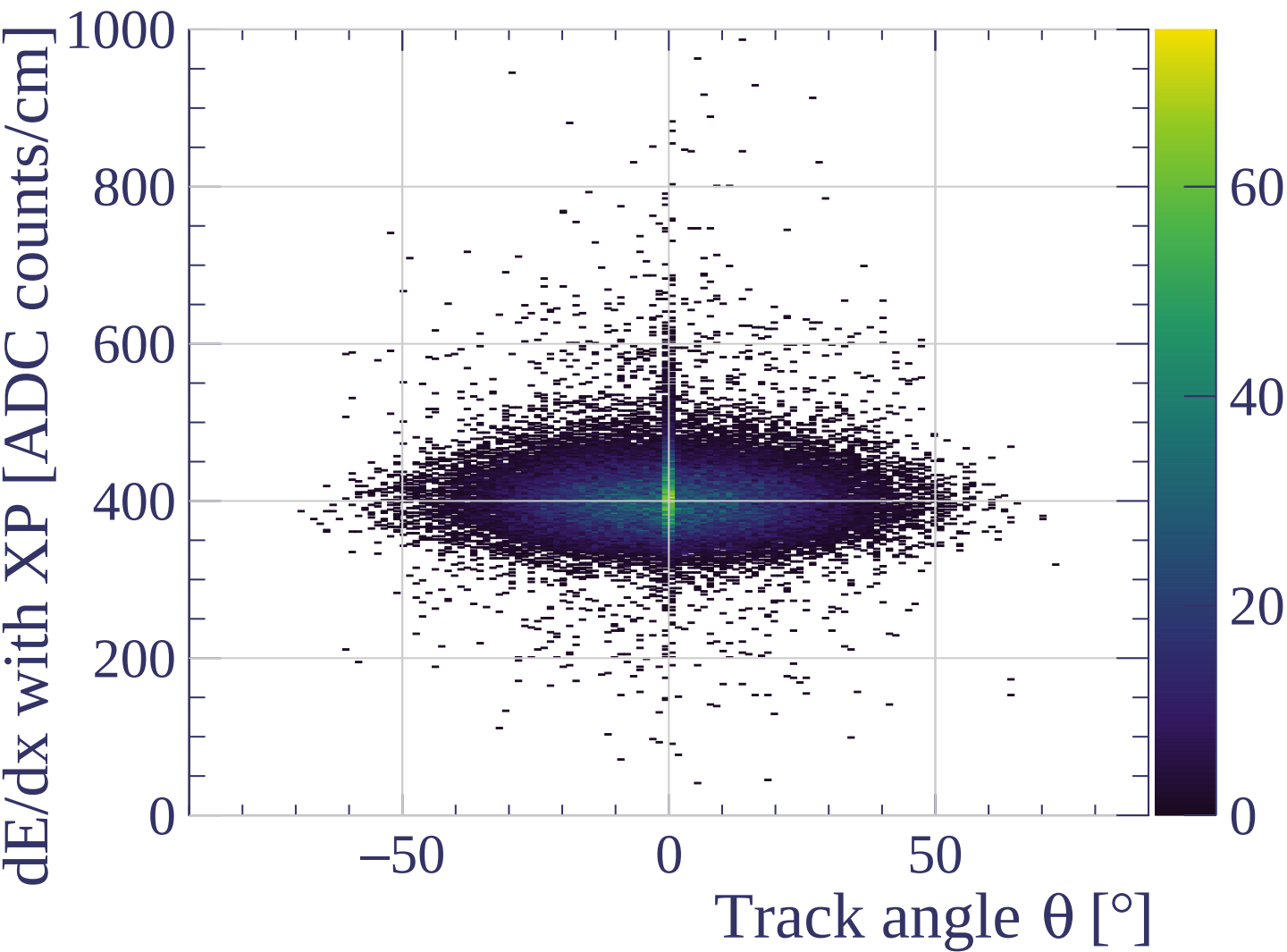




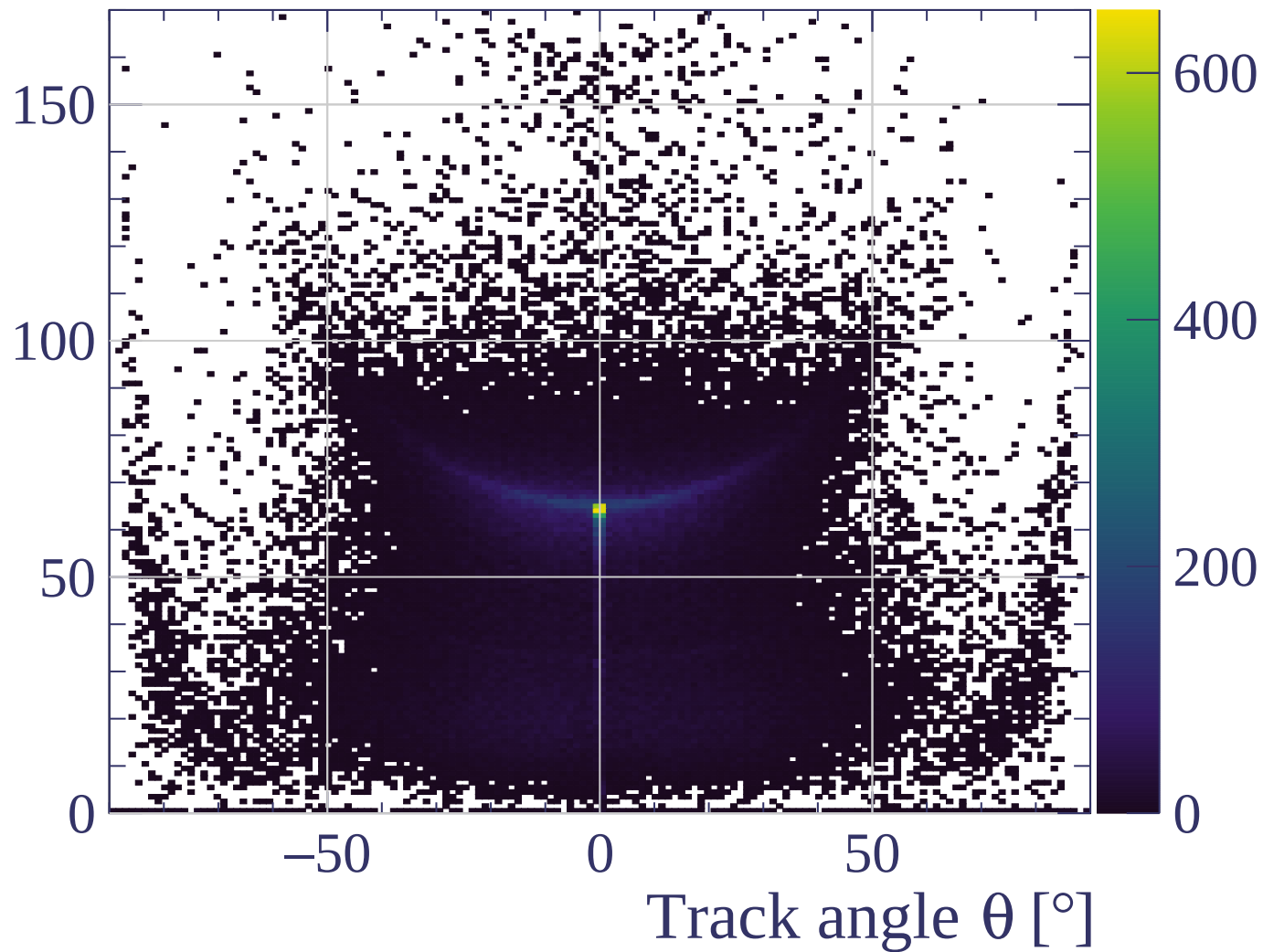


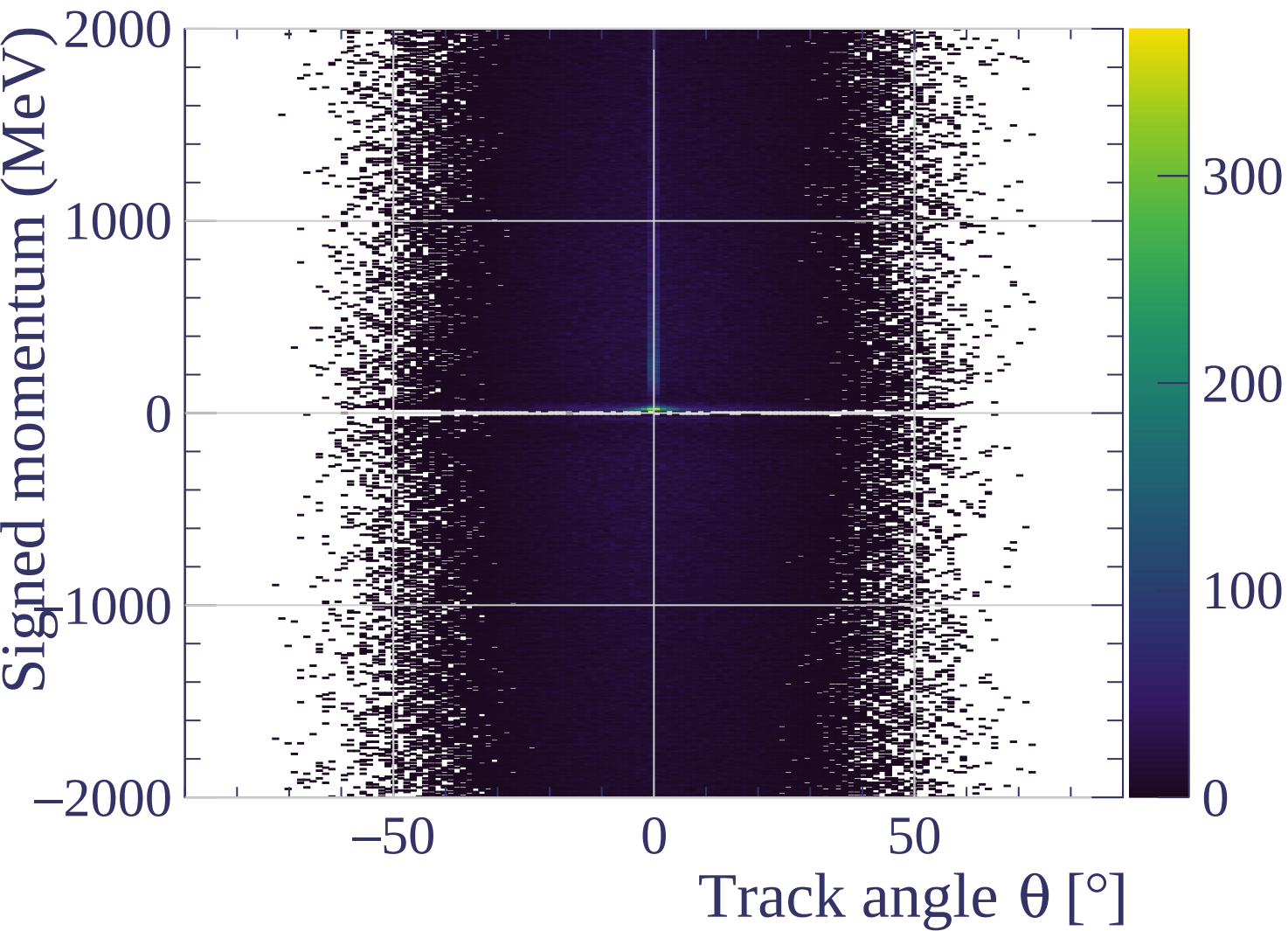


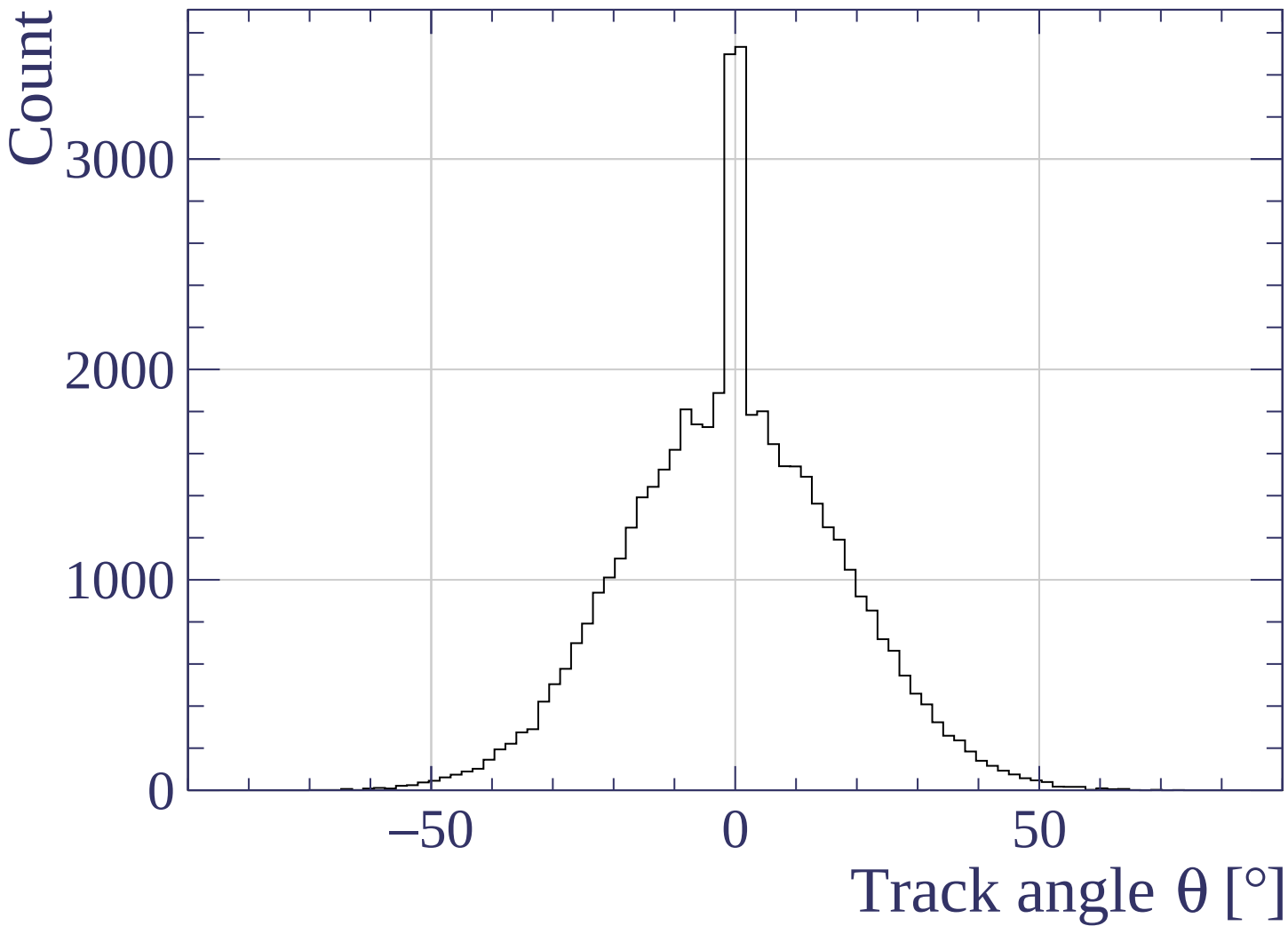




Track length [cm]

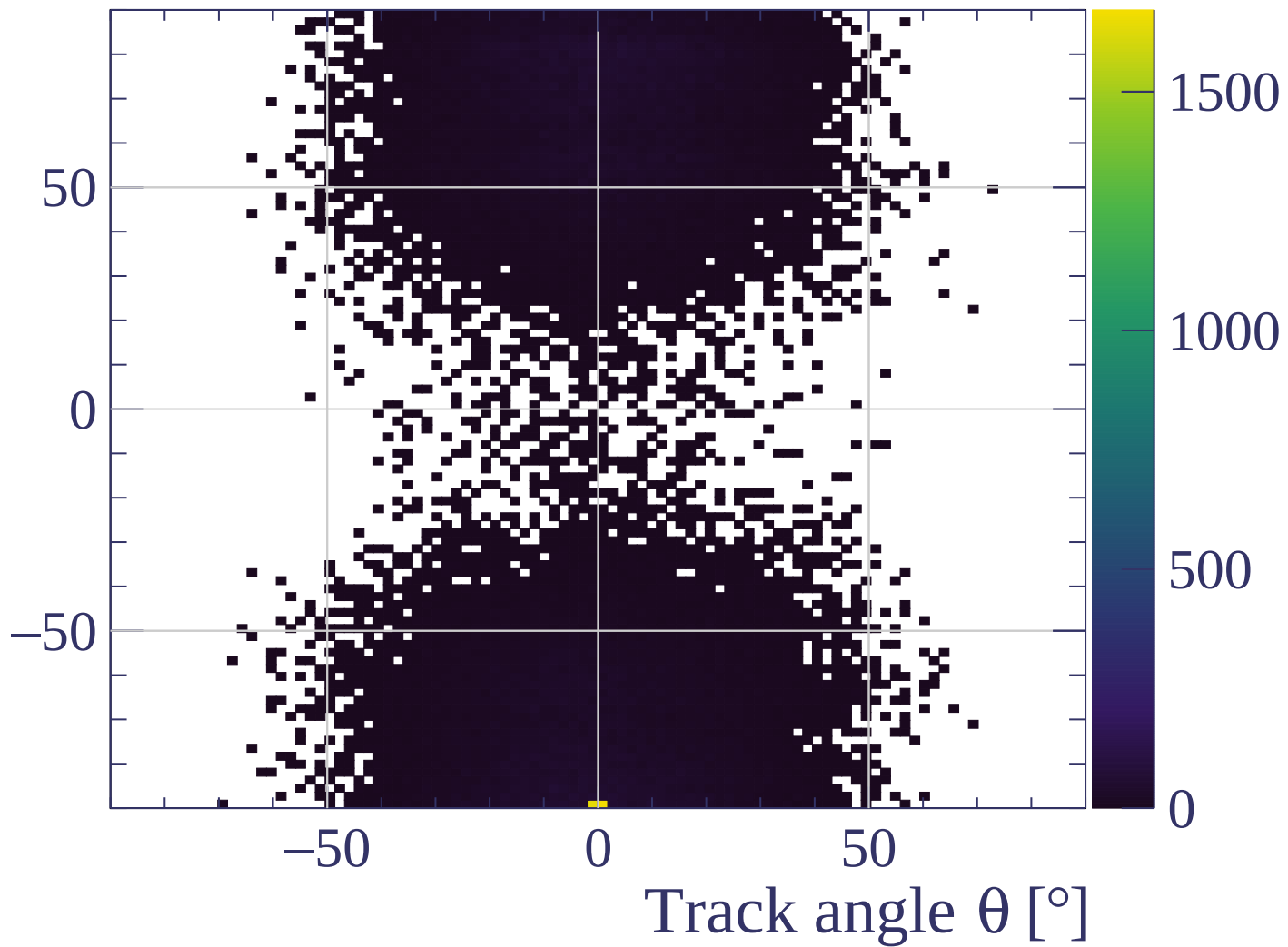




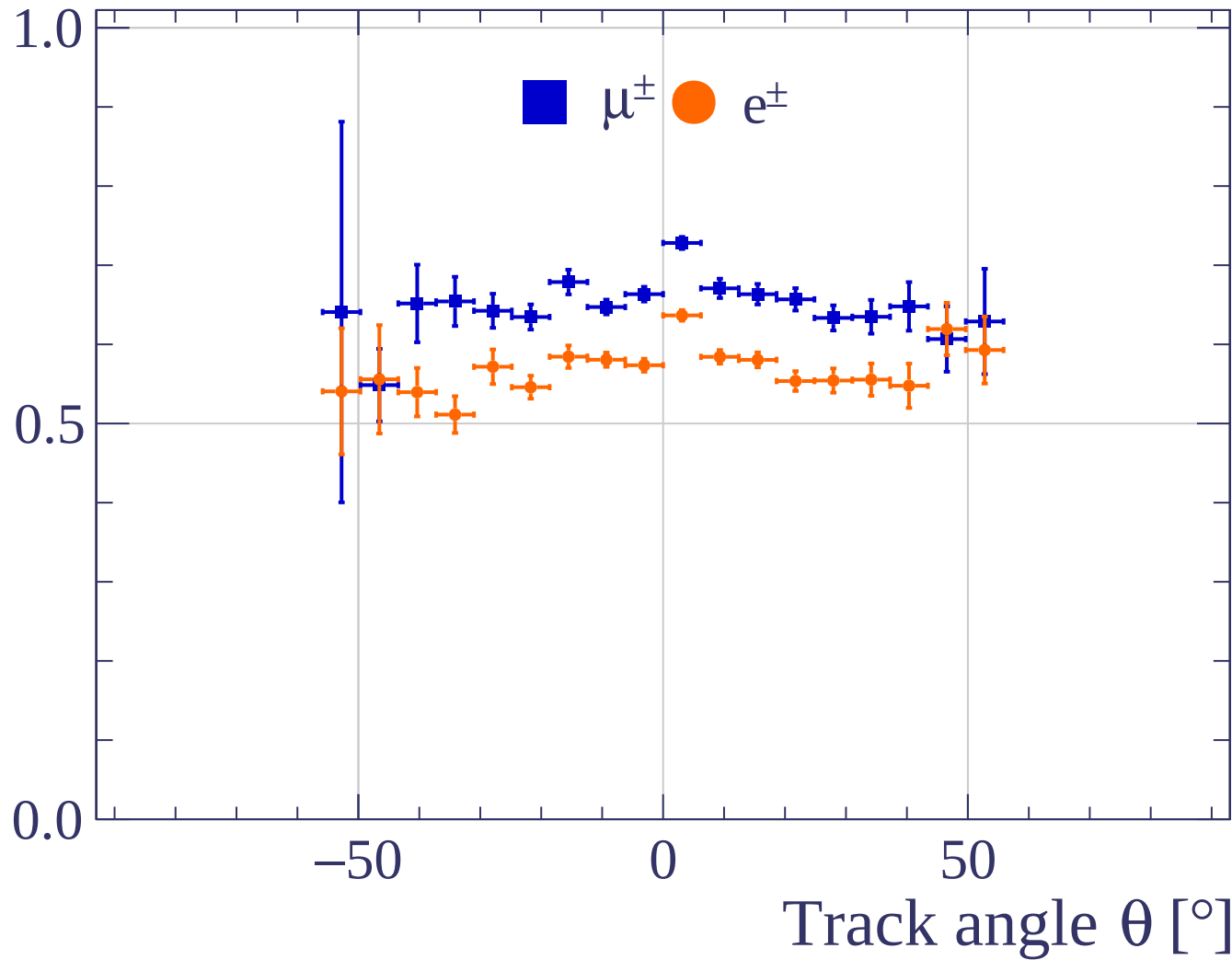




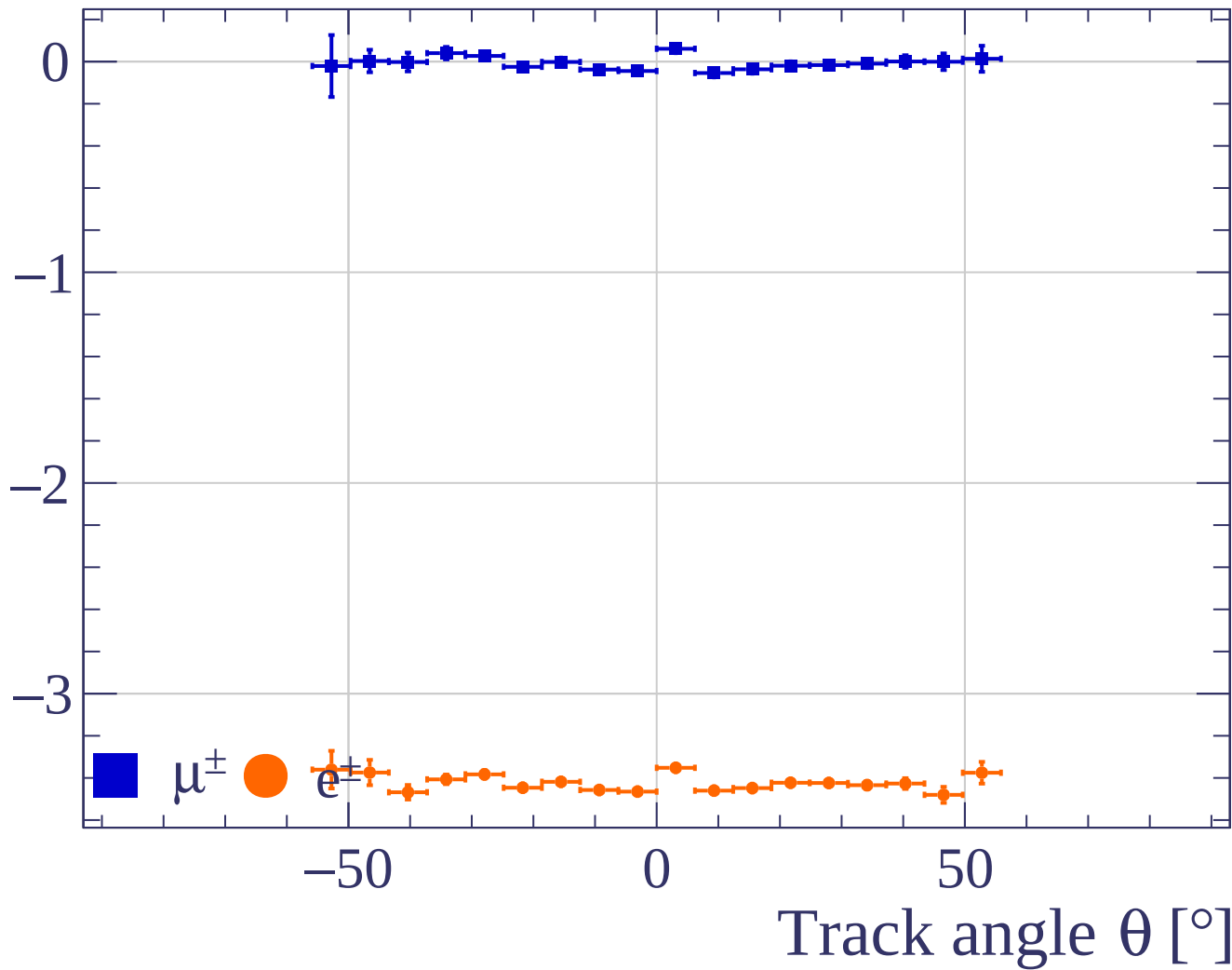
Track angle  $\phi$  [°]



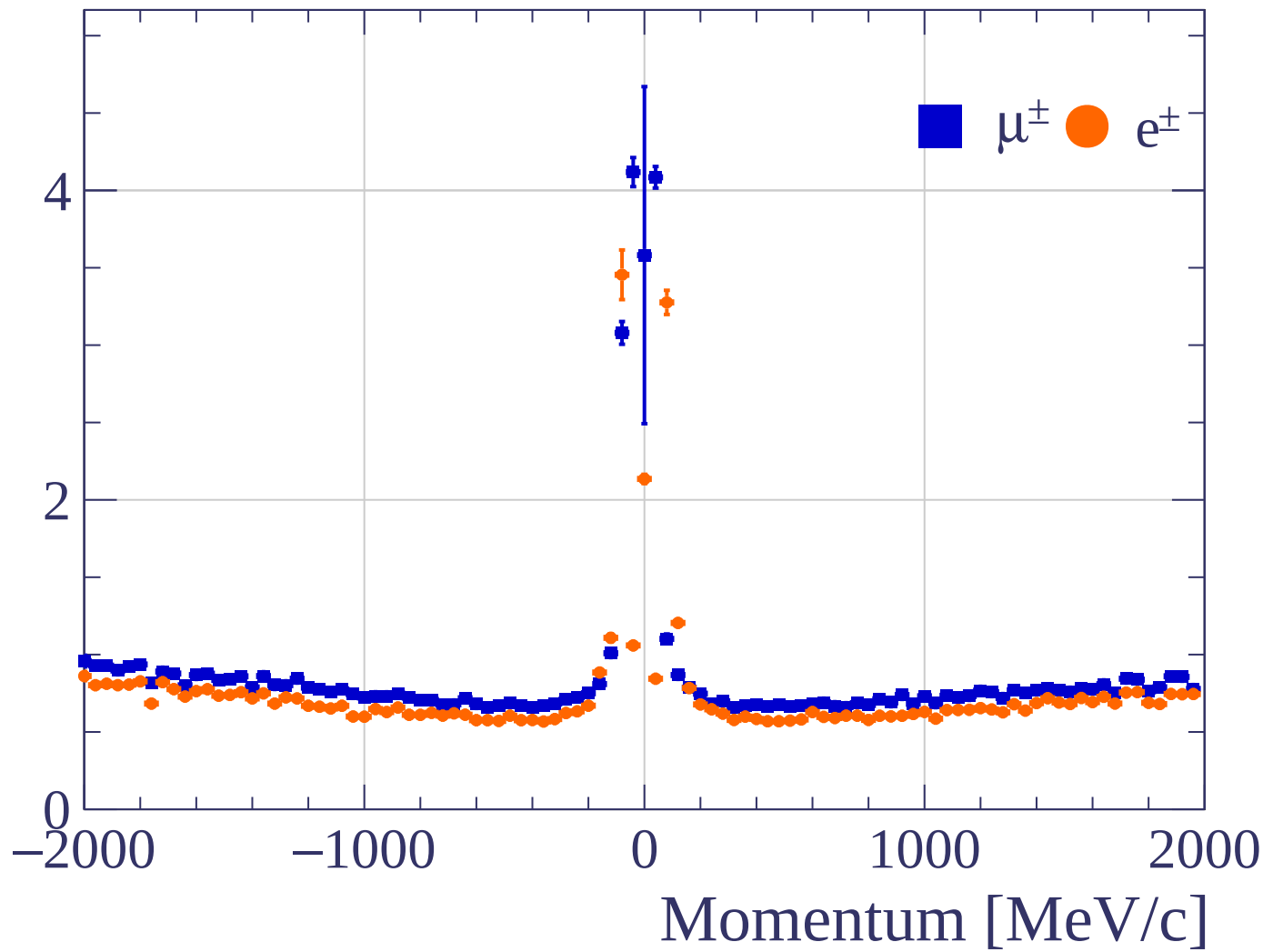
Pulls standard deviation



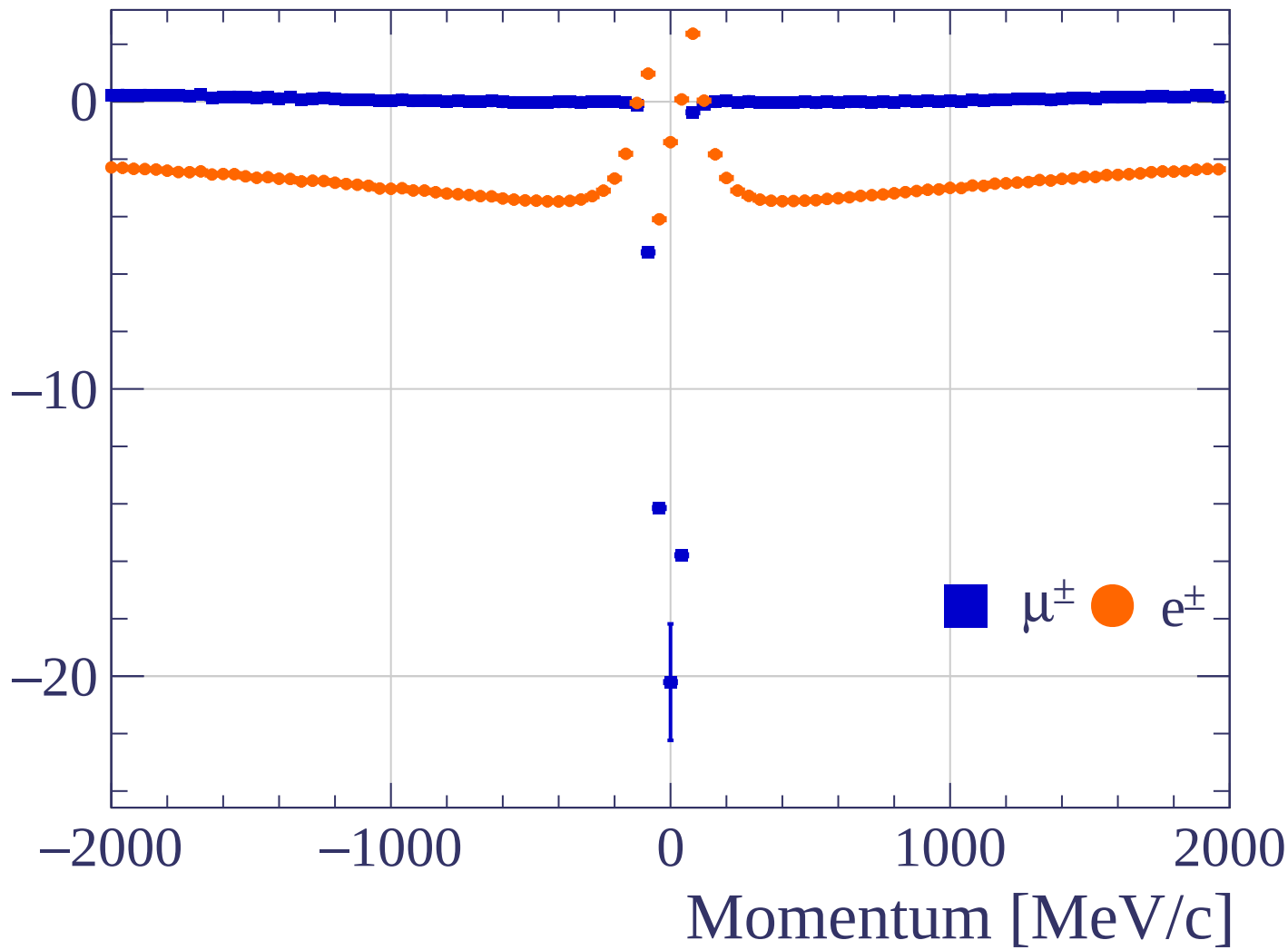
Pulls mean

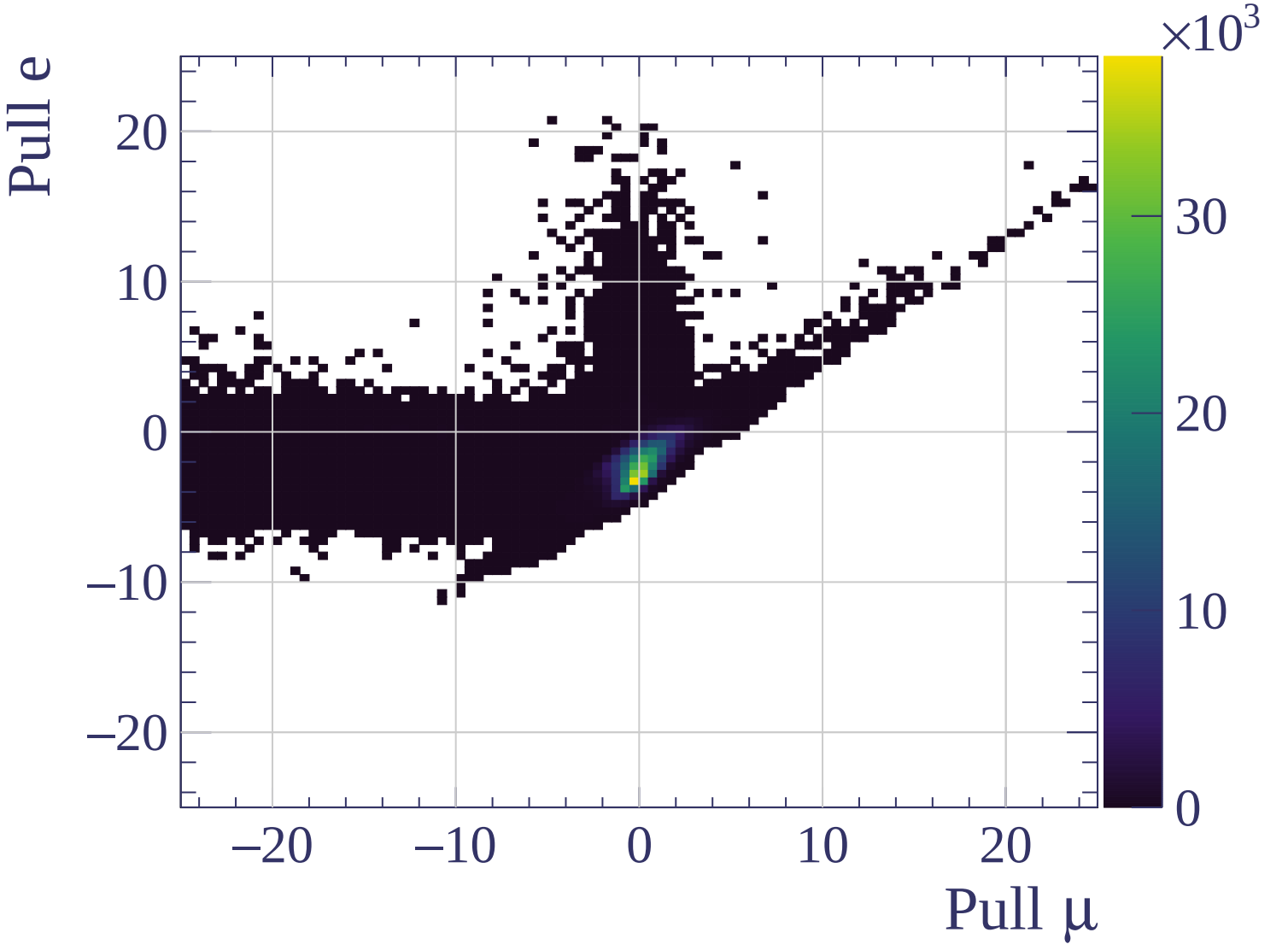


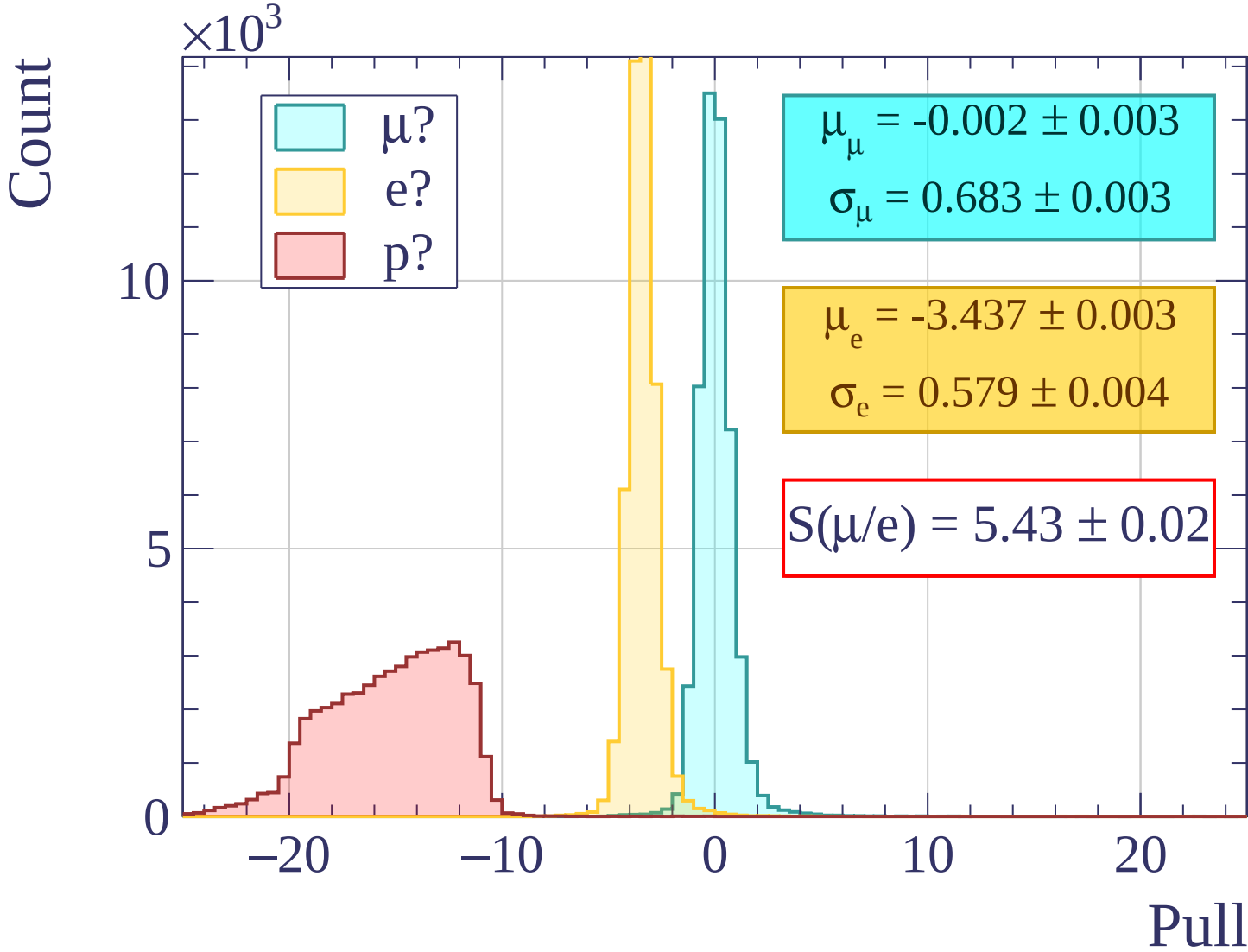
Pulls standard deviation

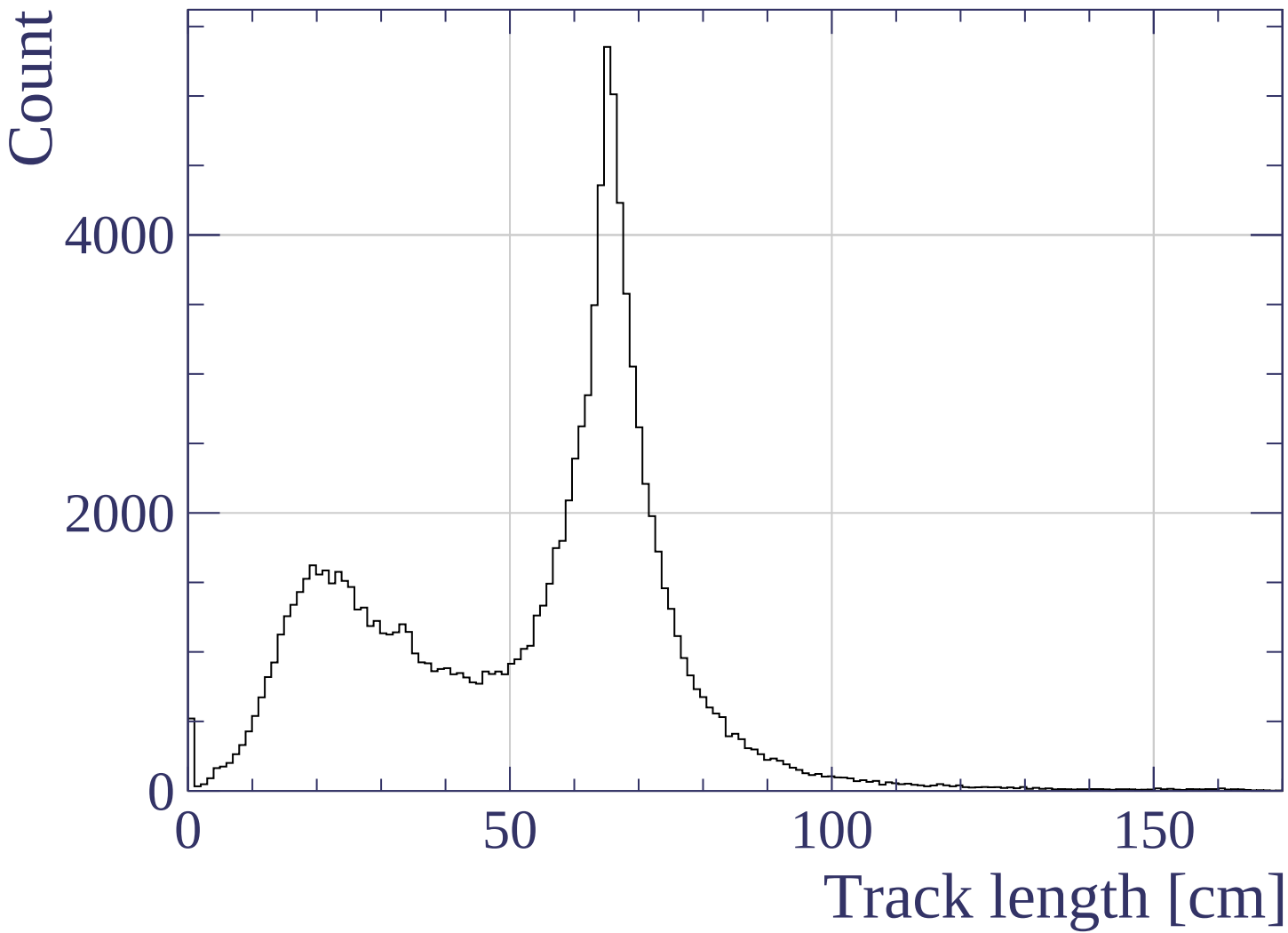


Pulls mean



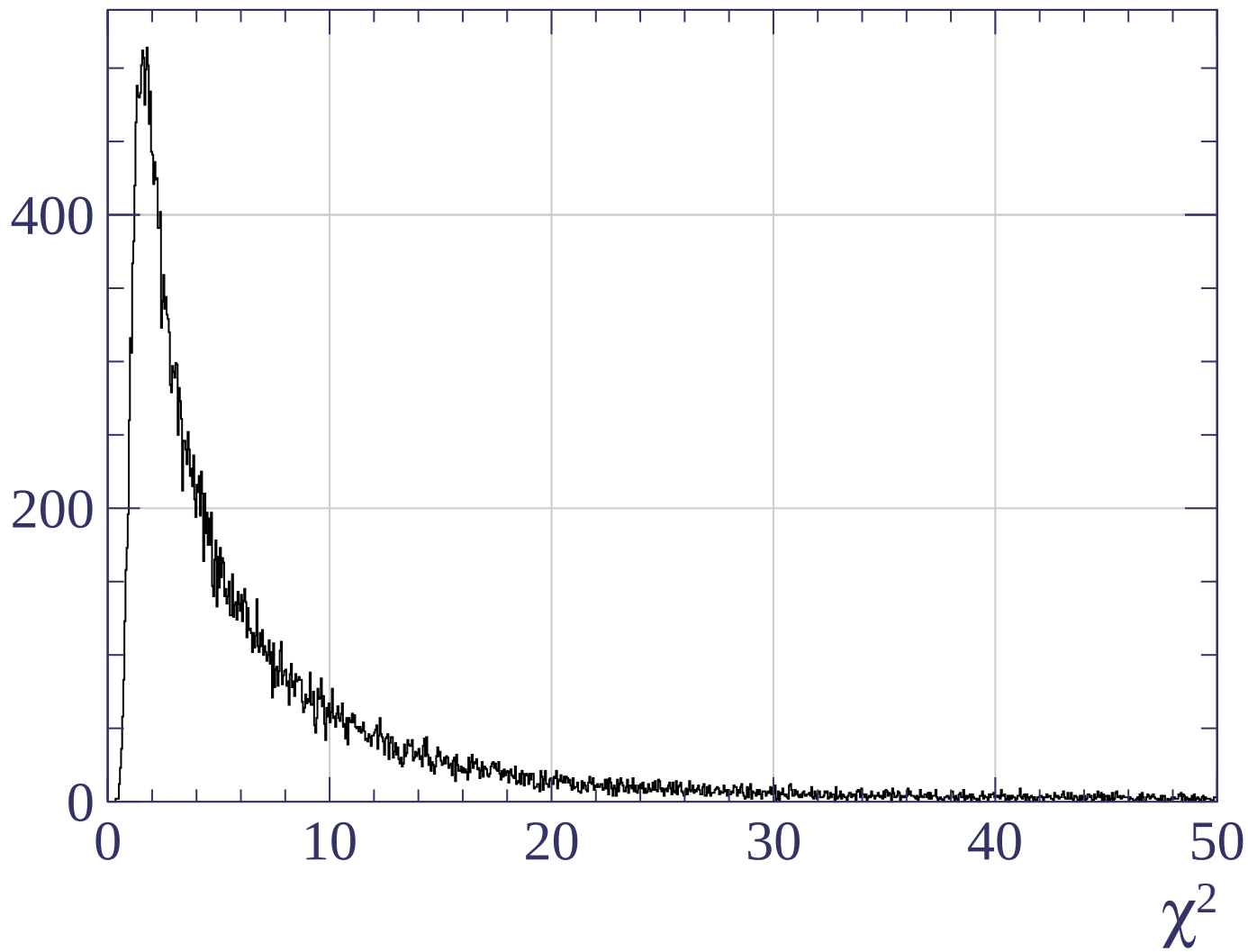




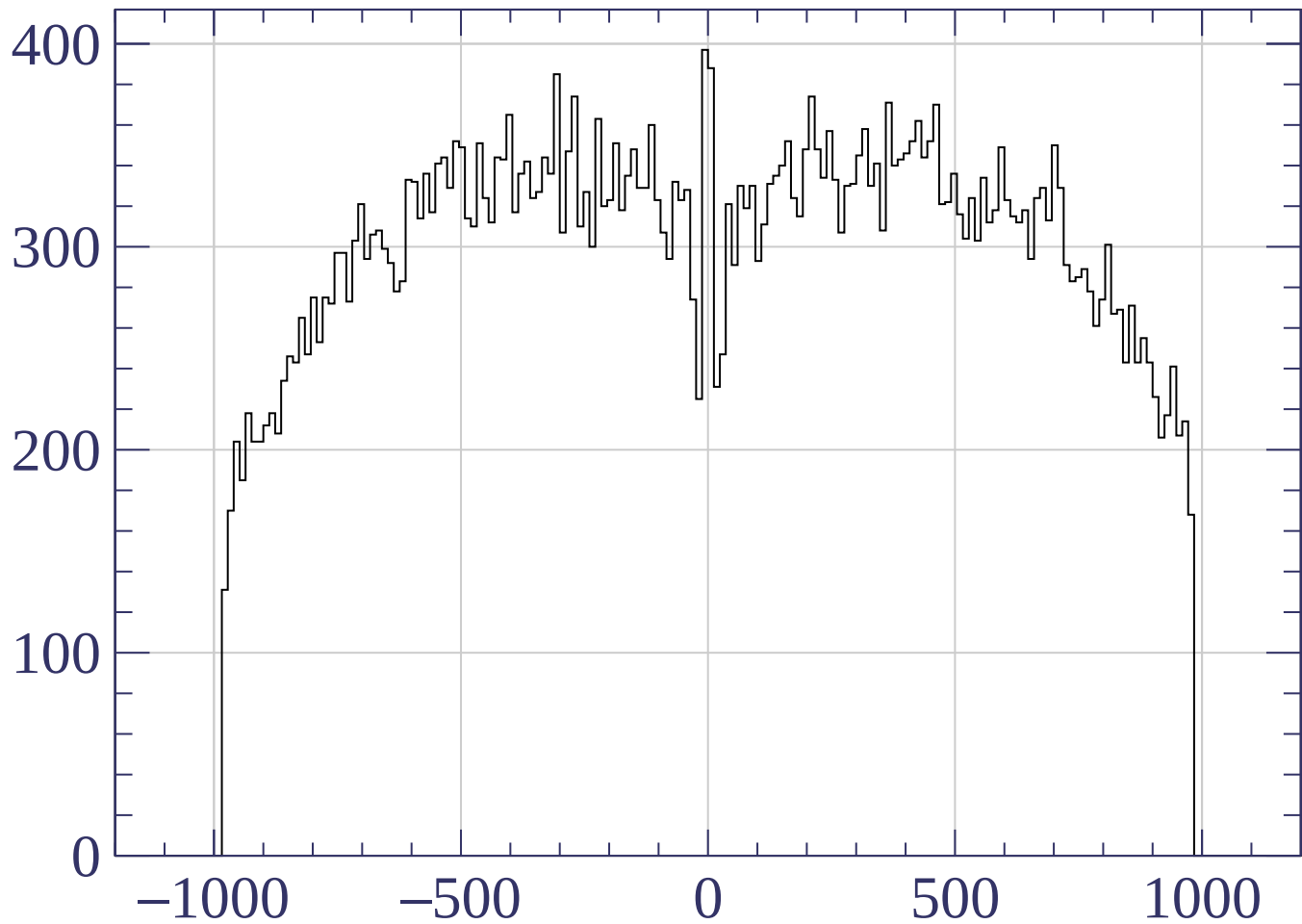




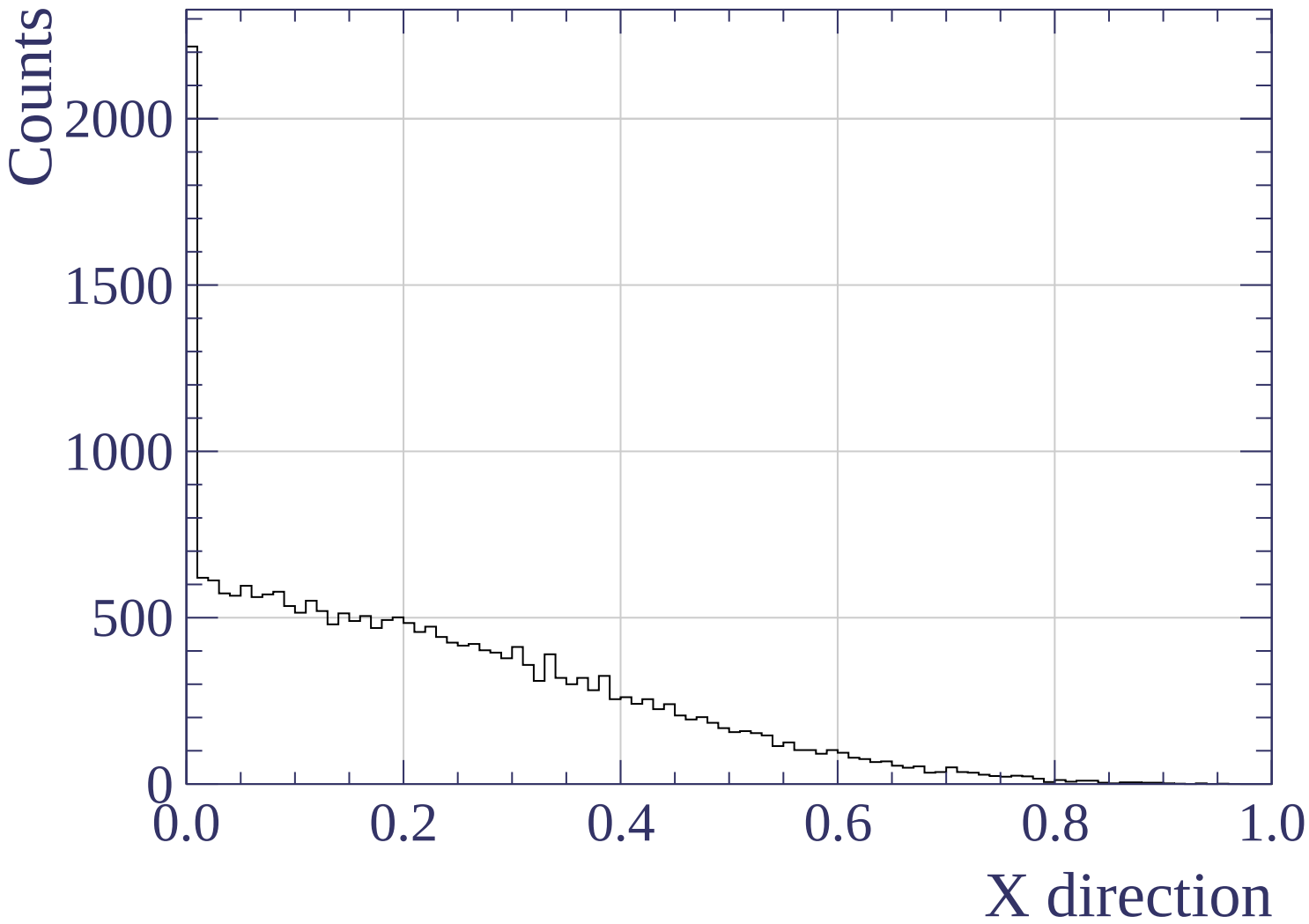
Count

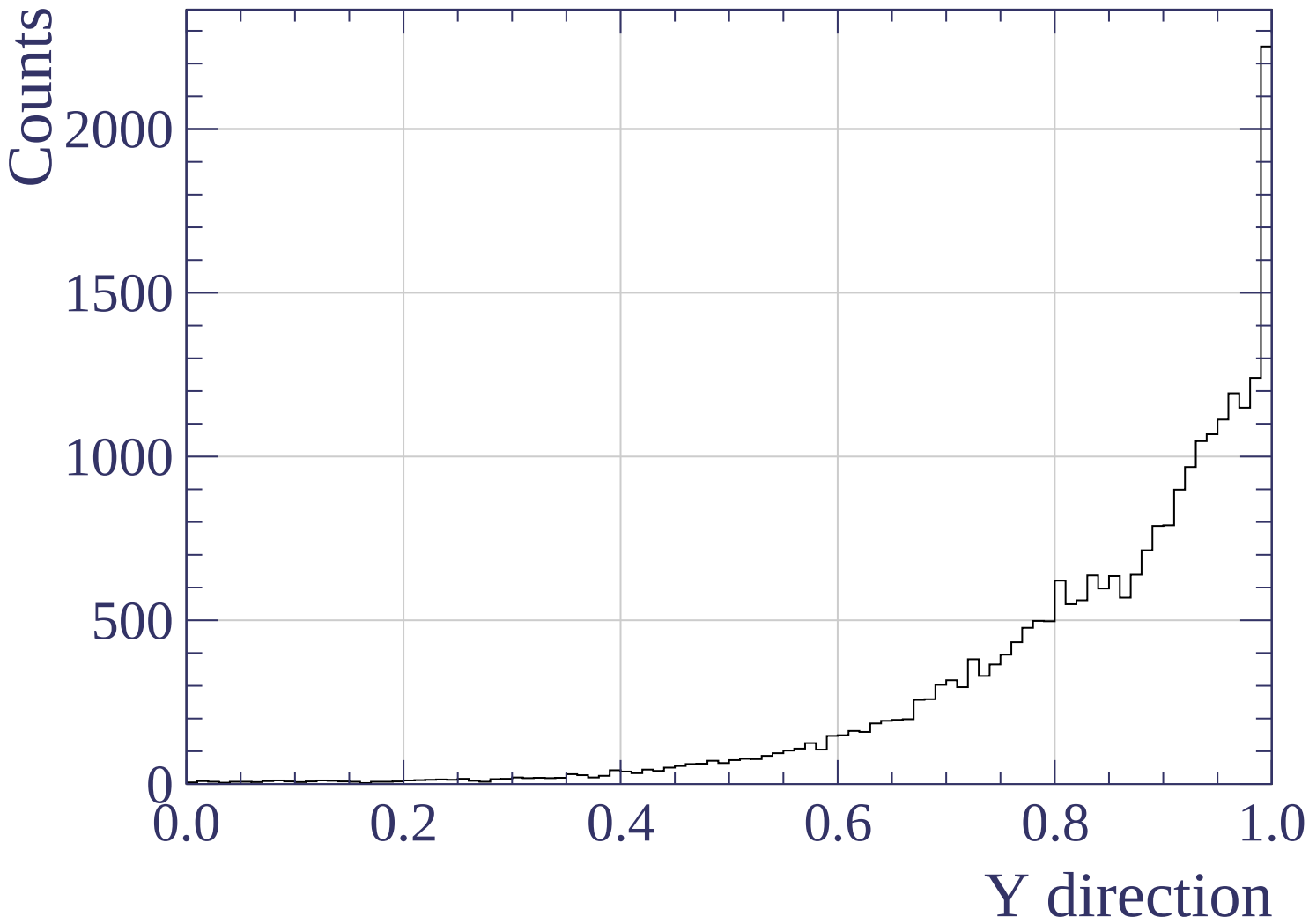


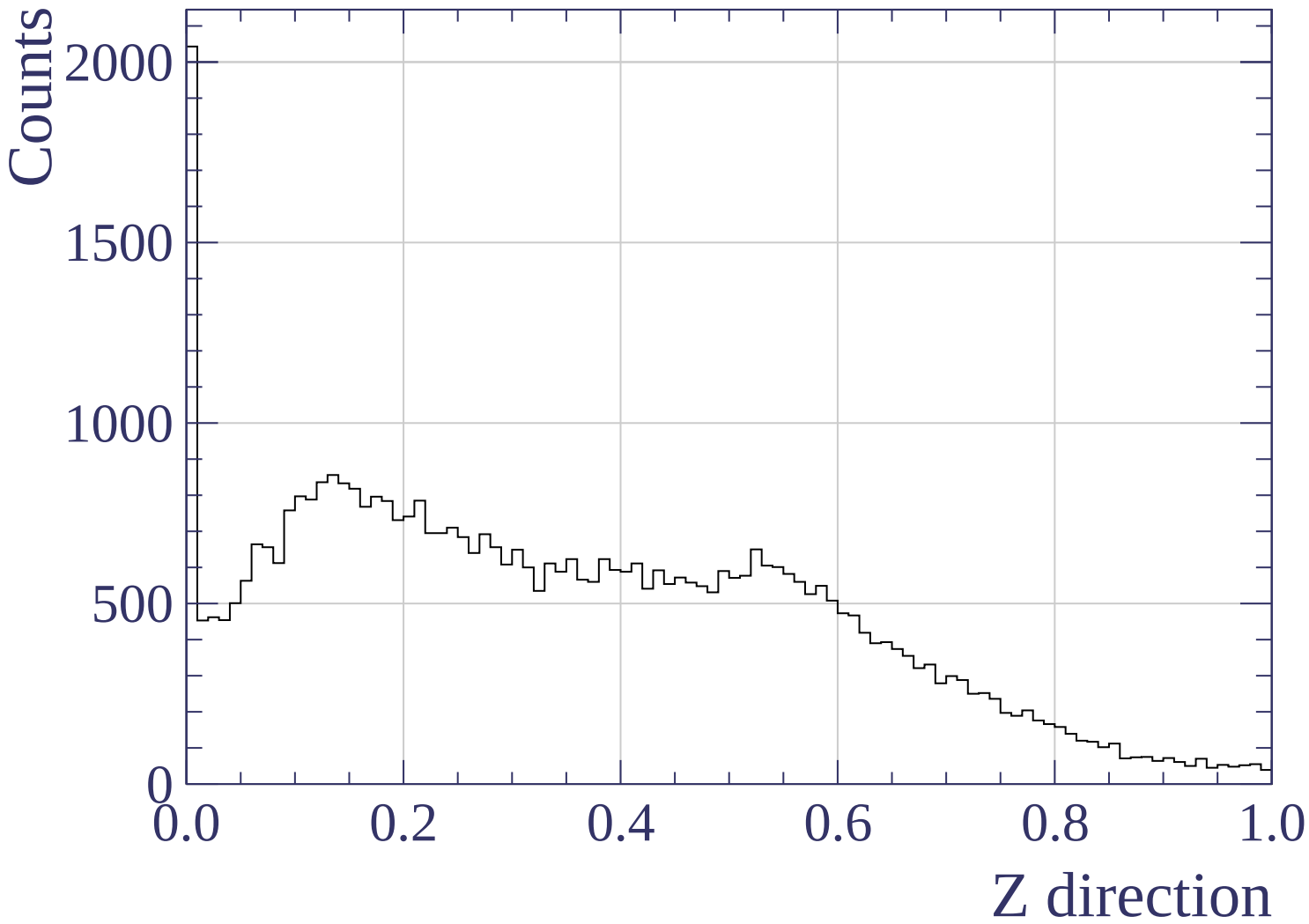
Count

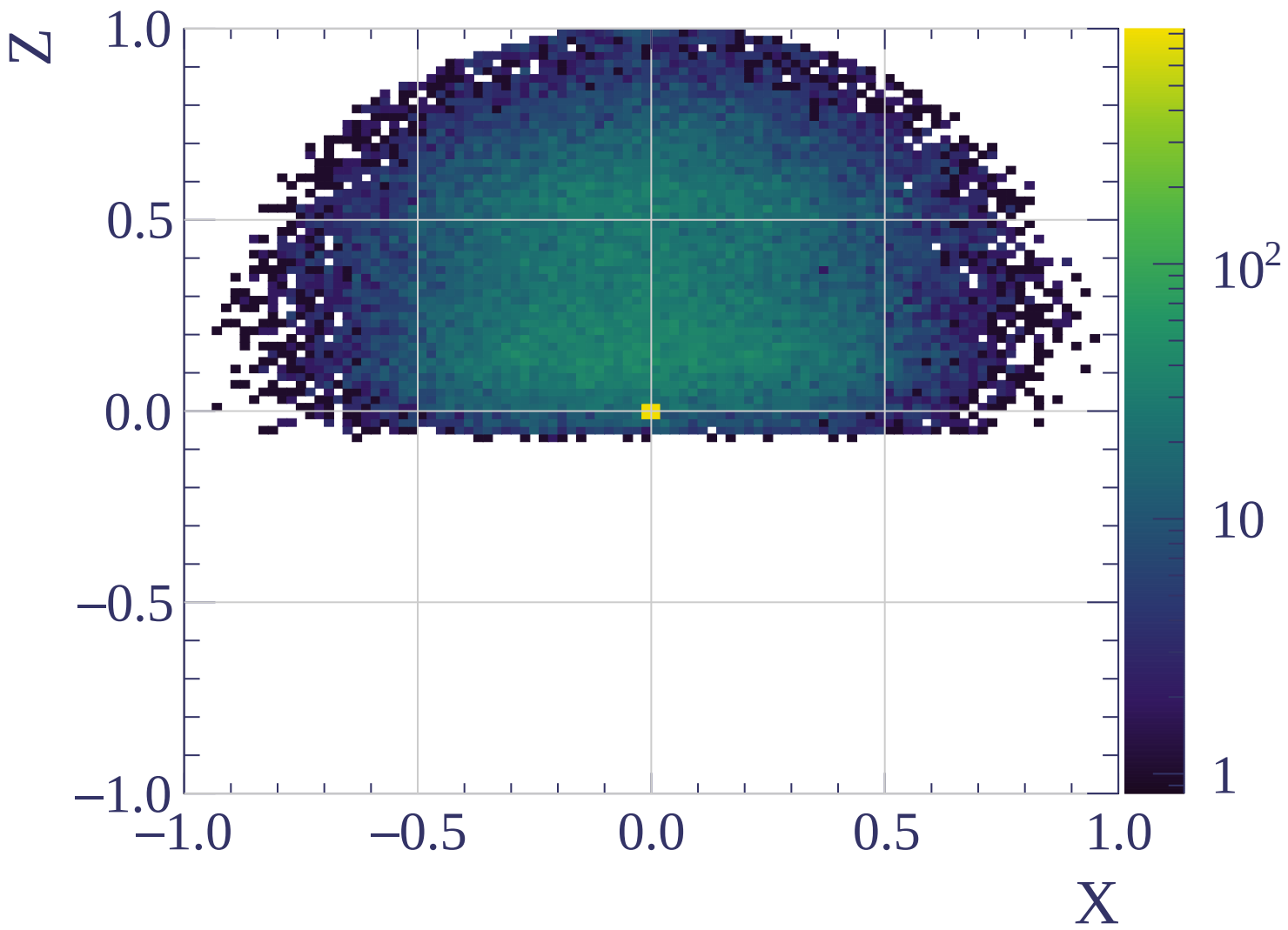


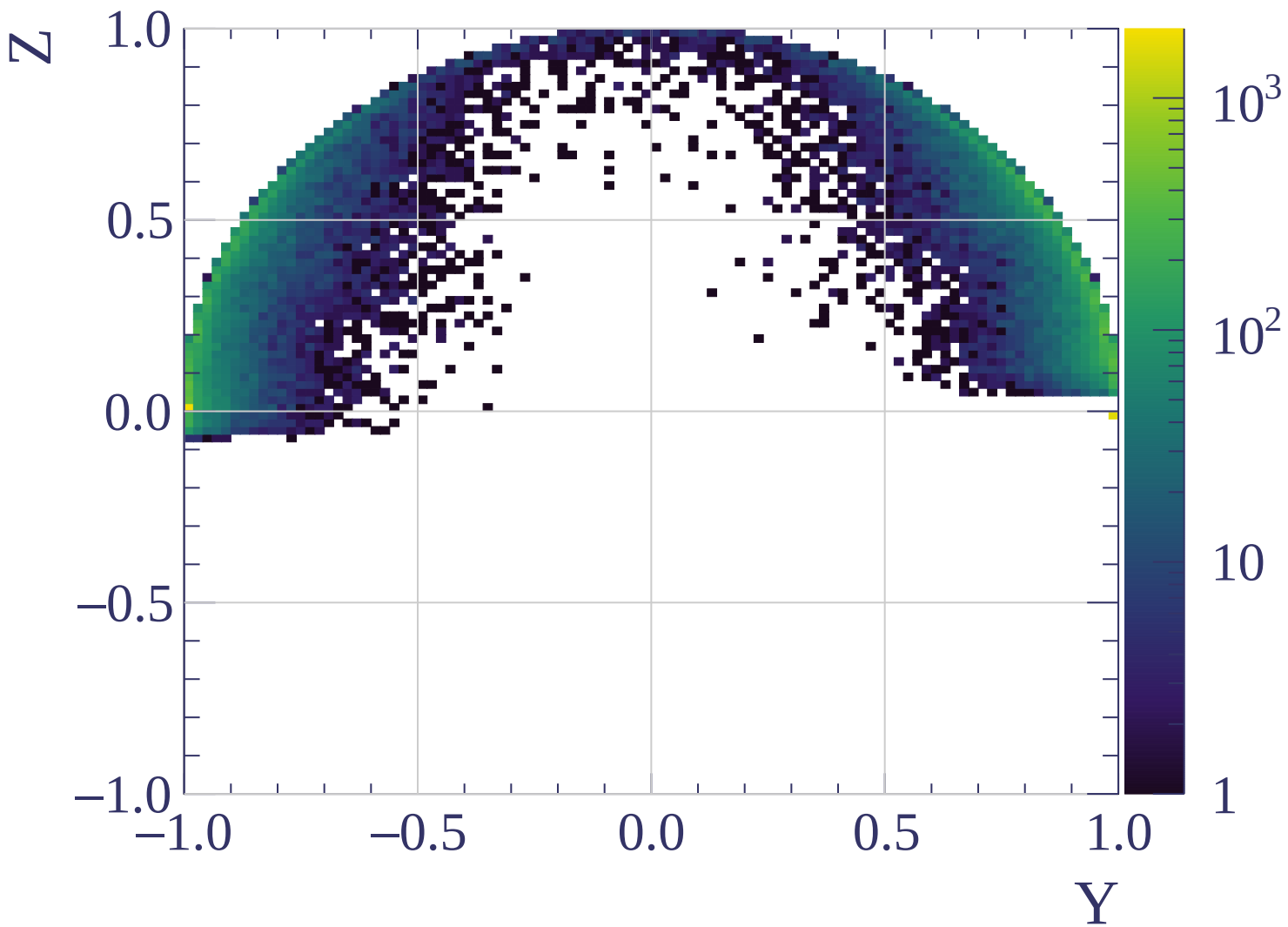
Track X position [mm]

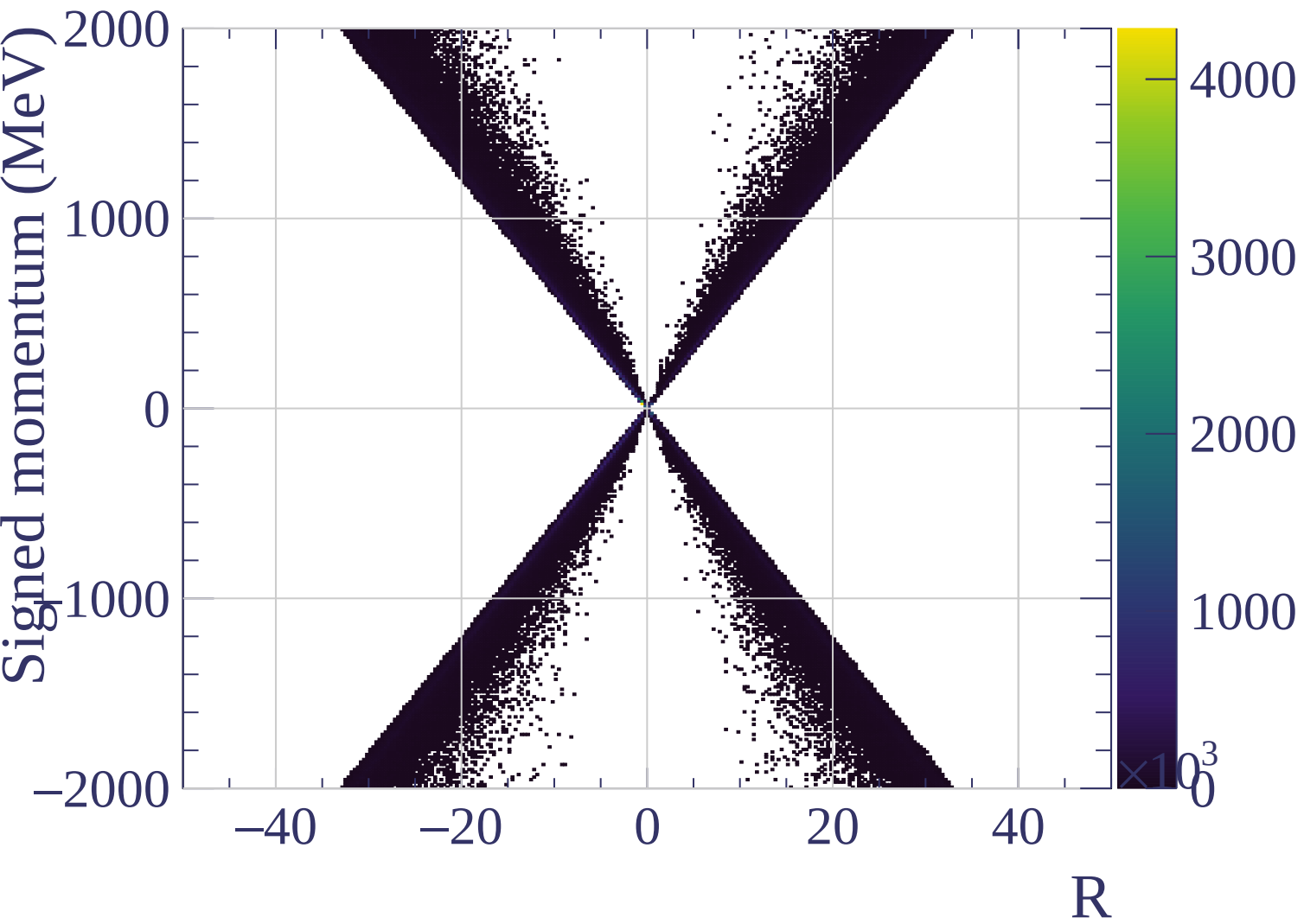




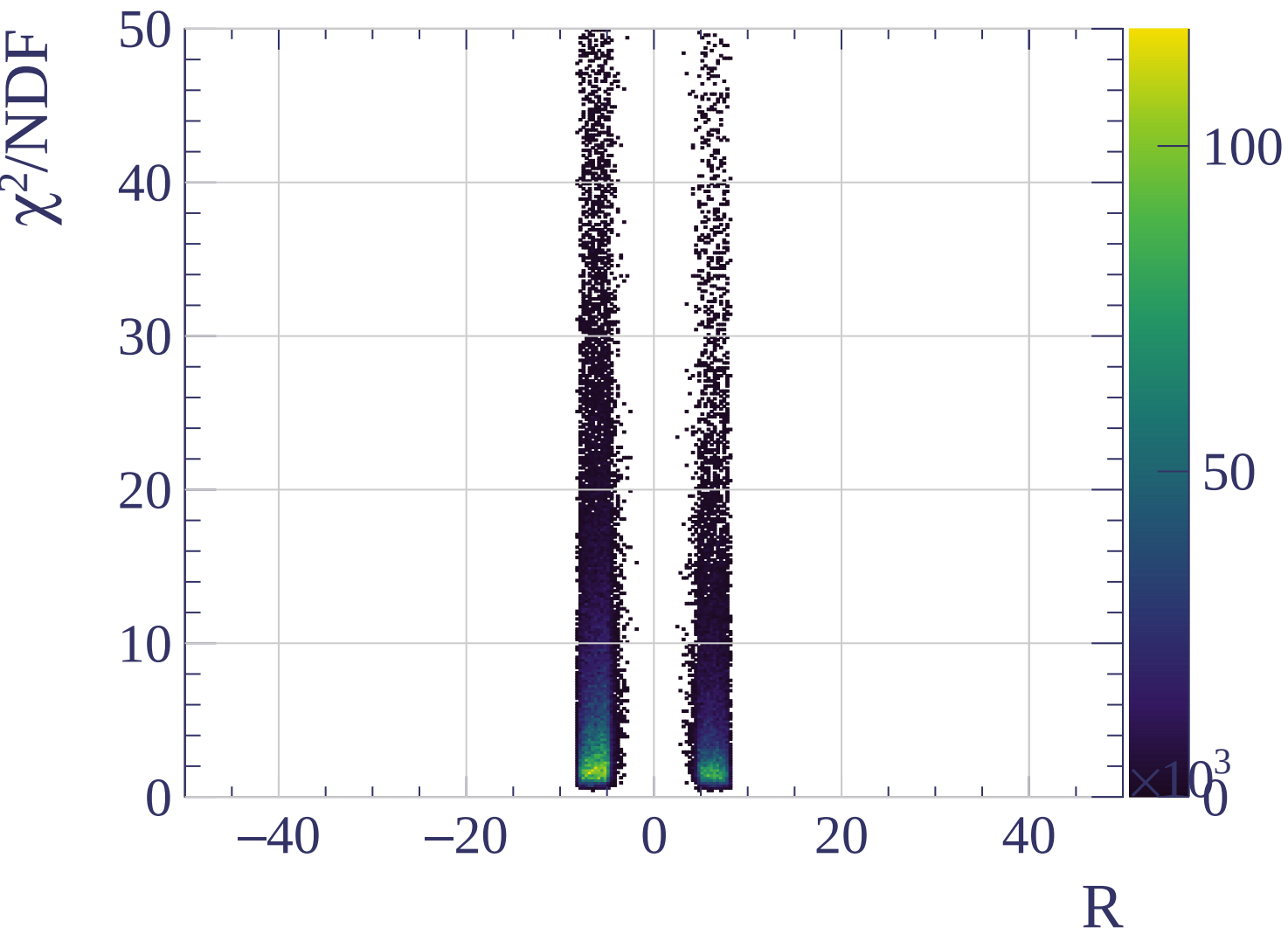












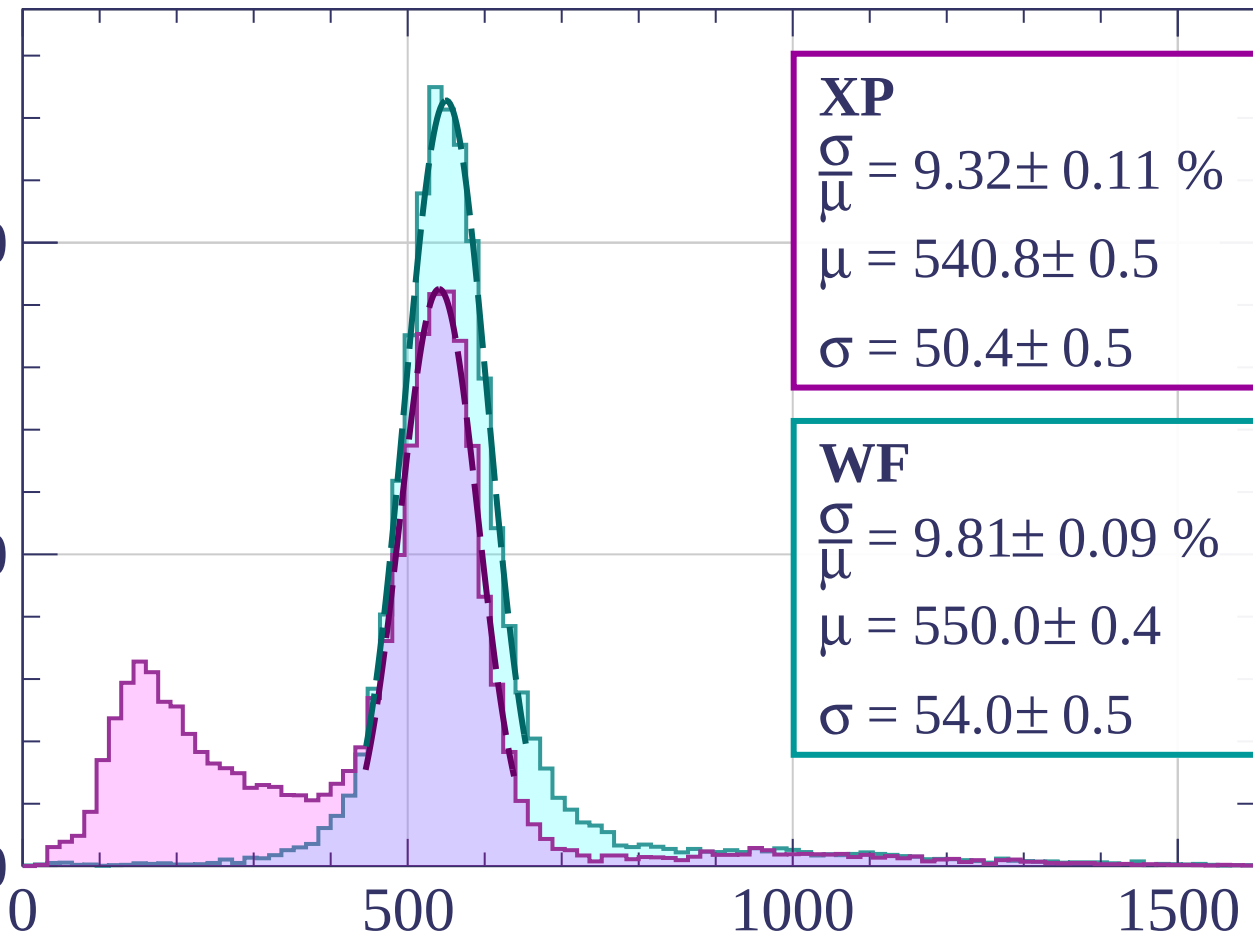
Energy loss |  $0 < p < 80$

Count

2000

1000

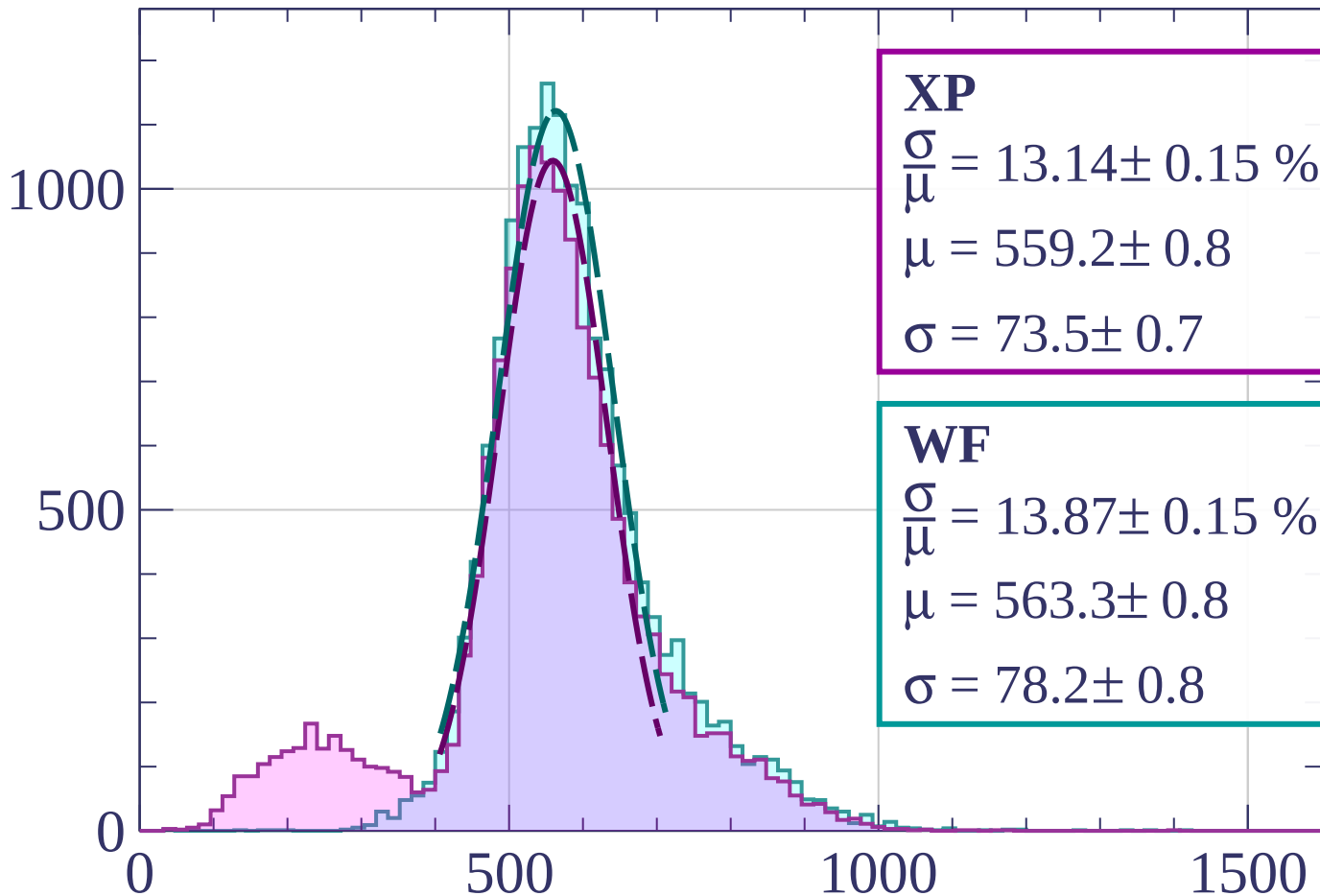
0



dE/dx [ADC counts/cm]

Energy loss |  $80 < p < 160$

Count



**XP**

$$\frac{\sigma}{\mu} = 13.14 \pm 0.15 \%$$

$$\mu = 559.2 \pm 0.8$$

$$\sigma = 73.5 \pm 0.7$$

**WF**

$$\frac{\sigma}{\mu} = 13.87 \pm 0.15 \%$$

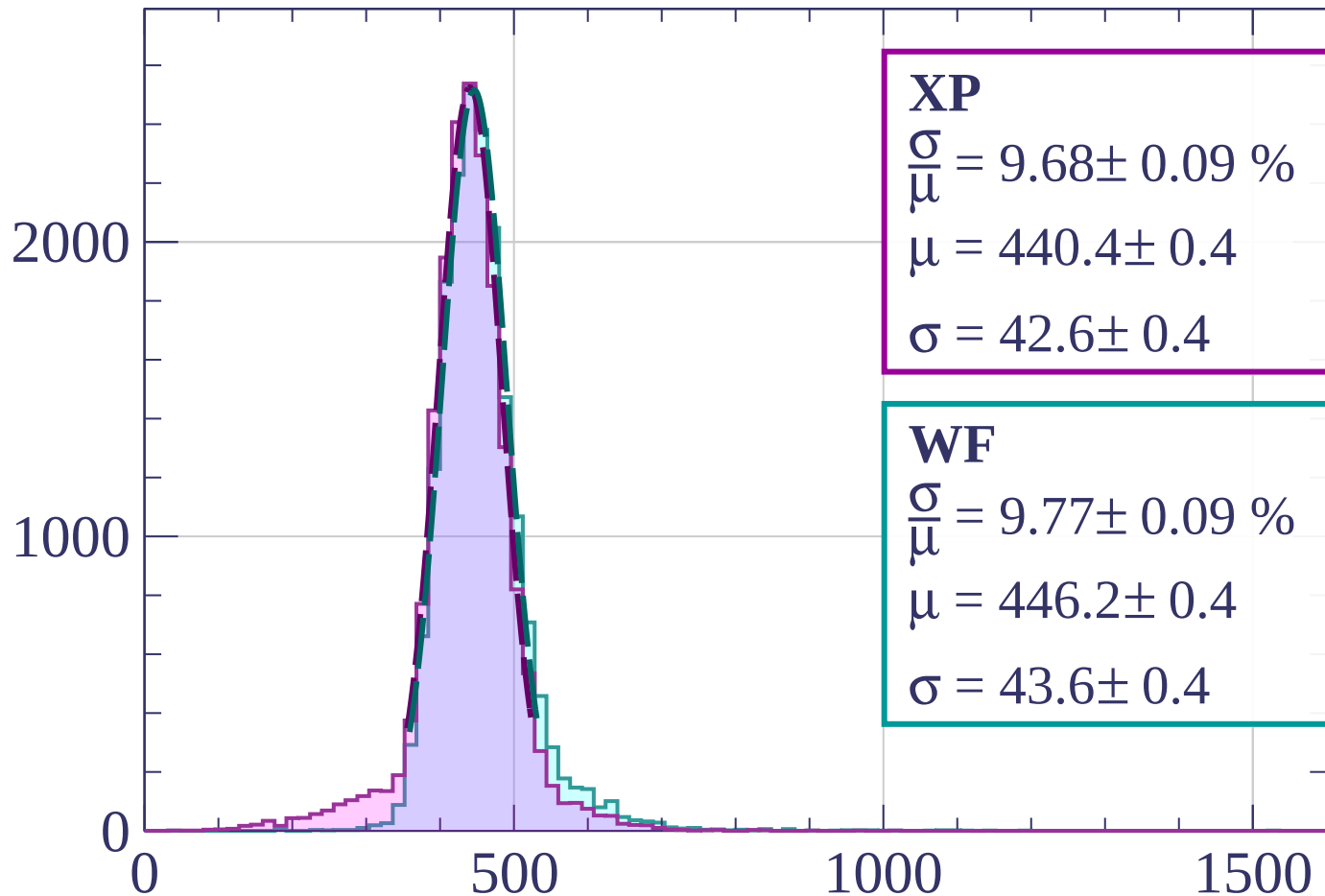
$$\mu = 563.3 \pm 0.8$$

$$\sigma = 78.2 \pm 0.8$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $160 < p < 240$

Count



**XP**

$$\frac{\sigma}{\mu} = 9.68 \pm 0.09 \%$$

$$\mu = 440.4 \pm 0.4$$

$$\sigma = 42.6 \pm 0.4$$

**WF**

$$\frac{\sigma}{\mu} = 9.77 \pm 0.09 \%$$

$$\mu = 446.2 \pm 0.4$$

$$\sigma = 43.6 \pm 0.4$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $240 < p < 320$

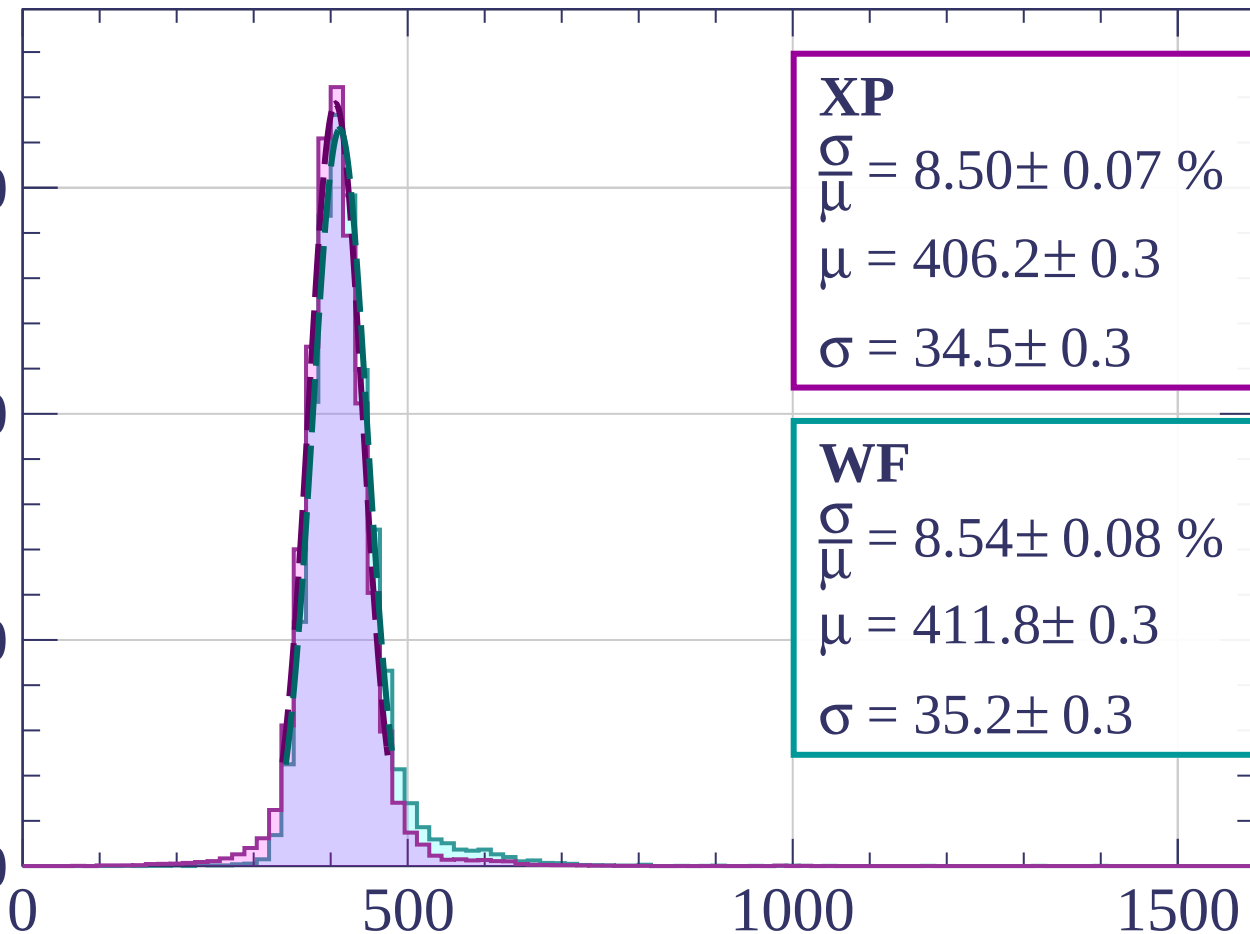
Count

3000

2000

1000

0



**XP**

$$\frac{\sigma}{\mu} = 8.50 \pm 0.07 \%$$

$$\mu = 406.2 \pm 0.3$$

$$\sigma = 34.5 \pm 0.3$$

**WF**

$$\frac{\sigma}{\mu} = 8.54 \pm 0.08 \%$$

$$\mu = 411.8 \pm 0.3$$

$$\sigma = 35.2 \pm 0.3$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $320 < p < 400$

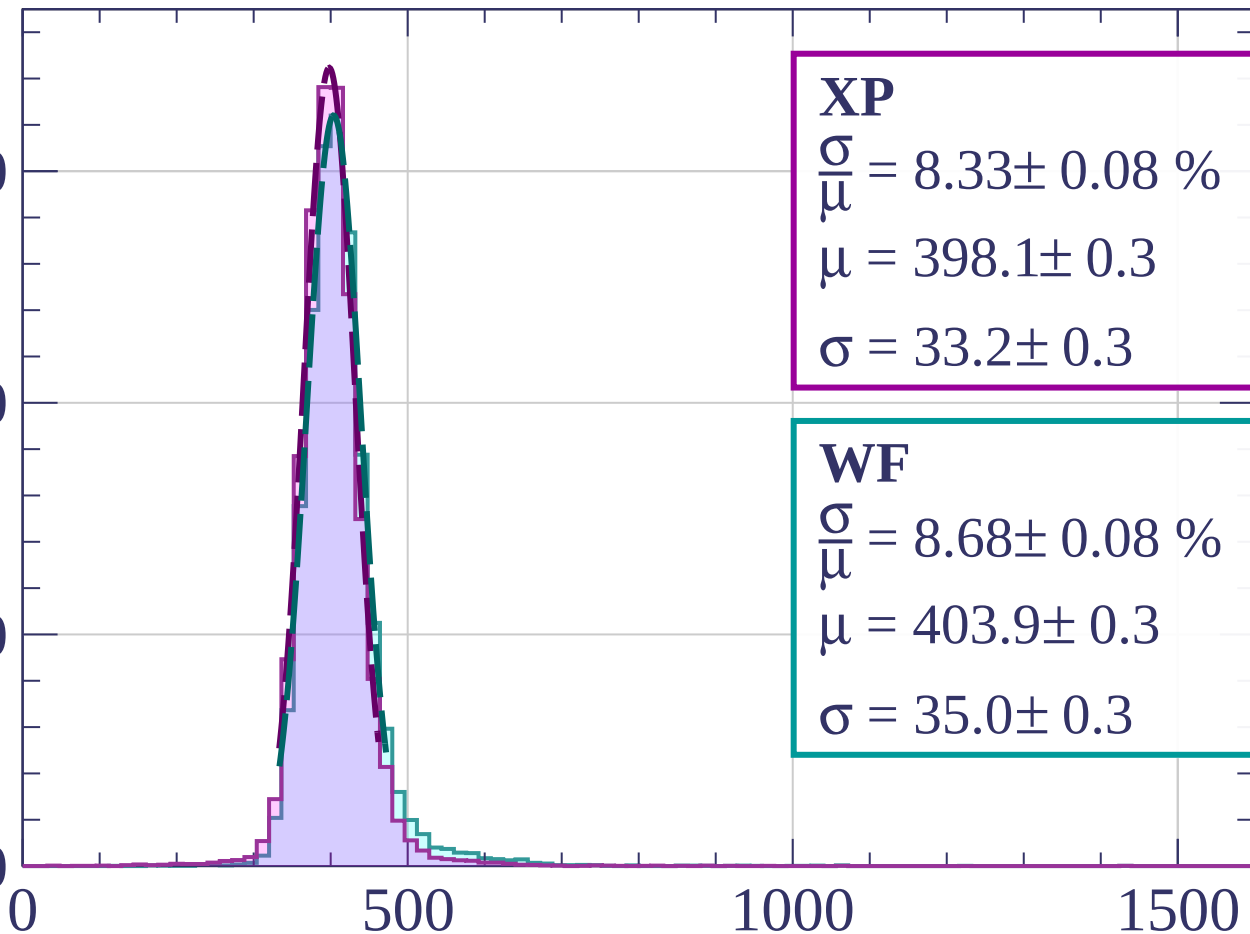
Count

3000

2000

1000

0



**XP**

$$\frac{\sigma}{\mu} = 8.33 \pm 0.08 \%$$

$$\mu = 398.1 \pm 0.3$$

$$\sigma = 33.2 \pm 0.3$$

**WF**

$$\frac{\sigma}{\mu} = 8.68 \pm 0.08 \%$$

$$\mu = 403.9 \pm 0.3$$

$$\sigma = 35.0 \pm 0.3$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $400 < p < 480$

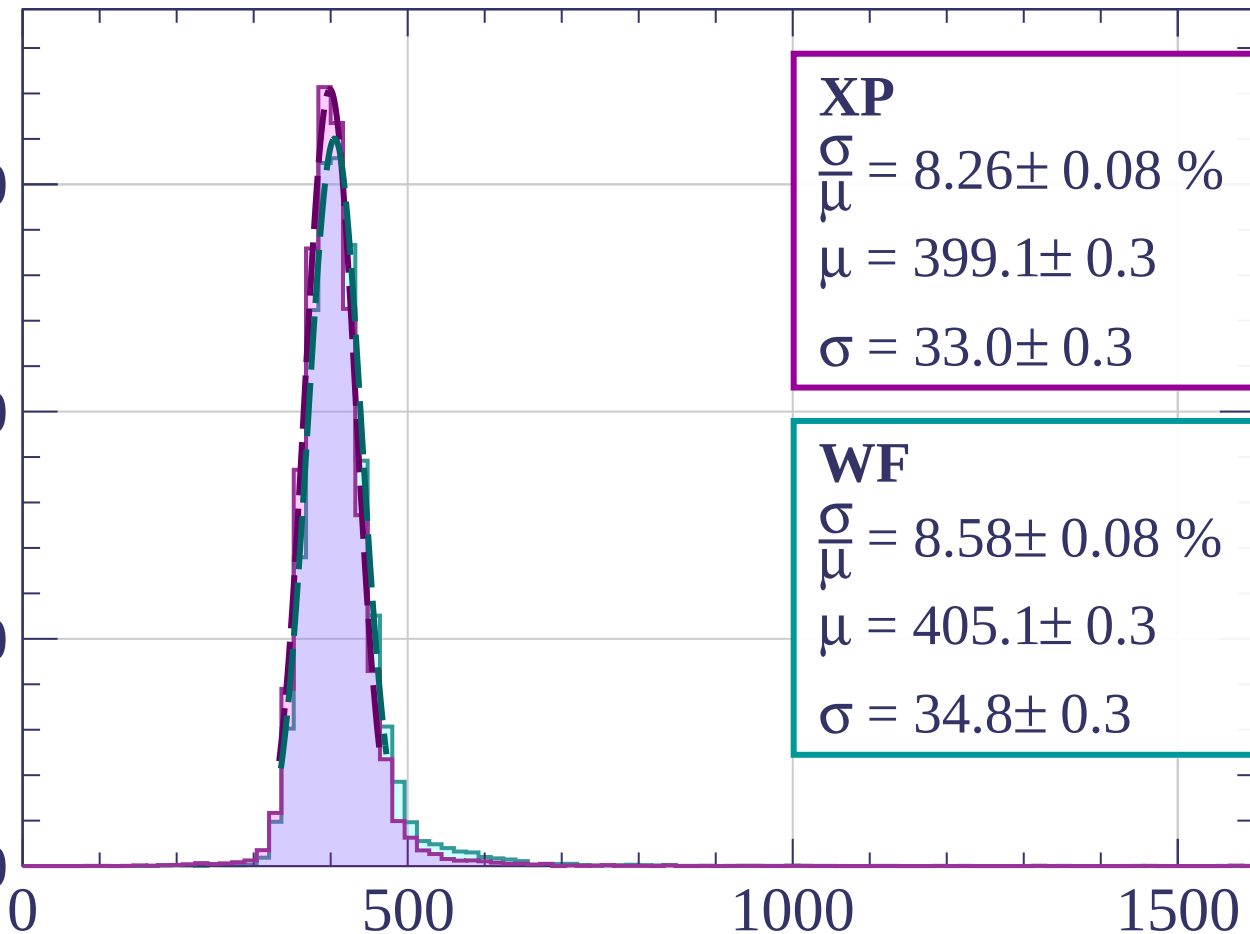
Count

3000

2000

1000

0



**XP**

$$\frac{\sigma}{\mu} = 8.26 \pm 0.08 \%$$

$$\mu = 399.1 \pm 0.3$$

$$\sigma = 33.0 \pm 0.3$$

**WF**

$$\frac{\sigma}{\mu} = 8.58 \pm 0.08 \%$$

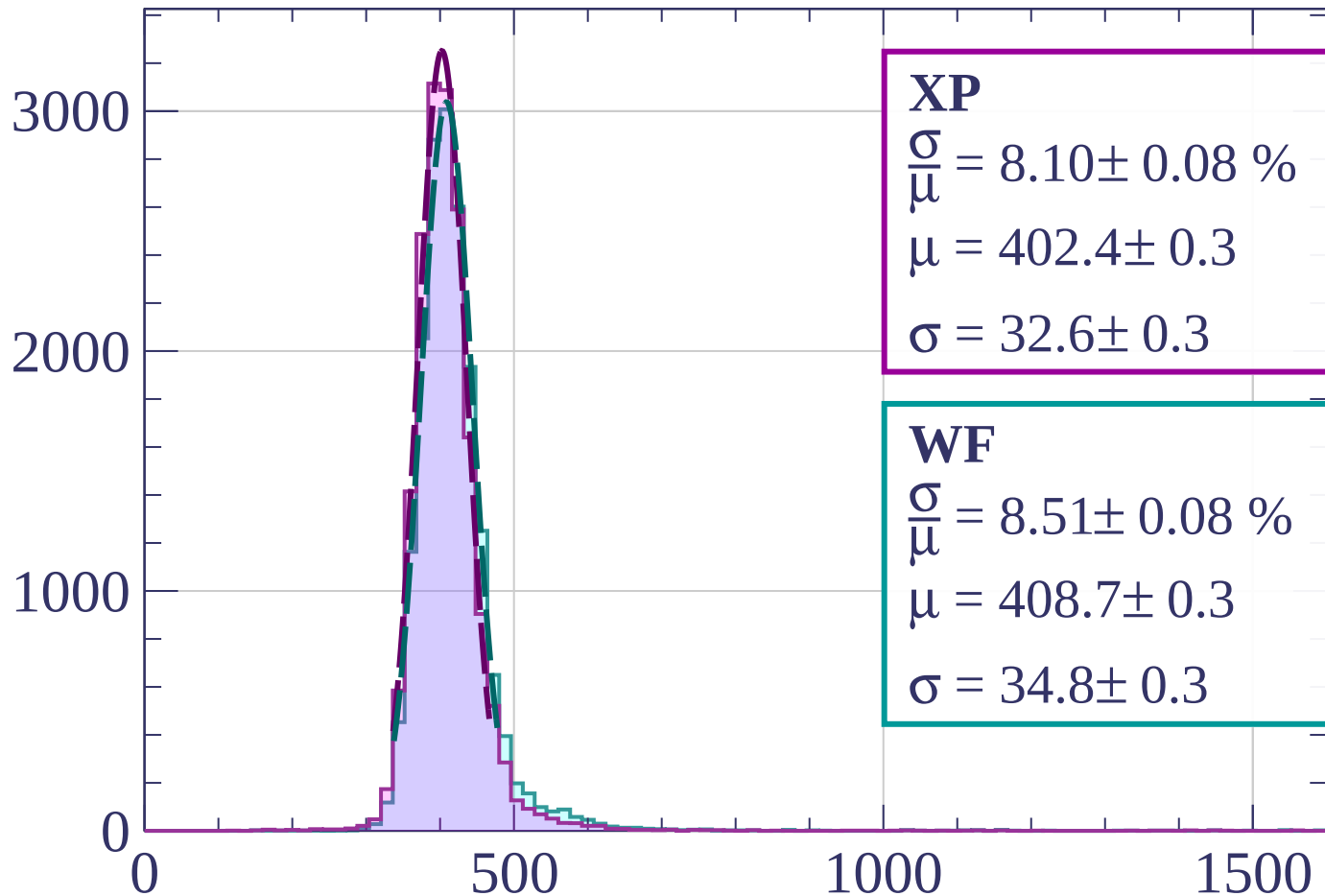
$$\mu = 405.1 \pm 0.3$$

$$\sigma = 34.8 \pm 0.3$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $480 < p < 560$

Count



$dE/dx$  [ADC counts/cm]



Energy loss |  $560 < p < 640$

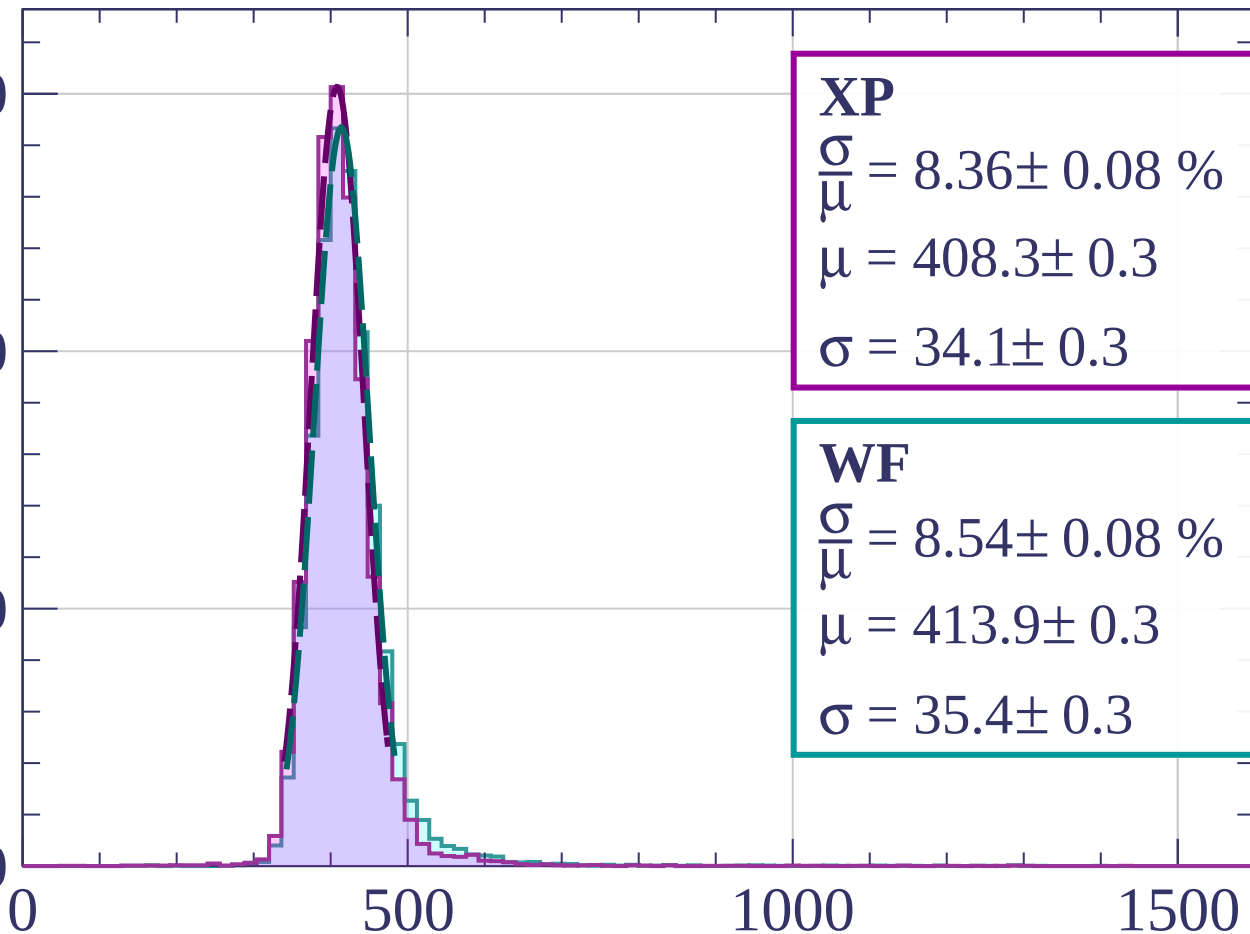
Count

3000

2000

1000

0



**XP**

$$\frac{\sigma}{\mu} = 8.36 \pm 0.08 \%$$

$$\mu = 408.3 \pm 0.3$$

$$\sigma = 34.1 \pm 0.3$$

**WF**

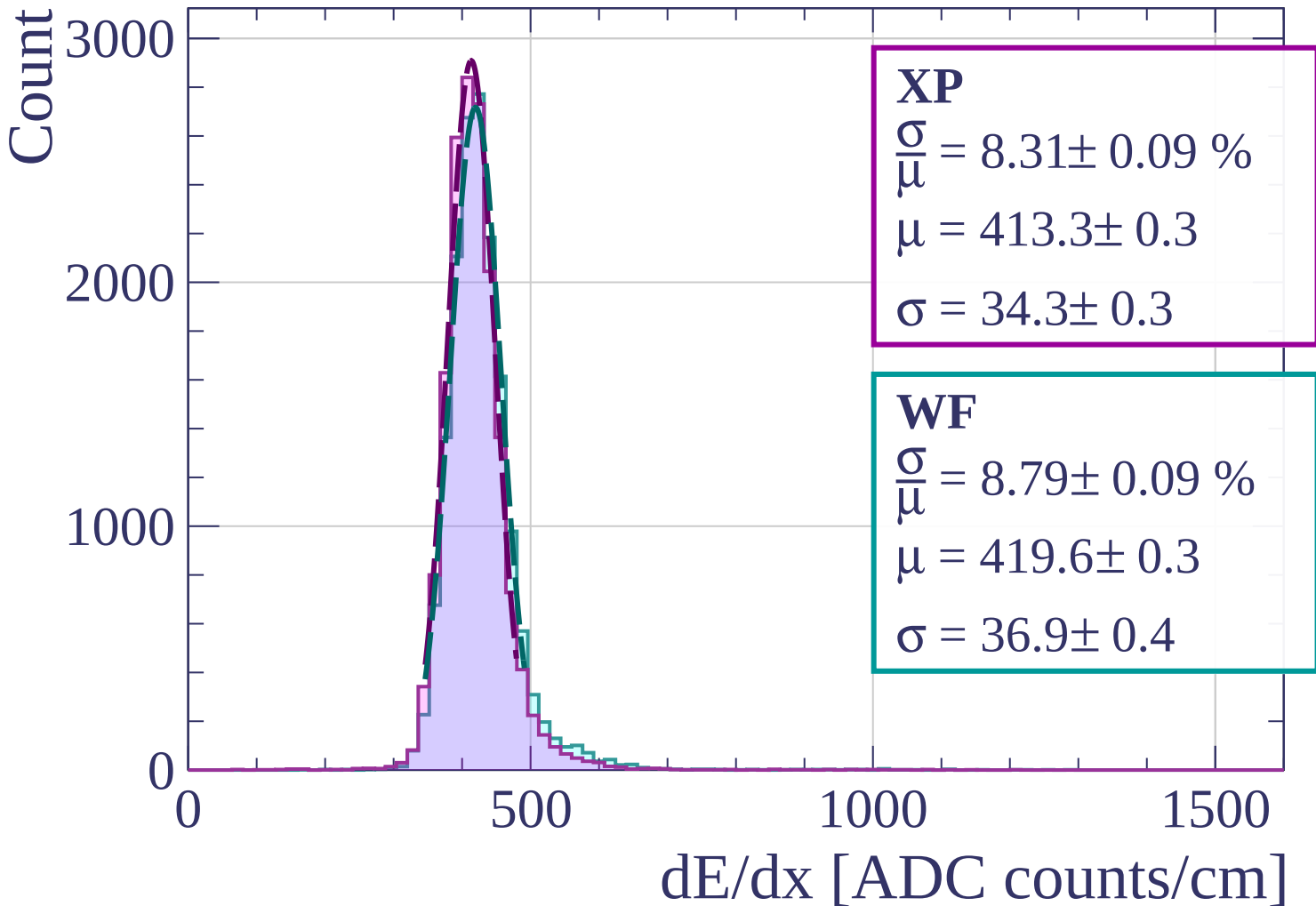
$$\frac{\sigma}{\mu} = 8.54 \pm 0.08 \%$$

$$\mu = 413.9 \pm 0.3$$

$$\sigma = 35.4 \pm 0.3$$

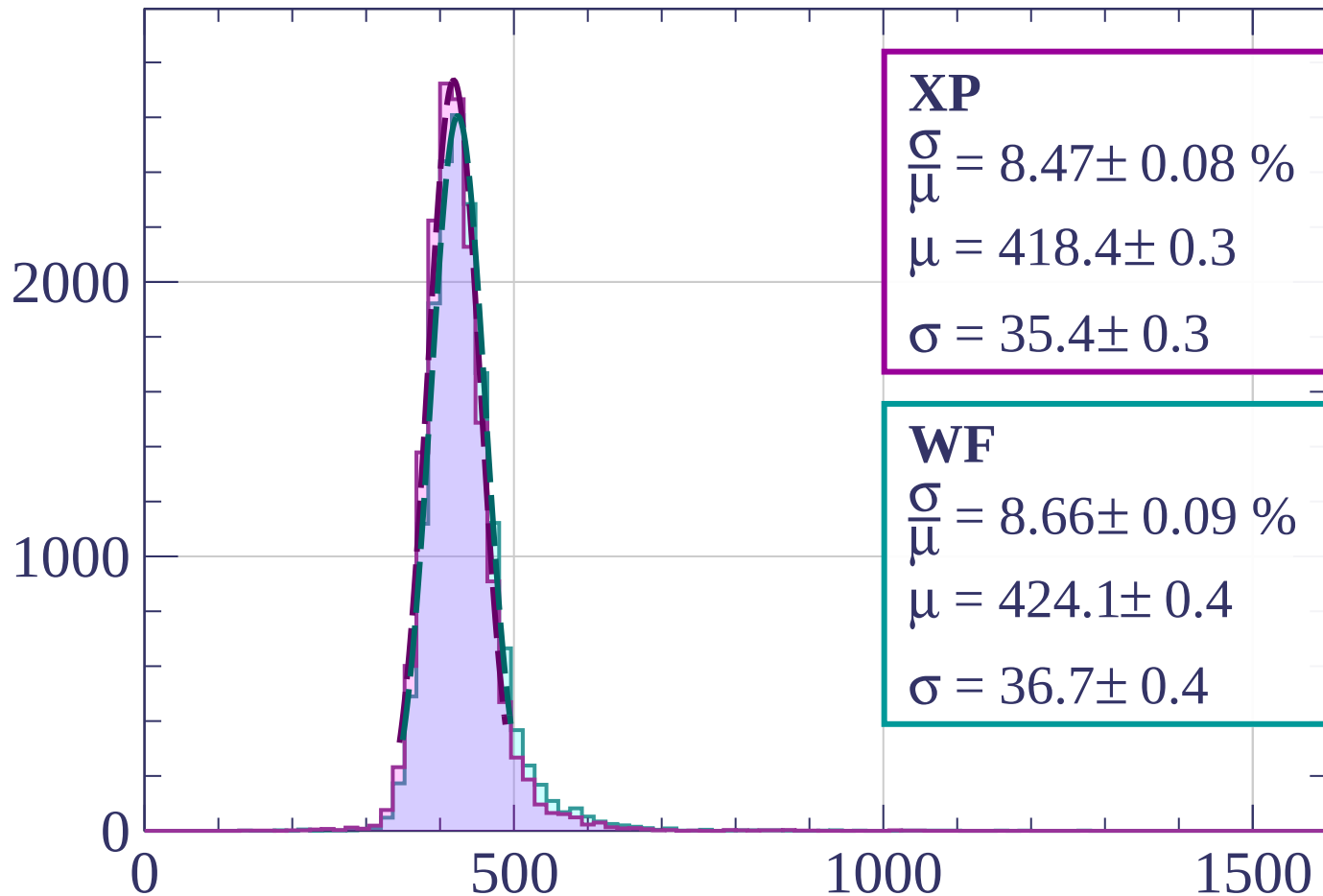
$dE/dx$  [ADC counts/cm]

Energy loss |  $640 < p < 720$



Energy loss |  $720 < p < 800$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.47 \pm 0.08 \%$$

$$\mu = 418.4 \pm 0.3$$

$$\sigma = 35.4 \pm 0.3$$

**WF**

$$\frac{\sigma}{\mu} = 8.66 \pm 0.09 \%$$

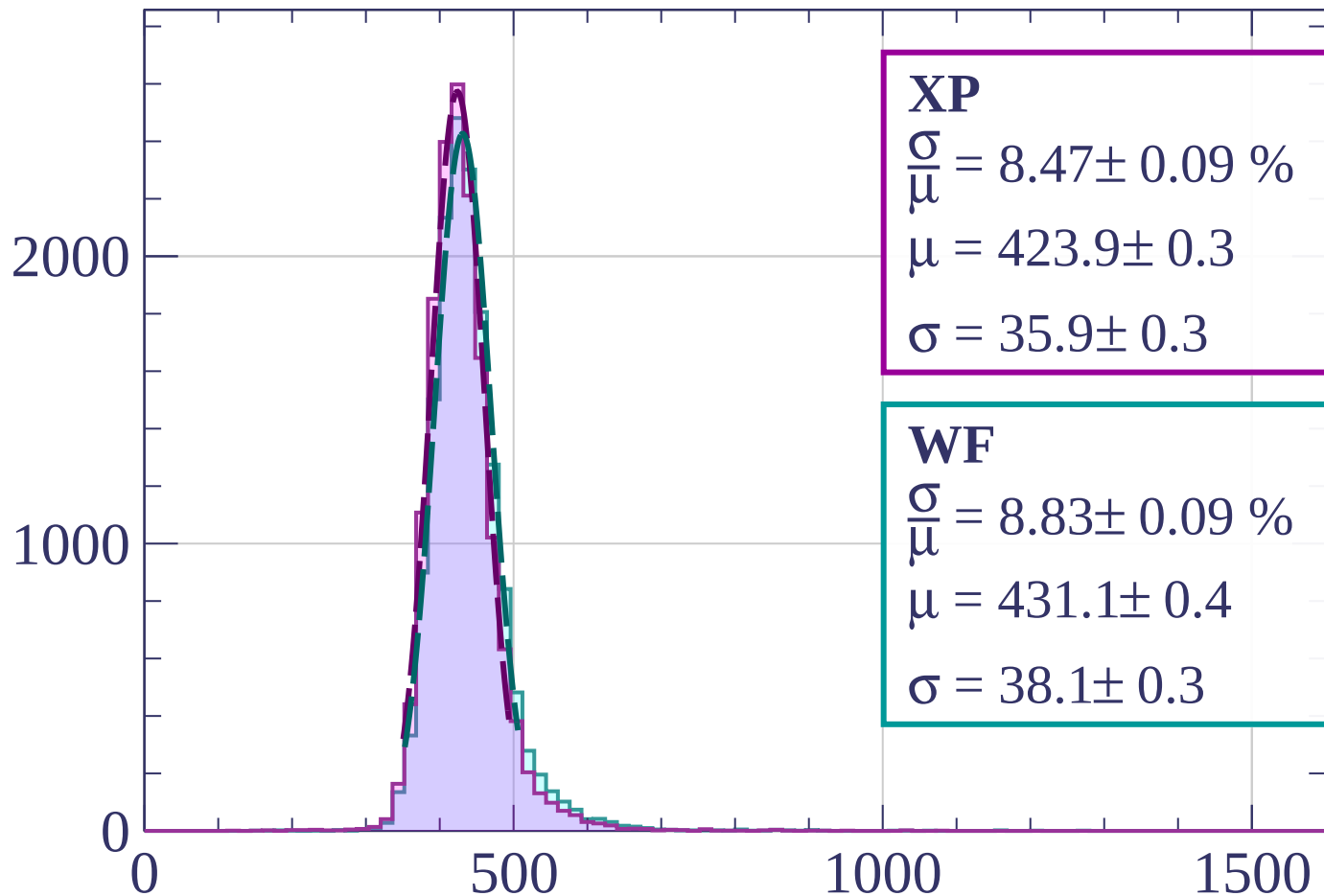
$$\mu = 424.1 \pm 0.4$$

$$\sigma = 36.7 \pm 0.4$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $800 < p < 880$

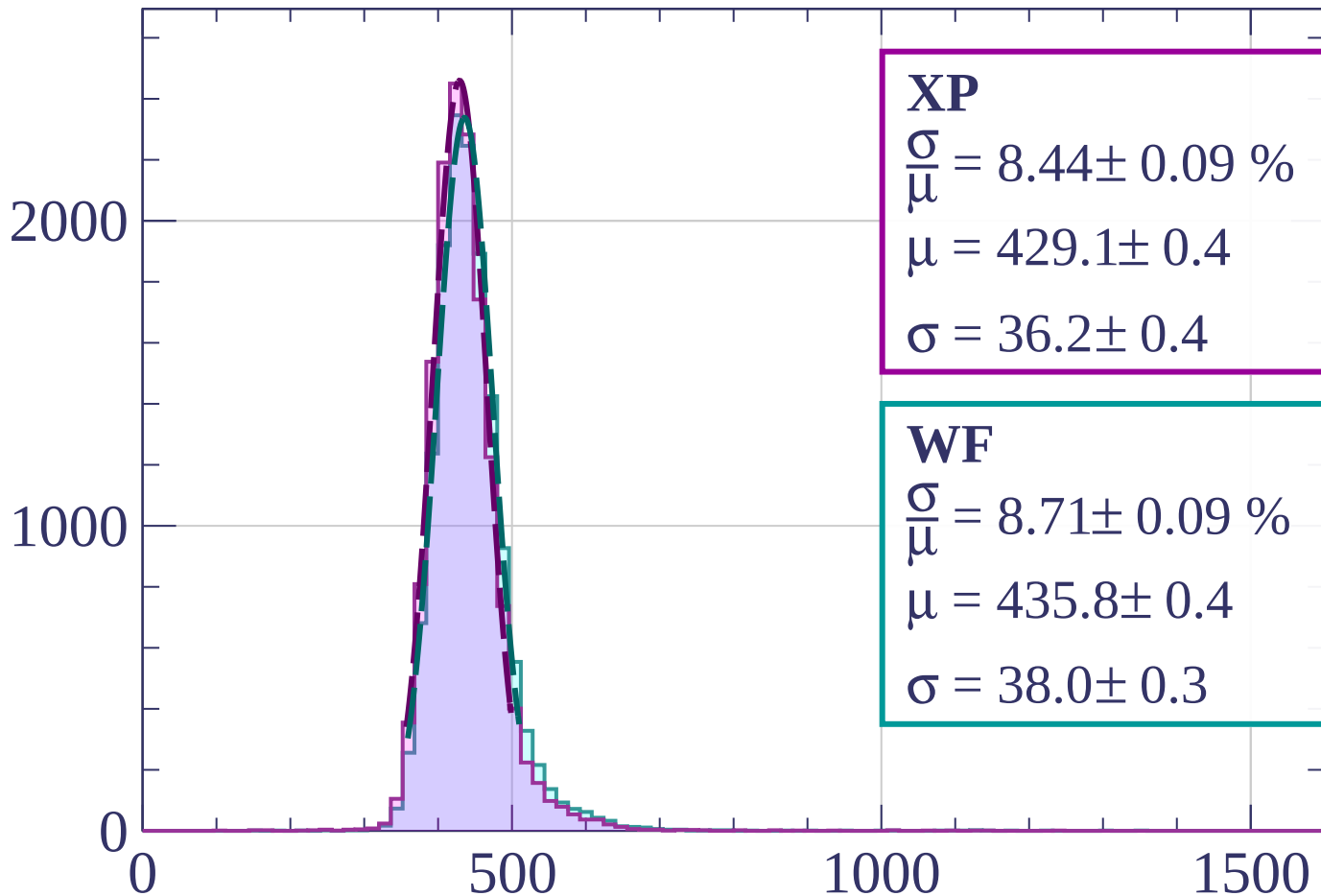
Count



$dE/dx$  [ADC counts/cm]

Energy loss |  $880 < p < 960$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.44 \pm 0.09 \%$$

$$\mu = 429.1 \pm 0.4$$

$$\sigma = 36.2 \pm 0.4$$

**WF**

$$\frac{\sigma}{\mu} = 8.71 \pm 0.09 \%$$

$$\mu = 435.8 \pm 0.4$$

$$\sigma = 38.0 \pm 0.3$$

$dE/dx$  [ADC counts/cm]

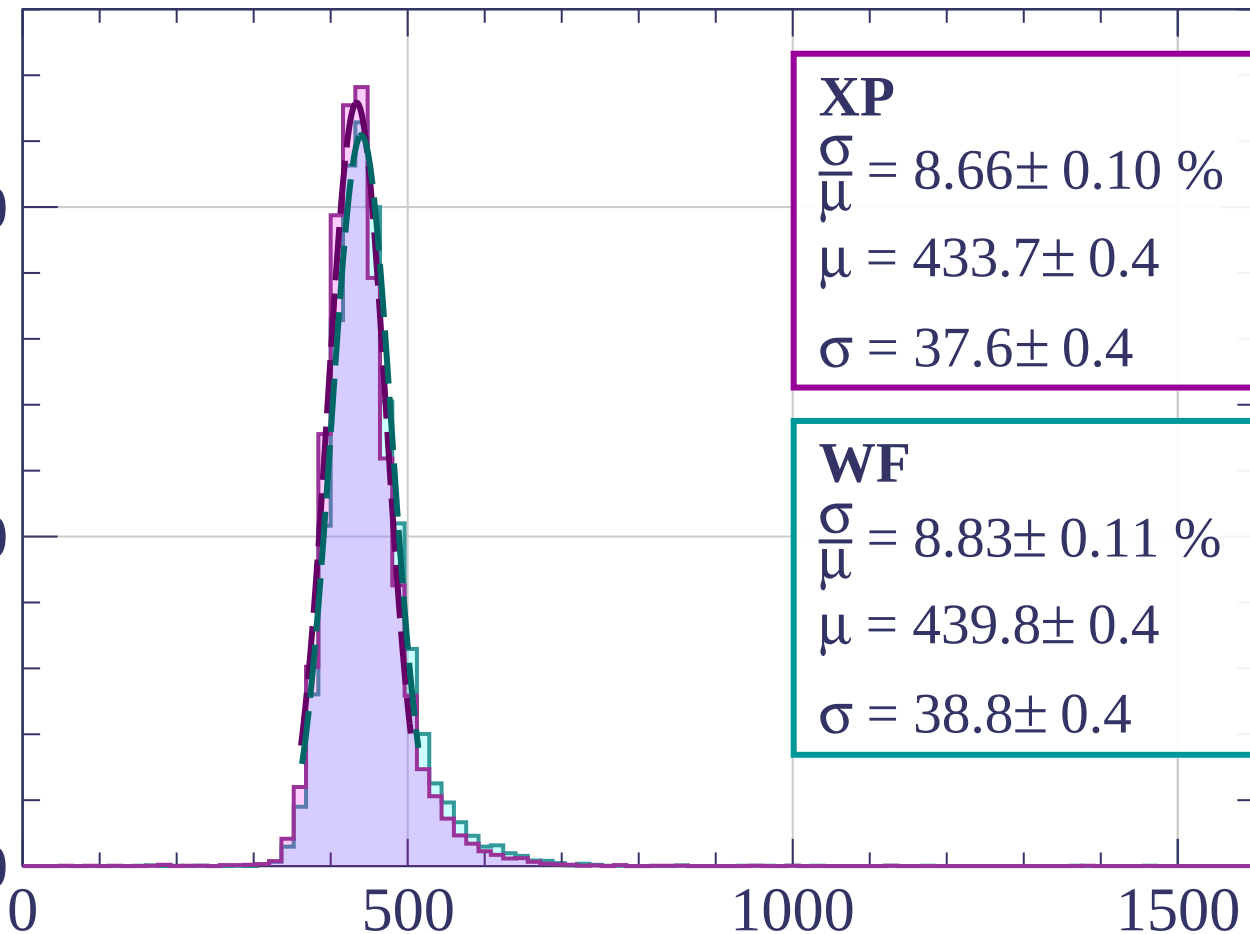
Energy loss |  $960 < p < 1040$

Count

2000

1000

0



**XP**

$$\frac{\sigma}{\mu} = 8.66 \pm 0.10 \%$$

$$\mu = 433.7 \pm 0.4$$

$$\sigma = 37.6 \pm 0.4$$

**WF**

$$\frac{\sigma}{\mu} = 8.83 \pm 0.11 \%$$

$$\mu = 439.8 \pm 0.4$$

$$\sigma = 38.8 \pm 0.4$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $1040 < p < 1120$

Count

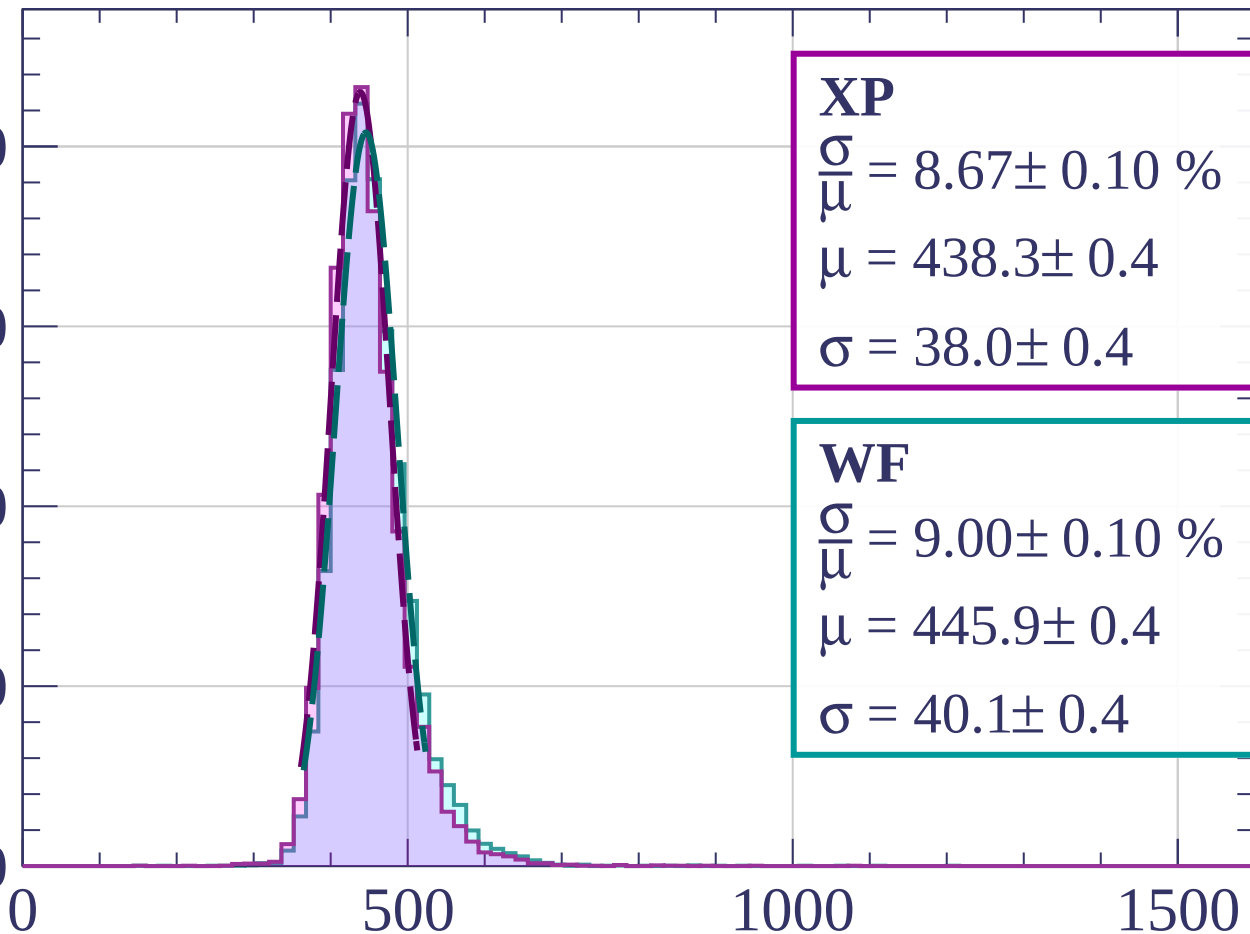
2000

1500

1000

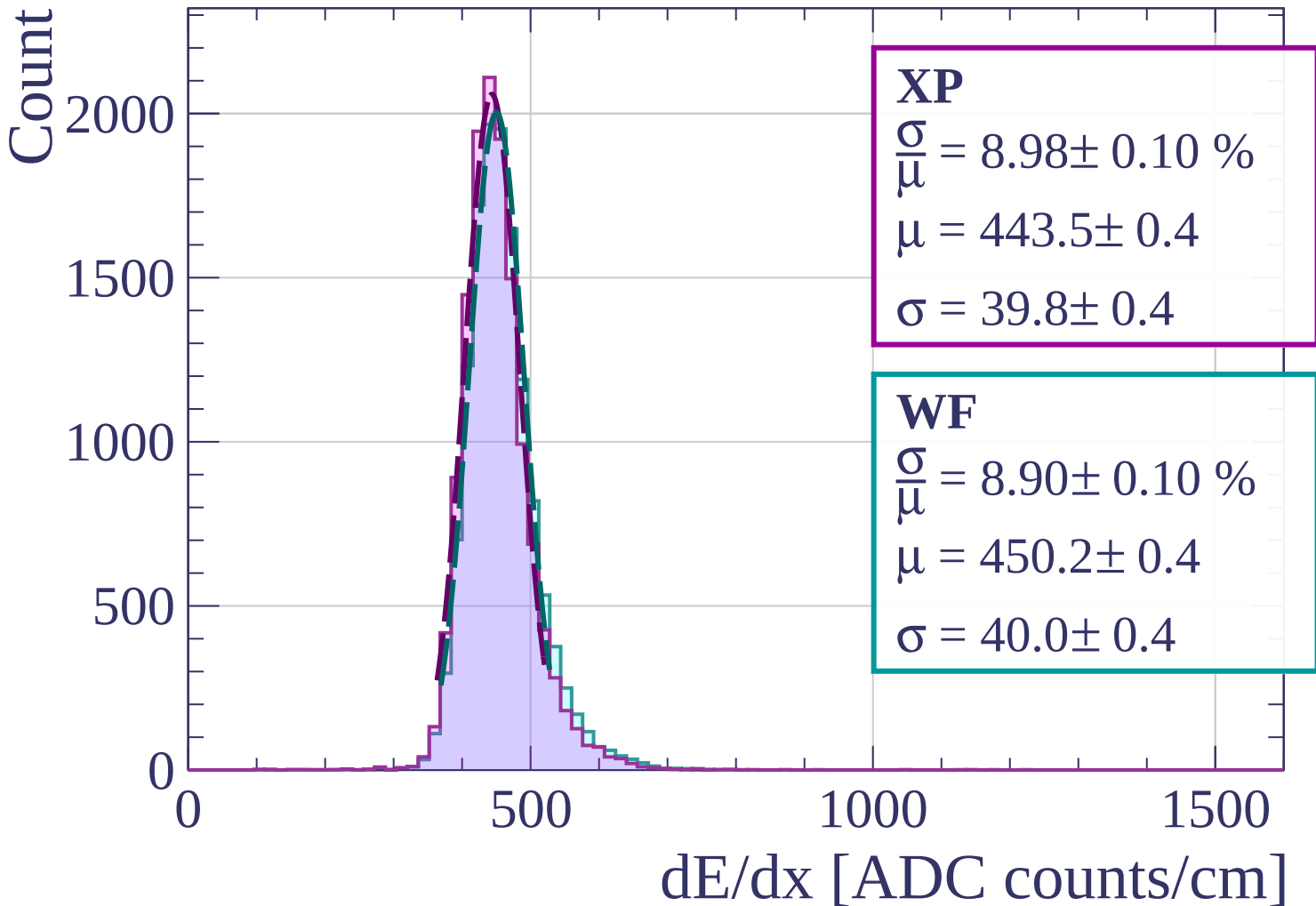
500

0



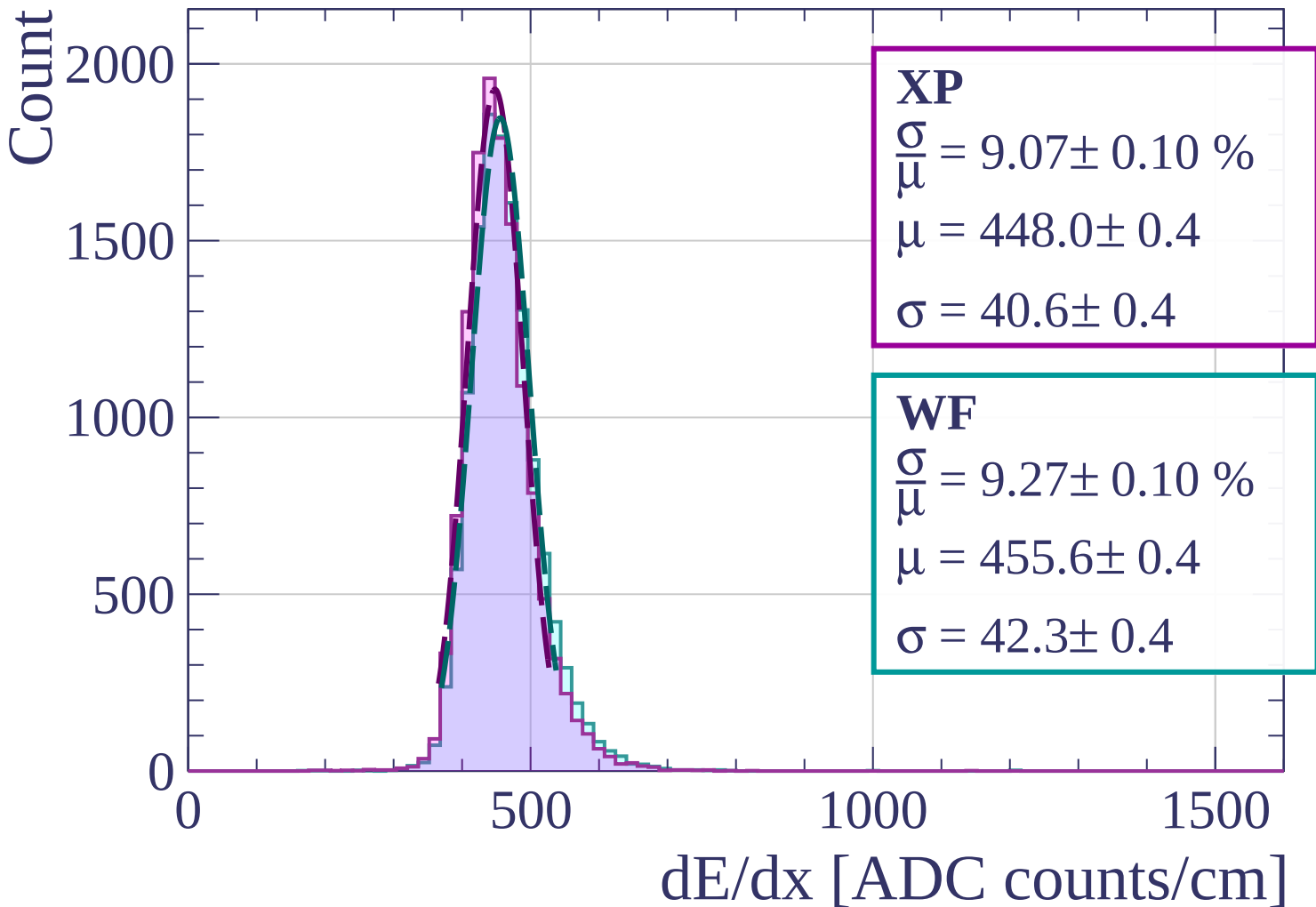
dE/dx [ADC counts/cm]

Energy loss |  $1120 < p < 1200$

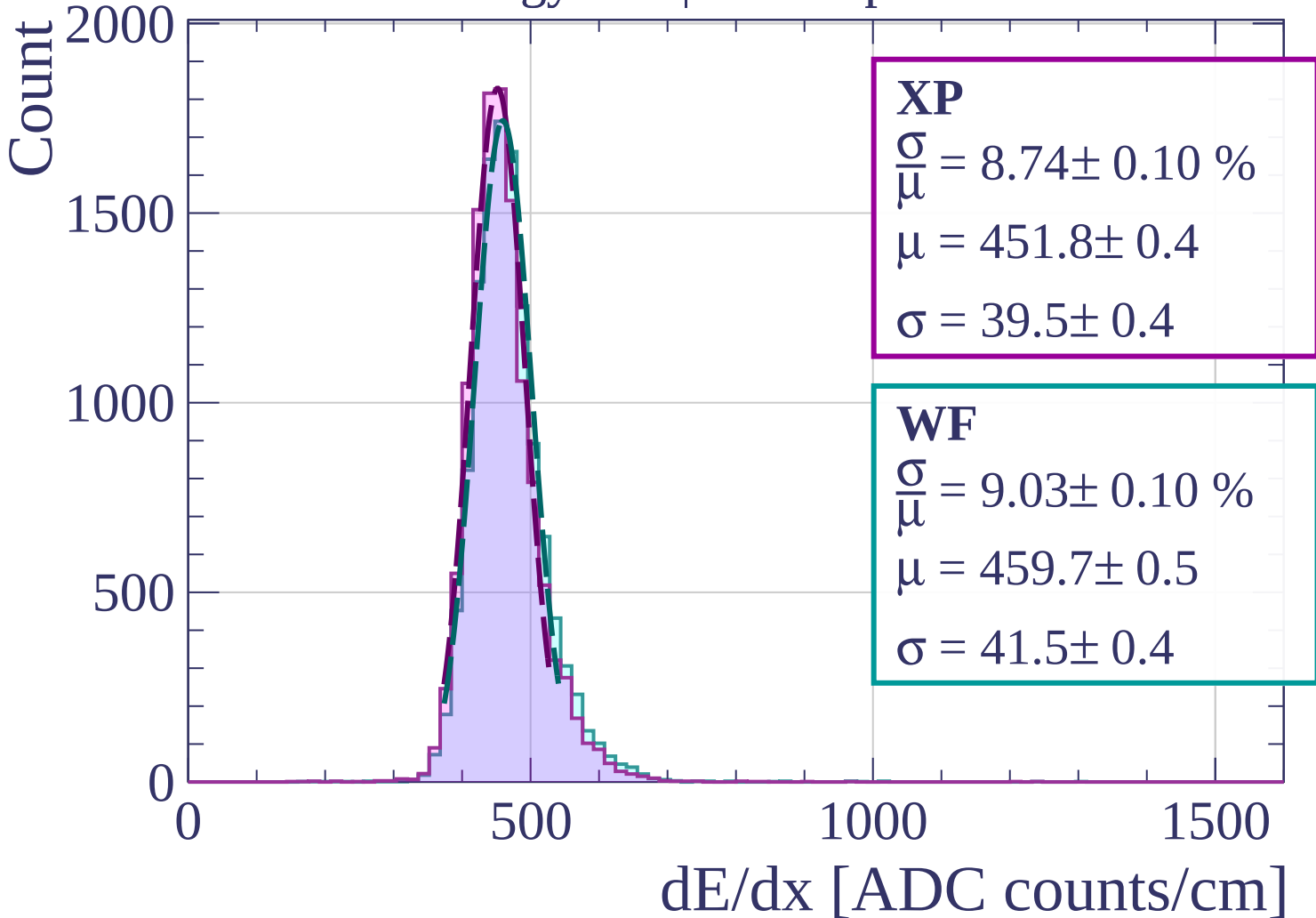




Energy loss |  $1200 < p < 1280$

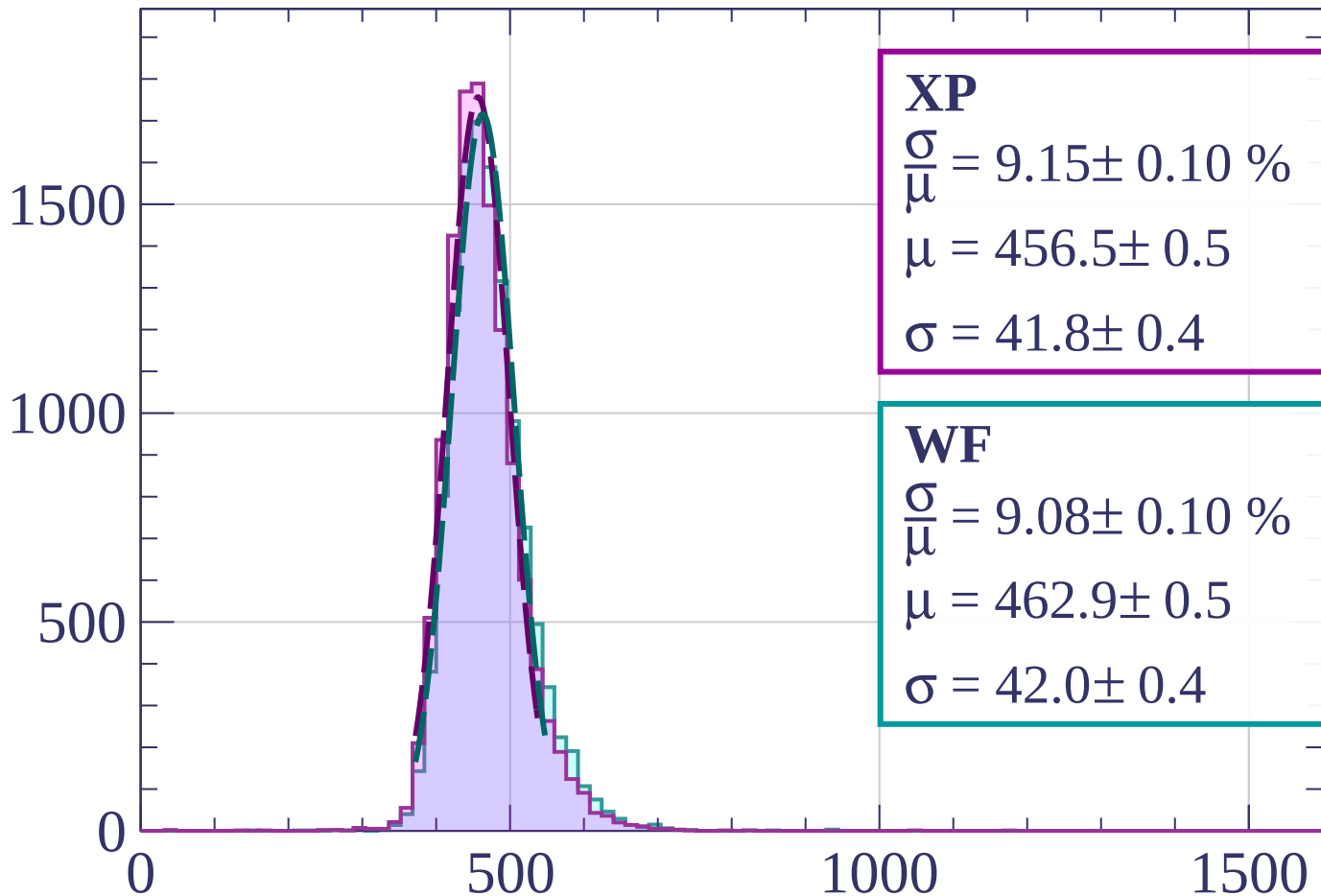


Energy loss |  $1280 < p < 1360$



Energy loss |  $1360 < p < 1440$

Count



**XP**

$$\frac{\sigma}{\mu} = 9.15 \pm 0.10 \%$$

$$\mu = 456.5 \pm 0.5$$

$$\sigma = 41.8 \pm 0.4$$

**WF**

$$\frac{\sigma}{\mu} = 9.08 \pm 0.10 \%$$

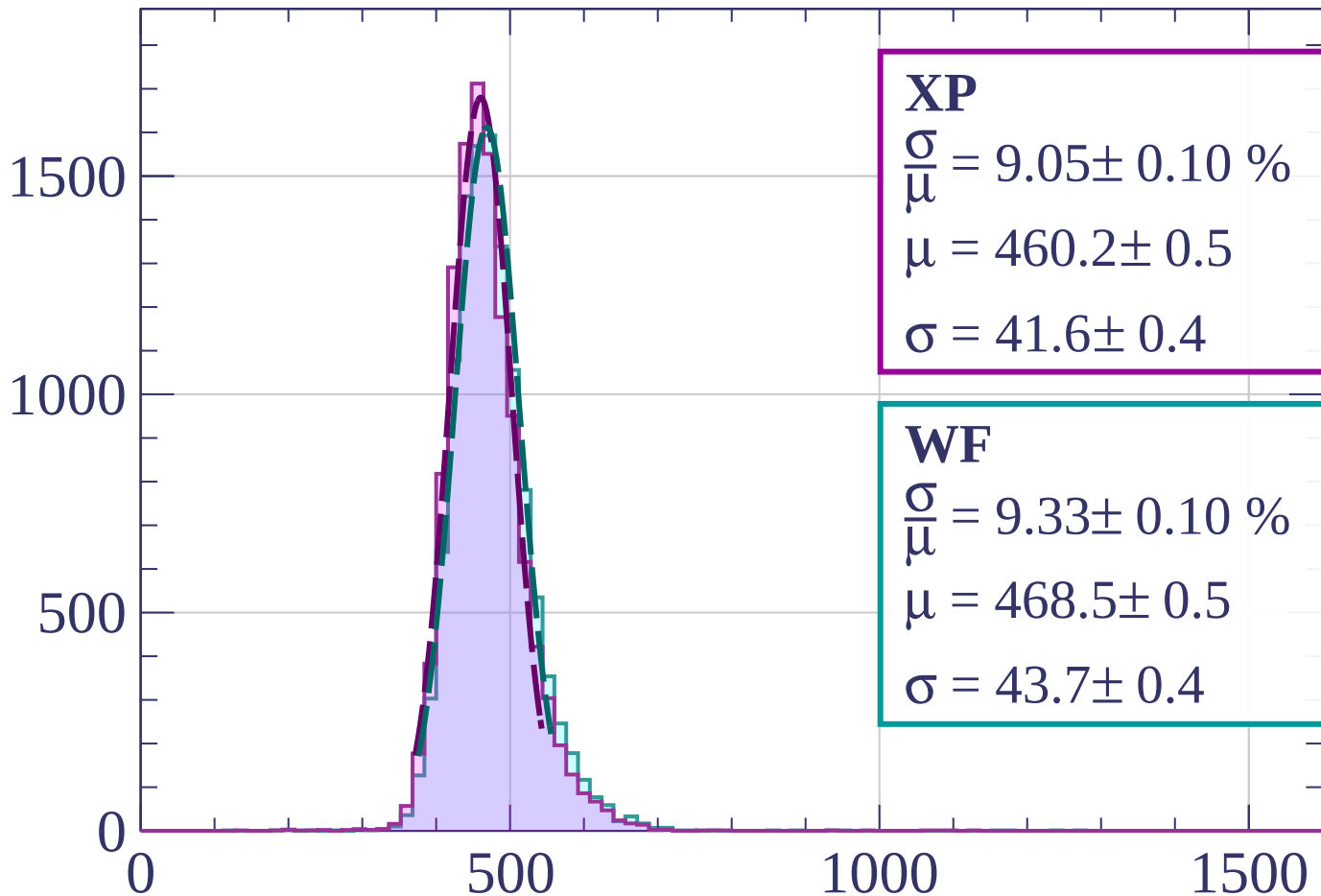
$$\mu = 462.9 \pm 0.5$$

$$\sigma = 42.0 \pm 0.4$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $1440 < p < 1520$

Count



**XP**

$$\frac{\sigma}{\mu} = 9.05 \pm 0.10 \%$$

$$\mu = 460.2 \pm 0.5$$

$$\sigma = 41.6 \pm 0.4$$

**WF**

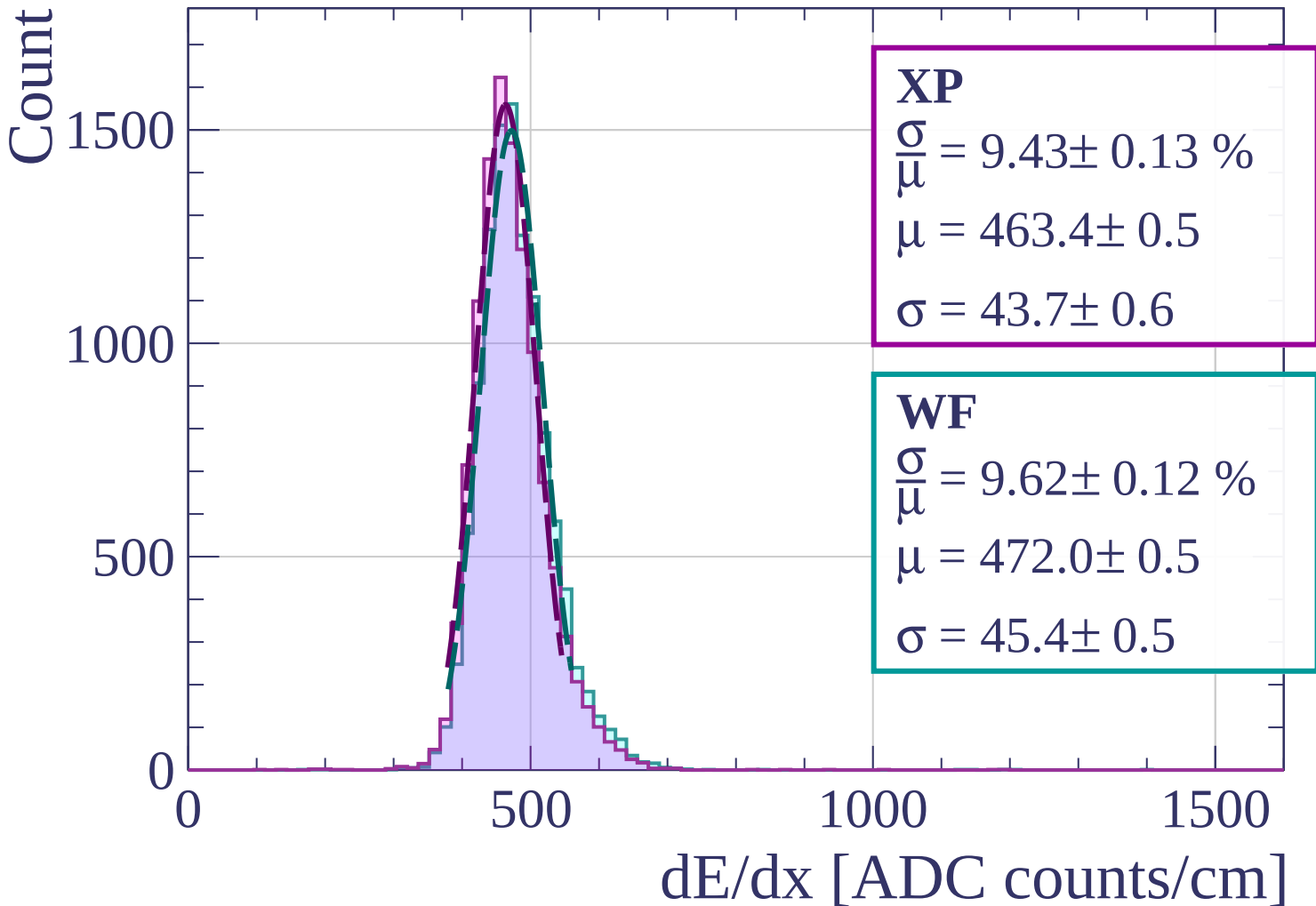
$$\frac{\sigma}{\mu} = 9.33 \pm 0.10 \%$$

$$\mu = 468.5 \pm 0.5$$

$$\sigma = 43.7 \pm 0.4$$

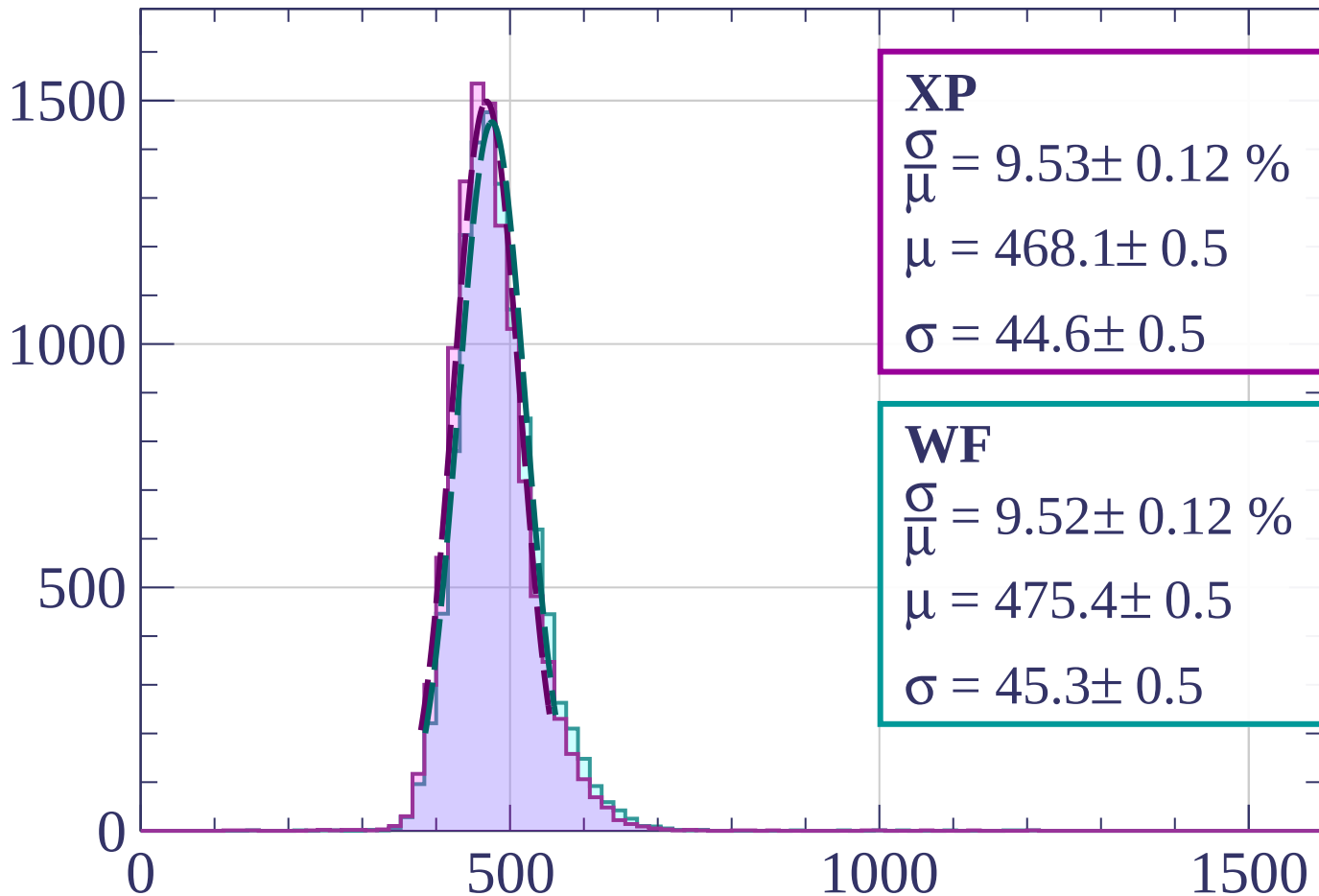
$dE/dx$  [ADC counts/cm]

Energy loss |  $1520 < p < 1600$



Energy loss |  $1600 < p < 1680$

Count



**XP**

$$\frac{\sigma}{\mu} = 9.53 \pm 0.12 \%$$

$$\mu = 468.1 \pm 0.5$$

$$\sigma = 44.6 \pm 0.5$$

**WF**

$$\frac{\sigma}{\mu} = 9.52 \pm 0.12 \%$$

$$\mu = 475.4 \pm 0.5$$

$$\sigma = 45.3 \pm 0.5$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $1680 < p < 1760$

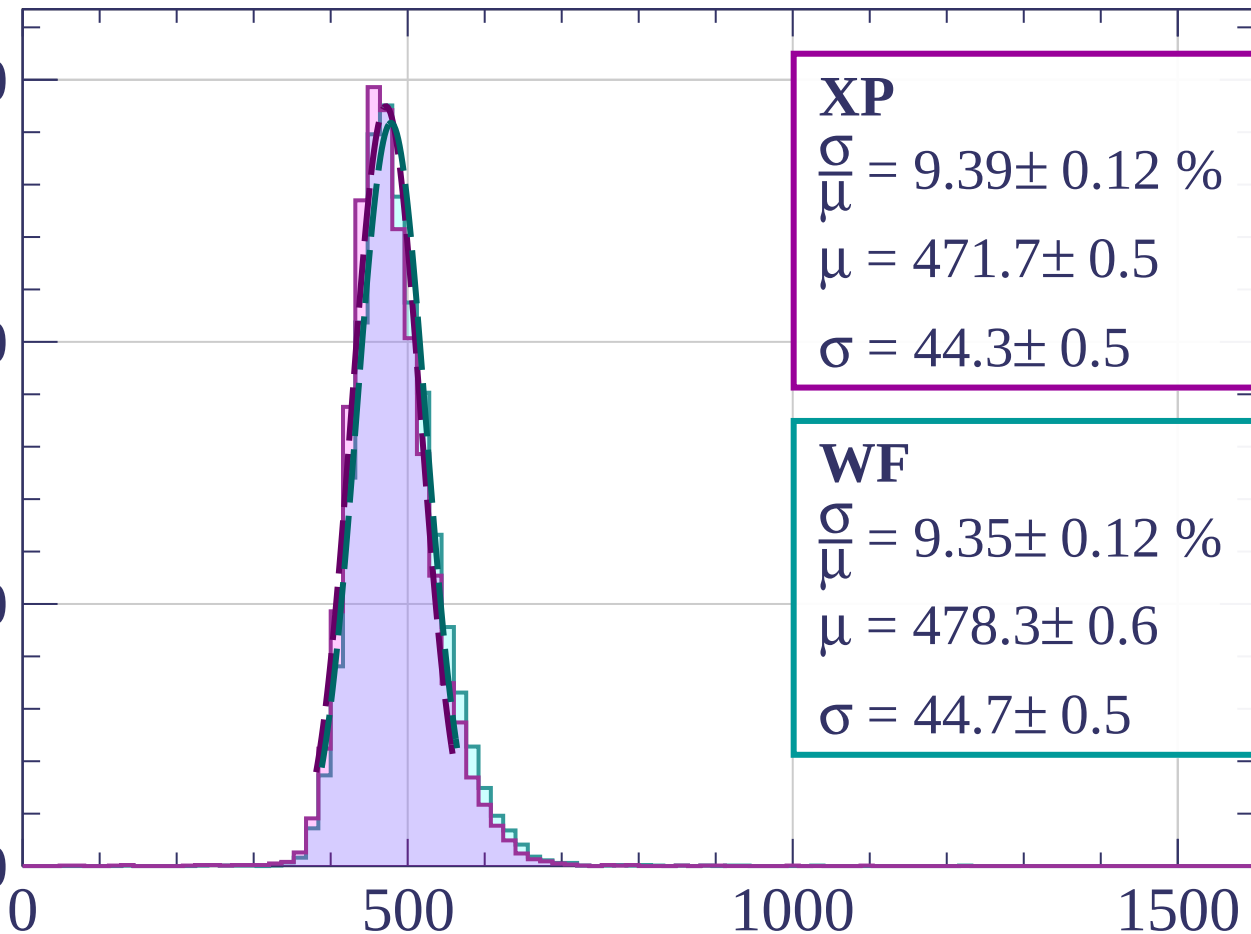
Count

1500

1000

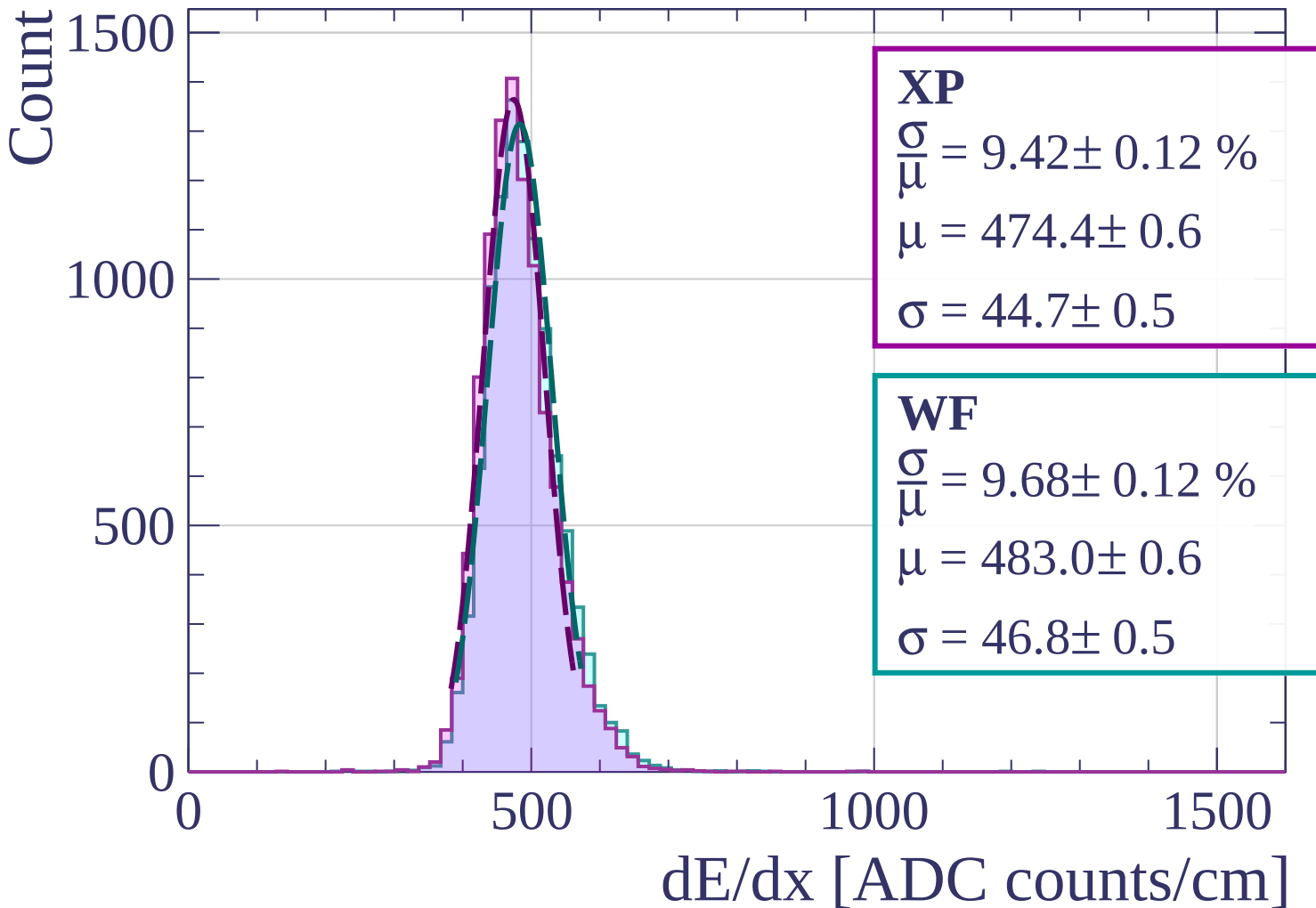
500

0



dE/dx [ADC counts/cm]

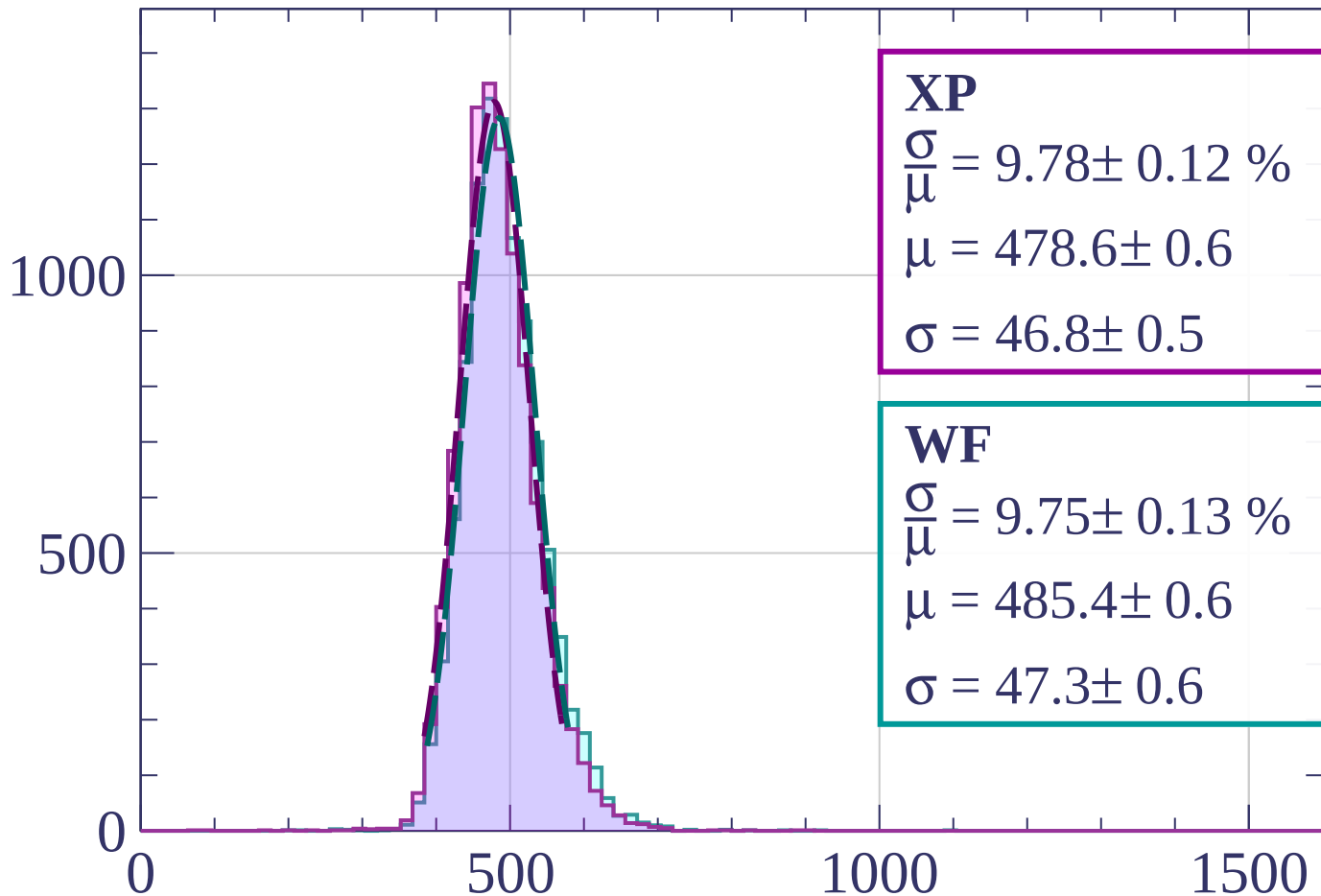
Energy loss |  $1760 < p < 1840$





Energy loss |  $1840 < p < 1920$

Count



**XP**

$$\frac{\sigma}{\mu} = 9.78 \pm 0.12 \%$$

$$\mu = 478.6 \pm 0.6$$

$$\sigma = 46.8 \pm 0.5$$

**WF**

$$\frac{\sigma}{\mu} = 9.75 \pm 0.13 \%$$

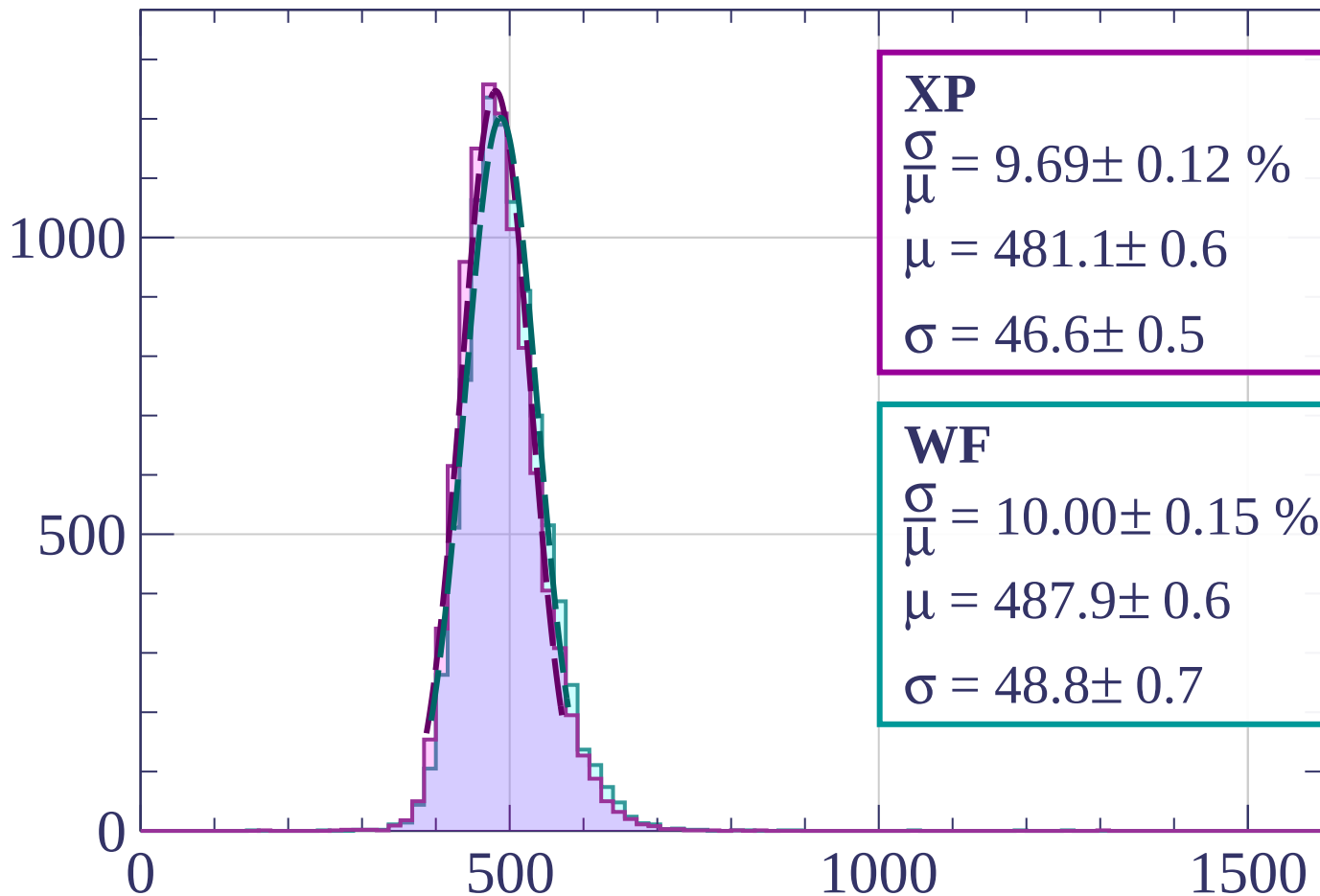
$$\mu = 485.4 \pm 0.6$$

$$\sigma = 47.3 \pm 0.6$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $1920 < p < 2000$

Count



**XP**

$$\frac{\sigma}{\mu} = 9.69 \pm 0.12 \%$$

$$\mu = 481.1 \pm 0.6$$

$$\sigma = 46.6 \pm 0.5$$

**WF**

$$\frac{\sigma}{\mu} = 10.00 \pm 0.15 \%$$

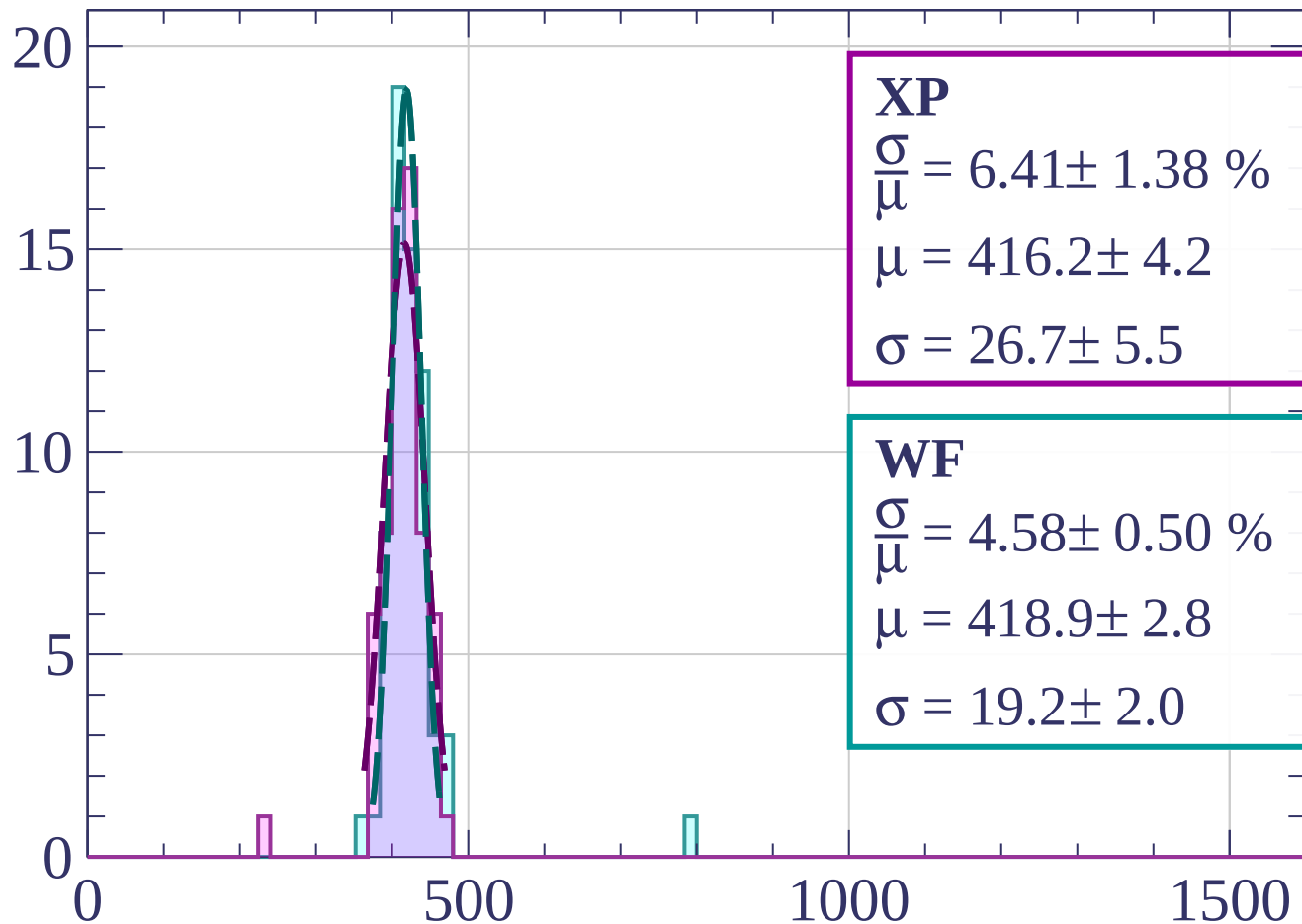
$$\mu = 487.9 \pm 0.6$$

$$\sigma = 48.8 \pm 0.7$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $0 < \phi < 3$

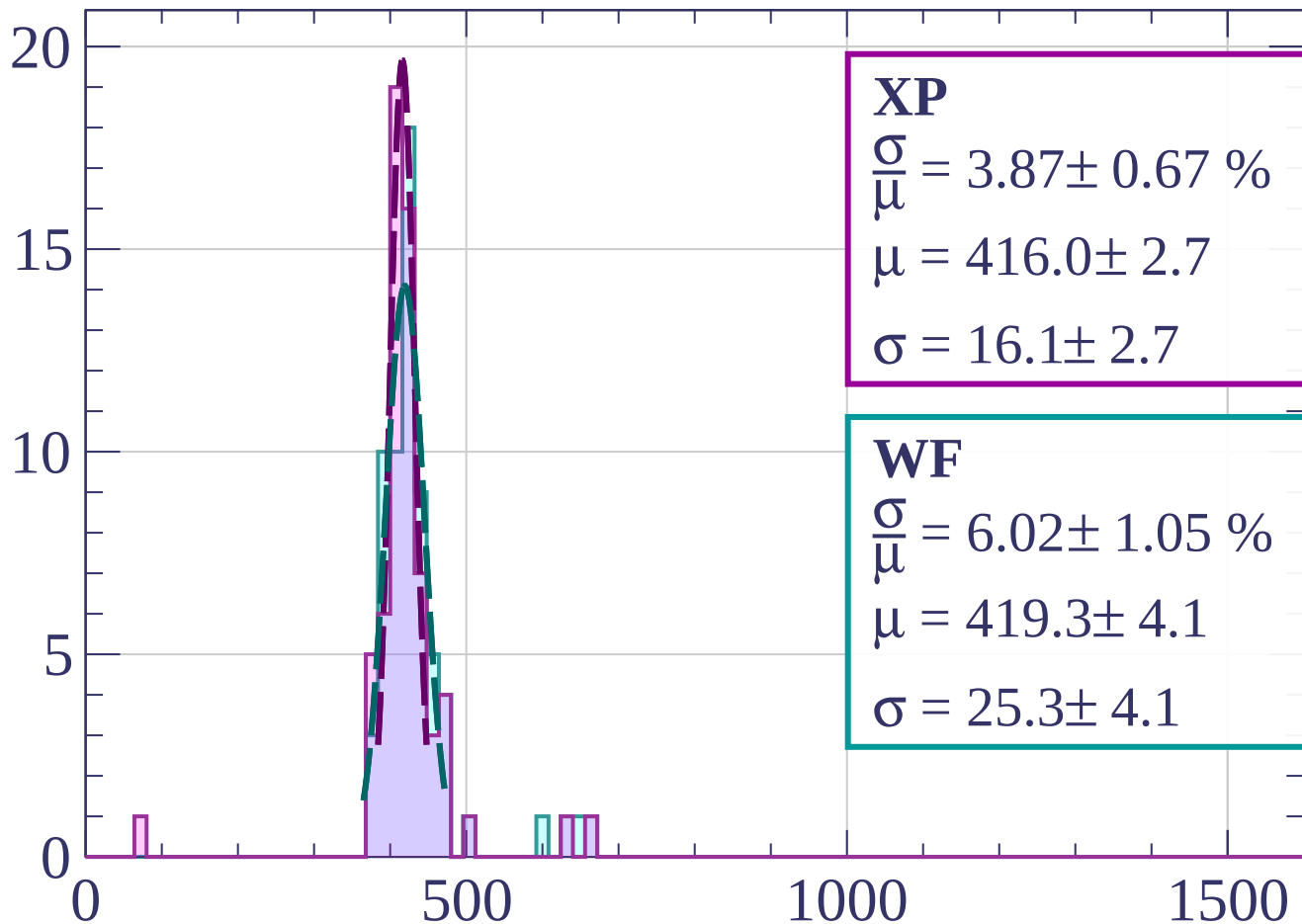
Count



dE/dx [ADC counts/cm]

Energy loss |  $3 < \phi < 6$

Count



**XP**

$$\frac{\sigma}{\mu} = 3.87 \pm 0.67 \%$$

$$\mu = 416.0 \pm 2.7$$

$$\sigma = 16.1 \pm 2.7$$

**WF**

$$\frac{\sigma}{\mu} = 6.02 \pm 1.05 \%$$

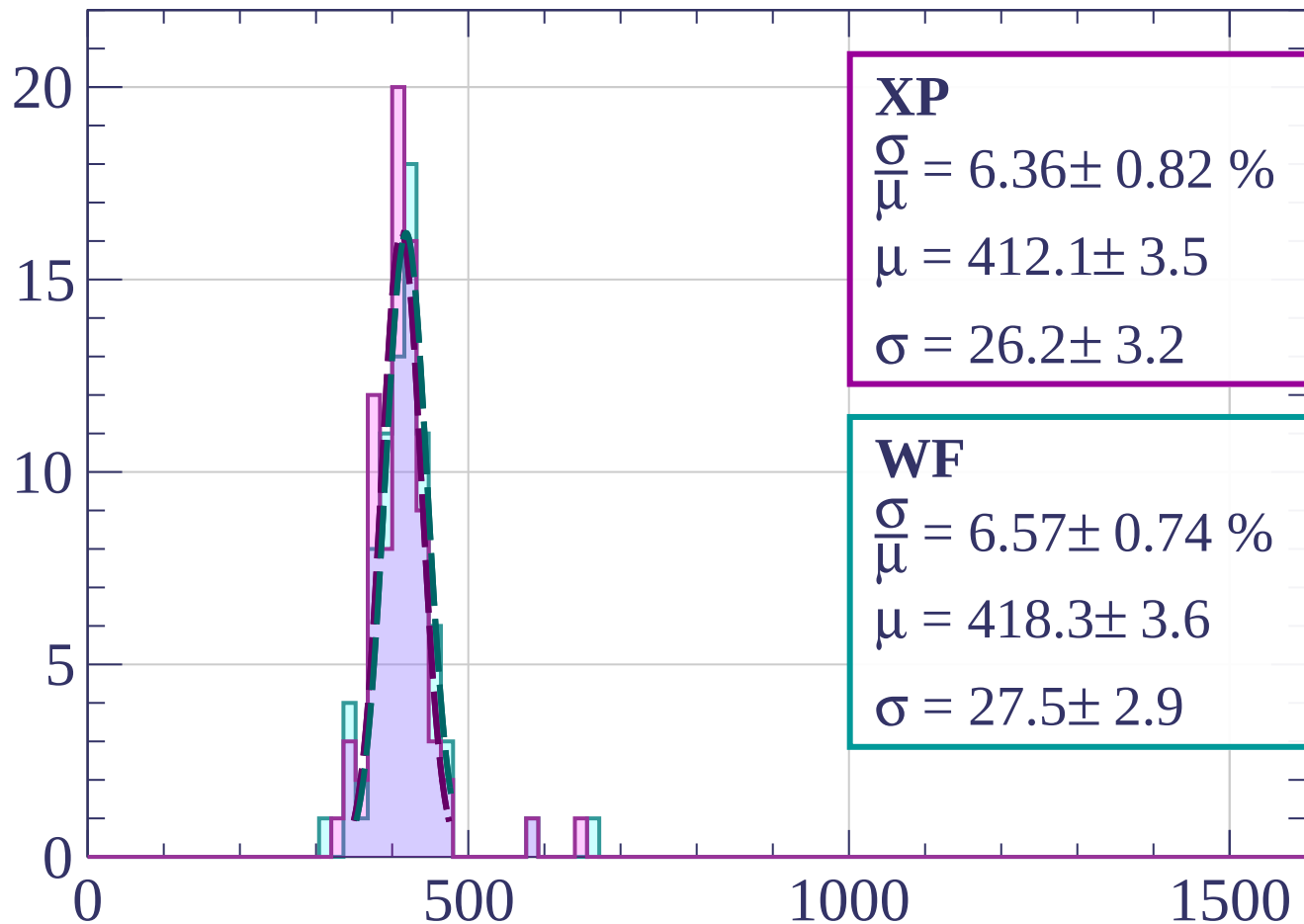
$$\mu = 419.3 \pm 4.1$$

$$\sigma = 25.3 \pm 4.1$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $6 < \phi < 9$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.36 \pm 0.82 \%$$

$$\mu = 412.1 \pm 3.5$$

$$\sigma = 26.2 \pm 3.2$$

**WF**

$$\frac{\sigma}{\mu} = 6.57 \pm 0.74 \%$$

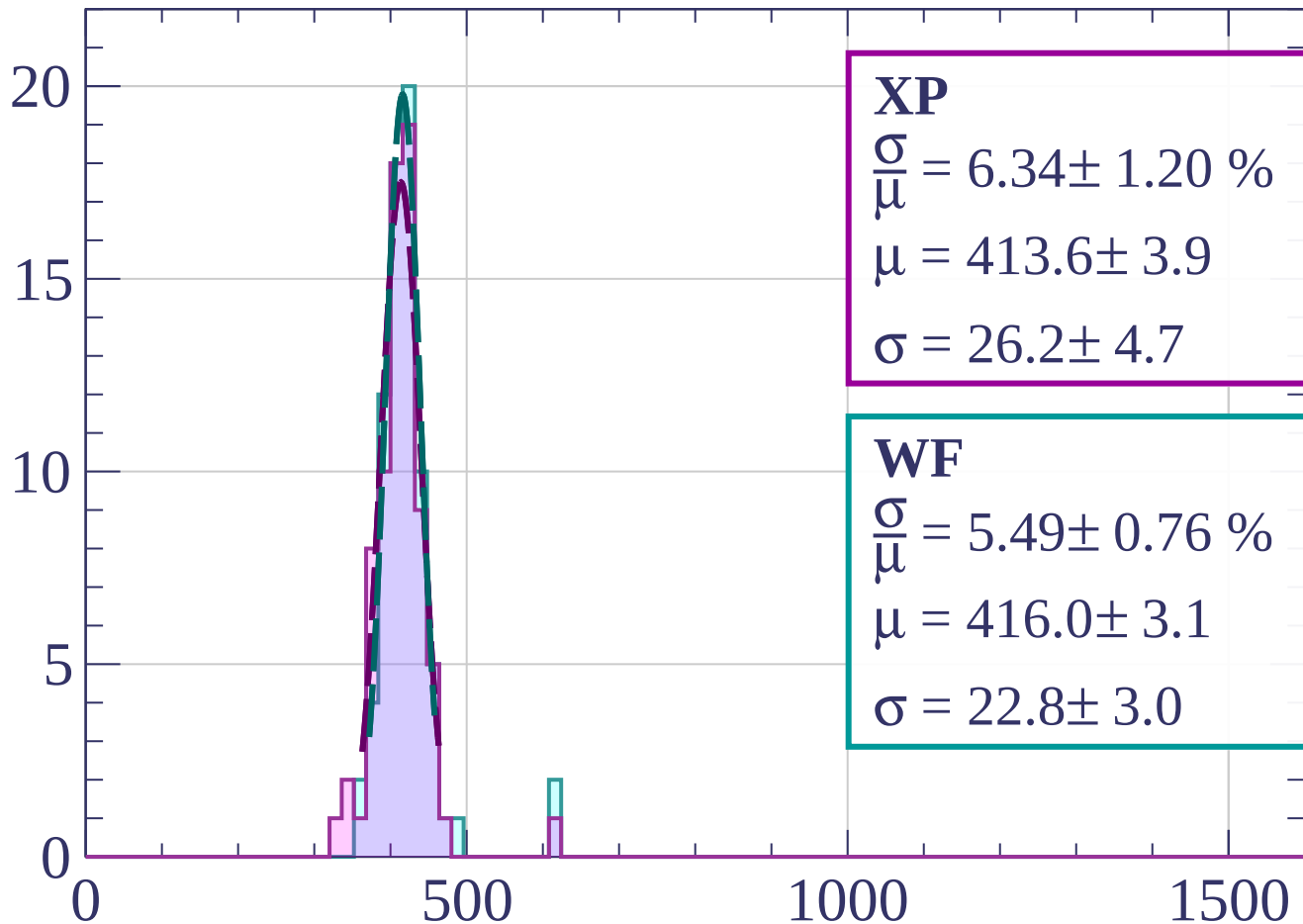
$$\mu = 418.3 \pm 3.6$$

$$\sigma = 27.5 \pm 2.9$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $9 < \phi < 12$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.34 \pm 1.20 \%$$

$$\mu = 413.6 \pm 3.9$$

$$\sigma = 26.2 \pm 4.7$$

**WF**

$$\frac{\sigma}{\mu} = 5.49 \pm 0.76 \%$$

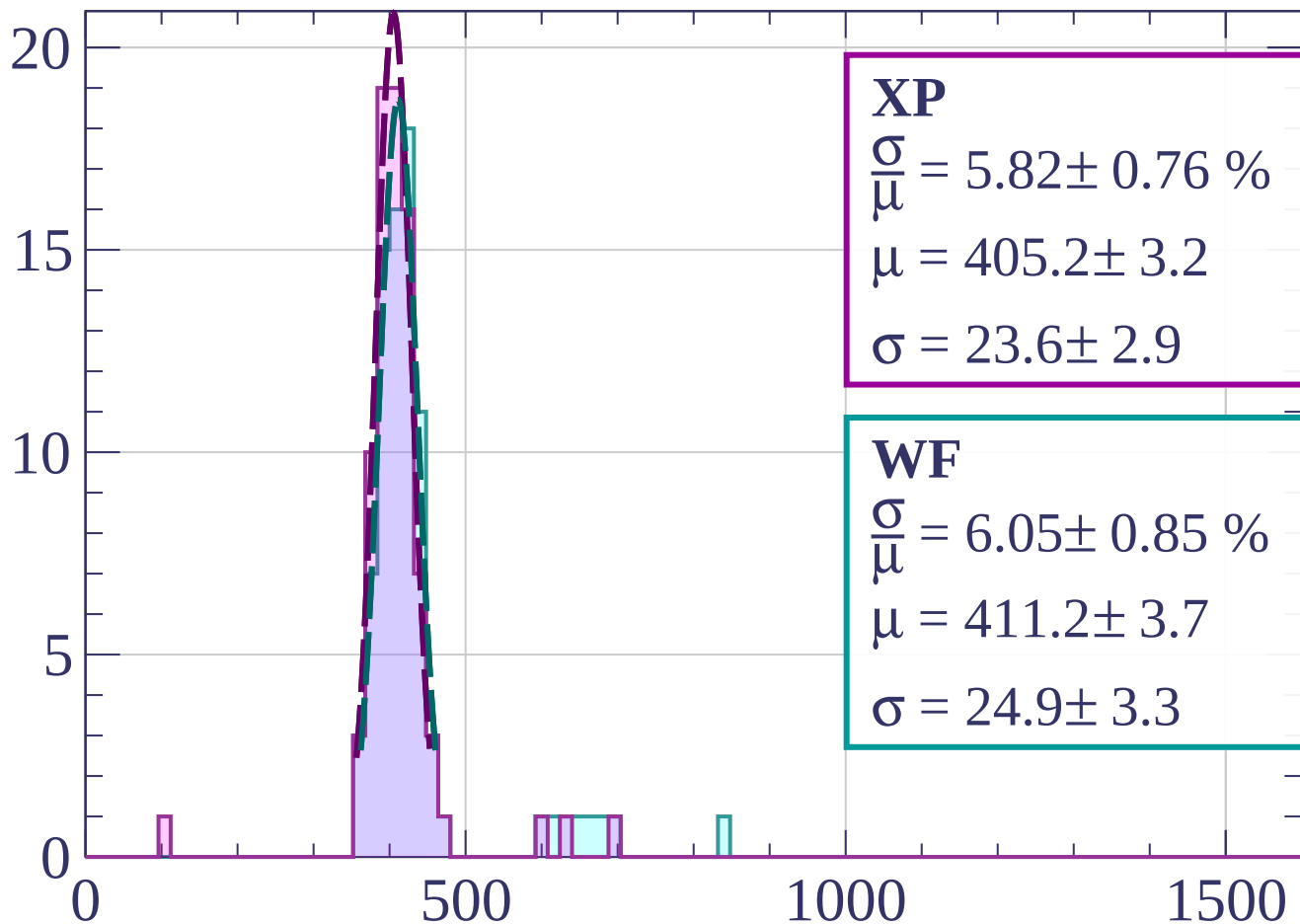
$$\mu = 416.0 \pm 3.1$$

$$\sigma = 22.8 \pm 3.0$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $12 < \phi < 15$

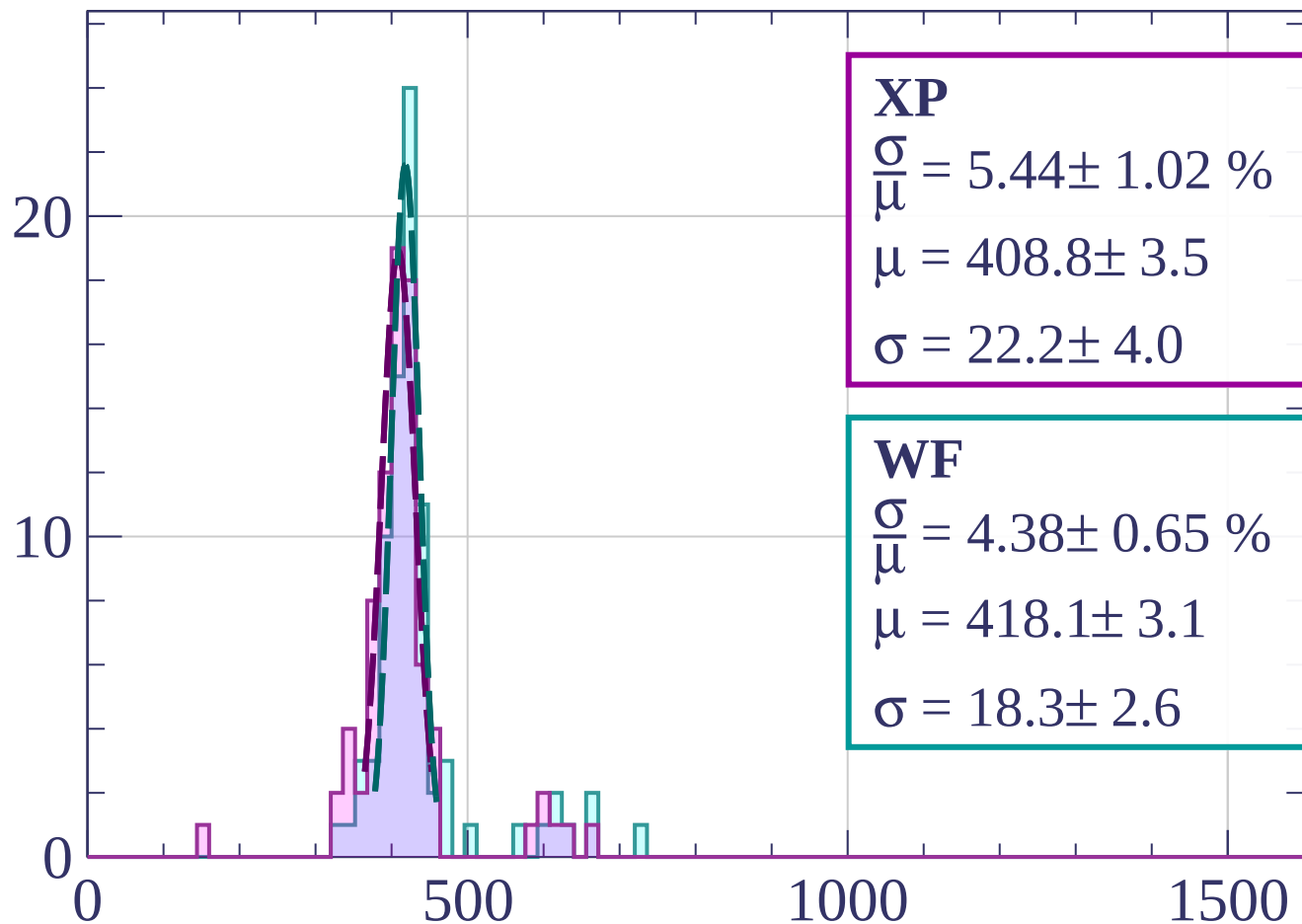
Count



dE/dx [ADC counts/cm]

Energy loss |  $15 < \phi < 18$

Count



**XP**

$$\frac{\sigma}{\mu} = 5.44 \pm 1.02 \%$$

$$\mu = 408.8 \pm 3.5$$

$$\sigma = 22.2 \pm 4.0$$

**WF**

$$\frac{\sigma}{\mu} = 4.38 \pm 0.65 \%$$

$$\mu = 418.1 \pm 3.1$$

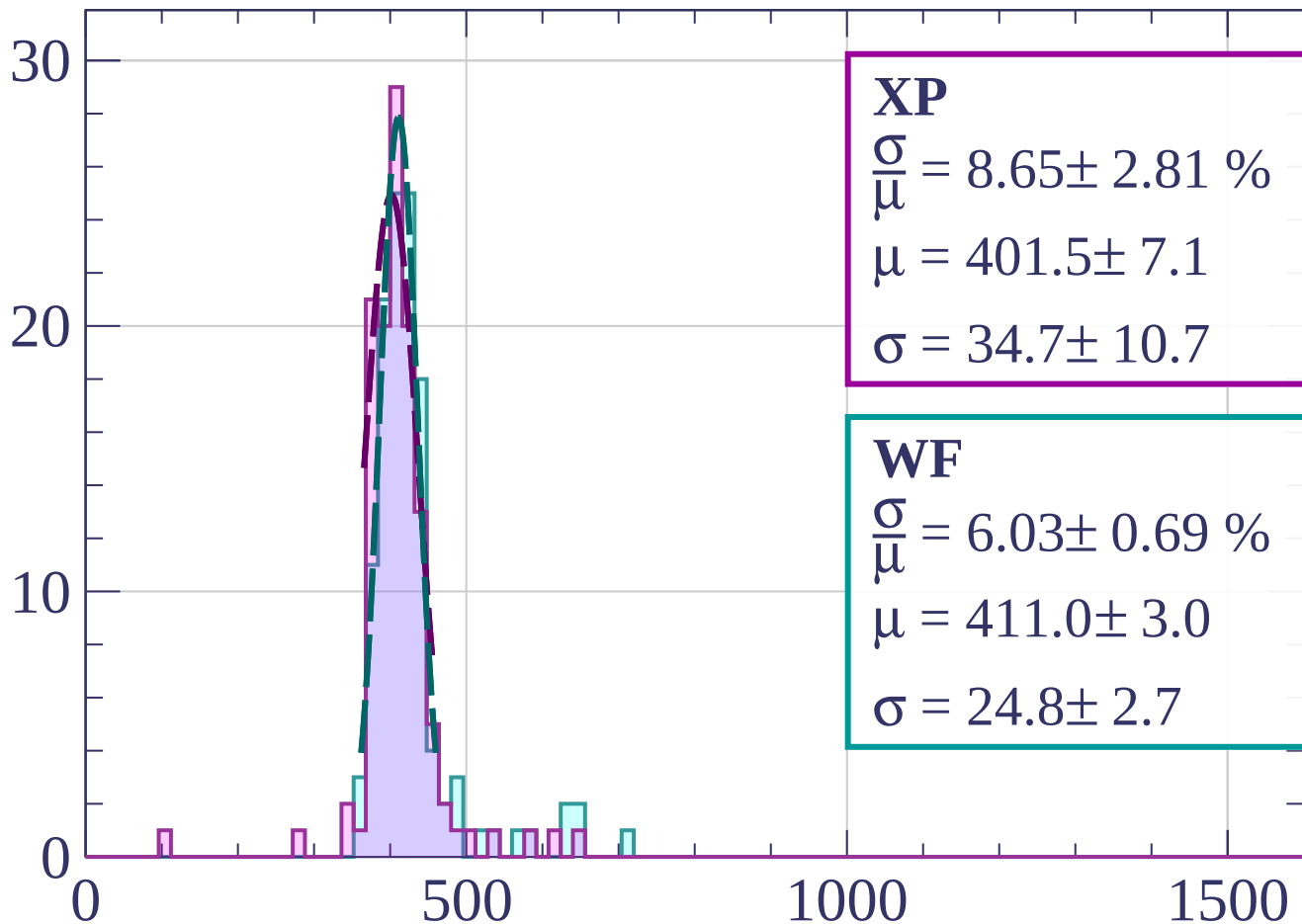
$$\sigma = 18.3 \pm 2.6$$

$dE/dx$  [ADC counts/cm]



Energy loss |  $18 < \phi < 21$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.65 \pm 2.81 \%$$

$$\mu = 401.5 \pm 7.1$$

$$\sigma = 34.7 \pm 10.7$$

**WF**

$$\frac{\sigma}{\mu} = 6.03 \pm 0.69 \%$$

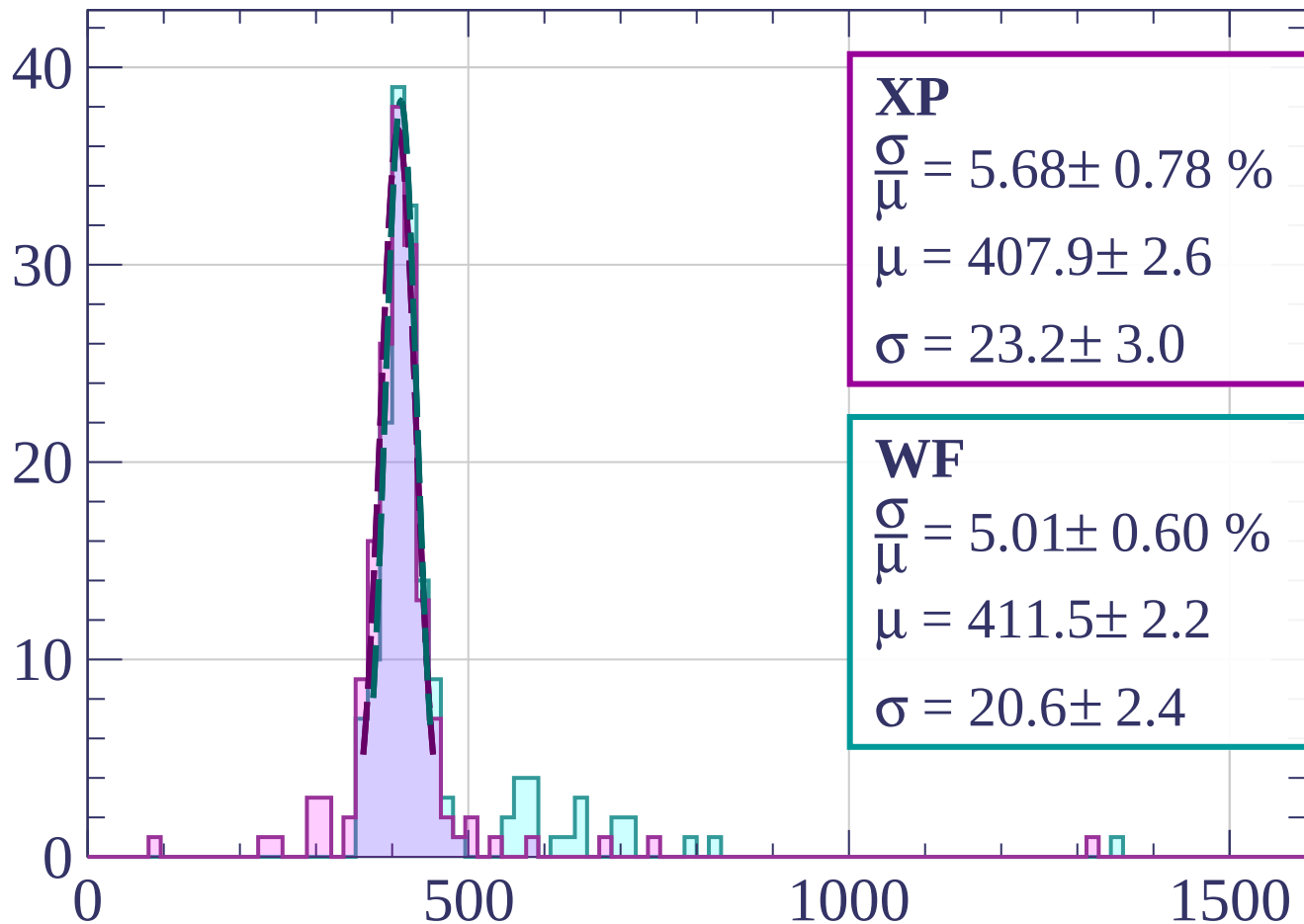
$$\mu = 411.0 \pm 3.0$$

$$\sigma = 24.8 \pm 2.7$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $21 < \phi < 24$

Count



**XP**

$$\frac{\sigma}{\mu} = 5.68 \pm 0.78 \%$$

$$\mu = 407.9 \pm 2.6$$

$$\sigma = 23.2 \pm 3.0$$

**WF**

$$\frac{\sigma}{\mu} = 5.01 \pm 0.60 \%$$

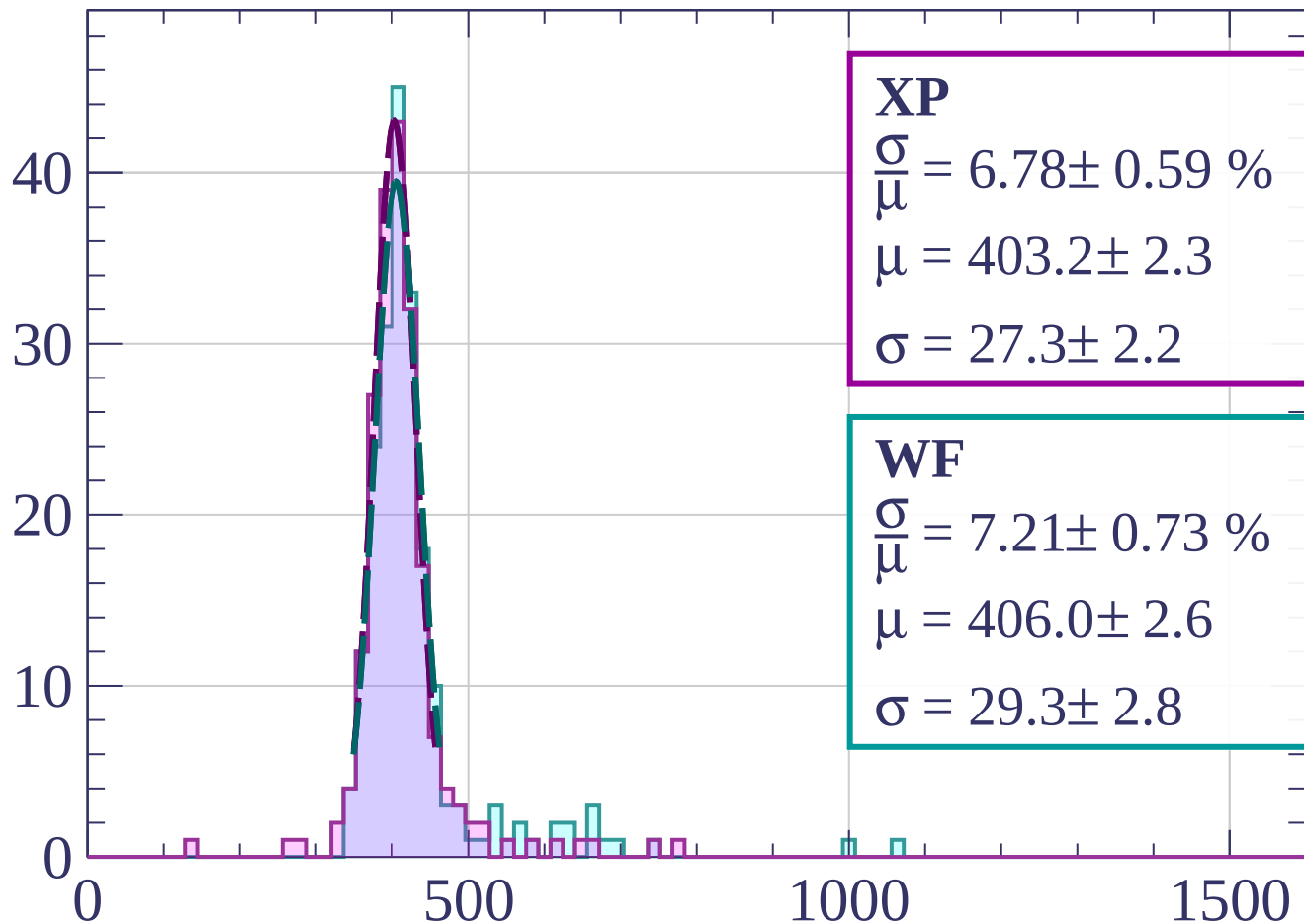
$$\mu = 411.5 \pm 2.2$$

$$\sigma = 20.6 \pm 2.4$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $24 < \phi < 27$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.78 \pm 0.59 \%$$

$$\mu = 403.2 \pm 2.3$$

$$\sigma = 27.3 \pm 2.2$$

**WF**

$$\frac{\sigma}{\mu} = 7.21 \pm 0.73 \%$$

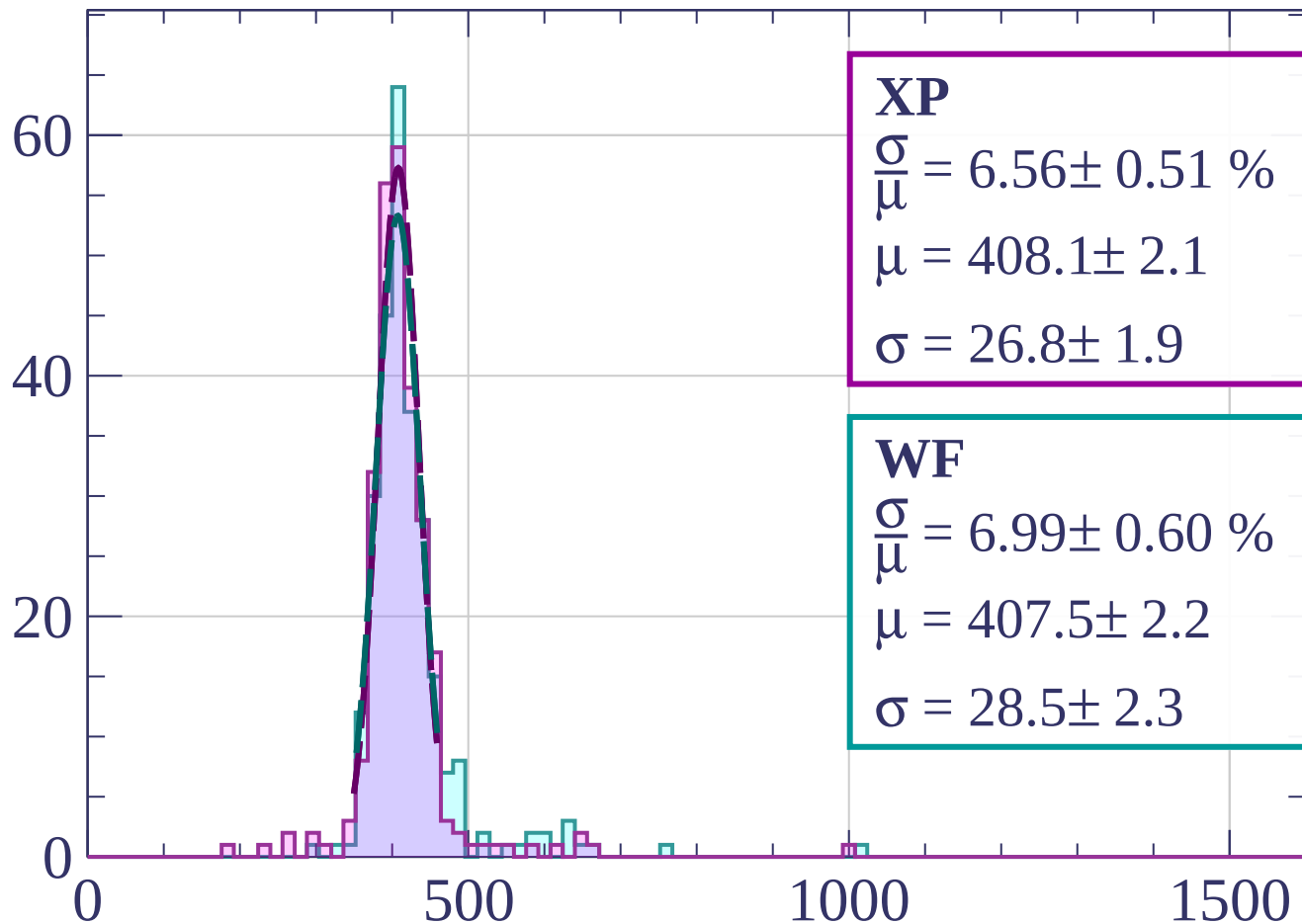
$$\mu = 406.0 \pm 2.6$$

$$\sigma = 29.3 \pm 2.8$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $27 < \phi < 30$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.56 \pm 0.51 \%$$

$$\mu = 408.1 \pm 2.1$$

$$\sigma = 26.8 \pm 1.9$$

**WF**

$$\frac{\sigma}{\mu} = 6.99 \pm 0.60 \%$$

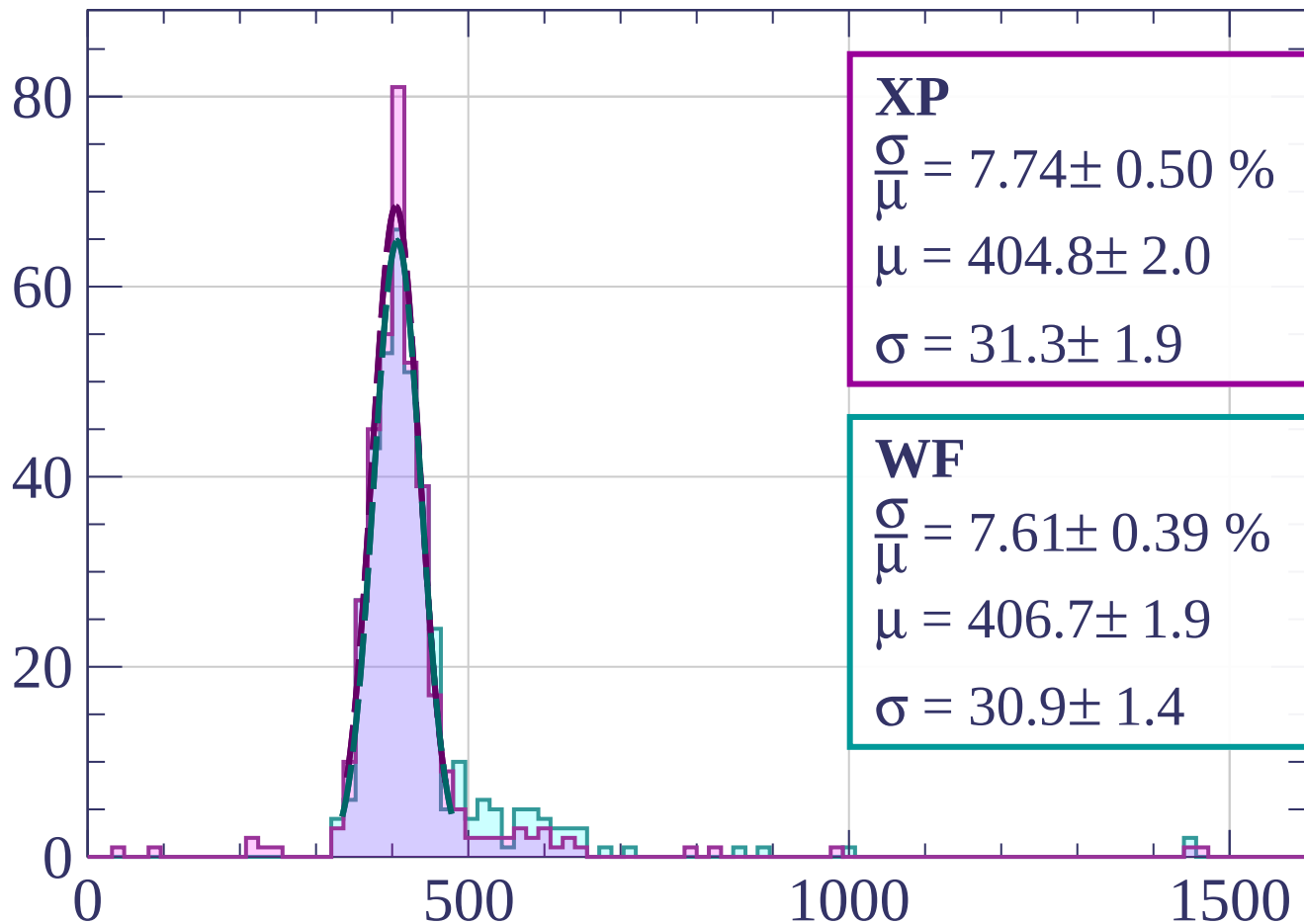
$$\mu = 407.5 \pm 2.2$$

$$\sigma = 28.5 \pm 2.3$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $30 < \phi < 33$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.74 \pm 0.50 \%$$

$$\mu = 404.8 \pm 2.0$$

$$\sigma = 31.3 \pm 1.9$$

**WF**

$$\frac{\sigma}{\mu} = 7.61 \pm 0.39 \%$$

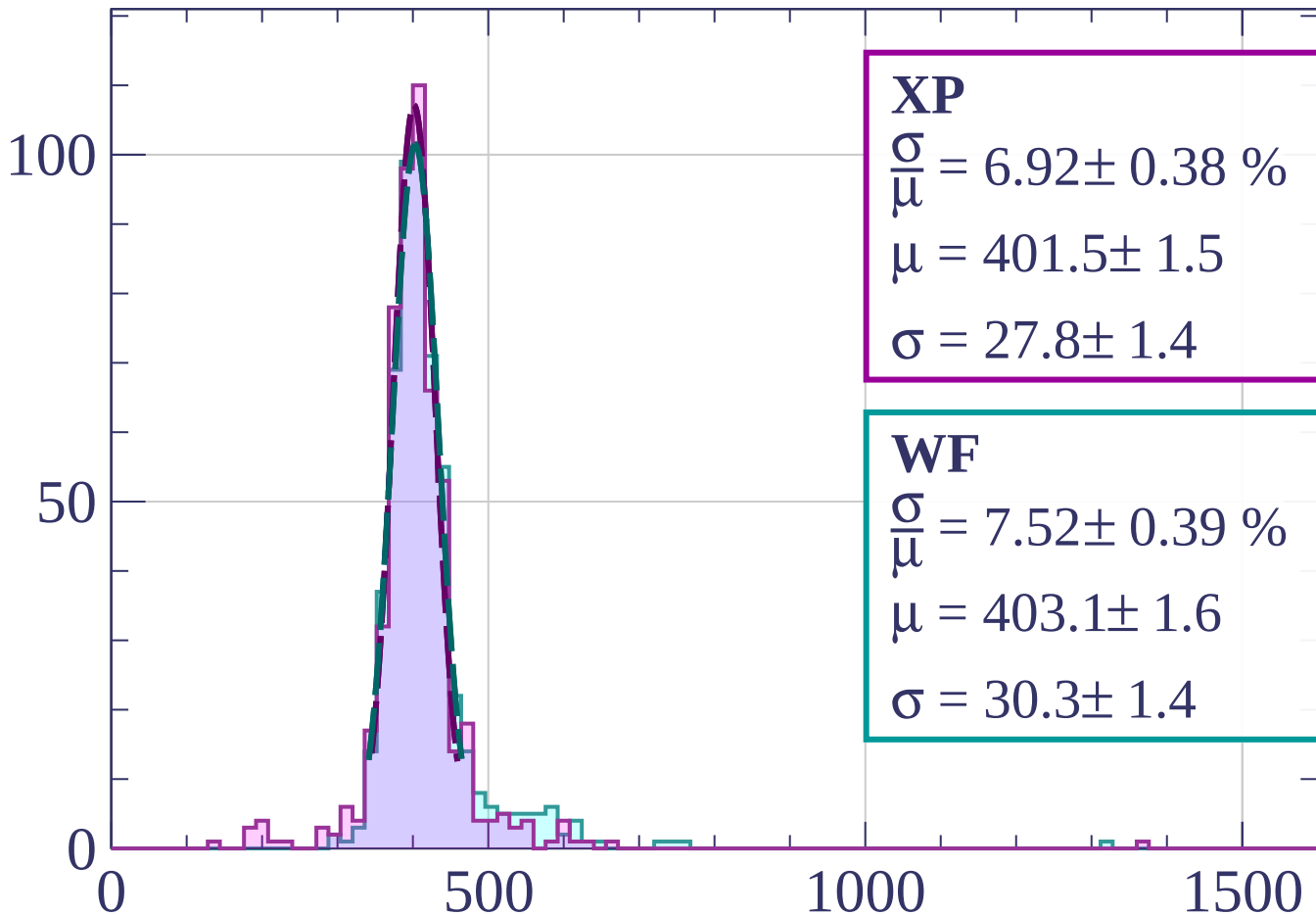
$$\mu = 406.7 \pm 1.9$$

$$\sigma = 30.9 \pm 1.4$$

dE/dx [ADC counts/cm]

Energy loss |  $33 < \phi < 36$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.92 \pm 0.38 \%$$

$$\mu = 401.5 \pm 1.5$$

$$\sigma = 27.8 \pm 1.4$$

**WF**

$$\frac{\sigma}{\mu} = 7.52 \pm 0.39 \%$$

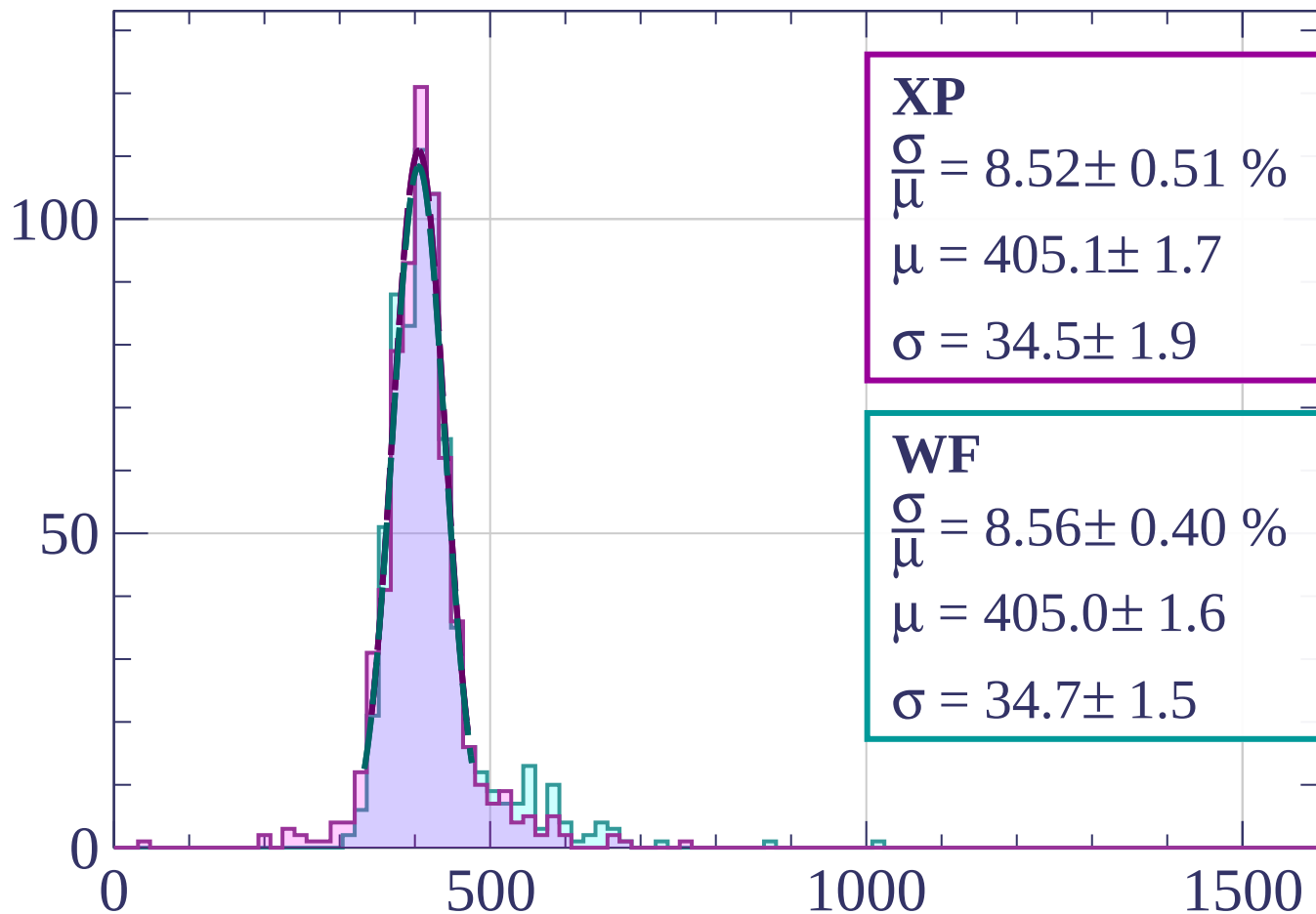
$$\mu = 403.1 \pm 1.6$$

$$\sigma = 30.3 \pm 1.4$$

dE/dx [ADC counts/cm]

Energy loss |  $36 < \phi < 39$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.52 \pm 0.51 \%$$

$$\mu = 405.1 \pm 1.7$$

$$\sigma = 34.5 \pm 1.9$$

**WF**

$$\frac{\sigma}{\mu} = 8.56 \pm 0.40 \%$$

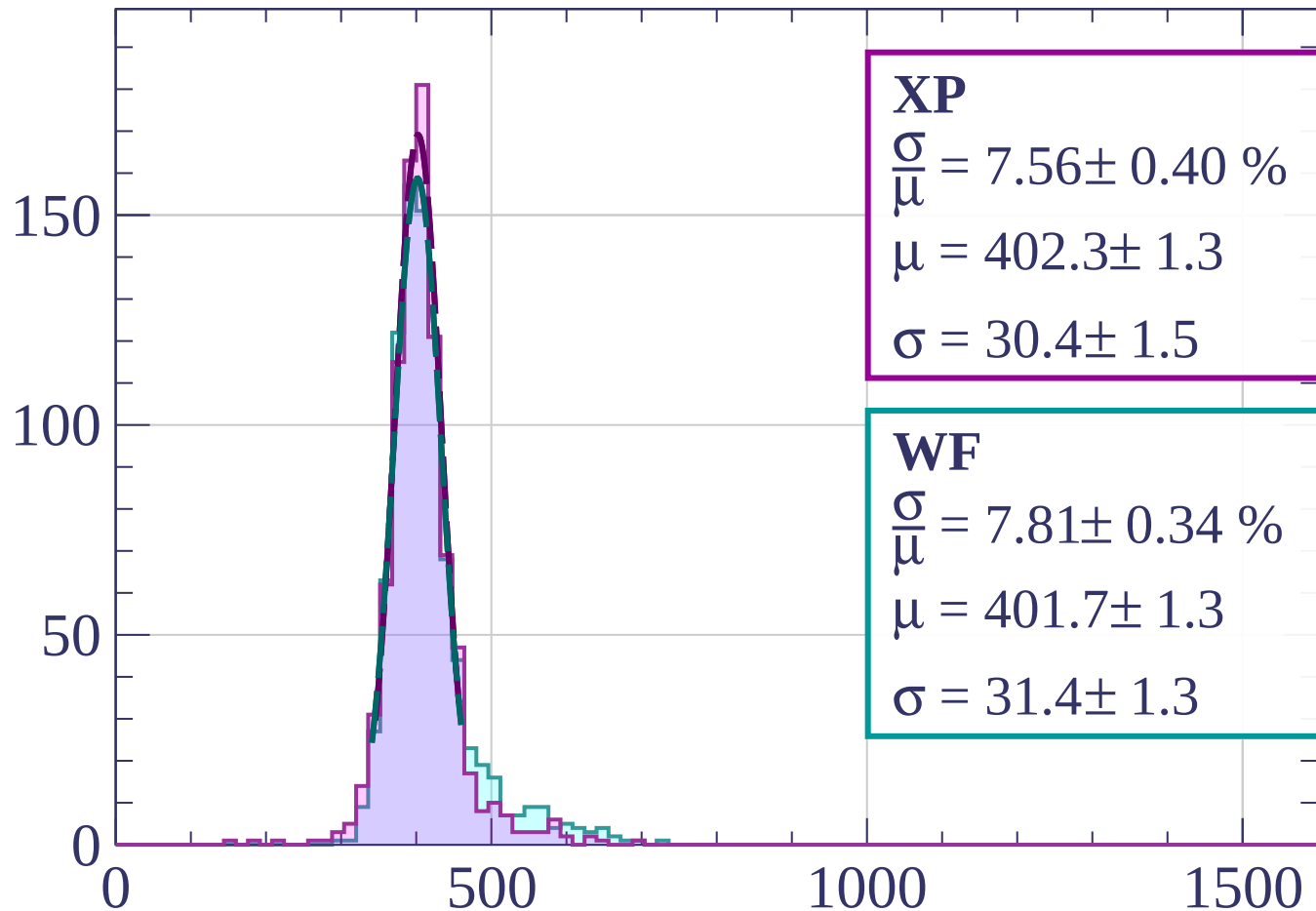
$$\mu = 405.0 \pm 1.6$$

$$\sigma = 34.7 \pm 1.5$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $39 < \phi < 42$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.56 \pm 0.40 \%$$

$$\mu = 402.3 \pm 1.3$$

$$\sigma = 30.4 \pm 1.5$$

**WF**

$$\frac{\sigma}{\mu} = 7.81 \pm 0.34 \%$$

$$\mu = 401.7 \pm 1.3$$

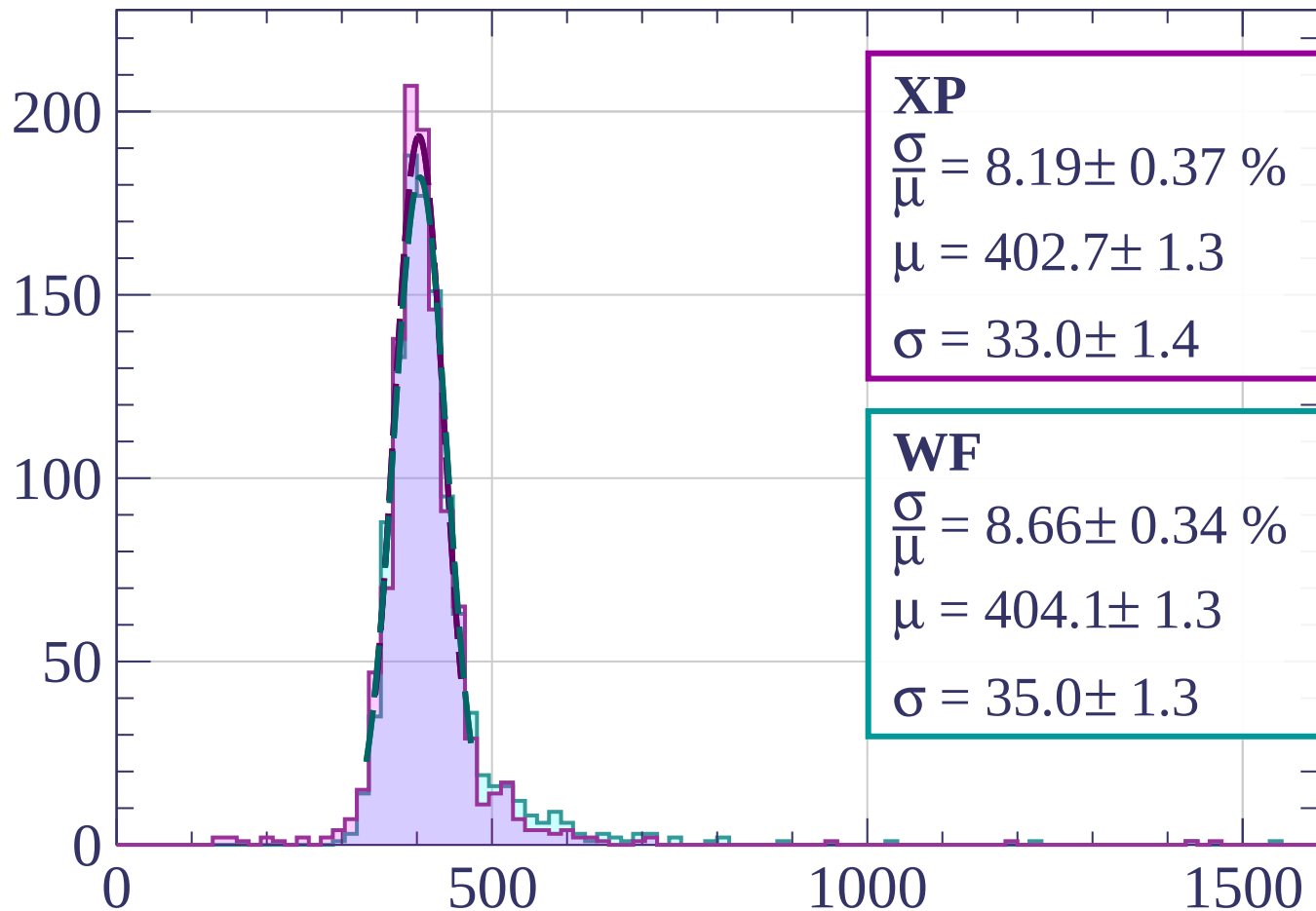
$$\sigma = 31.4 \pm 1.3$$

$dE/dx$  [ADC counts/cm]



Energy loss |  $42 < \phi < 45$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.19 \pm 0.37 \%$$

$$\mu = 402.7 \pm 1.3$$

$$\sigma = 33.0 \pm 1.4$$

**WF**

$$\frac{\sigma}{\mu} = 8.66 \pm 0.34 \%$$

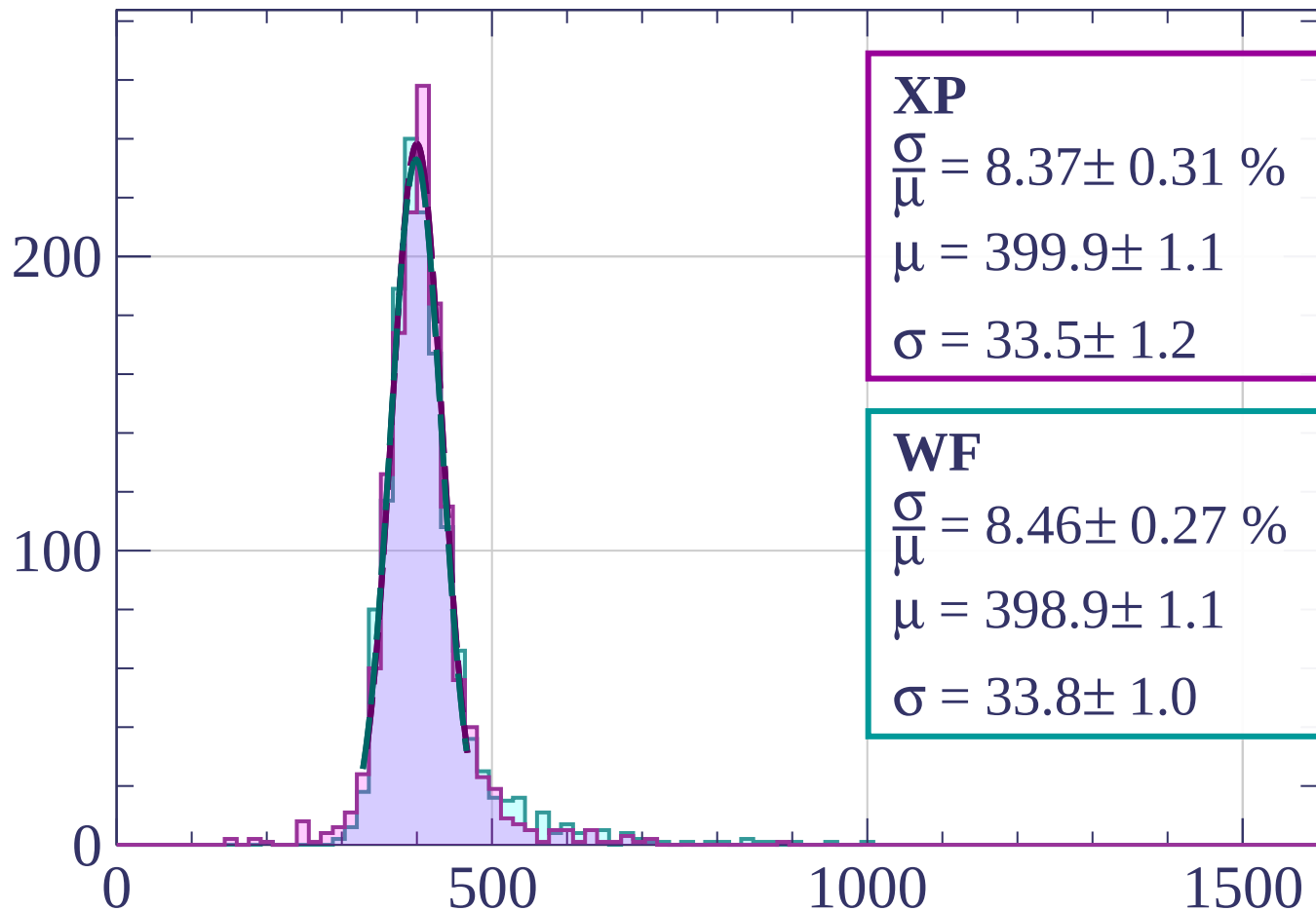
$$\mu = 404.1 \pm 1.3$$

$$\sigma = 35.0 \pm 1.3$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $45 < \phi < 48$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.37 \pm 0.31 \%$$

$$\mu = 399.9 \pm 1.1$$

$$\sigma = 33.5 \pm 1.2$$

**WF**

$$\frac{\sigma}{\mu} = 8.46 \pm 0.27 \%$$

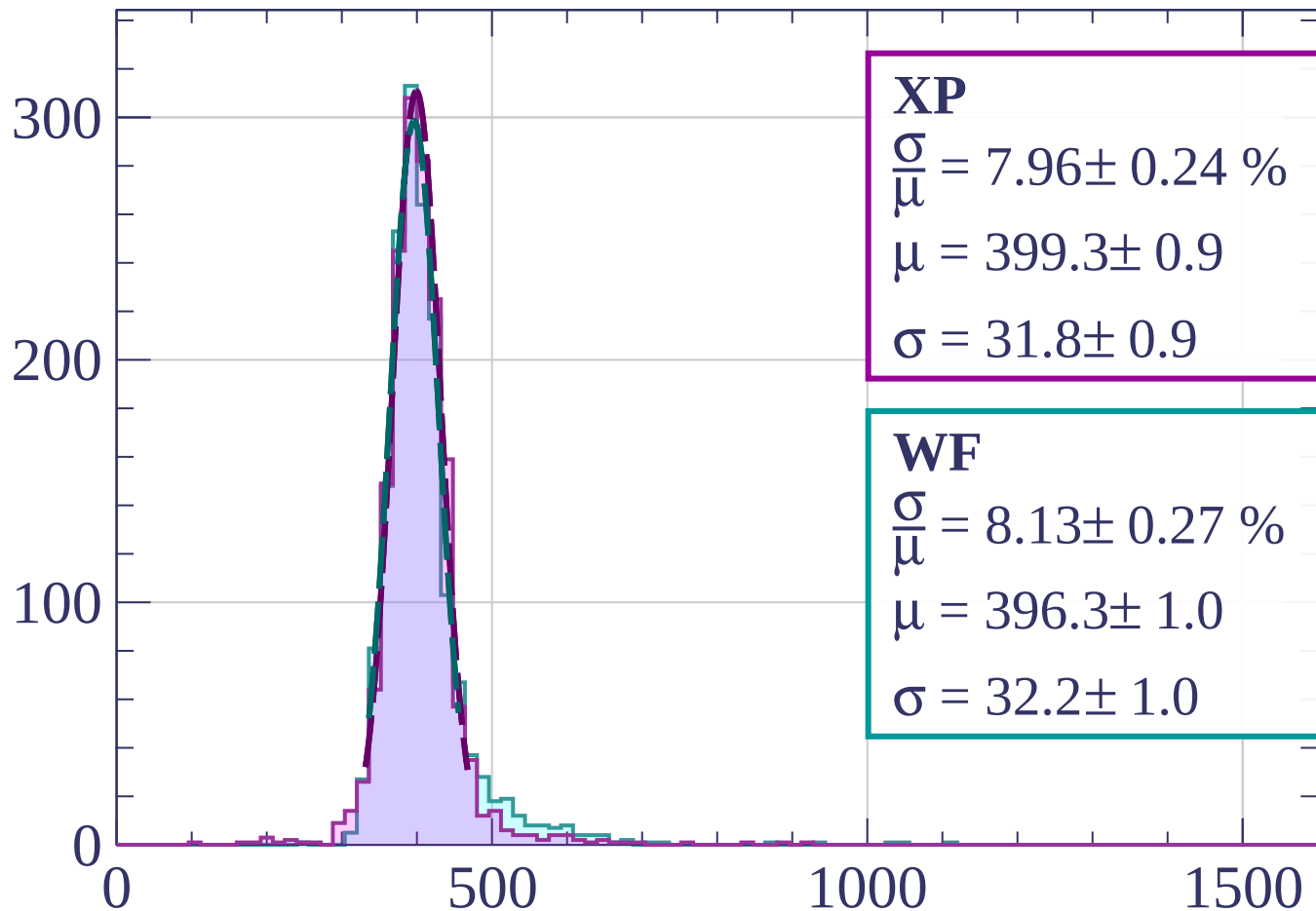
$$\mu = 398.9 \pm 1.1$$

$$\sigma = 33.8 \pm 1.0$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $48 < \phi < 51$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.96 \pm 0.24 \%$$

$$\mu = 399.3 \pm 0.9$$

$$\sigma = 31.8 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 8.13 \pm 0.27 \%$$

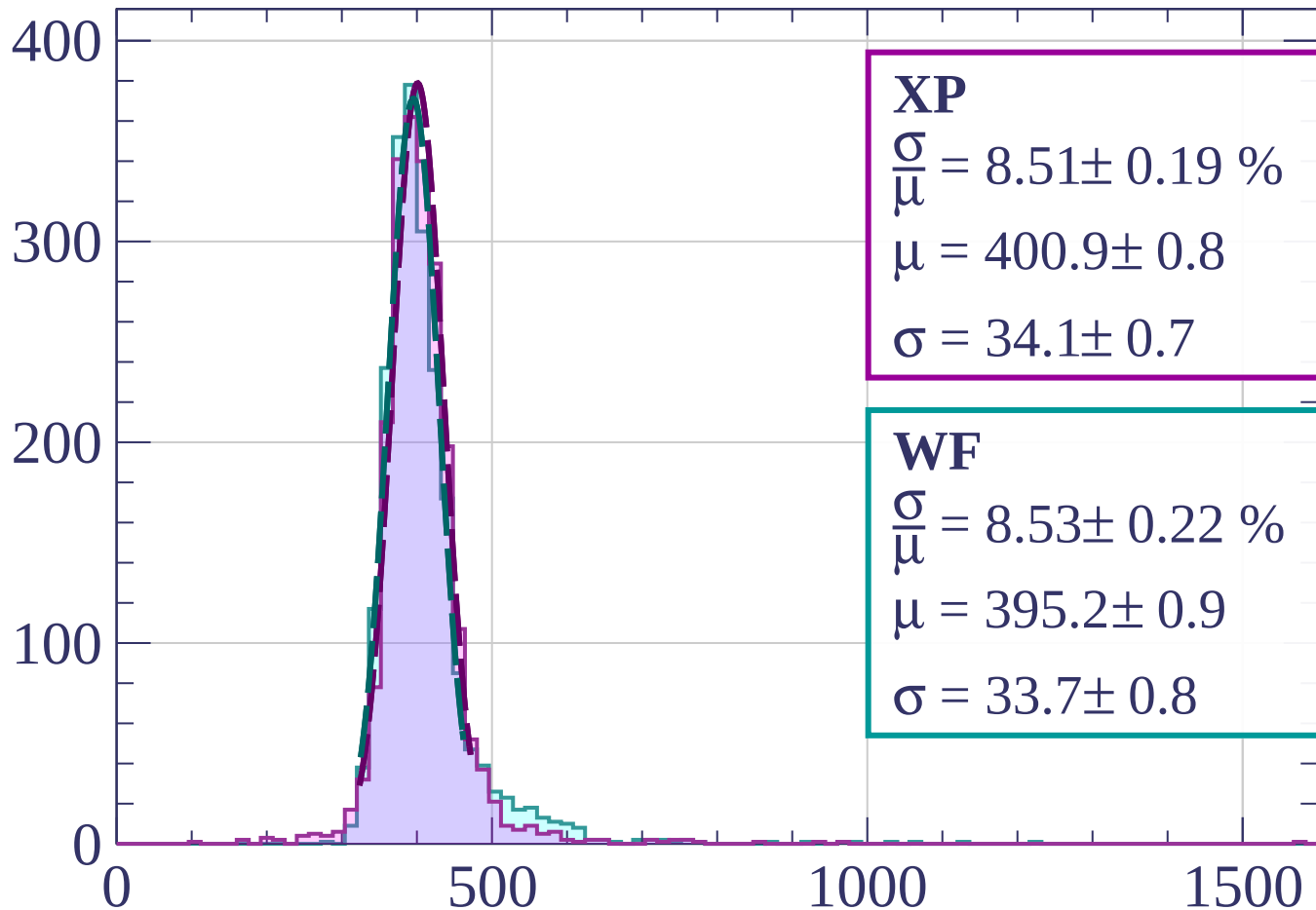
$$\mu = 396.3 \pm 1.0$$

$$\sigma = 32.2 \pm 1.0$$

dE/dx [ADC counts/cm]

Energy loss |  $51 < \phi < 54$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.51 \pm 0.19 \%$$

$$\mu = 400.9 \pm 0.8$$

$$\sigma = 34.1 \pm 0.7$$

**WF**

$$\frac{\sigma}{\mu} = 8.53 \pm 0.22 \%$$

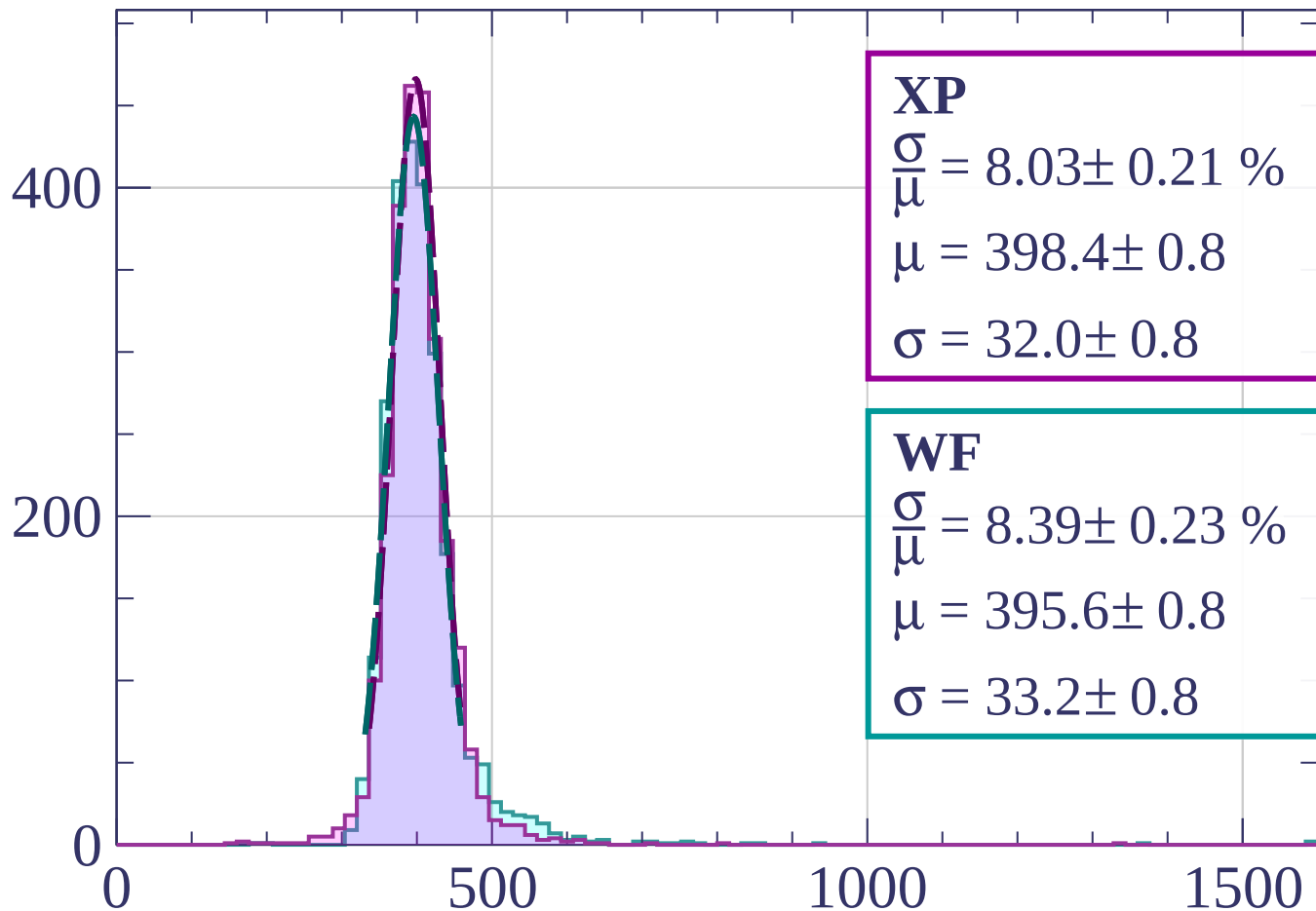
$$\mu = 395.2 \pm 0.9$$

$$\sigma = 33.7 \pm 0.8$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $54 < \phi < 57$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.03 \pm 0.21 \%$$

$$\mu = 398.4 \pm 0.8$$

$$\sigma = 32.0 \pm 0.8$$

**WF**

$$\frac{\sigma}{\mu} = 8.39 \pm 0.23 \%$$

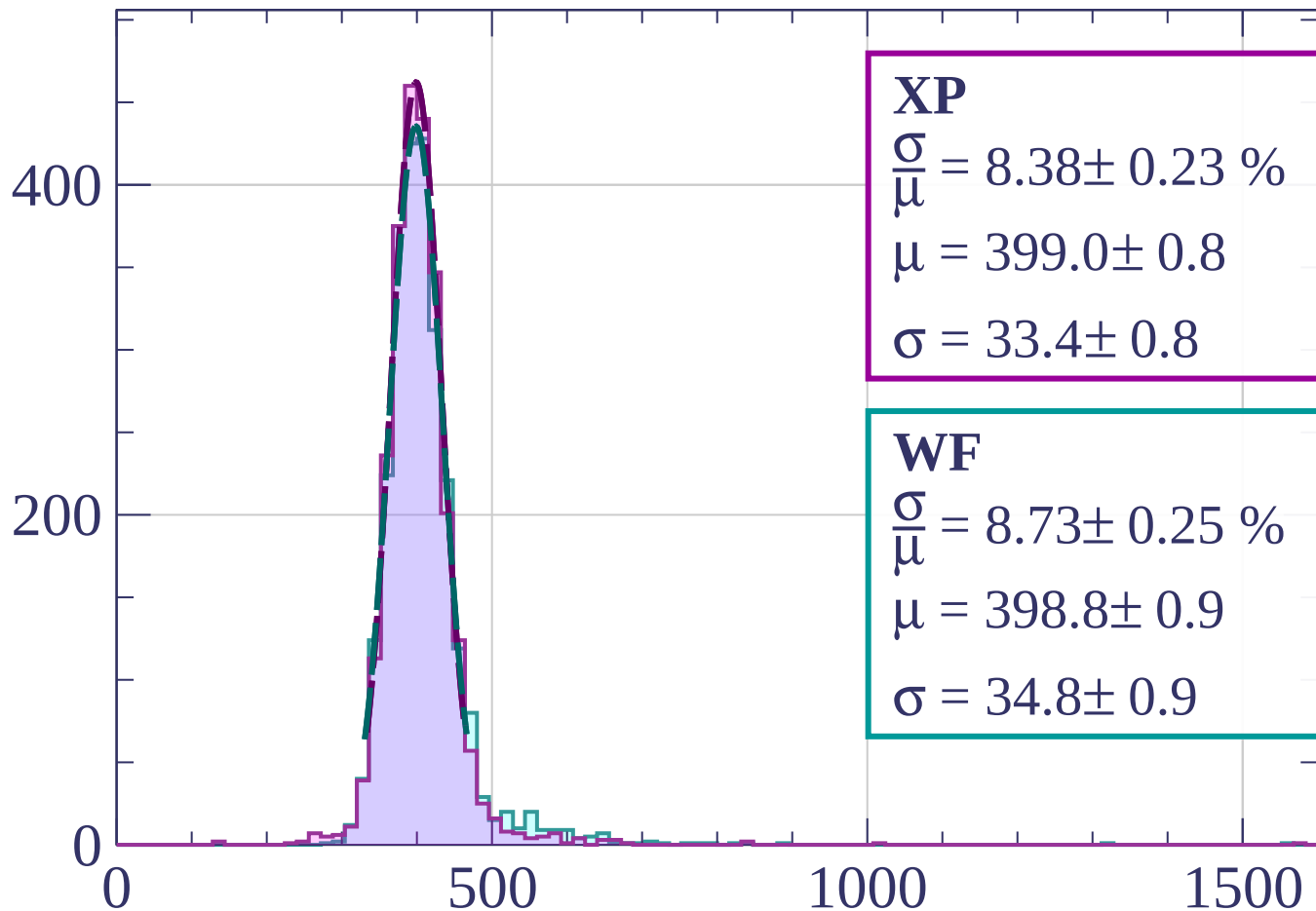
$$\mu = 395.6 \pm 0.8$$

$$\sigma = 33.2 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss |  $57 < \phi < 60$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.38 \pm 0.23 \%$$

$$\mu = 399.0 \pm 0.8$$

$$\sigma = 33.4 \pm 0.8$$

**WF**

$$\frac{\sigma}{\mu} = 8.73 \pm 0.25 \%$$

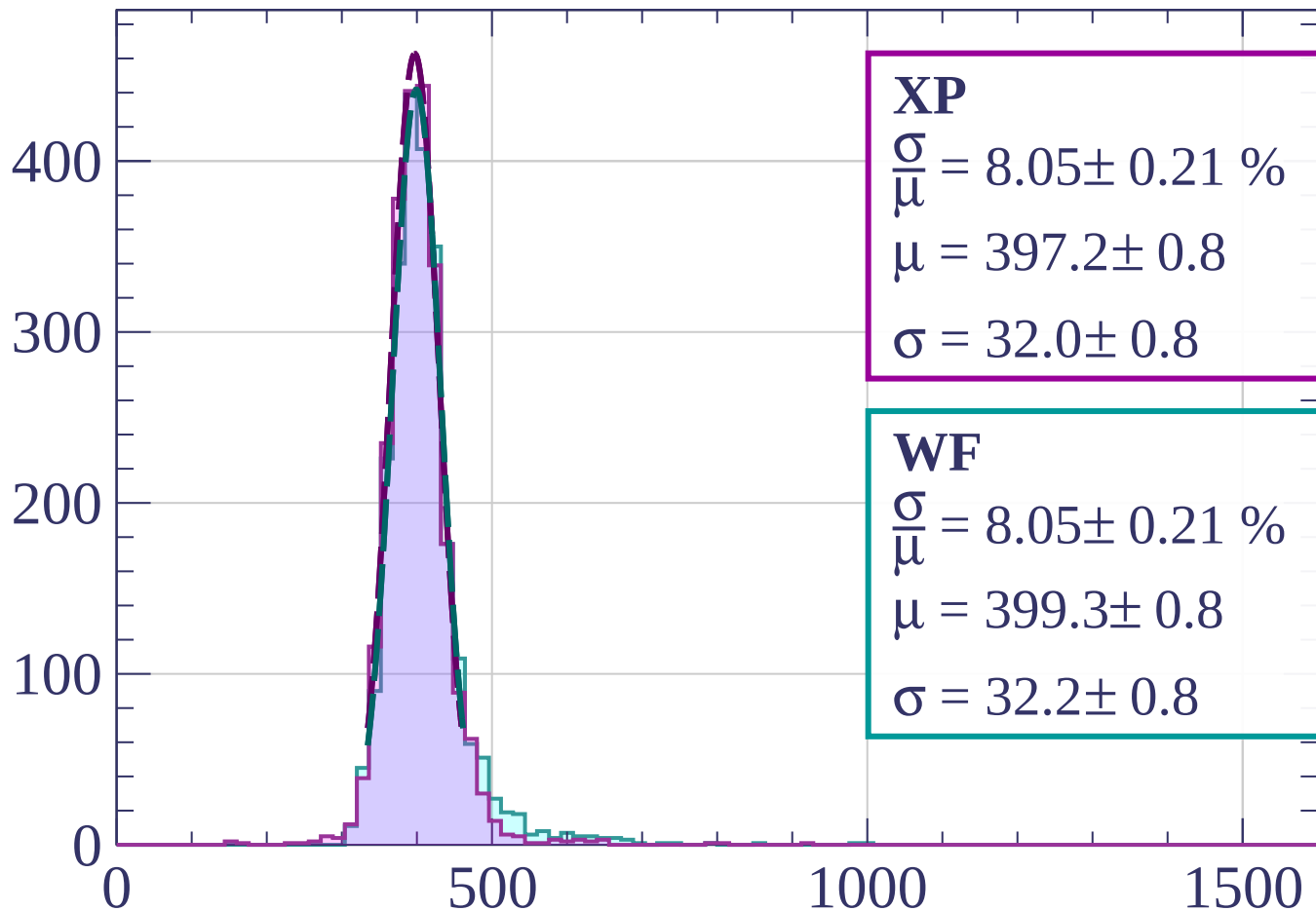
$$\mu = 398.8 \pm 0.9$$

$$\sigma = 34.8 \pm 0.9$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $60 < \phi < 63$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.05 \pm 0.21 \%$$

$$\mu = 397.2 \pm 0.8$$

$$\sigma = 32.0 \pm 0.8$$

**WF**

$$\frac{\sigma}{\mu} = 8.05 \pm 0.21 \%$$

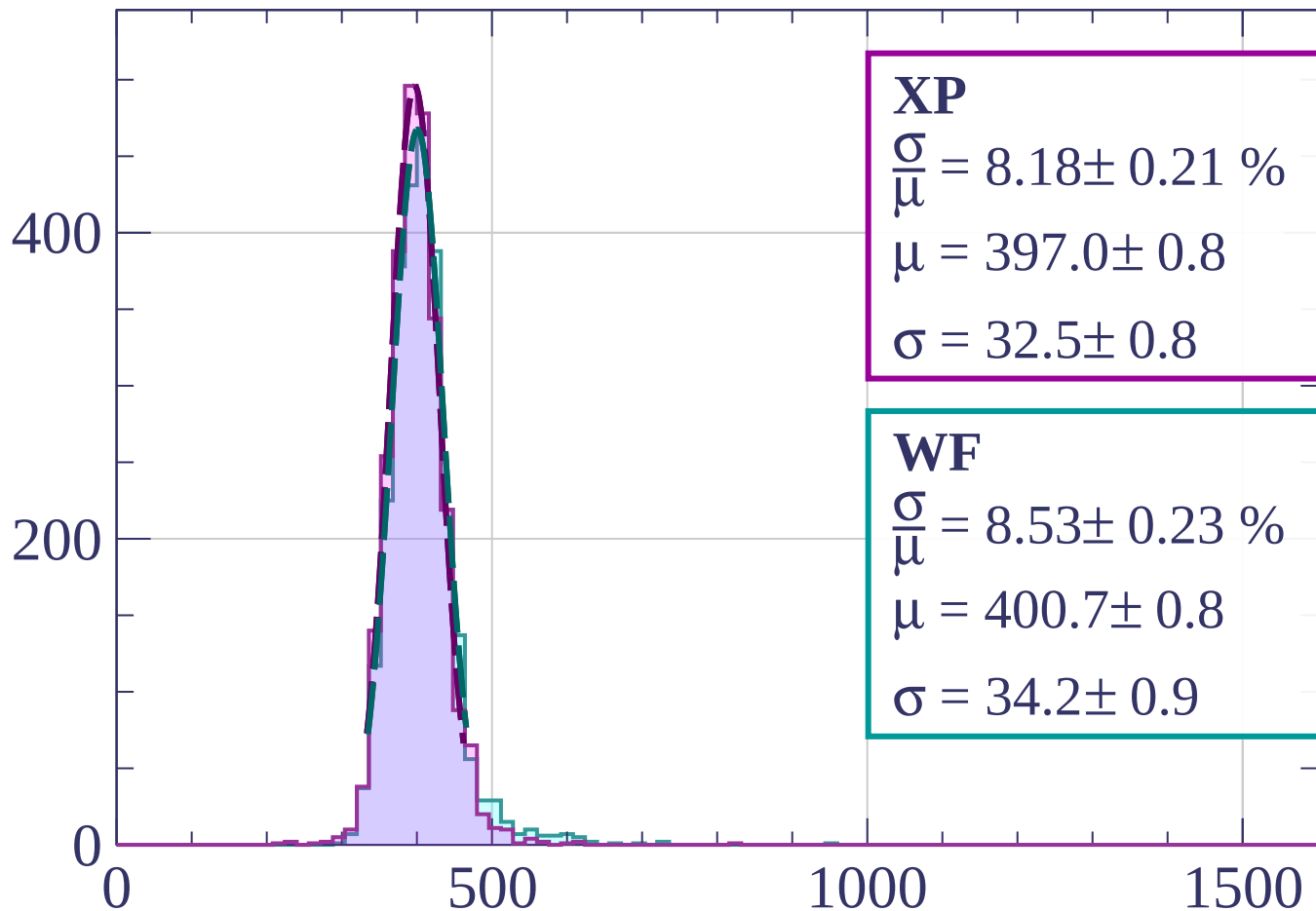
$$\mu = 399.3 \pm 0.8$$

$$\sigma = 32.2 \pm 0.8$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $63 < \phi < 66$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.18 \pm 0.21 \%$$

$$\mu = 397.0 \pm 0.8$$

$$\sigma = 32.5 \pm 0.8$$

**WF**

$$\frac{\sigma}{\mu} = 8.53 \pm 0.23 \%$$

$$\mu = 400.7 \pm 0.8$$

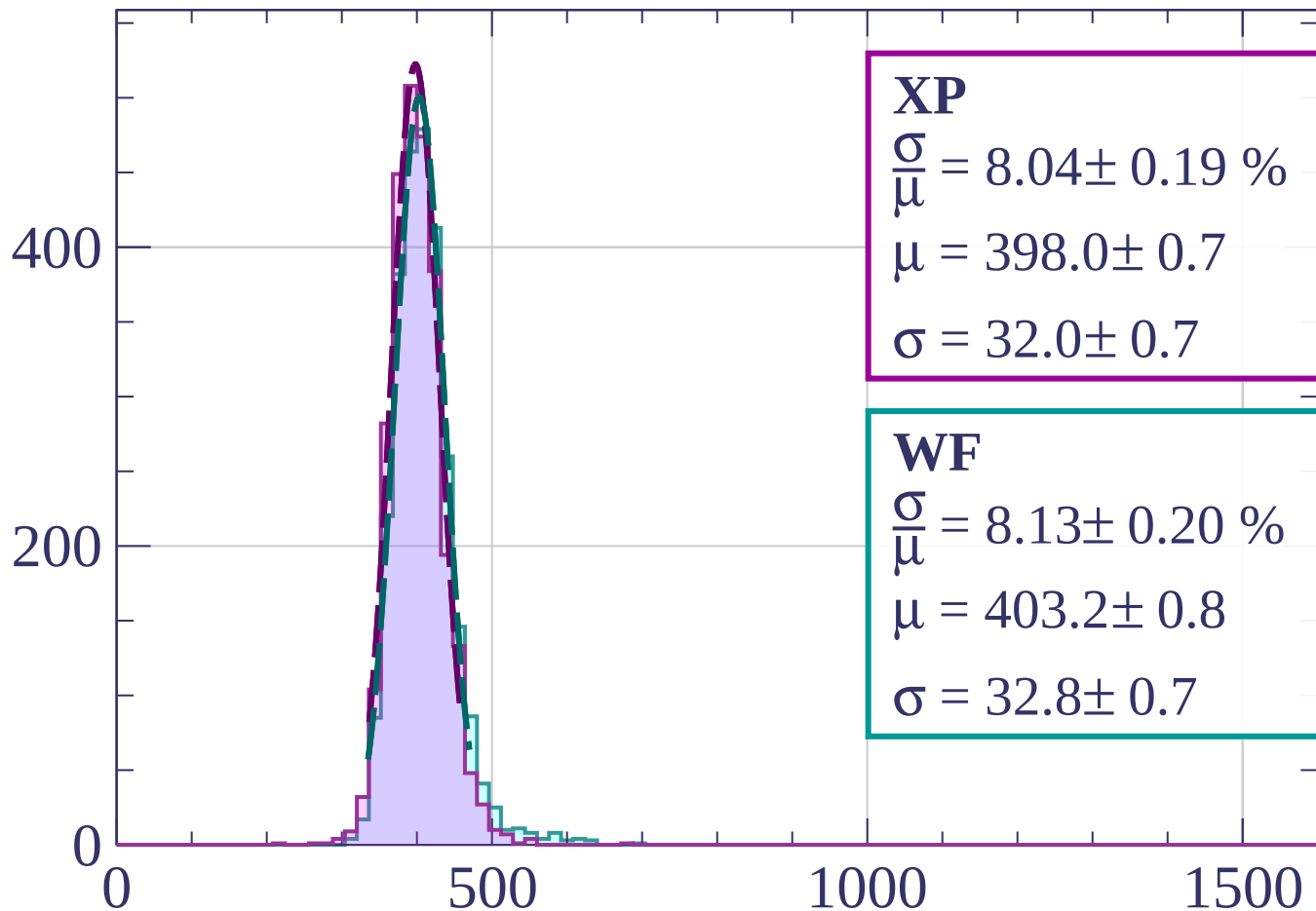
$$\sigma = 34.2 \pm 0.9$$

$dE/dx$  [ADC counts/cm]



Energy loss |  $66 < \phi < 69$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.04 \pm 0.19 \%$$

$$\mu = 398.0 \pm 0.7$$

$$\sigma = 32.0 \pm 0.7$$

**WF**

$$\frac{\sigma}{\mu} = 8.13 \pm 0.20 \%$$

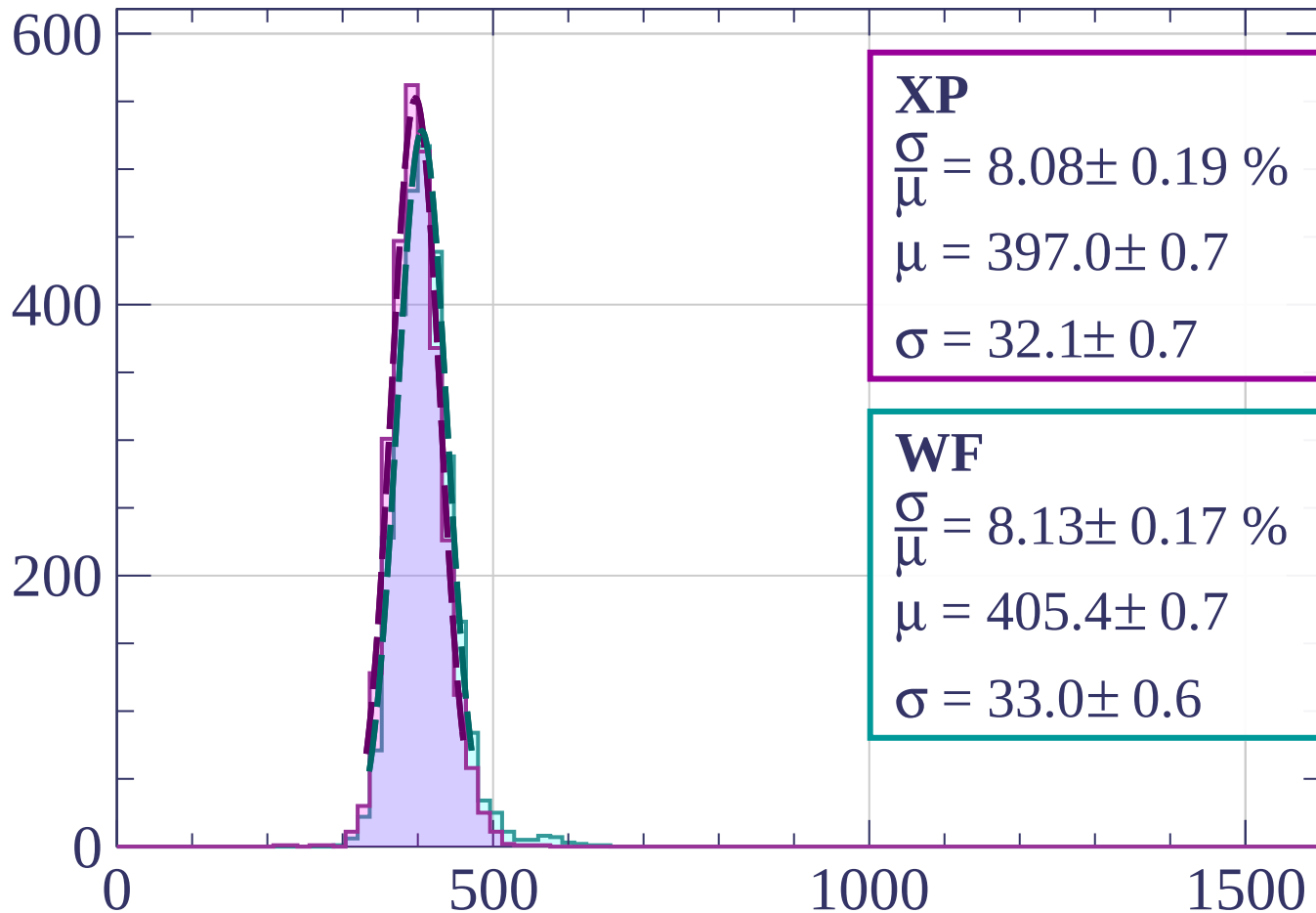
$$\mu = 403.2 \pm 0.8$$

$$\sigma = 32.8 \pm 0.7$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $69 < \phi < 72$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.08 \pm 0.19 \%$$

$$\mu = 397.0 \pm 0.7$$

$$\sigma = 32.1 \pm 0.7$$

**WF**

$$\frac{\sigma}{\mu} = 8.13 \pm 0.17 \%$$

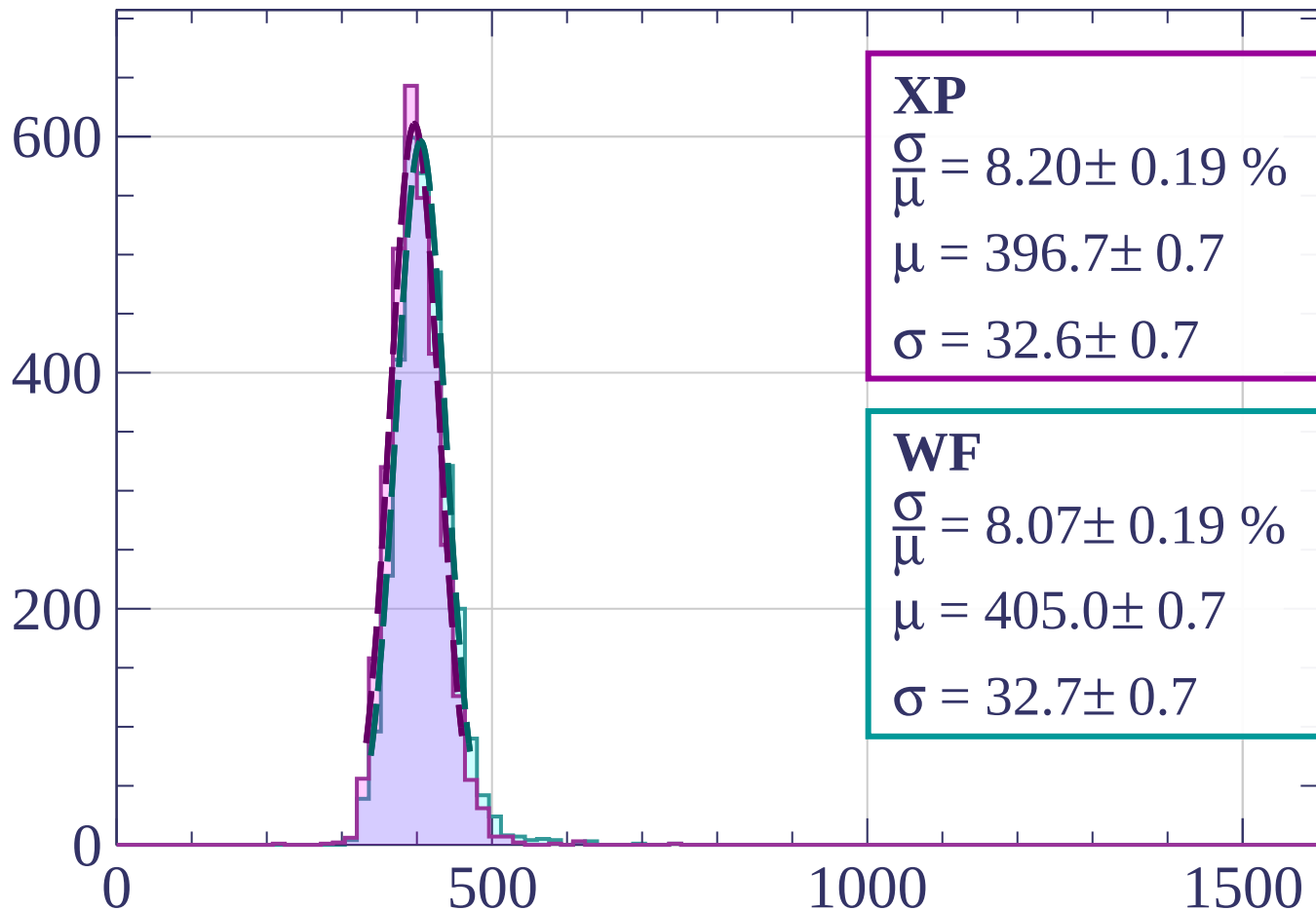
$$\mu = 405.4 \pm 0.7$$

$$\sigma = 33.0 \pm 0.6$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $72 < \phi < 75$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.20 \pm 0.19 \%$$

$$\mu = 396.7 \pm 0.7$$

$$\sigma = 32.6 \pm 0.7$$

**WF**

$$\frac{\sigma}{\mu} = 8.07 \pm 0.19 \%$$

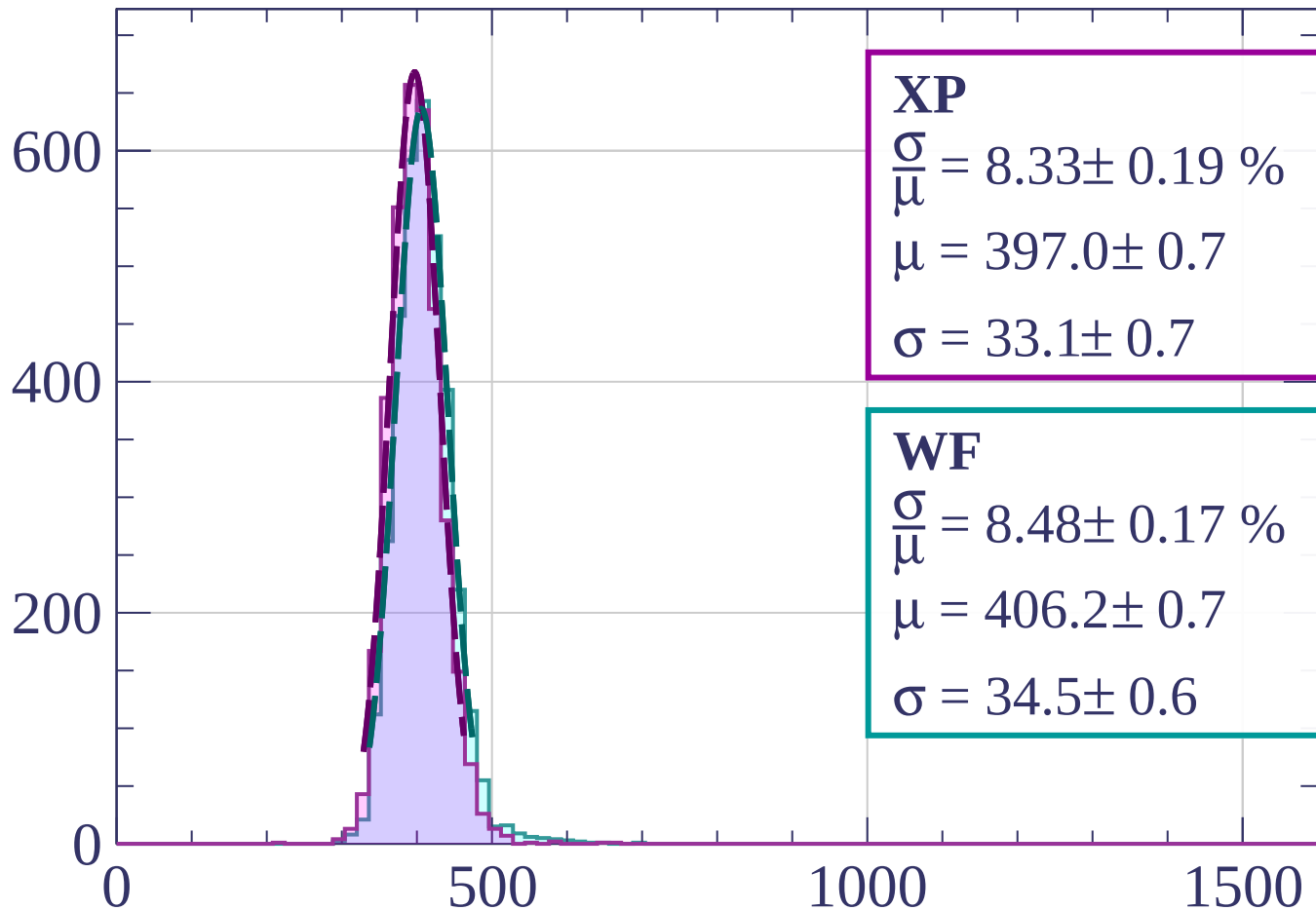
$$\mu = 405.0 \pm 0.7$$

$$\sigma = 32.7 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss |  $75 < \phi < 78$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.33 \pm 0.19 \%$$

$$\mu = 397.0 \pm 0.7$$

$$\sigma = 33.1 \pm 0.7$$

**WF**

$$\frac{\sigma}{\mu} = 8.48 \pm 0.17 \%$$

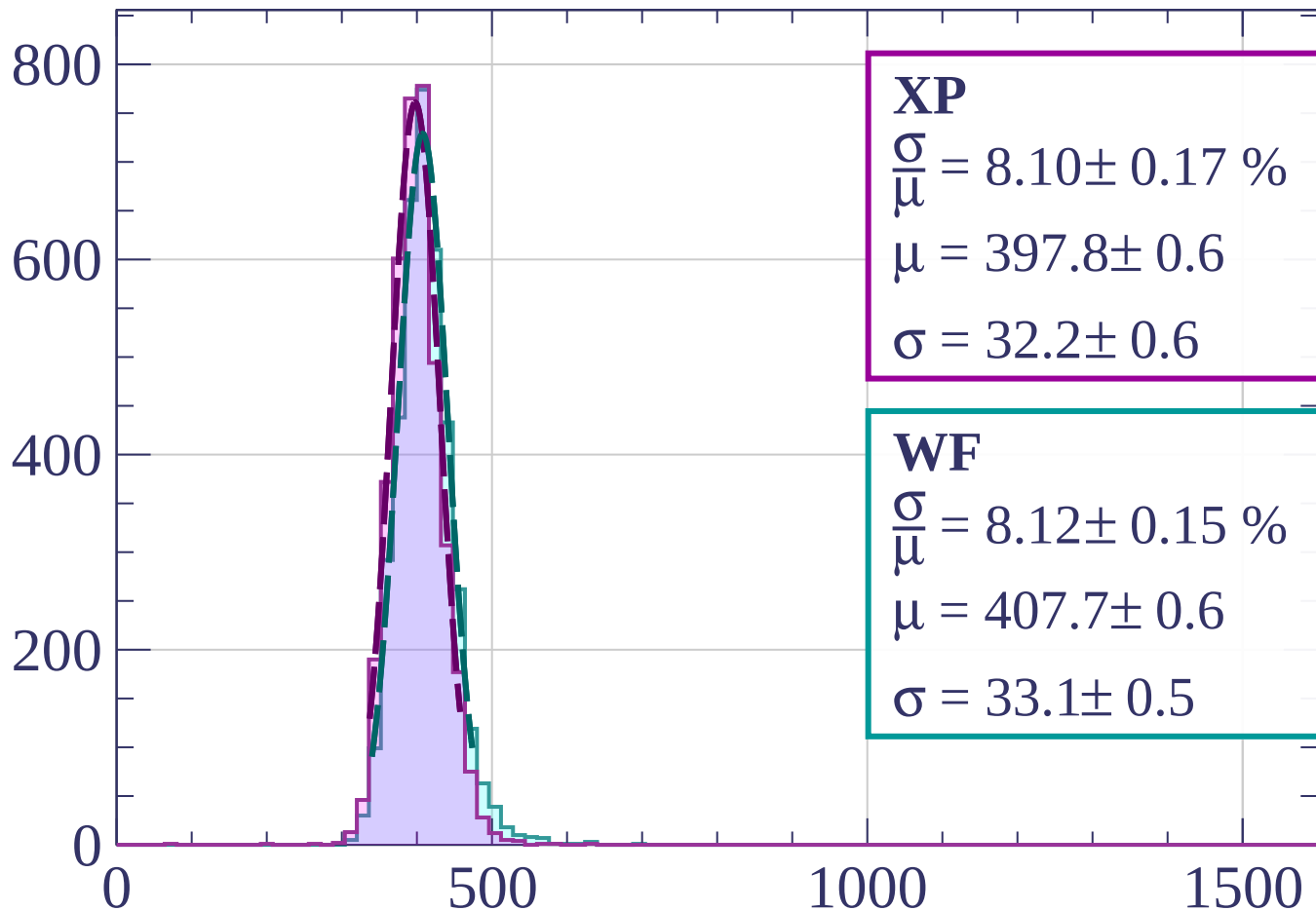
$$\mu = 406.2 \pm 0.7$$

$$\sigma = 34.5 \pm 0.6$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $78 < \phi < 81$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.10 \pm 0.17 \%$$

$$\mu = 397.8 \pm 0.6$$

$$\sigma = 32.2 \pm 0.6$$

**WF**

$$\frac{\sigma}{\mu} = 8.12 \pm 0.15 \%$$

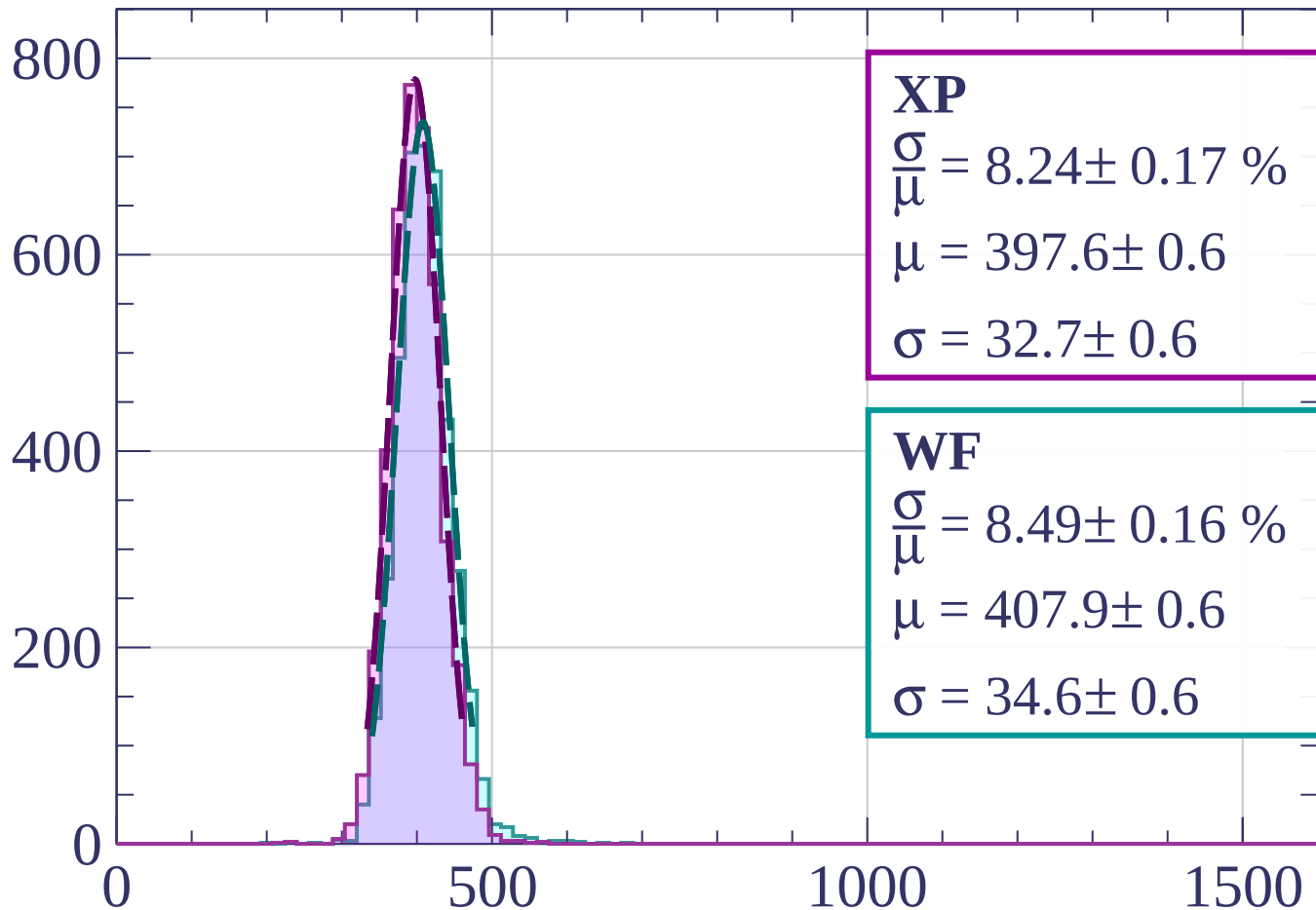
$$\mu = 407.7 \pm 0.6$$

$$\sigma = 33.1 \pm 0.5$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $81 < \phi < 84$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.24 \pm 0.17 \%$$

$$\mu = 397.6 \pm 0.6$$

$$\sigma = 32.7 \pm 0.6$$

**WF**

$$\frac{\sigma}{\mu} = 8.49 \pm 0.16 \%$$

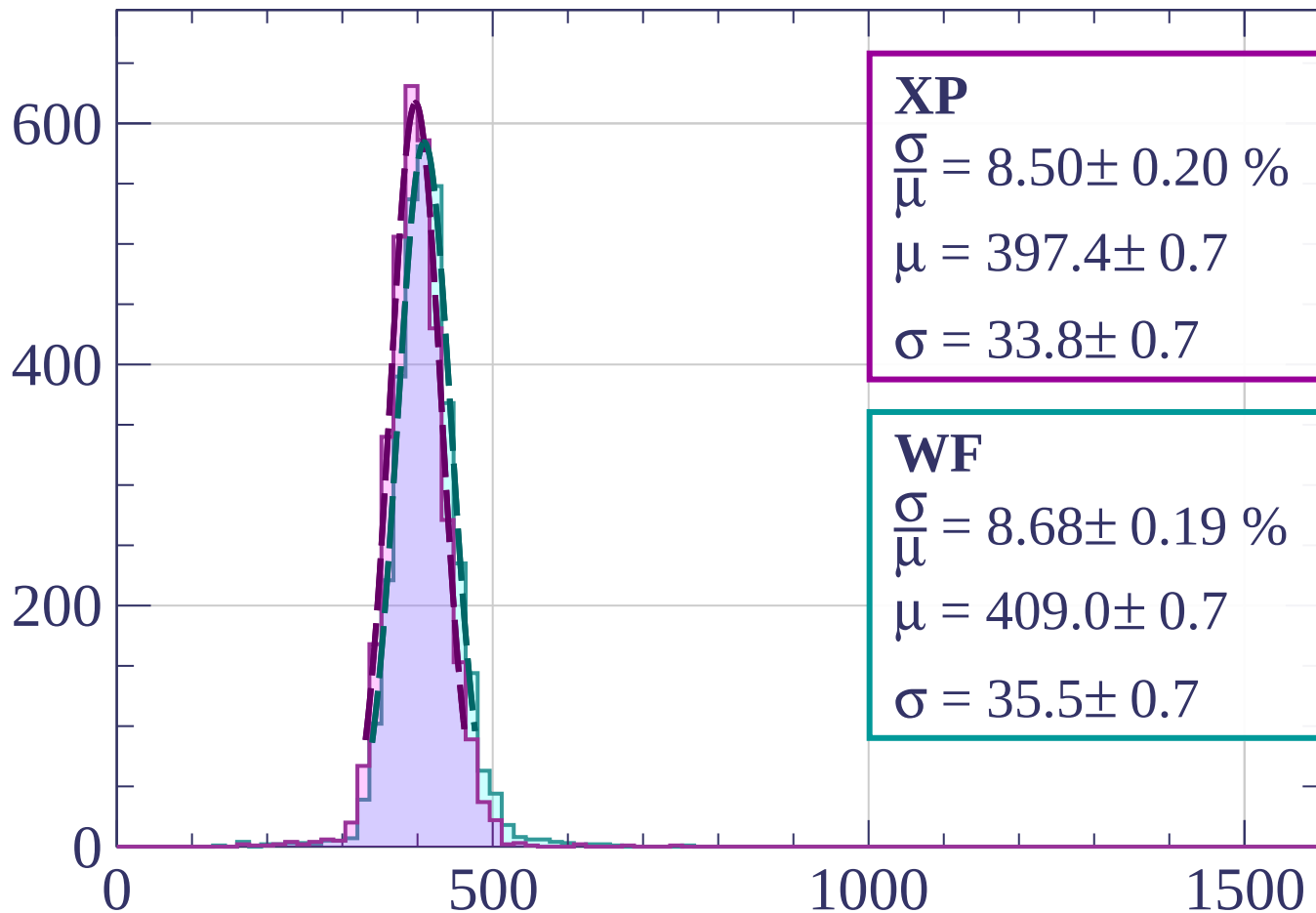
$$\mu = 407.9 \pm 0.6$$

$$\sigma = 34.6 \pm 0.6$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $84 < \phi < 87$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.50 \pm 0.20 \%$$

$$\mu = 397.4 \pm 0.7$$

$$\sigma = 33.8 \pm 0.7$$

**WF**

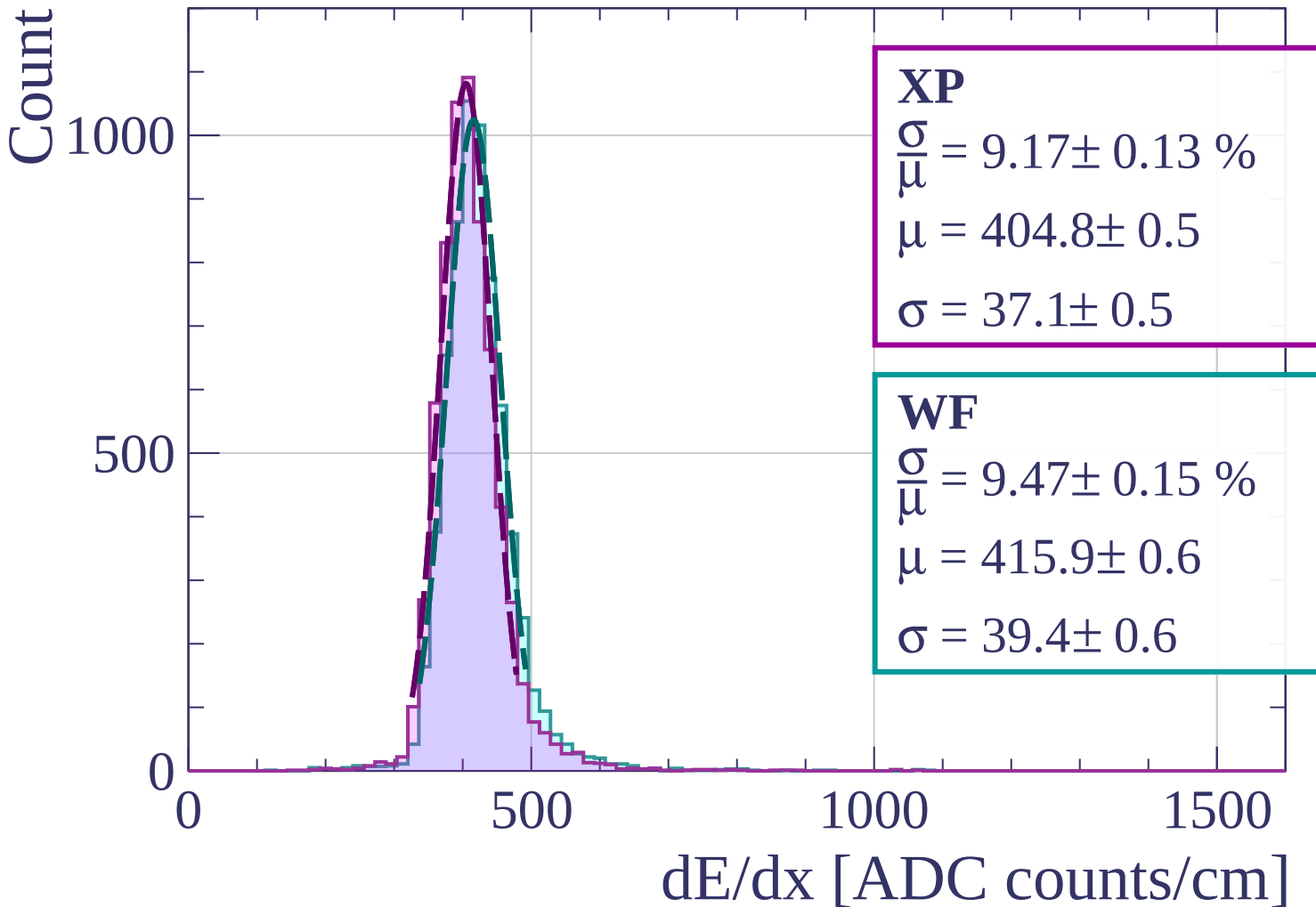
$$\frac{\sigma}{\mu} = 8.68 \pm 0.19 \%$$

$$\mu = 409.0 \pm 0.7$$

$$\sigma = 35.5 \pm 0.7$$

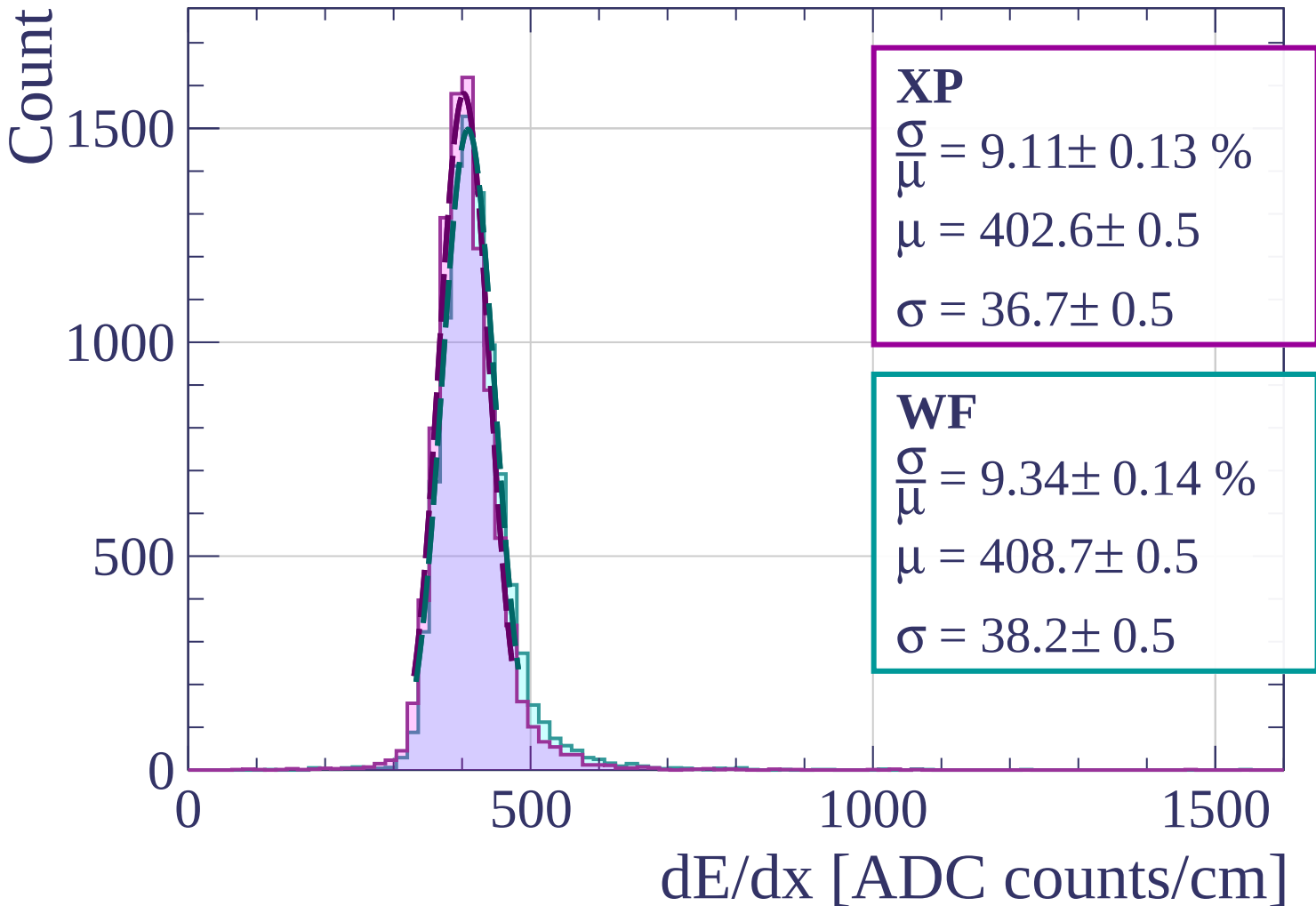
$dE/dx$  [ADC counts/cm]

Energy loss |  $87 < \phi < 90$

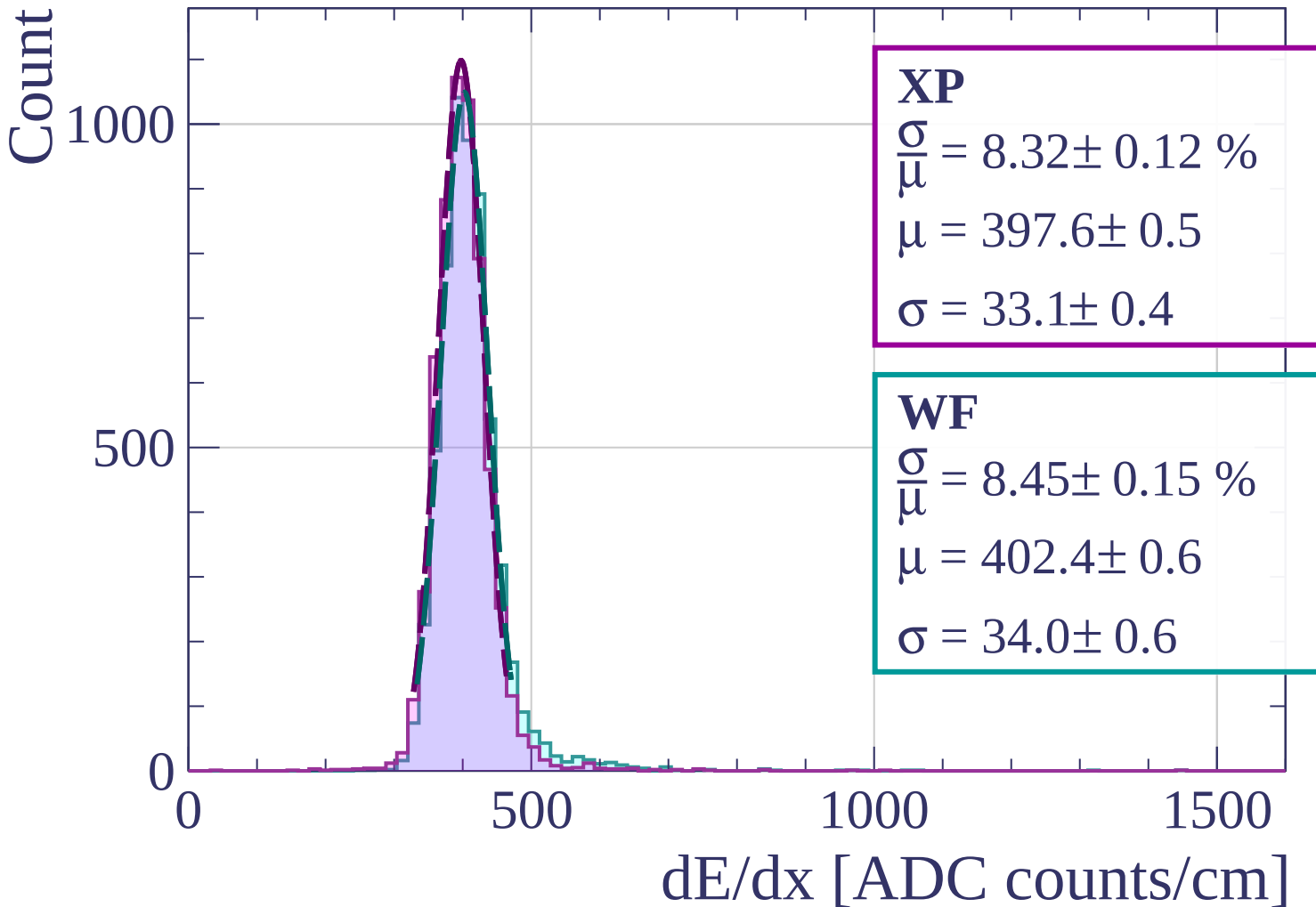




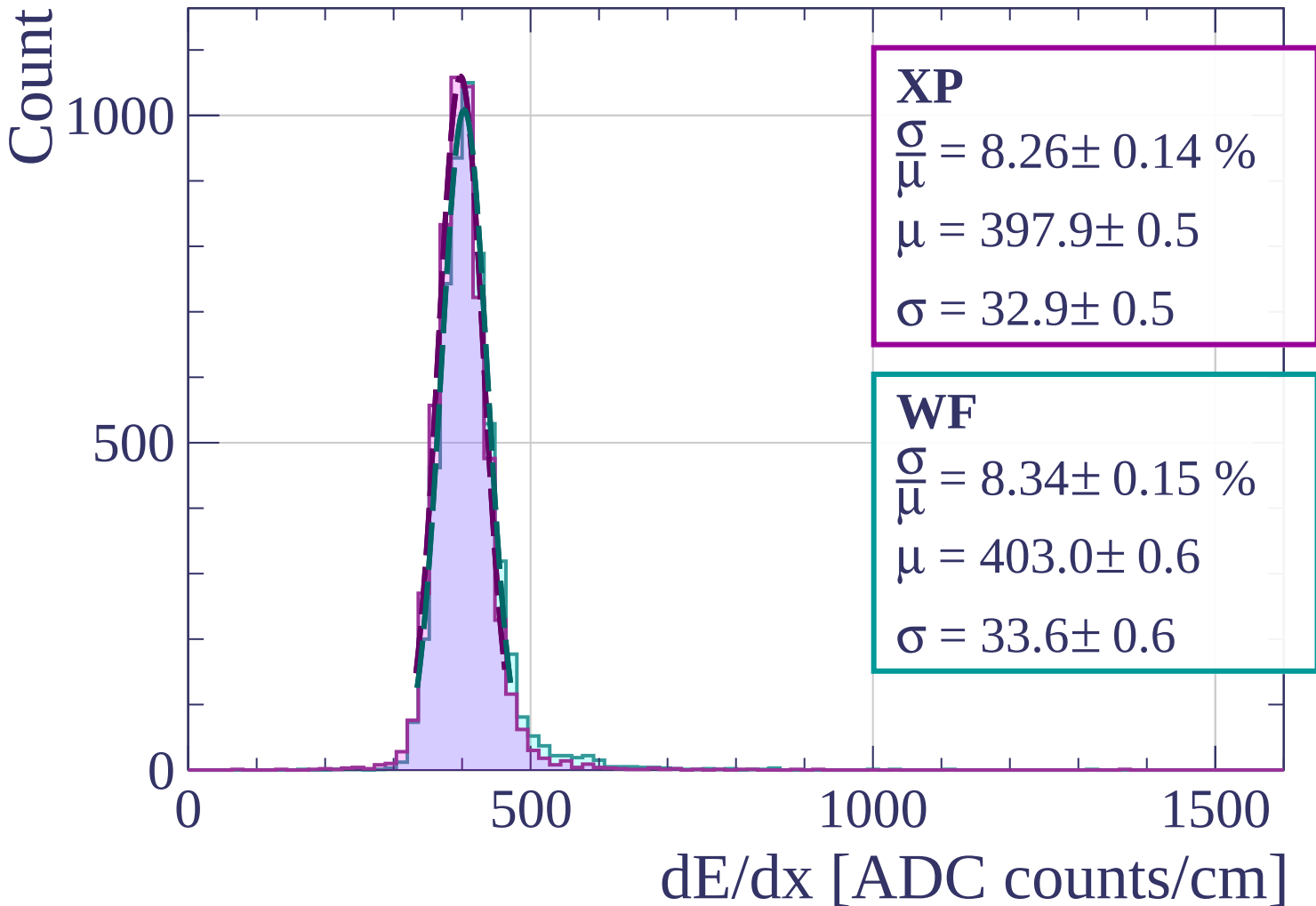
Energy loss |  $0 < \theta < 3$



Energy loss |  $3 < \theta < 6$



Energy loss |  $6 < \theta < 9$



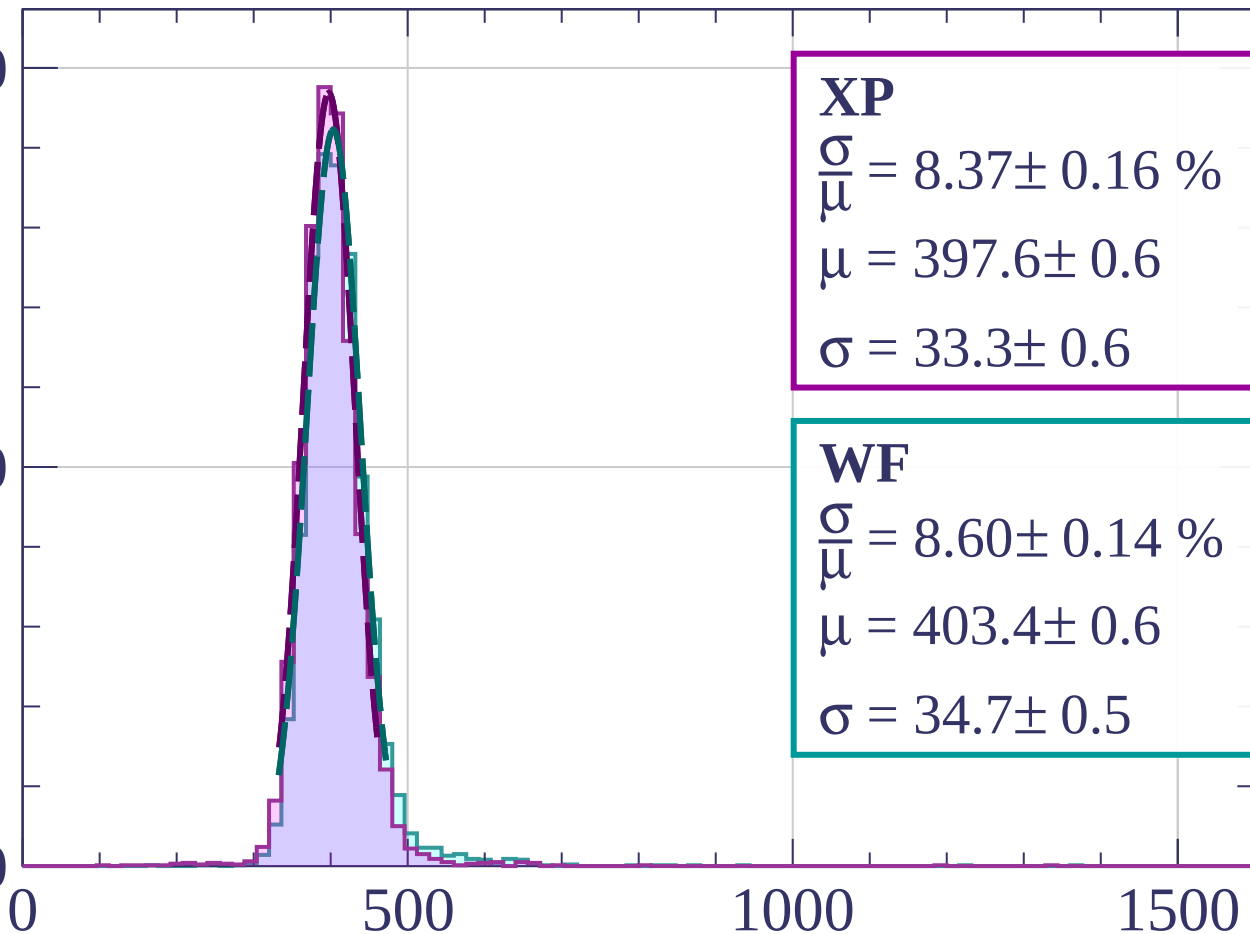
Energy loss |  $9 < \theta < 12$

Count

1000

500

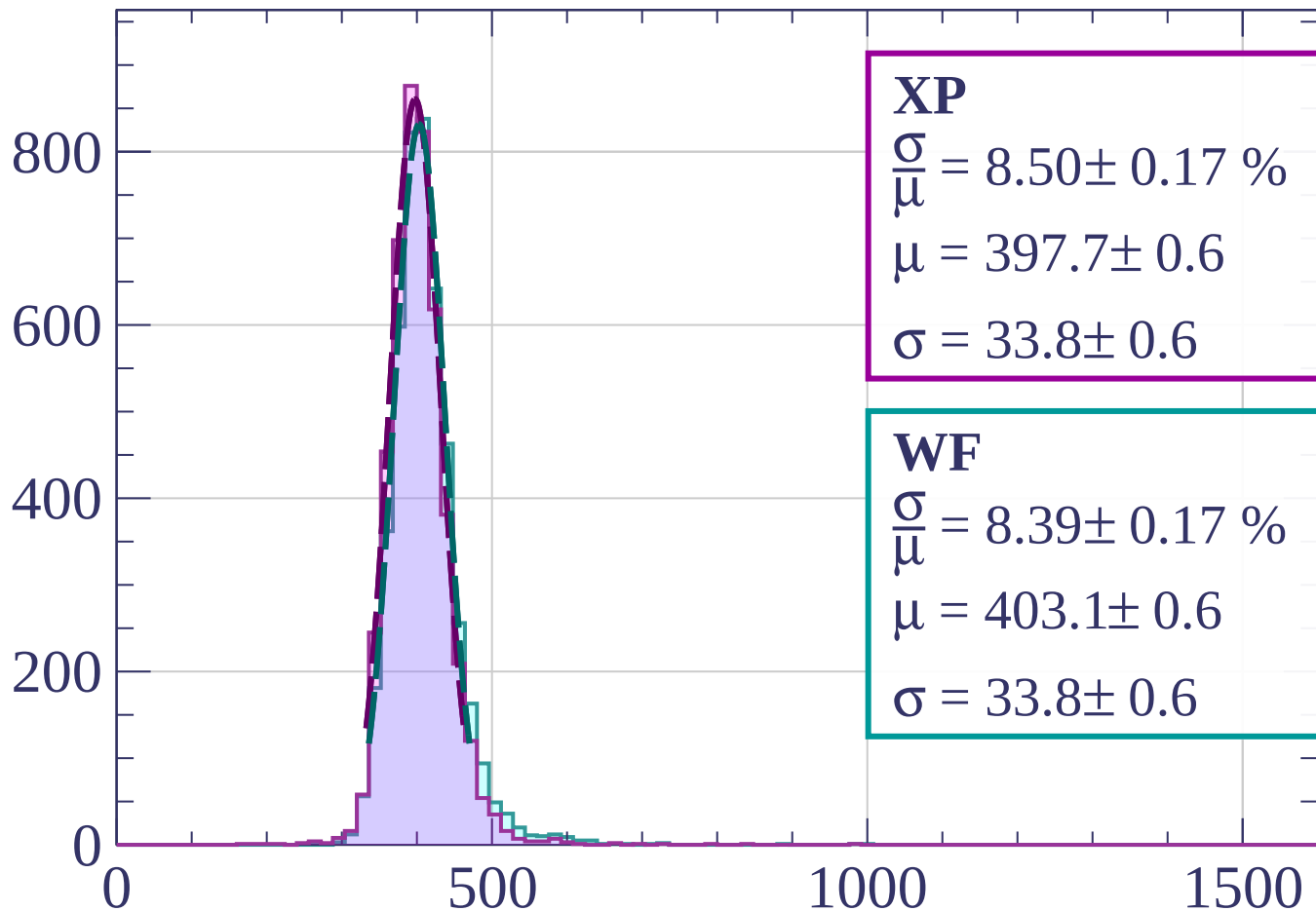
0



dE/dx [ADC counts/cm]

Energy loss |  $12 < \theta < 15$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.50 \pm 0.17 \%$$

$$\mu = 397.7 \pm 0.6$$

$$\sigma = 33.8 \pm 0.6$$

**WF**

$$\frac{\sigma}{\mu} = 8.39 \pm 0.17 \%$$

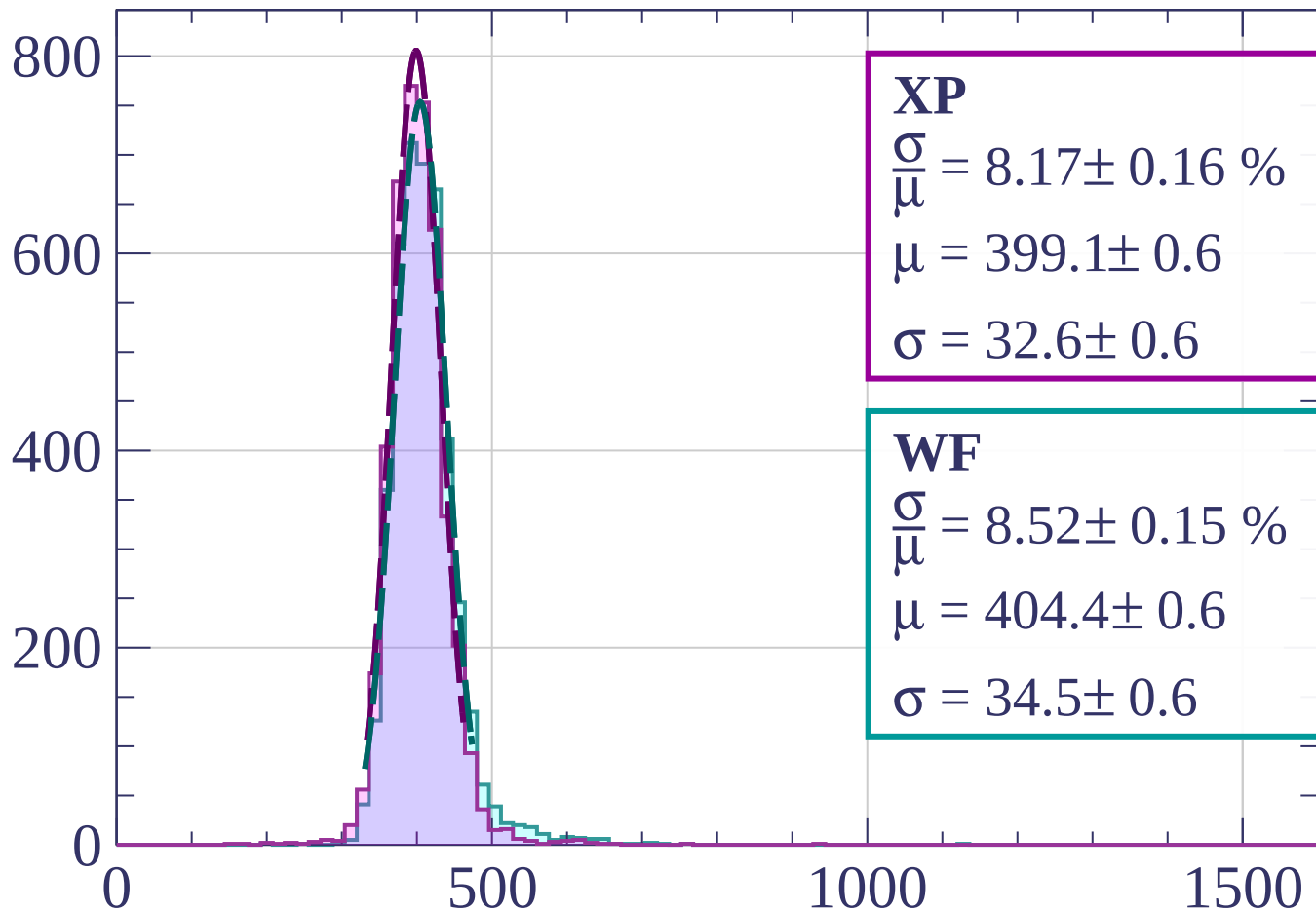
$$\mu = 403.1 \pm 0.6$$

$$\sigma = 33.8 \pm 0.6$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $15 < \theta < 18$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.17 \pm 0.16 \%$$

$$\mu = 399.1 \pm 0.6$$

$$\sigma = 32.6 \pm 0.6$$

**WF**

$$\frac{\sigma}{\mu} = 8.52 \pm 0.15 \%$$

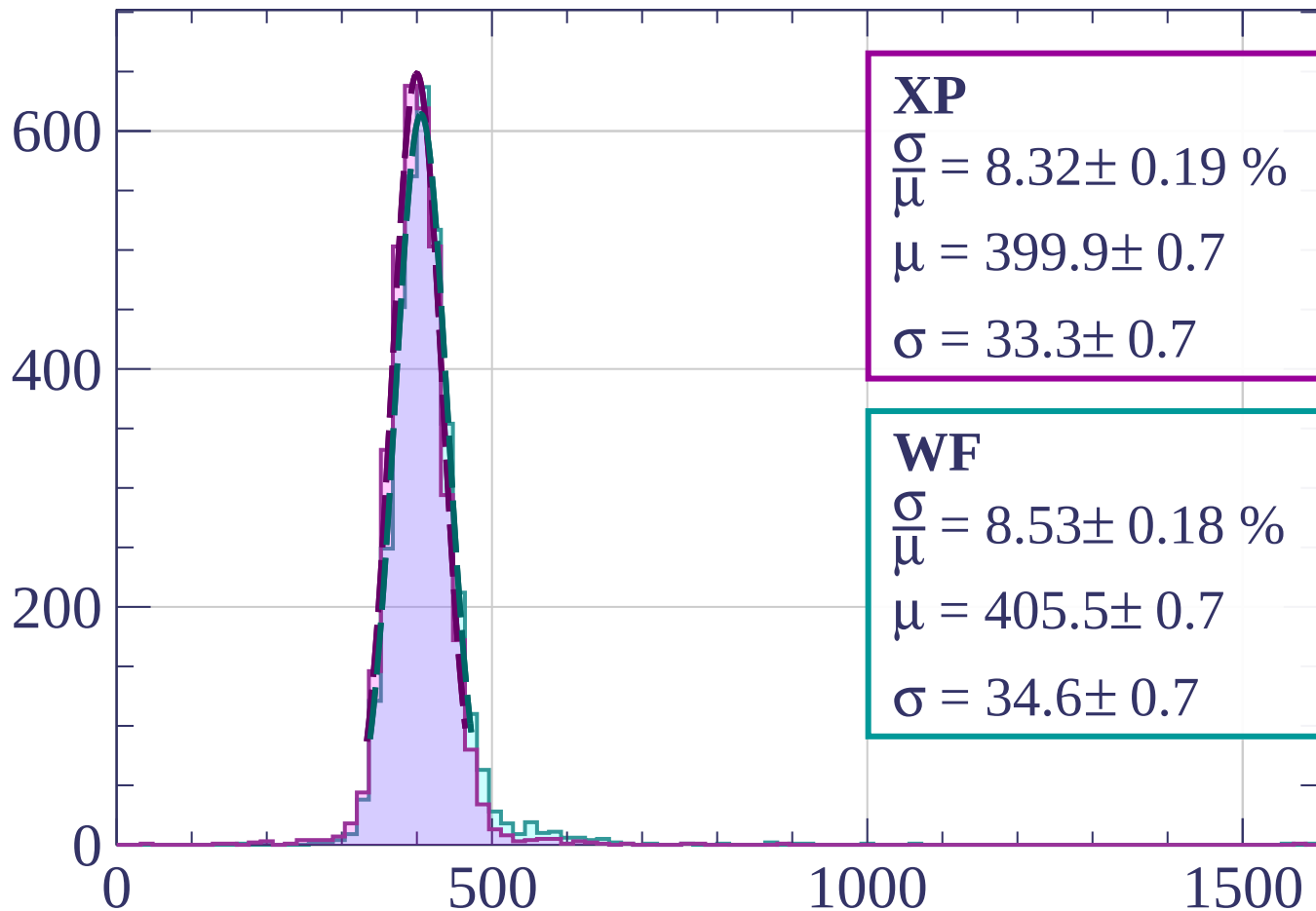
$$\mu = 404.4 \pm 0.6$$

$$\sigma = 34.5 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss |  $18 < \theta < 21$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.32 \pm 0.19 \%$$

$$\mu = 399.9 \pm 0.7$$

$$\sigma = 33.3 \pm 0.7$$

**WF**

$$\frac{\sigma}{\mu} = 8.53 \pm 0.18 \%$$

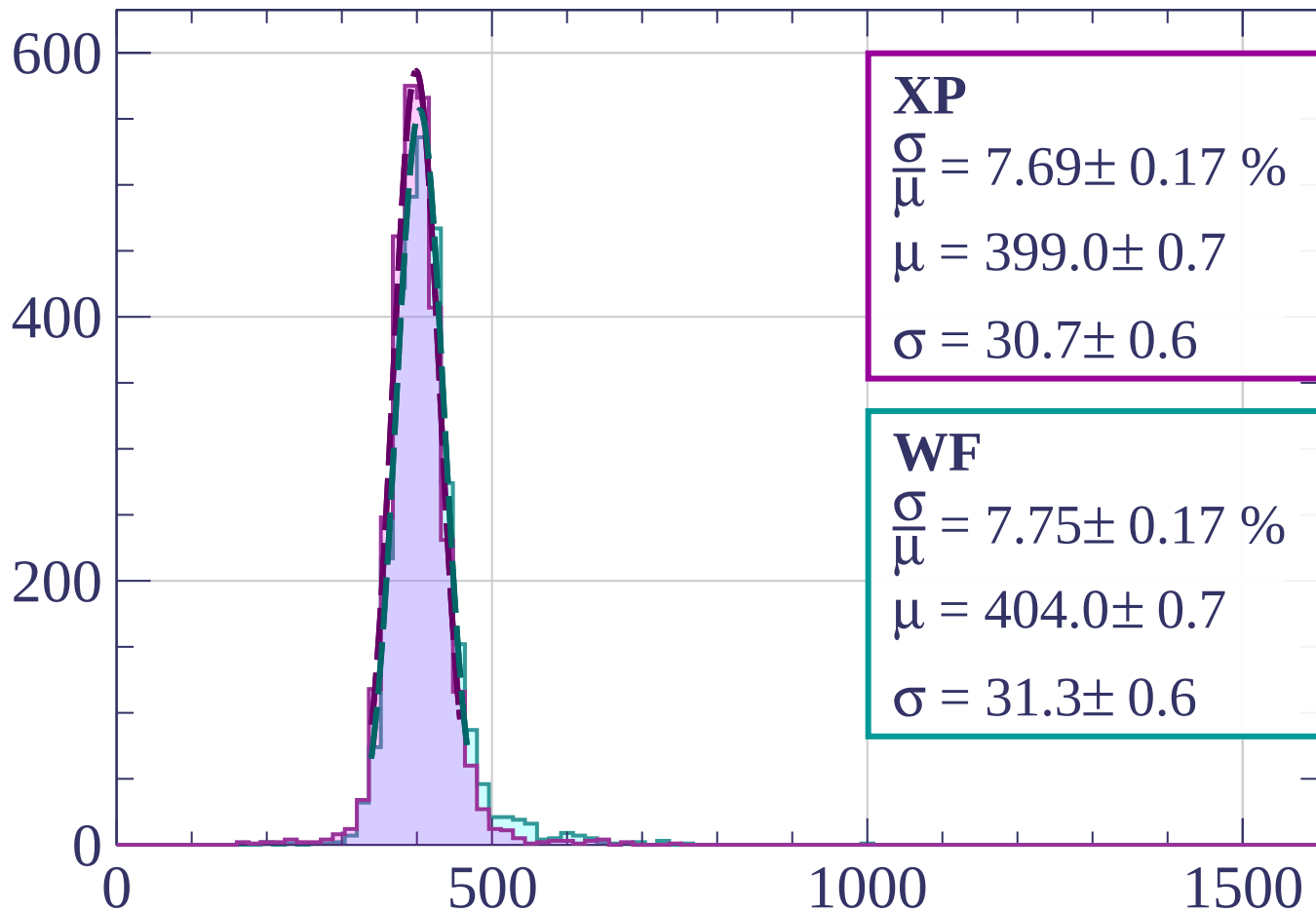
$$\mu = 405.5 \pm 0.7$$

$$\sigma = 34.6 \pm 0.7$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $21 < \theta < 24$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.69 \pm 0.17 \%$$

$$\mu = 399.0 \pm 0.7$$

$$\sigma = 30.7 \pm 0.6$$

**WF**

$$\frac{\sigma}{\mu} = 7.75 \pm 0.17 \%$$

$$\mu = 404.0 \pm 0.7$$

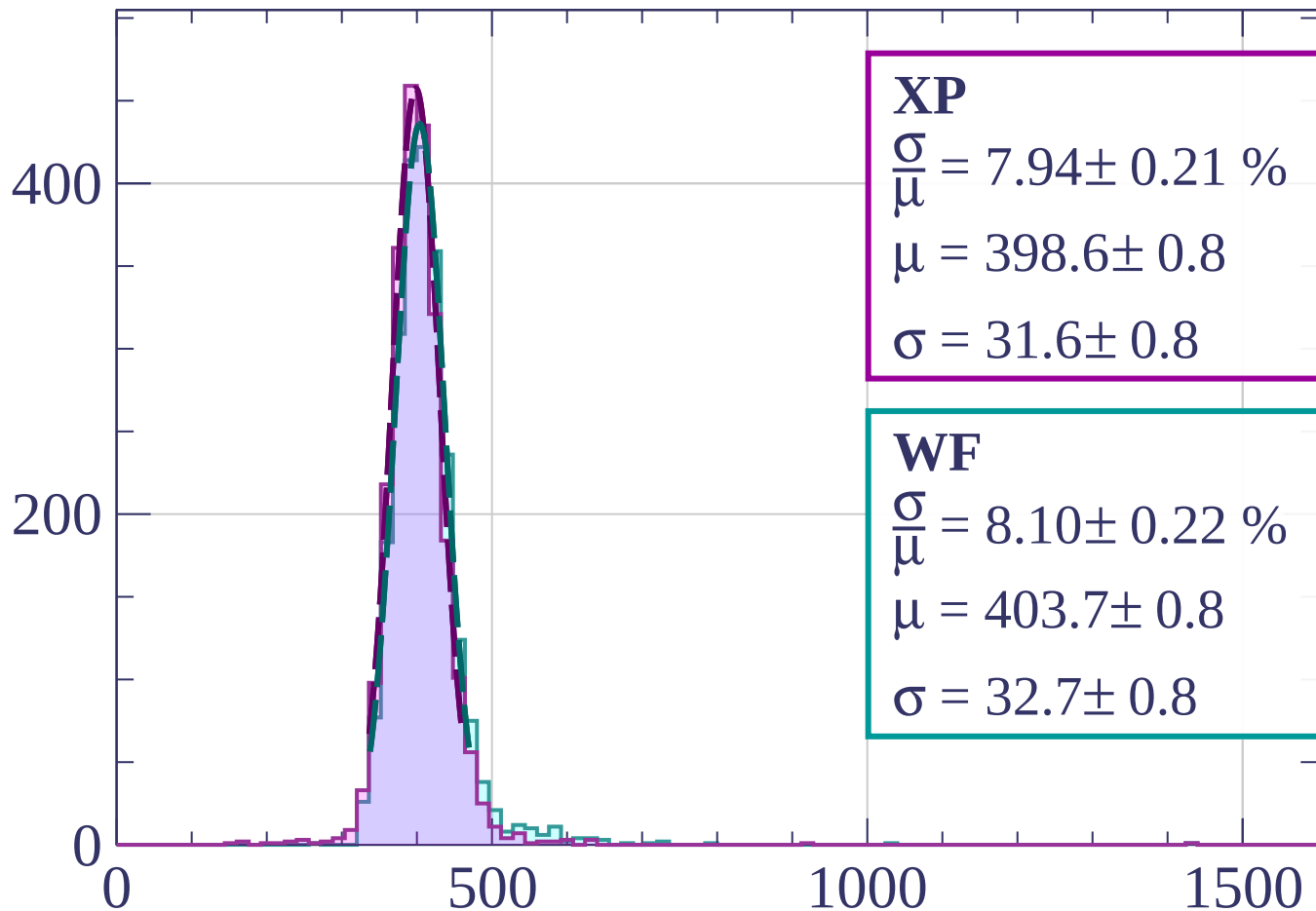
$$\sigma = 31.3 \pm 0.6$$

dE/dx [ADC counts/cm]



Energy loss |  $24 < \theta < 27$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.94 \pm 0.21 \%$$

$$\mu = 398.6 \pm 0.8$$

$$\sigma = 31.6 \pm 0.8$$

**WF**

$$\frac{\sigma}{\mu} = 8.10 \pm 0.22 \%$$

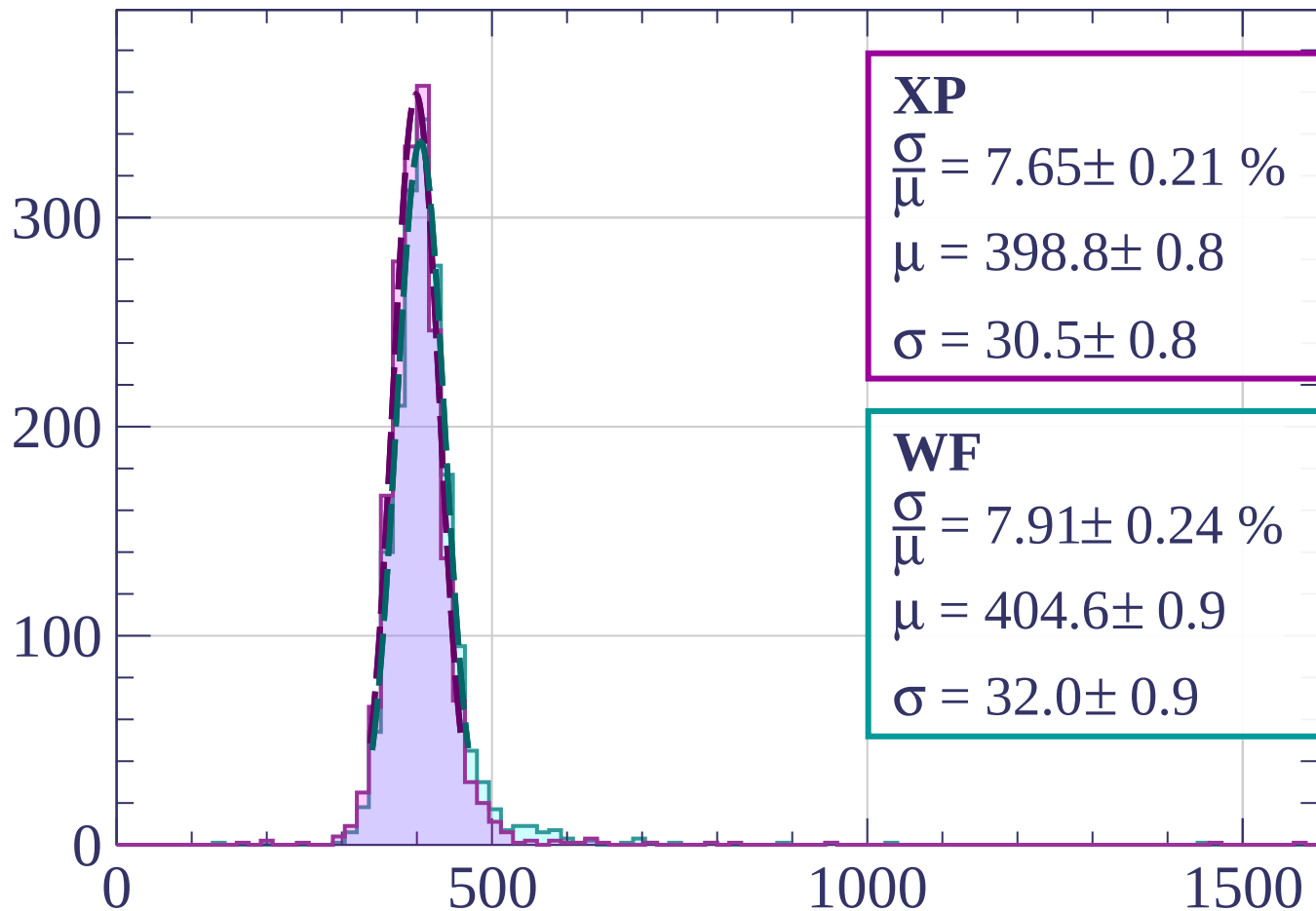
$$\mu = 403.7 \pm 0.8$$

$$\sigma = 32.7 \pm 0.8$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $27 < \theta < 30$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.65 \pm 0.21 \%$$

$$\mu = 398.8 \pm 0.8$$

$$\sigma = 30.5 \pm 0.8$$

**WF**

$$\frac{\sigma}{\mu} = 7.91 \pm 0.24 \%$$

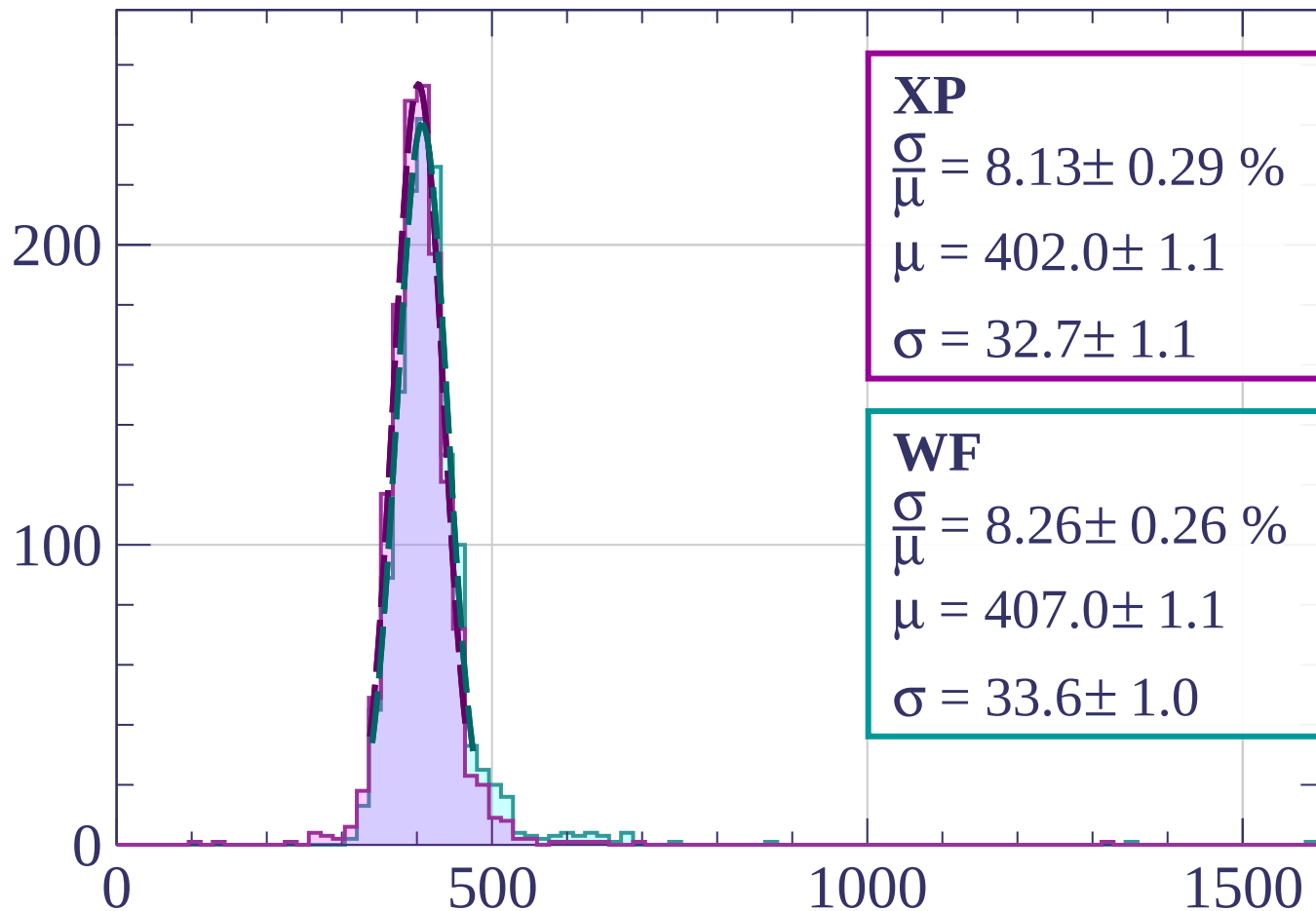
$$\mu = 404.6 \pm 0.9$$

$$\sigma = 32.0 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss |  $30 < \theta < 33$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.13 \pm 0.29 \%$$

$$\mu = 402.0 \pm 1.1$$

$$\sigma = 32.7 \pm 1.1$$

**WF**

$$\frac{\sigma}{\mu} = 8.26 \pm 0.26 \%$$

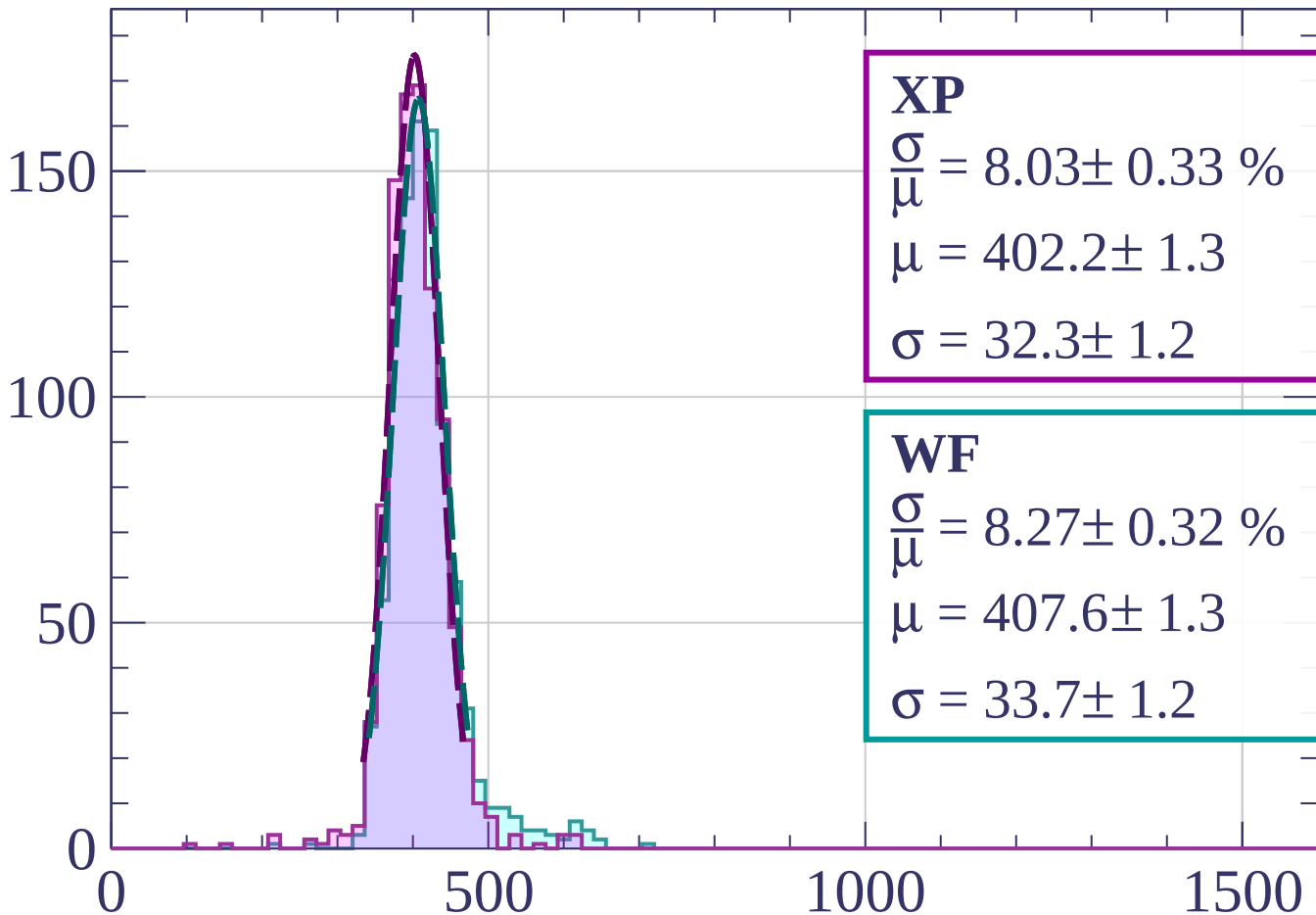
$$\mu = 407.0 \pm 1.1$$

$$\sigma = 33.6 \pm 1.0$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $33 < \theta < 36$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.03 \pm 0.33 \%$$

$$\mu = 402.2 \pm 1.3$$

$$\sigma = 32.3 \pm 1.2$$

**WF**

$$\frac{\sigma}{\mu} = 8.27 \pm 0.32 \%$$

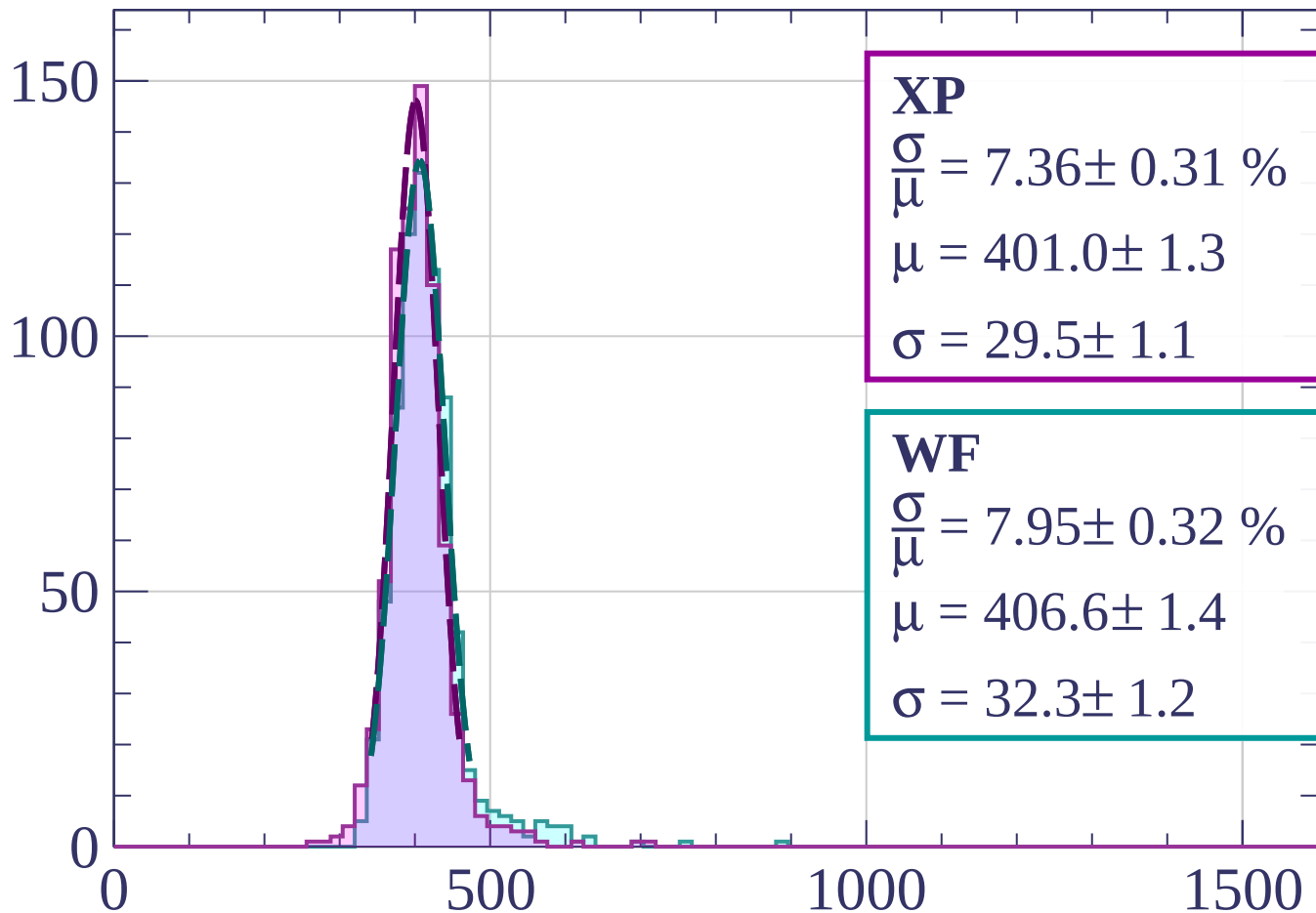
$$\mu = 407.6 \pm 1.3$$

$$\sigma = 33.7 \pm 1.2$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $36 < \theta < 39$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.36 \pm 0.31 \%$$

$$\mu = 401.0 \pm 1.3$$

$$\sigma = 29.5 \pm 1.1$$

**WF**

$$\frac{\sigma}{\mu} = 7.95 \pm 0.32 \%$$

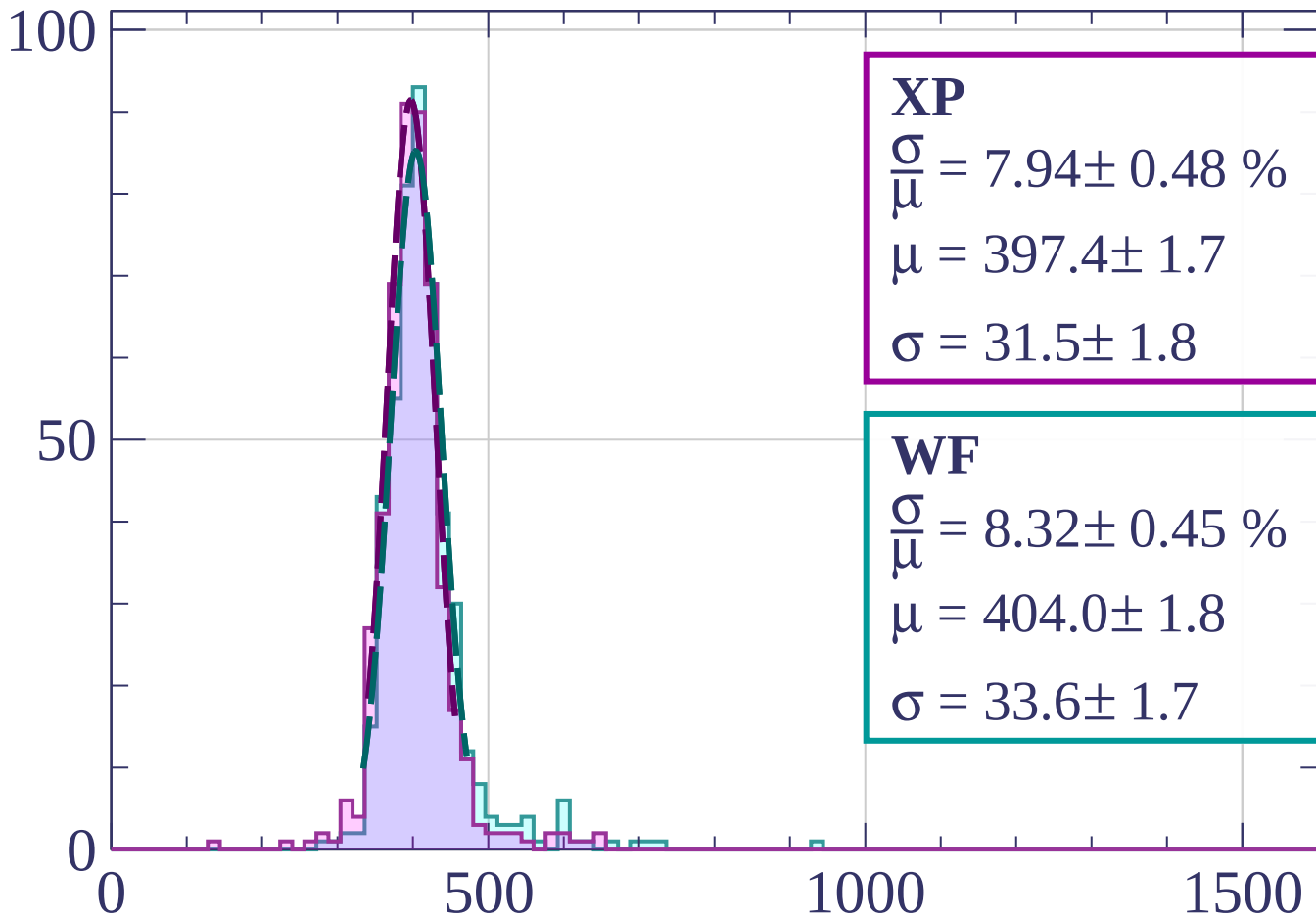
$$\mu = 406.6 \pm 1.4$$

$$\sigma = 32.3 \pm 1.2$$

dE/dx [ADC counts/cm]

Energy loss |  $39 < \theta < 42$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.94 \pm 0.48 \%$$

$$\mu = 397.4 \pm 1.7$$

$$\sigma = 31.5 \pm 1.8$$

**WF**

$$\frac{\sigma}{\mu} = 8.32 \pm 0.45 \%$$

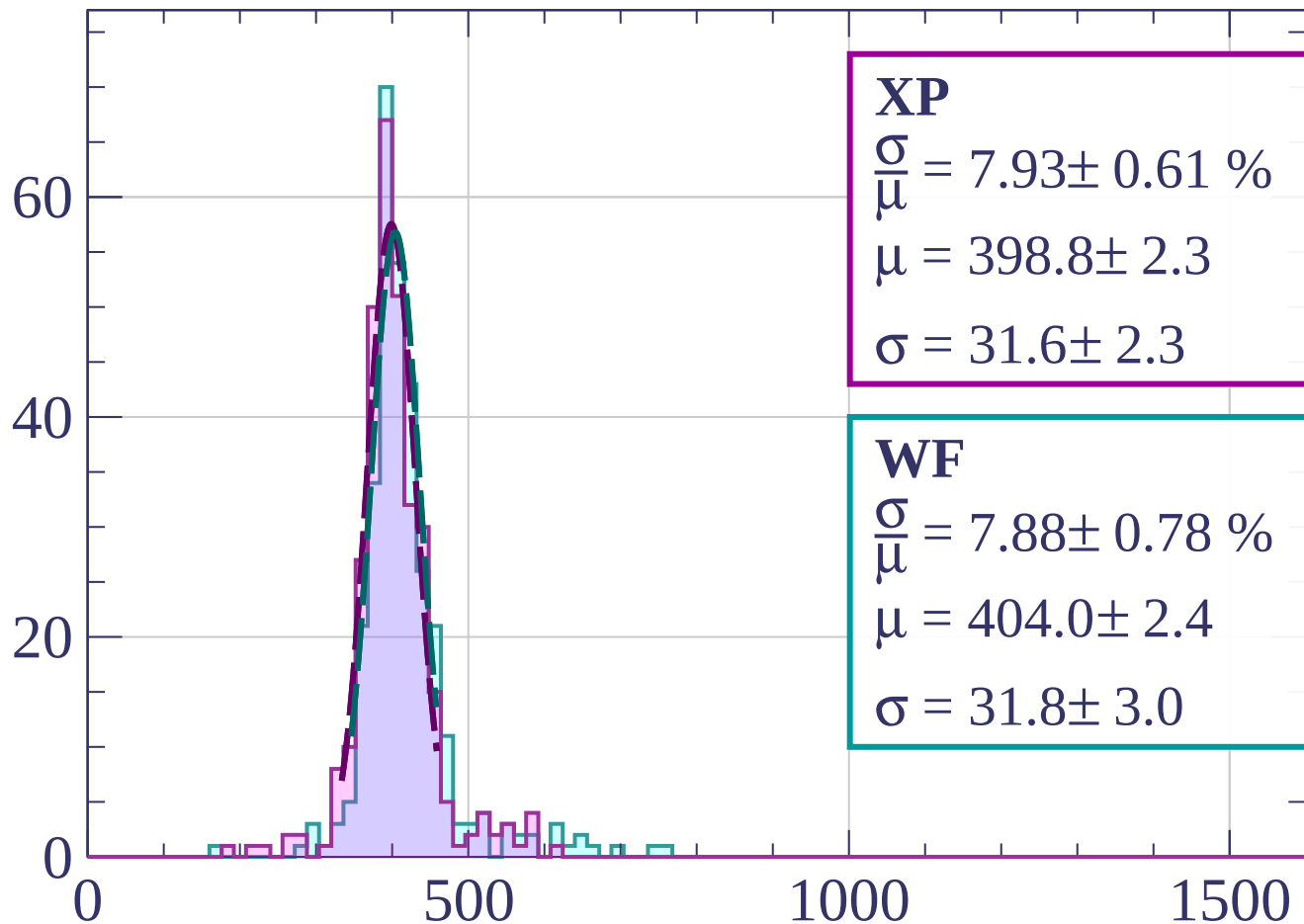
$$\mu = 404.0 \pm 1.8$$

$$\sigma = 33.6 \pm 1.7$$

dE/dx [ADC counts/cm]

Energy loss |  $42 < \theta < 45$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.93 \pm 0.61 \%$$

$$\mu = 398.8 \pm 2.3$$

$$\sigma = 31.6 \pm 2.3$$

**WF**

$$\frac{\sigma}{\mu} = 7.88 \pm 0.78 \%$$

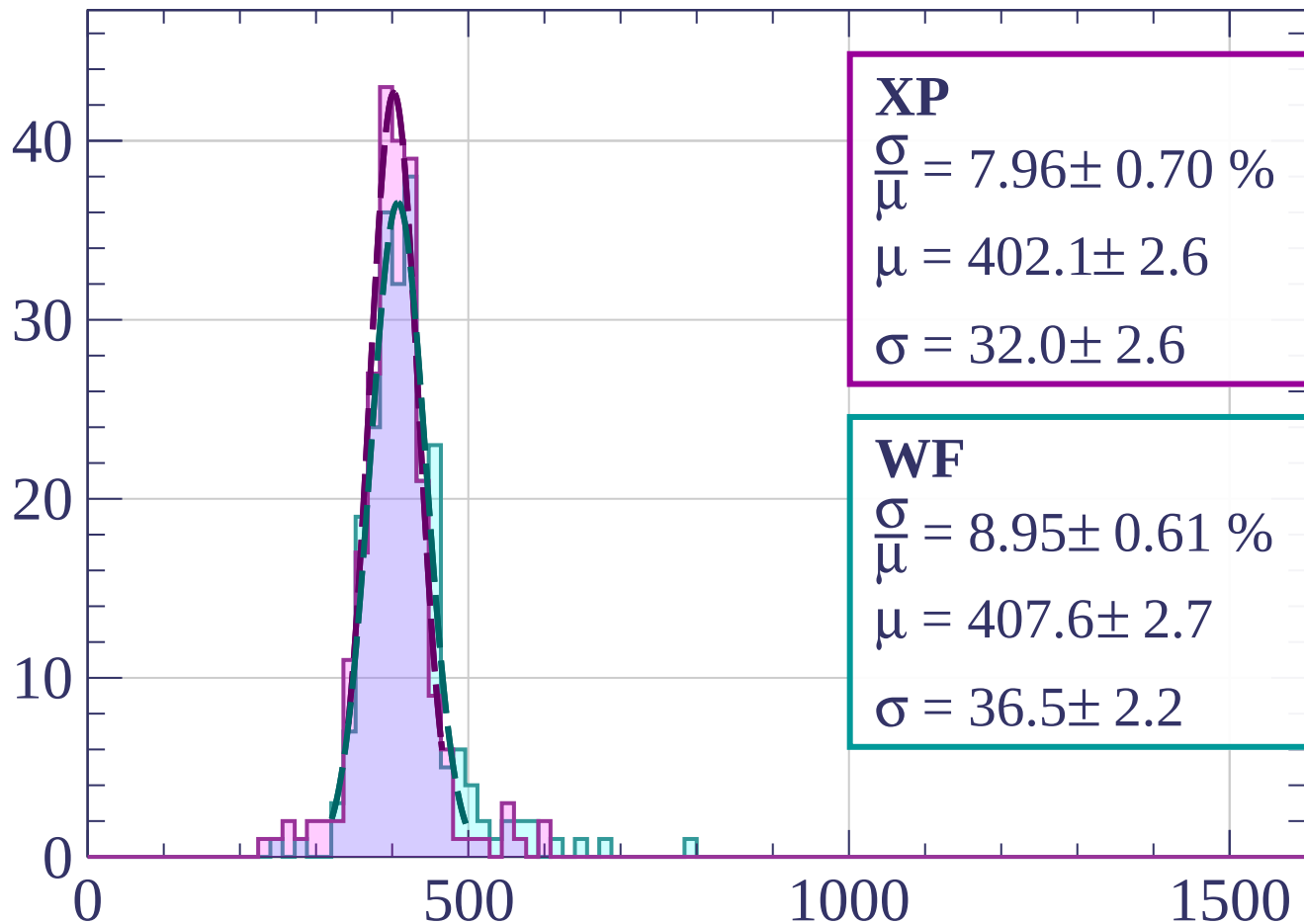
$$\mu = 404.0 \pm 2.4$$

$$\sigma = 31.8 \pm 3.0$$

dE/dx [ADC counts/cm]

Energy loss |  $45 < \theta < 48$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.96 \pm 0.70 \%$$

$$\mu = 402.1 \pm 2.6$$

$$\sigma = 32.0 \pm 2.6$$

**WF**

$$\frac{\sigma}{\mu} = 8.95 \pm 0.61 \%$$

$$\mu = 407.6 \pm 2.7$$

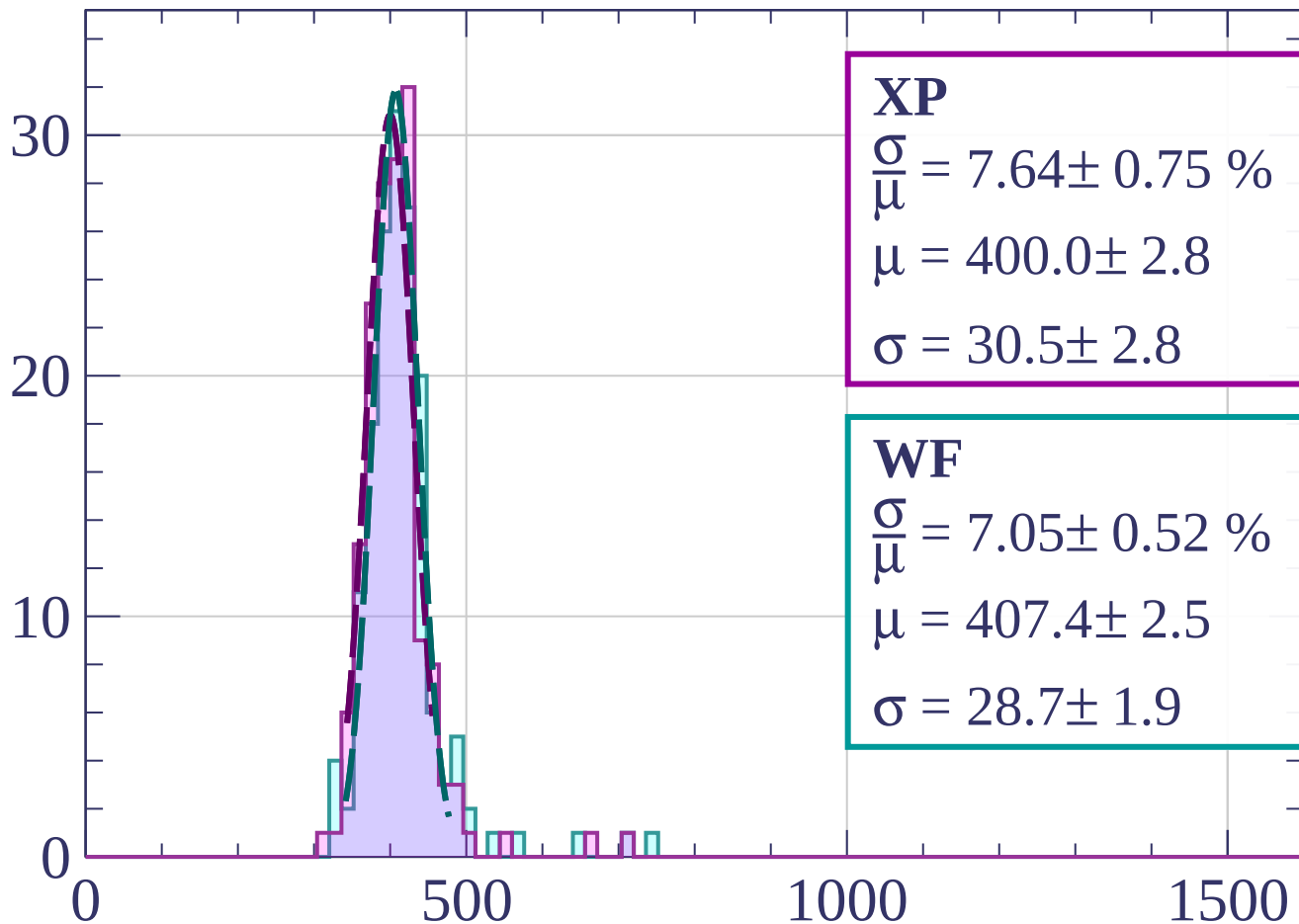
$$\sigma = 36.5 \pm 2.2$$

$dE/dx$  [ADC counts/cm]



Energy loss |  $48 < \theta < 51$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.64 \pm 0.75 \%$$

$$\mu = 400.0 \pm 2.8$$

$$\sigma = 30.5 \pm 2.8$$

**WF**

$$\frac{\sigma}{\mu} = 7.05 \pm 0.52 \%$$

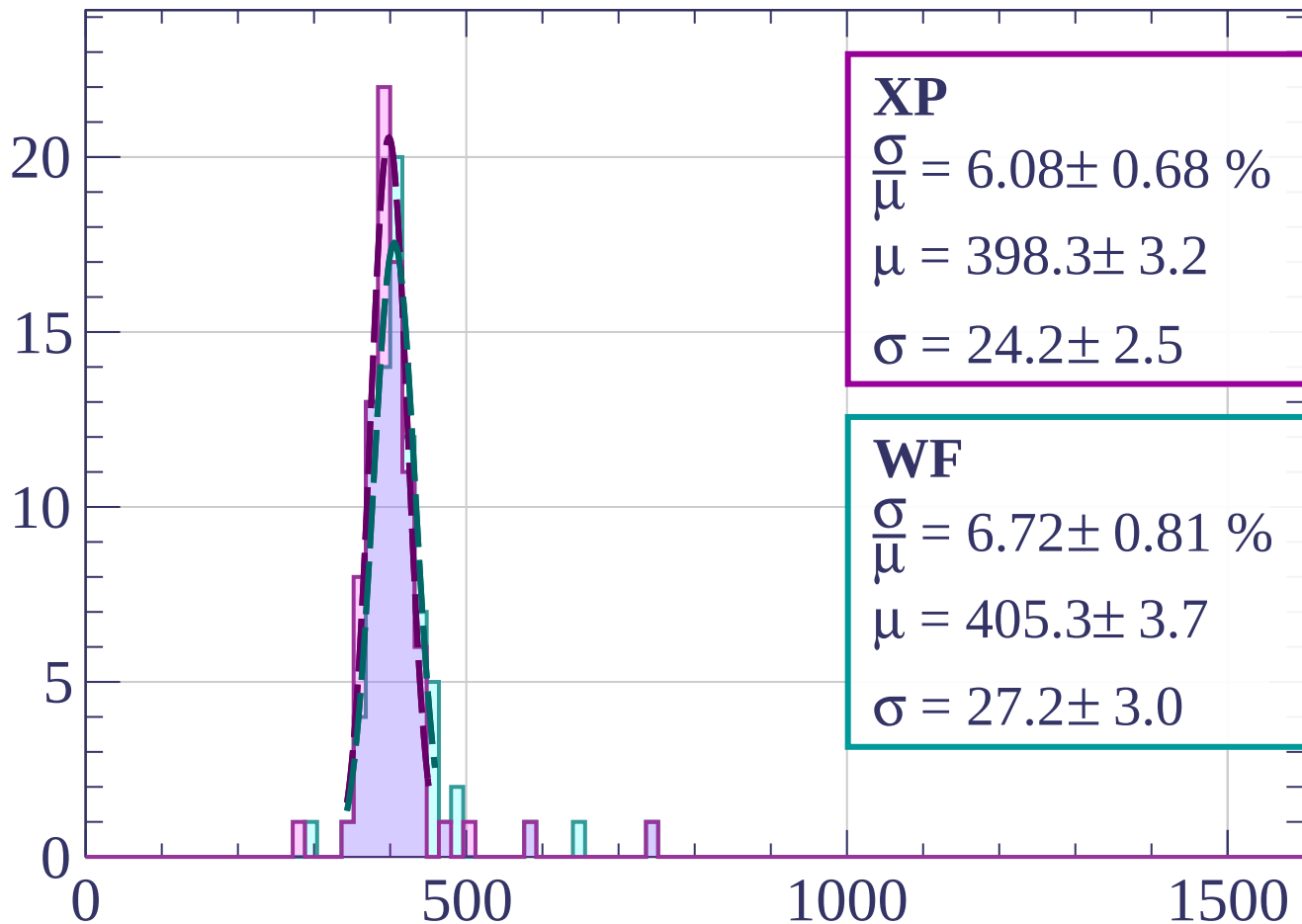
$$\mu = 407.4 \pm 2.5$$

$$\sigma = 28.7 \pm 1.9$$

dE/dx [ADC counts/cm]

Energy loss |  $51 < \theta < 54$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.08 \pm 0.68 \%$$

$$\mu = 398.3 \pm 3.2$$

$$\sigma = 24.2 \pm 2.5$$

**WF**

$$\frac{\sigma}{\mu} = 6.72 \pm 0.81 \%$$

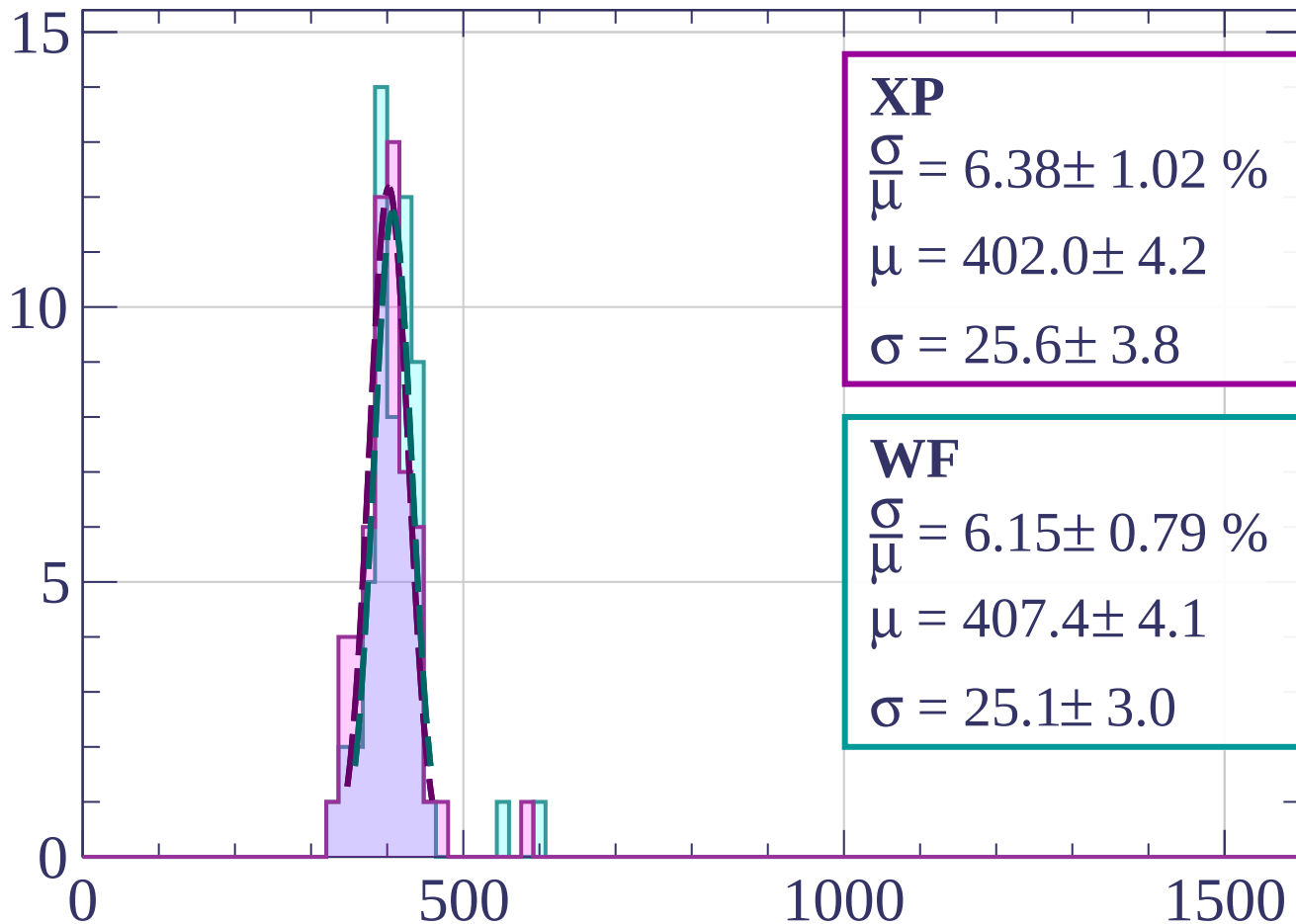
$$\mu = 405.3 \pm 3.7$$

$$\sigma = 27.2 \pm 3.0$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $54 < \theta < 57$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.38 \pm 1.02 \%$$

$$\mu = 402.0 \pm 4.2$$

$$\sigma = 25.6 \pm 3.8$$

**WF**

$$\frac{\sigma}{\mu} = 6.15 \pm 0.79 \%$$

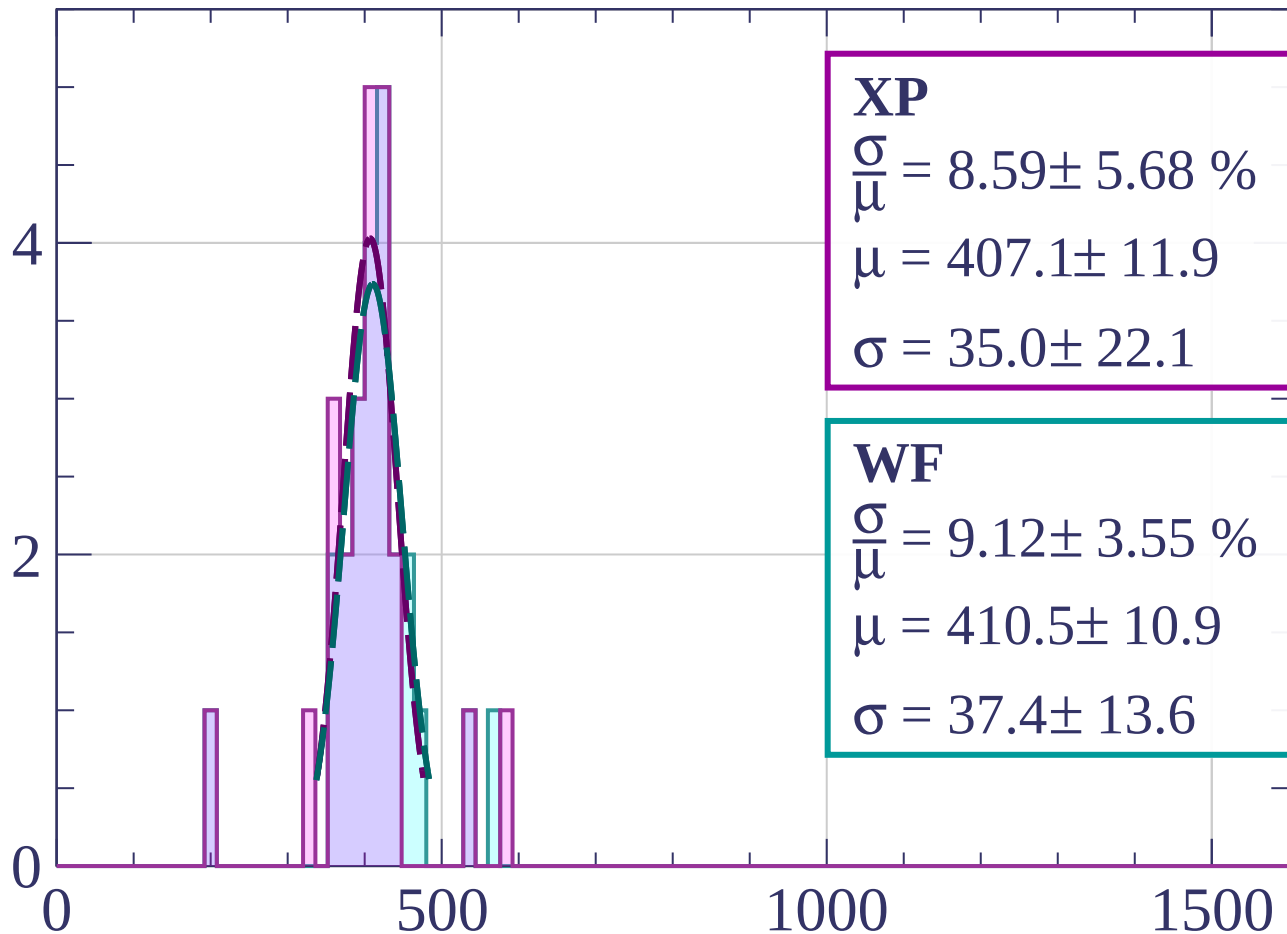
$$\mu = 407.4 \pm 4.1$$

$$\sigma = 25.1 \pm 3.0$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $57 < \theta < 60$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.59 \pm 5.68 \%$$

$$\mu = 407.1 \pm 11.9$$

$$\sigma = 35.0 \pm 22.1$$

**WF**

$$\frac{\sigma}{\mu} = 9.12 \pm 3.55 \%$$

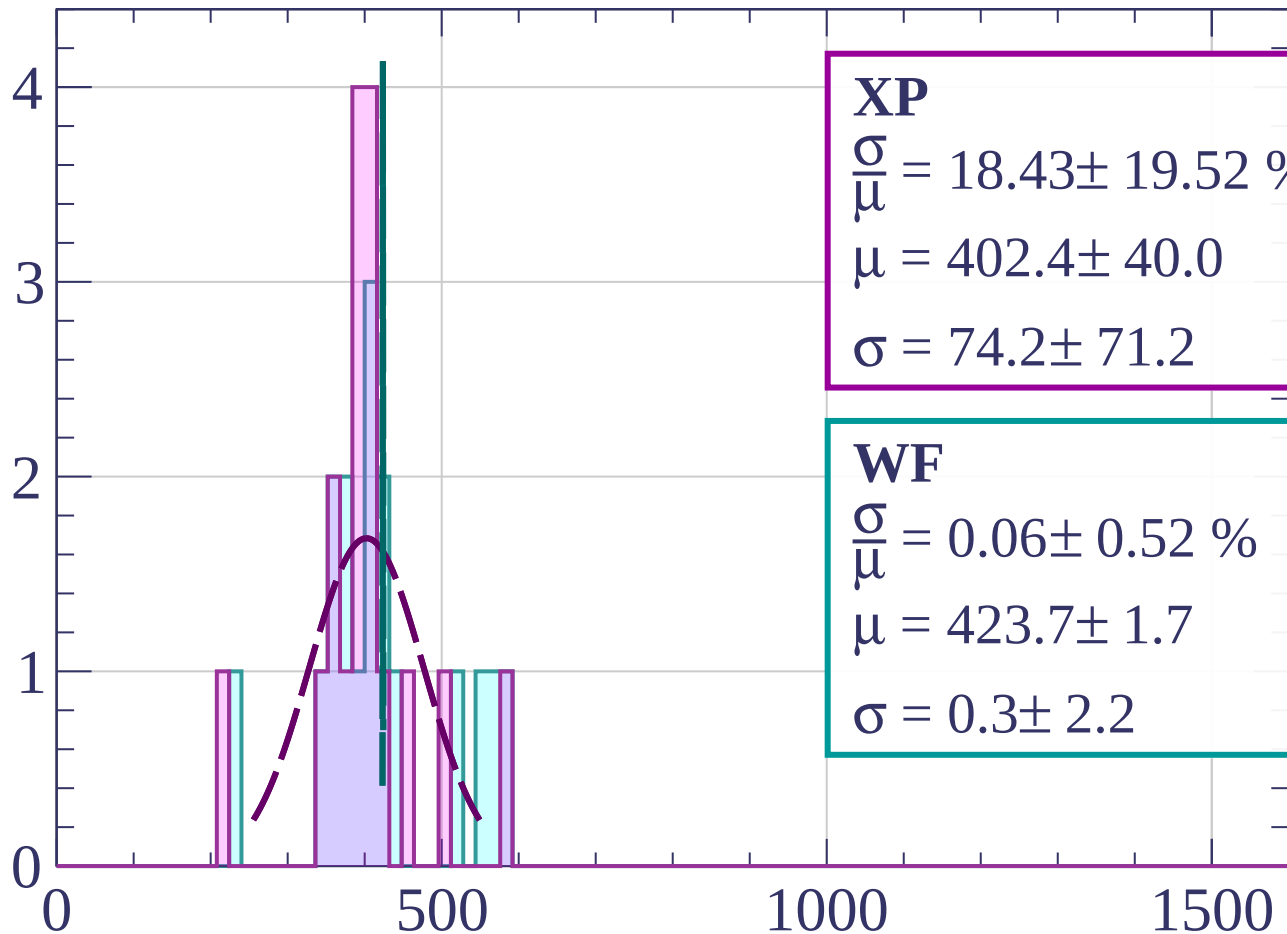
$$\mu = 410.5 \pm 10.9$$

$$\sigma = 37.4 \pm 13.6$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $60 < \theta < 63$

Count



**XP**

$$\frac{\sigma}{\mu} = 18.43 \pm 19.52 \%$$

$$\mu = 402.4 \pm 40.0$$

$$\sigma = 74.2 \pm 71.2$$

**WF**

$$\frac{\sigma}{\mu} = 0.06 \pm 0.52 \%$$

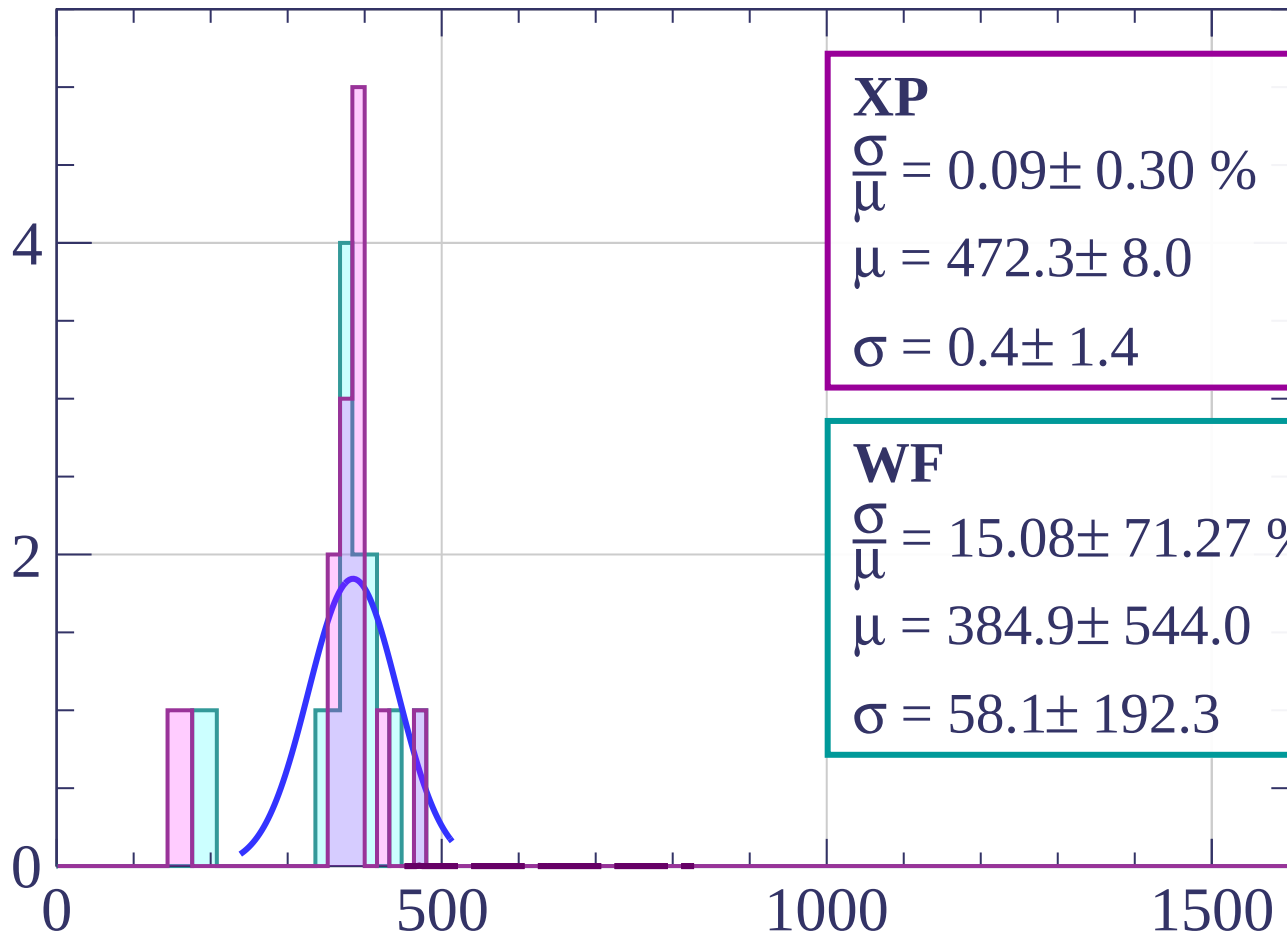
$$\mu = 423.7 \pm 1.7$$

$$\sigma = 0.3 \pm 2.2$$

dE/dx [ADC counts/cm]

Energy loss |  $63 < \theta < 66$

Count



**XP**

$$\frac{\sigma}{\mu} = 0.09 \pm 0.30 \%$$

$$\mu = 472.3 \pm 8.0$$

$$\sigma = 0.4 \pm 1.4$$

**WF**

$$\frac{\sigma}{\mu} = 15.08 \pm 71.27 \%$$

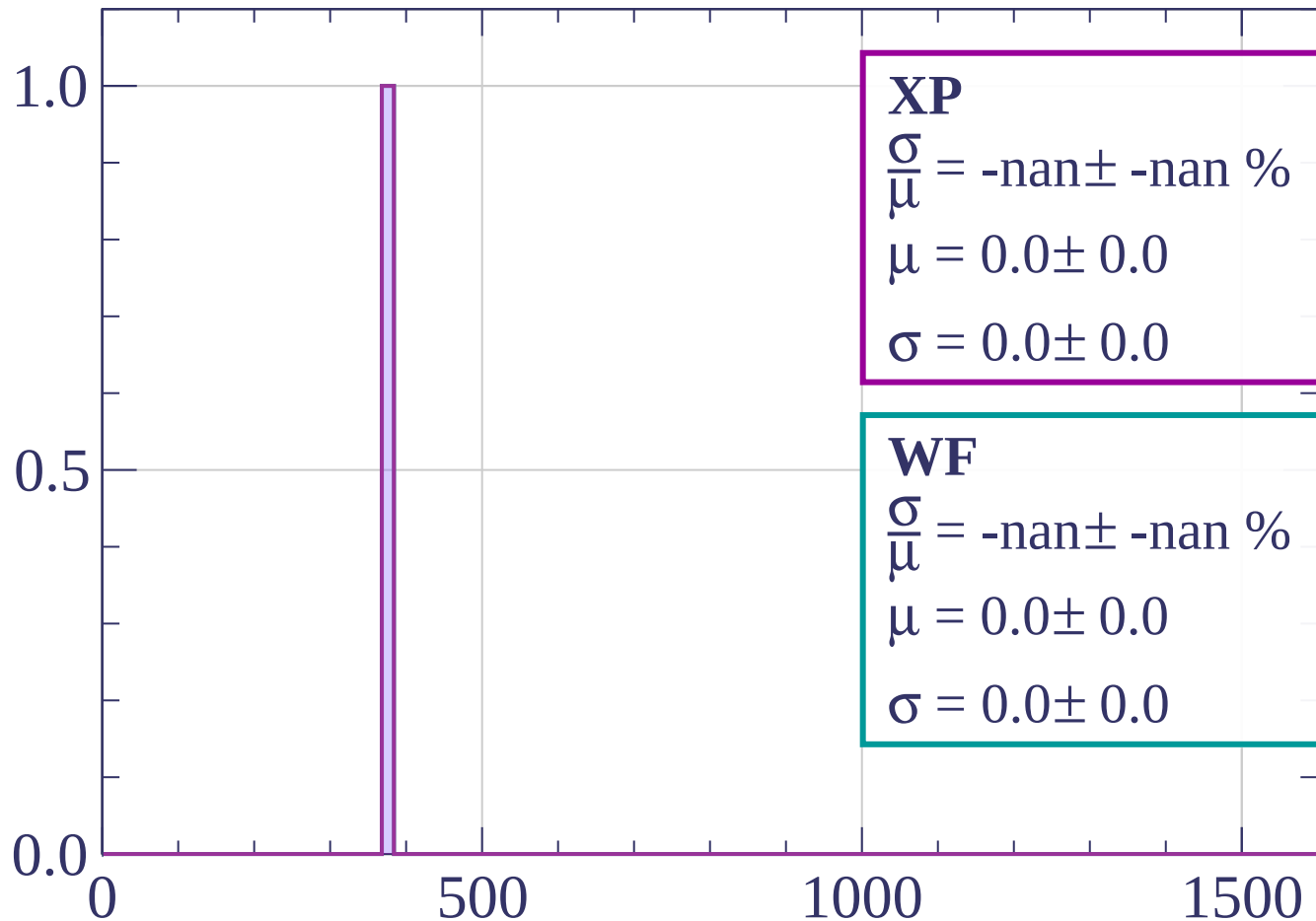
$$\mu = 384.9 \pm 544.0$$

$$\sigma = 58.1 \pm 192.3$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $66 < \theta < 69$

Count



**XP**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

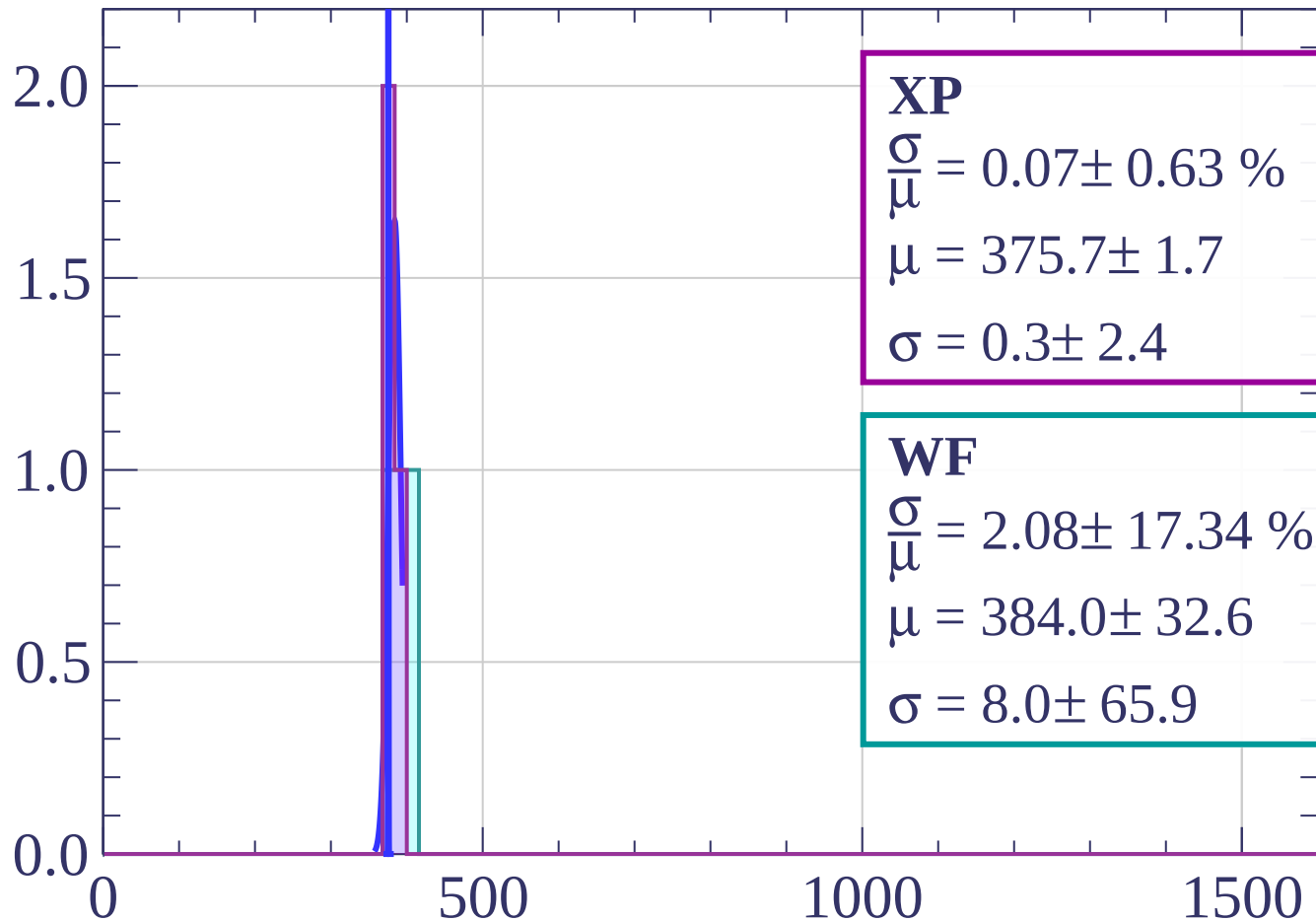
$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

$dE/dx$  [ADC counts/cm]

Energy loss |  $69 < \theta < 72$

Count

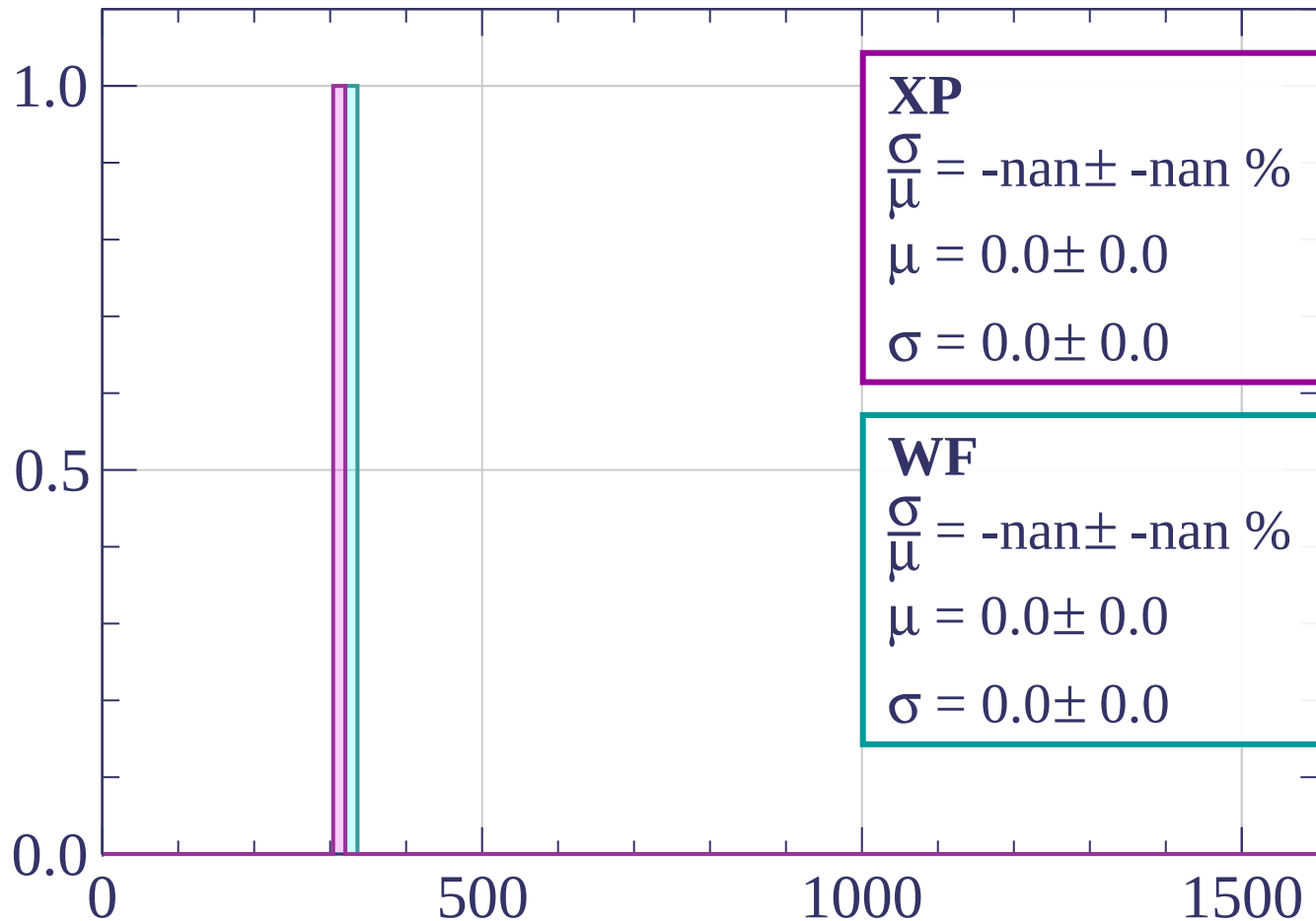


dE/dx [ADC counts/cm]



Energy loss |  $72 < \theta < 75$

Count



**XP**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

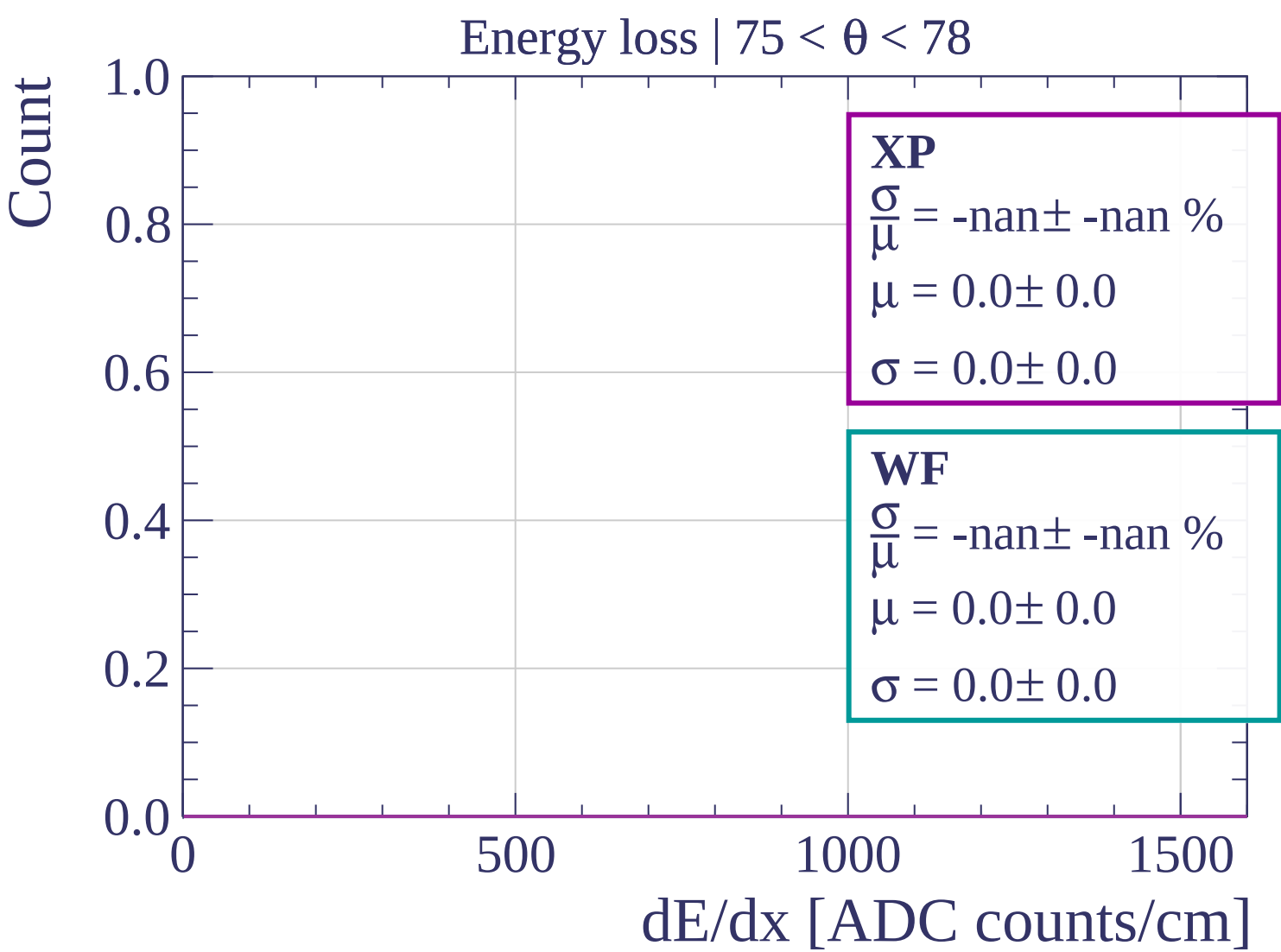
**WF**

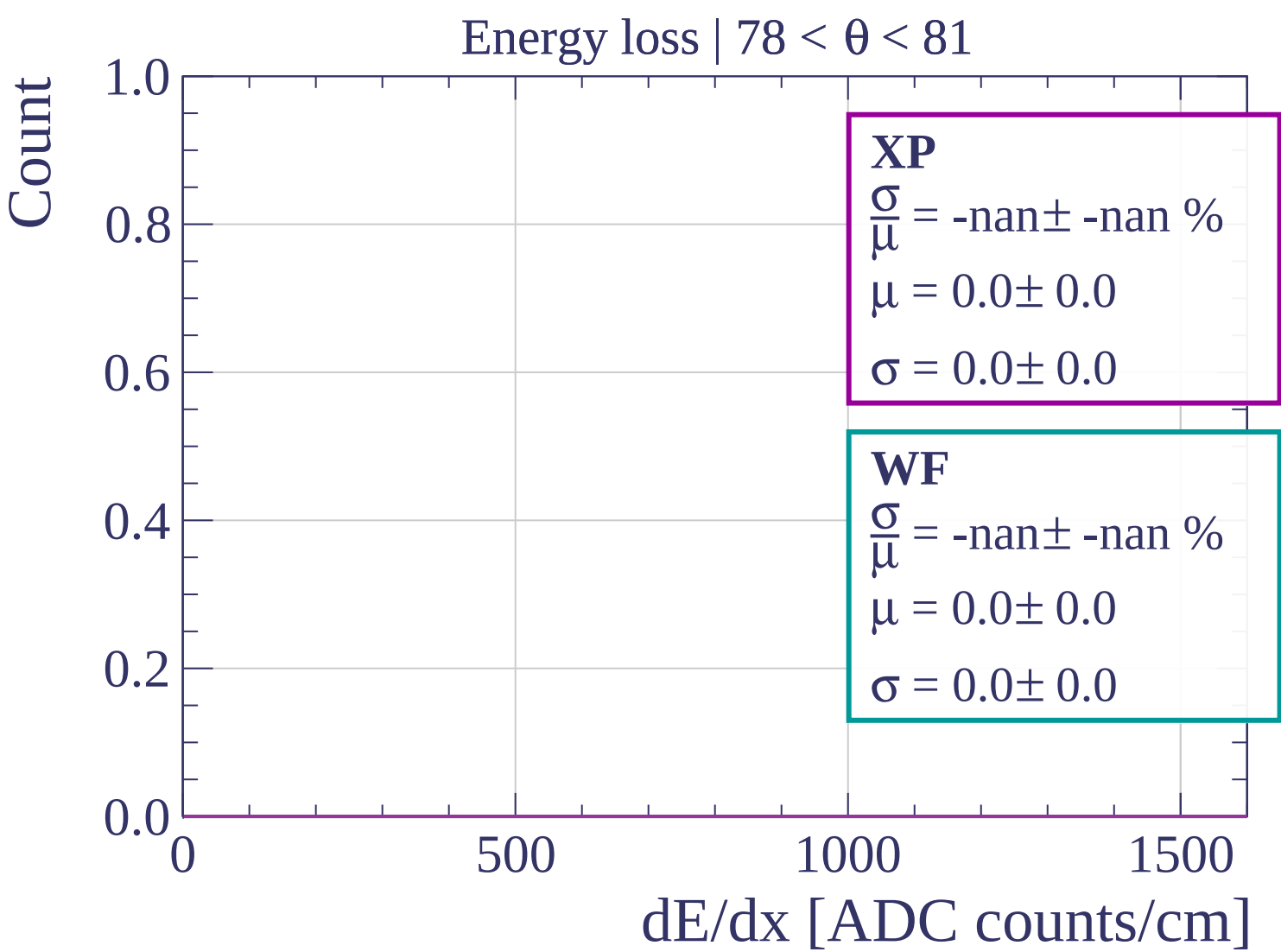
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

$dE/dx$  [ADC counts/cm]





Count

Energy loss |  $81 < \theta < 84$

**XP**

$$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$$

$$\mu = 0.0 \pm 0.0$$

$$\sigma = 0.0 \pm 0.0$$

**WF**

$$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$$

$$\mu = 0.0 \pm 0.0$$

$$\sigma = 0.0 \pm 0.0$$

1.0  
0.8  
0.6  
0.4  
0.2  
0.0

0

500

1000

1500

dE/dx [ADC counts/cm]

Count

Energy loss |  $84 < \theta < 87$

**XP**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.8

0.6

0.4

0.2

0.0

0

500

1000

1500

dE/dx [ADC counts/cm]

Count

Energy loss |  $87 < \theta < 90$

**XP**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0  
0.8  
0.6  
0.4  
0.2  
0.0

0

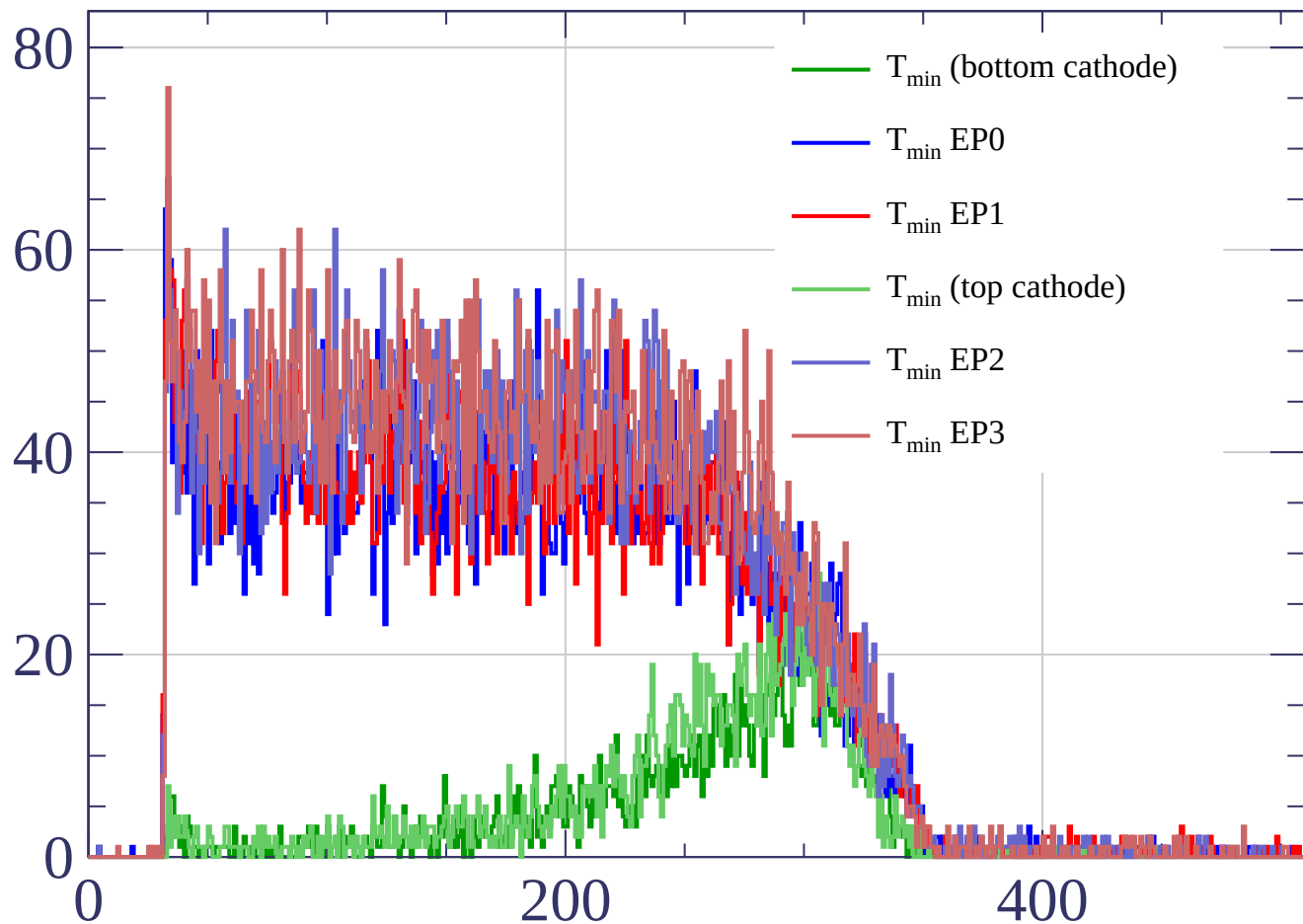
500

1000

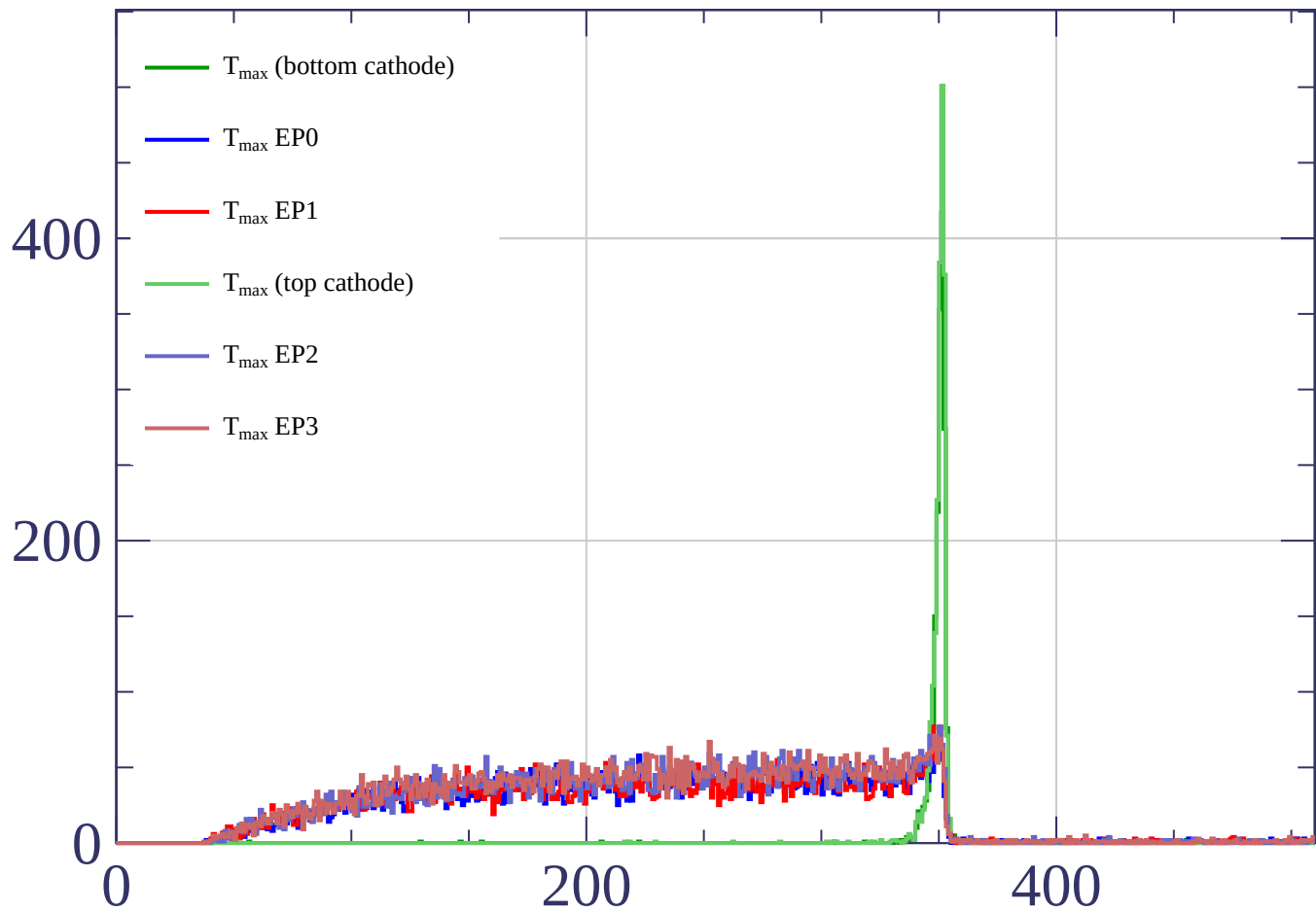
1500

dE/dx [ADC counts/cm]

Count



Count



time bin